Something is happening

Occurrence Theory

Finitary Occurrent Actualism, Mathematical Intrinsic Occurrence, Evolutionary Cosmogenesis, and Conscious Actuality

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Version 2.367, 13 October 2026. DOI: 10.5281/zenodo.22875288
Working paper.

This paper develops a speculative metaphysical framework. Its claims and speculations carry claim grades, set out in the note after the theory map.

Abstract

This paper asks what it might mean for reality to be *actually happening*. Any complete conscious experience raises two questions. How did its structure, including the information in a brain and body, come to be there? And what makes that structure actual, undergone *now*, not merely described somewhere as being undergone?

Occurrence theory, proposed in this paper, offers an answer to each, from one starting point: *something is actually happening.*

Finitary Occurrent Actualism, its core, takes as an axiom that at least one base process simply occurs, without beginning or end. If everything that exists is mathematical, the base is a mathematical structure (a finite rule and its states), and its happening is that structure running forever, needing no driving force and meeting no resistance. Assuming nothing infinite ever exists as a finished whole, the base's existence is not a separate fact from its happening; the paper calls this aspect of the core **Mathematical Intrinsic Occurrence**. The base has a "now" but no clock: procedural change is endlessly taking place. A related but independent mathematical result, **structured nullity**, shows that something can add up to exactly zero without being empty. It is suggested that this may go some way towards explaining why there is anything at all.

Richer structure is proposed to emerge from this occurring process. Reality is proposed to be a stack of increasingly complex levels beginning with a simple base layer, each carried by the activity below it; our universe would be one world among many at one of those higher levels. A base in endless occurrence could supply quasi-infinite abundance and change that never stops: enough random variation for selection to build structures for copying, memory and repair. What selection builds would then ultimately go on to make new worlds, blindly or by design, which the paper calls **evolutionary cosmogenesis**.

Finally, it is proposed that the same occurring makes things actual, giving us our deepest sense that there really is a now. Finite happenings, our lives included, are organised parts of that occurring base. Structure fixes what an experience is like, but being part of the base's occurring is what makes it actually undergone, which this paper names **conscious actuality**.

Keywords: occurrence theory; finitary occurrent actualism; mathematical intrinsic occurrence; evolutionary cosmogenesis; conscious actuality; strict potentialism; occurrence; mathematical structure; cellular automata; evolution; process measure; block universe; consciousness; structured nullity; nothingness

Theory map

What follows is the theory map: eleven steps, in the order the paper approaches them, each carrying the grade it is claimed at, and drawn as a figure in Section 1.1, where the two gaps are set out. What the paper claims, suspects and leaves open is set out in the resulting claims ladder in Appendix A.

1. **The first-person datum.** At least one experience is being undergone. Even an illusion is something undergone. That something actual is going on now, the fact the title names, is the paper's starting datum, not an inference from cosmology. It is conscious actuality met in the one case where each of us meets it directly, our own. [Grade: first-person premise.]
2. **Conditional mathematical monism.** If mathematical reality is all that exists, whatever is occurring must be mathematical in the relevant ontological sense. [Grade: conditional argument.]
3. **Strict potentialism.** No completed infinite object exists in the ontology, but continuation never runs out. This is similar to Aristotle's potential infinity (*Physics* III.6), taken as a commitment about what exists. With it the paper adopts a further commitment, the bounded base: finite approximation may be refined as far as the base's resources reach and no further: the base is mathematically extensible, quasi-infinite but not infinite, and physically bounded (Section 2.3, which states the bounded base and defines quasi-infinite scale, and Section 10.4, which prices the debt a finite physics owes). [Grade: metaphysical postulate.]
4. **Primitive base occurrence.** The central postulate of Finitary Occurrent Actualism (FOA): at least one base process simply occurs, changing procedurally, without a first stage, last stage or completed past. Occurrence, not endlessness, supplies actuality. [Grade: metaphysical postulate.]
5. **Causal-Informational Parthood (CIP).** Finite happenings are causally organised regions of the base; their transitions are some of its occurring transitions. Informational coupling can incorporate those transitions into larger organisations, and reading is one role such an organisation may perform. Section 2.6 sets out the five requirements the bridge must meet and the compositional transitivity it still owes. [Grade: cross-level bridge, metaphysical, real-life counterpart (a whirlpool in a river; a player and a game character) and a formal research programme.]
6. **Mathematical Intrinsic Occurrence (MIO).** The base is posited as occurring, and no separate fact of its existing is posited alongside. Because the base never finishes, there is no completed, unrun version of it for it to have been instead, so occurrence at the base is not a further fact laid over a structure that might have lacked it. [Grade: metaphysical postulate, the content of item 4 (primitive base occurrence) as FOA reads it rather than a further postulate; that no completed version exists follows from items 3 and 4 (strict potentialism and primitive base occurrence).]
7. **Intelligibility substitution.** Global nullity is used as a precise surrogate for "nothing overall" because comparative claims about absolute non-being require a shared framework. Global nullity and absolute non-being are not identified. [Grade: methodological choice.]
8. **Structured nullity.** A finite object can carry arbitrarily rich structure while a selected additive valuation totals exactly zero. With the intelligibility substitution, it meets the structure gap at its narrowest; the reality stack and evolution above the base give the fuller answer, and nothing else in the argument depends on these two steps. [Grade: mathematical result; it shows compatibility only.]
9. **The reality stack.** The base's occurrence carries further organised activity, level above level, hosted, emergent or both, and finite worlds and lives are realised through, and encoded in, the levels below them. How many levels lie between the base and home

reality, our own universe as we perceive it, is left open (Section 5.7, on what the reality stack would add to a theory of everything). [Grade: metaphysical postulate and real-life counterpart (the worlds we host, carried by ours).]

10. **Evolution above the base.** It is proposed that the bridge from rare viable organisation to abundant organised process rests on a standing foundation, **mathematical abiogenesis** (abundance always supplies enough being tried for selection to work on), and runs through two overlapping regimes: **computational genesis**, where selection makes rare viable arrangements common, and **evolutionary cosmogenesis**, where developed processes produce new effective worlds. Selection, wherever variants are retained and continue unequally, is postulated to be more fundamental than forces like gravity. This addresses how much organised process there is, not what makes any of it actualised, and Section 7.1 proposes how to count the observers in it: lifetimes of human-like experience within one full cycle of the finite base. [Grade: research hypothesis and real-life counterpart (evolution, selection building organisation).]
11. **Conscious actuality.** Structure delivers conscious character; participation in base occurrence supplies conscious actuality. A theory of which organisations have conscious character is still required, because the parthood bridge of Section 2.5 settles only that an organised region of the base's occurring is actual, and not which organisations have any conscious character at all. [Grade: cross-level bridge, metaphysical, neutral among several consciousness theories.]

How claims are graded in this paper Note

This paper crosses mathematics, metaphysics, philosophy of mind, theoretical computer science, artificial life, foundations of physics and cosmology. A mathematical proof, a working model and an observation of nature establish different things. A useful toy model can establish formal possibility without necessarily proving that nature uses it. A metaphysical postulate can organise a problem without becoming an empirical discovery. A firsthand datum can motivate an ontology without uniquely selecting it. The paper therefore gives each claim one of five **claim grades**, which say what the claim rests on:

- **mathematical result:** follows from the stated definitions;
- **formal possibility:** implemented or demonstrated in an abstract model;
- **real-life counterpart:** a process or system observed in the real world, natural or made, that has the relevant structure without being the case proposed: symbiosis for a coupling that benefits both partners, evolution for selection building organisation, a game and its developer for a hosted world and its host;
- **metaphysical postulate:** a coherent ontological proposal not derived from standard mathematics;
- **physical conjecture:** a possible application to our universe without unique empirical support.

Derived labels identify narrower roles: **methodological choice**, **research hypothesis**, **cross-level bridge**, **conditional argument** (a conclusion drawn from stated premises, adding no postulate of its own), **first-person premise** (the datum that an experience is being undergone, met directly rather than posited), **historical comparison**, **summary** (claims graded elsewhere, gathered), **open problem** and **admitted limit**.

Speculation is separated into two tiers, and the separation is deliberate. **Signed speculation** states what the author suspects, labelled as such, without allowing it to carry the paper's argument. **Wild speculation** is a weaker thing again: a possibility Occurrence theory permits and the author finds interesting enough to put into the record, offered as food for the imagination rather than as an assertion. Longer passages of both kinds are set in ruled boxes in smaller type, blue for signed and orange for wild, so that a reader can see which is which at a glance, and skip over if so inclined; shorter ones are marked as signed or wild in the running text where they appear.

Most sections end with a square-bracketed **[Grade: …]** line that opens with the grade, label or tier of speculation just used. It is part of the paper, and lets a reader tell proved results from speculation where each appears.

The **claims ladder** in Appendix A lists what the paper claims, what the author suspects without claiming, and what it leaves open. Each line names its reality level (the levels are set out in Section 1.5), so an abstract result is not silently promoted into a claim about our universe or the base. It differs from the claim grades above and from the current concessions of Section 14.6, which list what Occurrence theory has not paid for.

1. Two gaps and one primitive occurrence

1.1 Two gaps and the question "why am I me, now?"

Leibniz's question, "Why is there something rather than nothing?" (Leibniz 1989 [1714]), is often framed as a search for a cause, law, necessity or selector. That framing can smuggle in a questionable concept: first there was nothing, then something happened to it. If time is part of the world being explained, there was no earlier interval in which non-being waited to change into being. Literal non-being is not a physical state with a duration, boundary or decay mechanism.

A timeless mathematical answer avoids that picture. Perhaps reality is mathematical structure rather than matter that had to be added to mathematics. A complete structure could contain spacetime, fields, brains, memories, decisions and every sentence ever written about consciousness. Yet this answer makes a second question hard to avoid. Why is anything actually happening now, live, within any of that structure? Why is a present sensation being undergone rather than merely represented by a complete pattern that includes the sentence "this sensation is being felt now"?

One profound question sits underneath everything that follows. Stated in the form in which it is actually asked, it is this: **Why am I me, now, and not me at another time, or someone else entirely?**

The phrase is often meant as a complaint or a lament; but here it is the one the more philosophically minded child asks out of simple, matter-of-fact curiosity. It is easy to dismiss with the observation that whoever asks it is necessarily the one asking. That reply is true and yet does not settle the question. Something is being undergone, here and now, at this point of a life rather than at another, and the demand for an account of that is not answered by pointing out that any account would be given from somewhere. A view on which every moment of every life is equally and timelessly actual does not answer it either. It dissolves it, at a cost that Section 8 sets out when it compares block, rewrite and growing architectures of occurrence, and that Section 11.3, on the two meanings of now, spells out: if every moment is actual in the same way then none is the one being undergone, and on this paper's reading the thing that needed explaining has been renamed rather than explained.

This paper's answer, assembled across the sections that follow, is that occurrence is not a property distributed evenly over a completed structure but something a process does, and that a finite life is actual by being part of what is doing it. Whether that answer succeeds is for the reader to judge. What it is trying to answer, at least, should be clear.

Three separate debts hide behind that question. There is the fact that there is anyone at all rather than no one. There is the fact that the someone is *this* one, with this history, rather than another. And there is the fact that it is at one apparent point of that history rather than any other. The first debt, whether there is anyone at all, is the actuality problem and is the paper's primary subject; Section 11 keeps it apart from conscious character throughout, because what an experience is like and that it is being undergone are two different explanatory jobs. The individual identity and temporal self-location debts receive proposals rather than full answers. For temporal self-location, Sections 11.3 and 8.2 distinguish represented chronology from the episode being undergone. For individual identity, Sections 11.5 and 12 propose individuation through integrated activity and maintained coupling, though the coupling law itself is not supplied. Self-location also calls for a measure: a rule for how much each kind of observer, or each stretch of experience, counts. The paper needs one wherever it argues from typicality, that observers like us are probably typical and so what we see tells us what reality is mostly like: Section 7 uses that shape of argument to expect observers who evolved rather than minds

assembled by pure chance. Physicists who work on the multiverse call the difficulty of finding such a rule the measure problem, and in an infinite universe it is hard at the root, because one infinity set against another gives no answer. A finite base changes that. One full cycle of it is finite, so everything in it can in principle be counted, and Section 7.1, on what a process measure must count, proposes the count: lifetimes of human-like experience, weighted by how long, rich and intense they are.

The two questions this section opened with, why there is something rather than nothing and why anything is actually happening now, mark two gaps, and the paper keeps them apart:

1. **The structure gap:** how can differentiated reality exist without being constructed from antecedent non-being? *Structure* is used throughout for whatever a complete description fixes, and *organisation* for that structure as it is arranged causally in a running system; a complete description fixes the organisation, so the two words never pick out different things, only different emphases.
2. **The actuality gap:** why is any mathematically describable process occurring or experienced now, live, rather than merely existing as a specification?

The fullest answer to the first gap is the layered, evolutionary account of Sections 5 and 6, which builds organised worlds level on top of level within one occurring base. Structured nullity meets only its narrowest part, and only after a declared substitution. It shows that a global zero need not be internally empty: the finite construction of Section 4.3 proves only that a finite object can total exactly zero under a declared valuation while its internal structure survives intact. Finitary Occurrent Actualism (FOA) addresses the second by accepting one primitive occurrence and identifying finite happenings as parts of it, as developed in Section 2.2, where the base occurrence is postulated, and Section 2.5, where parthood, coupling and reading are separated. Structured nullity can look as if it explains three things at once: how something with a rich interior can total nothing, why its states change, and why any changing is happening at all. Those are three different jobs, and nullity does only the first. Why the states change is a fact about a rule, settled by whatever dynamics the structure carries. That any changing is happening at all is the actuality gap, which nullity does not touch and which primitive occurrence is posited to answer. Neither structured nullity nor primitive occurrence derives the other. A universe could be globally non-zero and still have something going on in it. Structured nullity could remain purely abstract, an object with nothing happening in it. FOA does not need the universe, or the base, to total zero.

Finite lives are organised parts of one occurring base: Sections 2 and 3 develop occurrence, parthood and MIO, and Section 5 builds the reality stack on that core. Sections 6 and 7 propose how abundance and cumulative selection could build organised worlds within that occurrence, a route to richness without plenitude, and how observers are weighted, by counting lifetimes of human-like experience. The bounded base has only finitely many configurations (Section 2.3); that it traverses them as one exact, quasi-infinite cycle and spends nearly the whole of each pass inside a band where organised complexity holds together is signed speculation, and Section 9 carries it. A person's conscious character may be fixed by their own organisation while a deeper and longer-lived actualised entity is coupled to that organisation: that too is signed, and it is optional, in Section 12, where Gōsuto, Reikon and Sasayaki are defined. Copies, replacement, branching and recurrence are given candidate conditions for continuing one person rather than merely resembling one, again signed, in Sections 13.1 and 13.2. Interdimensional transit (passage between different worlds, Section 5.5), ascension to a host reality or otherwise escaping a reality, and god-like beings are given precise readings, marked wild, in Sections 13.4 and 13.5. They are possibilities Occurrence theory permits, and they

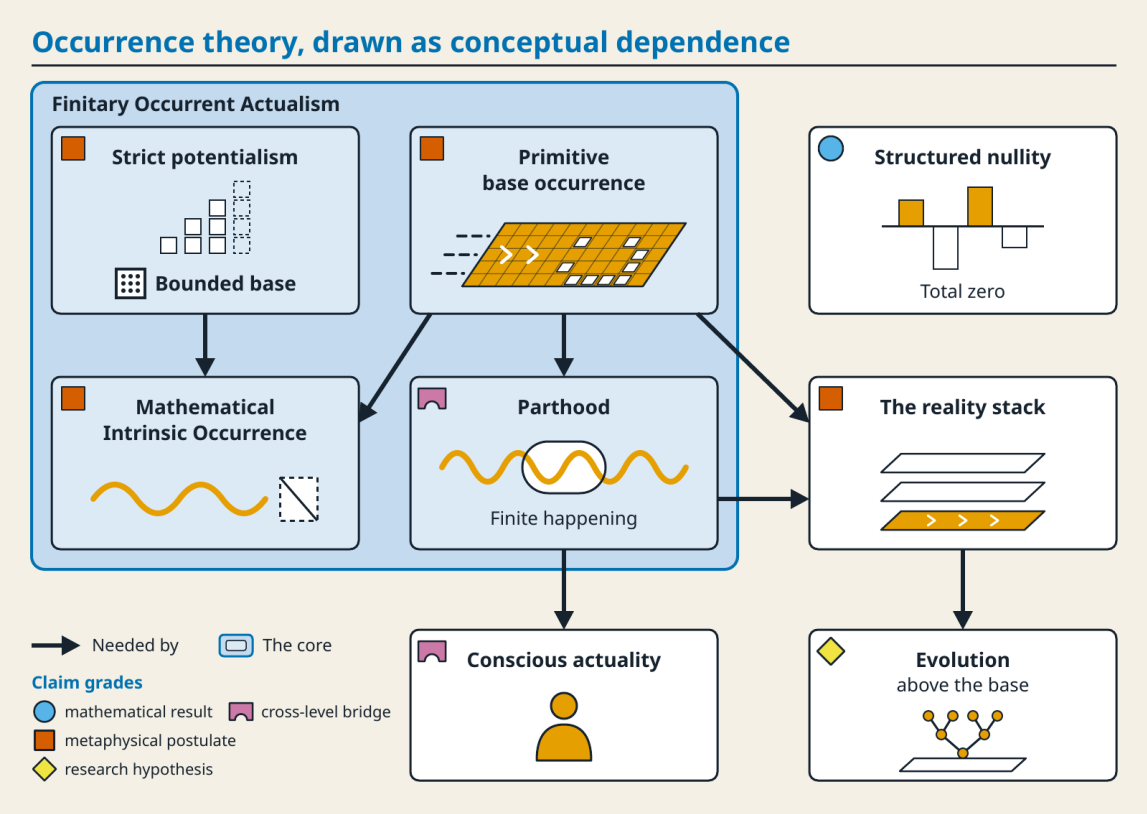


Figure 1: Occurrence theory, drawn as conceptual dependence. The shaded core is Finitary Occurrent Actualism; structured nullity stands apart, a related but independent mathematical result.

have clear modern real-life counterparts, a game developer's standing to their game and a program escaping its sandbox, set out where Section 13.4, on potential interdimensional transit and ascension, begins. These real-life counterparts motivate the author's suspicion that the cosmological versions may occur at our own level; whether they do remains open.

Two further strands accompany the metaphysical argument. The first is a set of exact consequences of finite deterministic dynamics, in Section 9.1, on recurrence and higher-level laws, and Section 10.2, on kinds of randomness and the information in histories. The second is the computational companion in Section 15.3, on organisation under a reversible signed rule. Both are graded separately from the reality stack's proposed extension below our physics, set out in Section 5.7, on the stack and a theory of everything.

Each formula's point is also stated in words. You can skip the notation and follow the argument in the surrounding prose.

[Grade: open problem and real-life counterpart (a game developer's standing to their game; a program escaping its sandbox); conceptual problem statement; neither gap is claimed solved here, and the answer sketched is owed by every section that follows.]

Occurrence theory's conceptual origin Signed note

The drive behind this paper is a fusion of two bodies of understanding, both relatively recent. One is evolution based on variation and selection, barely two hundred years old as a theory, and understood only much more recently, in the sense used here, as a process that applies far more widely than just that of biological life. The other is our experience over just the last few decades, of building computed worlds (especially procedurally generated ones) and watching how they play out. And finally sharply accelerated by the arrival of human-like artificial intelligences, which are also more grown than designed. This paper applies that understanding on the working assumption that what is true at one level, or in one situation, is very often echoed elsewhere in reality (Section 6.5, on uniceptals, recurring motifs across systems). It would have been very difficult for what is proposed here to have been fully conceived of two hundred years ago, and even harder to point to real-world examples of the principles in action (though Section 16, on antecedents, novelty and priority, records how people tried in the best ways they could at the time). Recently, though, connecting these insights has become much easier for someone like the author who builds digital worlds, works with artificial minds, and has a fairly solid understanding of evolutionary biology and psychology. The question this paper opens with, "why am I me, now?", is one the author has asked since childhood, and his work across these fields is what let him bring the pieces together.

1.2 The first-person datum

The paper begins from a narrower datum than a complete theory of consciousness. At least one experience is being undergone. The contents might be misrepresented. The apparent external world might be partly wrong. The subject's conceptual report might be confused. What cannot be removed while leaving an illusion in place is the occurrence of the seeming itself. An illusion is something undergone; an inscription reading "I am fooled" exists as text and fools nobody unless the seeming is undergone.

The datum establishes only that occurrence is not an empty word in the one case from which we know it directly. It is not inferred, the way one infers from a sound on the roof that it is raining. The datum is a premise met in the undergoing rather than a report offered as third-person evidence: it is the undergoing itself, not the sentence or the judgement that reports it, the distinction used in Section 14.3's report objection. And it concerns conscious actuality only, that an experience is live at all; it is not a full account of access consciousness or of what an experience is like, and Section 1.6, on the three meanings of consciousness, separates those three terms before any later section uses them.

[Grade: first-person premise; it establishes occurrence in one directly available case and supports no cosmology by itself.]

1.3 Conditional mathematical monism

The paper then asks what follows conditionally if everything that exists is mathematical. The argument is simple:

1. something is going on;
2. if everything that exists is mathematical, whatever is going on is mathematical;
3. therefore, under mathematical monism, some mathematics is actually occurring.

The conclusion does not prove mathematical monism. It says that a mathematical ontology adequate to firsthand actuality cannot consist only of abstract specification. It needs at least one occurrence fact.

A reader who reserves *mathematical* for abstract structure may read FOA as saying that reality has a structural form and an occurrent fact.

Mathematical plenitude, the proposal that every consistent mathematical structure exists, is already on the table (Tegmark 2008). It is worth naming at the outset for two reasons. It is a precedent for holding that something mathematical simply exists, with no event that manufactures it. It is also a measure of what is being asked for here, which is less. Where plenitude posits every structure, this paper needs only one occurrence. The reason it can ask for so much less is selection. Plenitude buys richness by positing every structure at once; an occurring base buys it by running, because once something is occurring, and variants are retained and continue unequally, selection can build organisation the base never had to contain, which is the argument of Section 6, on evolution and the production of organised worlds. Enough occurrence of that kind makes plenitude unnecessary. It posits no plenitude of its own, and it does not grant one in order to build its opponent either. Section 2.2 adopts one connected occurrence, rather than a separate happening for each coherent structure. Section 3.2 argues that the occurring base needs no separately posited completed counterpart; Section 3.4 sets out the structuralist alternative.

[Grade: conditional argument, not a proof of mathematical monism; its force rests on the datum of Section 1.2.]

1.4 The residual brute fact, stated as a question

The price is visible. FOA does not explain why the base occurs. A cause of occurrence would already have to occur, so ordinary production cannot explain the primitive without regress. It does not derive occurrence from non-termination, complexity, self-reference or nullity. It makes one brute postulate about actuality, because the alternative, on its reading of conscious actuality, is to leave the starting datum without a truthmaker, nothing in reality that makes it true. The paper's other postulates, strict potentialism, the bounded base and the reality stack among them, concern what exists and how it is arranged, not what makes it actual. A structuralist can deny that the datum adds anything beyond the complete structure. The disagreement is real and is returned to throughout the paper. A second price concerns infinity: FOA rejects completed infinite histories in both directions, and that restriction does not supply occurrence, as Section 2.1's strict potentialism makes explicit. The restriction itself is old: it is Aristotle's potential infinity, the infinite that exists as a day does, by going on (*Physics* III.6), taken here as a claim about what exists rather than about mathematics alone. The base's occurring might be necessary, necessary for reasons beyond our reach, or simply unexplained, or the question may be badly formed; none is established, and the author's loop (Section 9.5) may explain why the base continues to exist, though not why it occurs at all.

[Grade: open problem, metaphysical; the residual occurrence fact is postulated in Section 2.2.]

1.5 The levels of reality

The paper's claims are made about named levels of reality, and the levels are kept explicit from the start:

- **true base reality:** the lowest occurring process in the connected ancestry relevant to us;
- **familial base reality:** the first world of a family of worlds, the one whose physics the others share or clearly derive from; it is usually also the root host of those worlds, but hosting is not part of the definition;

- **home reality:** the effective world in which a finite observer appears to live;
- **cross-level:** a claim linking local processes to their base;
- **abstract, no level:** an atemporal mathematical pattern, neither changing nor settled.

The phrase "neither changing nor settled" matters. *Settled*, *quiescent* and *dormant* are used in this paper only of regions within an occurring base, never of abstract patterns. Section 9.5 applies this vocabulary to base regions.

Three of these, true base reality, familial base reality and home reality, are offered as a set in their own right. They are a serviceable vocabulary for elements of simulation theory, since simulations themselves are no longer hypothetical, though whether we live in one still is, and the discussion needs to distinguish clearly the level that runs a world from the level that resembles it, and both from the level an observer appears to live at.

The observer's world is called *home reality* rather than *local reality* deliberately. In physics *local* already carries a technical load, in locality constraints, no-signalling and Bell's theorem, all of

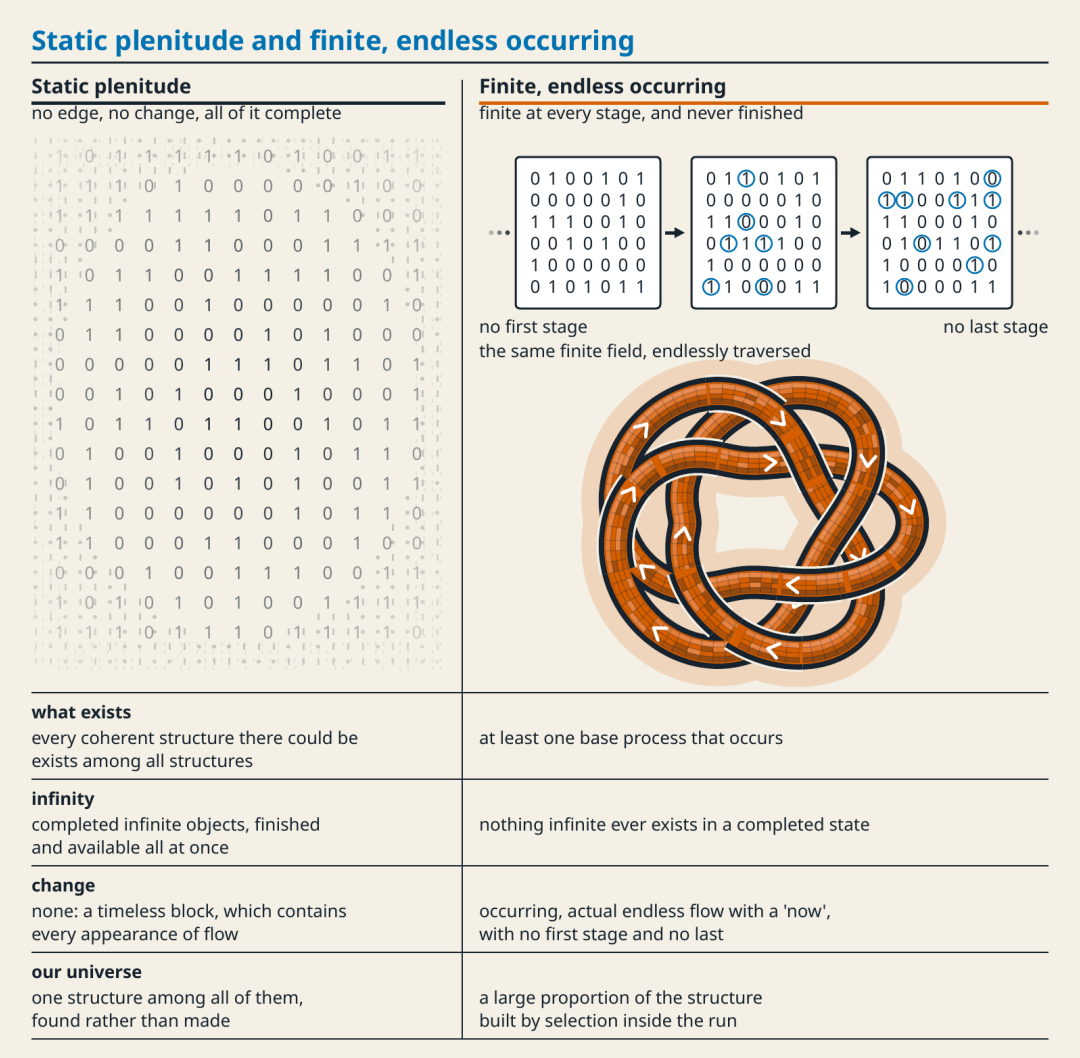


Figure 2: Two pictures of mathematics. On the left, static plenitude: every coherent structure exists, complete and unchanging, our world one structure among them. The field has no edge and fades rather than stopping, because on that view nothing lies outside it. On the right, Occurrence theory's picture: a bounded base whose state changes from one stage to the next, with no first stage and no last, and below it a closed knot, finite yet endlessly traversed, each circuit complete. The rows compare the two.

which this paper discusses; a level named local reality invites a reader to hear a claim about that. *Home* says what is meant, which is the world an inhabitant lives in.

Higher and lower in the **reality stack** refer to realisation: a host is below the world it carries, and everything in that world is encoded in the host's activity, as a game is encoded in the machine that runs it. Along the developmental route the author expects a broad rise in effective complexity, but a hosted descendant can instead simplify or approximately repeat its host's physics. Neither internal historical date, individual intelligence nor direction of control is fixed by that placement. Section 5.5 names reductive, peer and enriched hosted worlds; Section 5.6 separates the proposed complexity gradient from the stack's carrying relation.

A level is a depth in the stack, not a world. Many worlds can sit at one level, whether they share a host, as many files share one disk, or have different hosts, as the same kind of game can run on two different computers; and a world can carry further worlds inside it, as a file can hold other files. An emulator running inside a virtual machine on a physical computer is a stack three deep that anyone can set up. Our universe is therefore one world at its level, not necessarily the whole of it, and home reality names that world: the one an observer appears to live in, not everything that shares its level.

[Grade: methodological choice and real-life counterpart (running simulations, a game encoded in the machine that runs it; an emulator in a virtual machine, for a stack three deep); terminology; the level names are fixed here and used from Section 2 onwards, and how many levels there are is left open.]

1.6 Three meanings of consciousness

The word "consciousness" easily hides several questions, so three meanings are separated here from the beginning. They have three names but, for the purposes of this paper's arguments, effectively two sides, as the division between the structural and actuality sides matters far more than the distinction between access consciousness and conscious character within the structural side.

On one side, both settled by how a system is organised:

- **Access consciousness:** information available for report, memory, reasoning and control.
- **Conscious character:** what a particular experience is like, its content, including its qualities and organisation. The standard literature calls this phenomenal character.

These two are distinguished in the literature for good reasons, and this paper does not dispute them. But they are treated as one throughout, because this paper is not separating them from each other but separating both from conscious actuality. They stand or fall together on the only question this paper asks of them: a complete structural description fixes both. That is a grouping for this argument, not a claim that either requires the other in every system. So wherever this paper says that structure supplies character, access is included, and *character* covers both throughout (Section 11.2's organisation-to-character assumption). The claim is about what settles them, not about where they sit in a brain: they may well come apart anatomically or developmentally, and nothing here turns on whether they do.

On the other side, settled essentially non-structurally:

- **Conscious actuality:** that an experience is live at all, rather than merely structurally represented as live.

What the two sides make together is simply **a conscious experience**: one live episode, organised and being undergone. Conscious actuality is not a name for that whole, and not a third thing the factors manufacture between them. Participation in the base supplies the

generic fact that an episode's transitions are occurring rather than merely specified; the paper calls that fact **operative actuality**, and Sections 11.2 and 11.3, on organisational determination of conscious character and on actuality and participation, develop it. Felt from inside an organised episode, that same fact is conscious actuality; and the episode that has both organisation and actuality is a conscious experience.

This paper's target is **conscious actuality**, and it is not simply a third item beside the other two. No quantity of access or character settles it, which is what Section 3.3 argues when it sets a complete finite description beside the run it describes. Keeping actuality apart from the other two is the discipline the whole argument revolves around, because running them together is a common way this question gets dissolved instead of answered. The term is chosen to avoid double-booking "actuality" for every kind of occurrence. **Conscious illumination** is the lay shorthand for the same thing, the lights being on, and this paper's figures use it that way, drawing a conscious experience as a pocket of light. It is an explanatory metaphor only. Section 11.2 states the two-factor account. Section 11.4 extends it to a felt floor, the Ground of Experience Conjecture: since participation in base occurrence supplies conscious actuality, bare occurrence everywhere carries a minimal experiential aspect, however slight, and Occurrence theory holds this as a postulate.

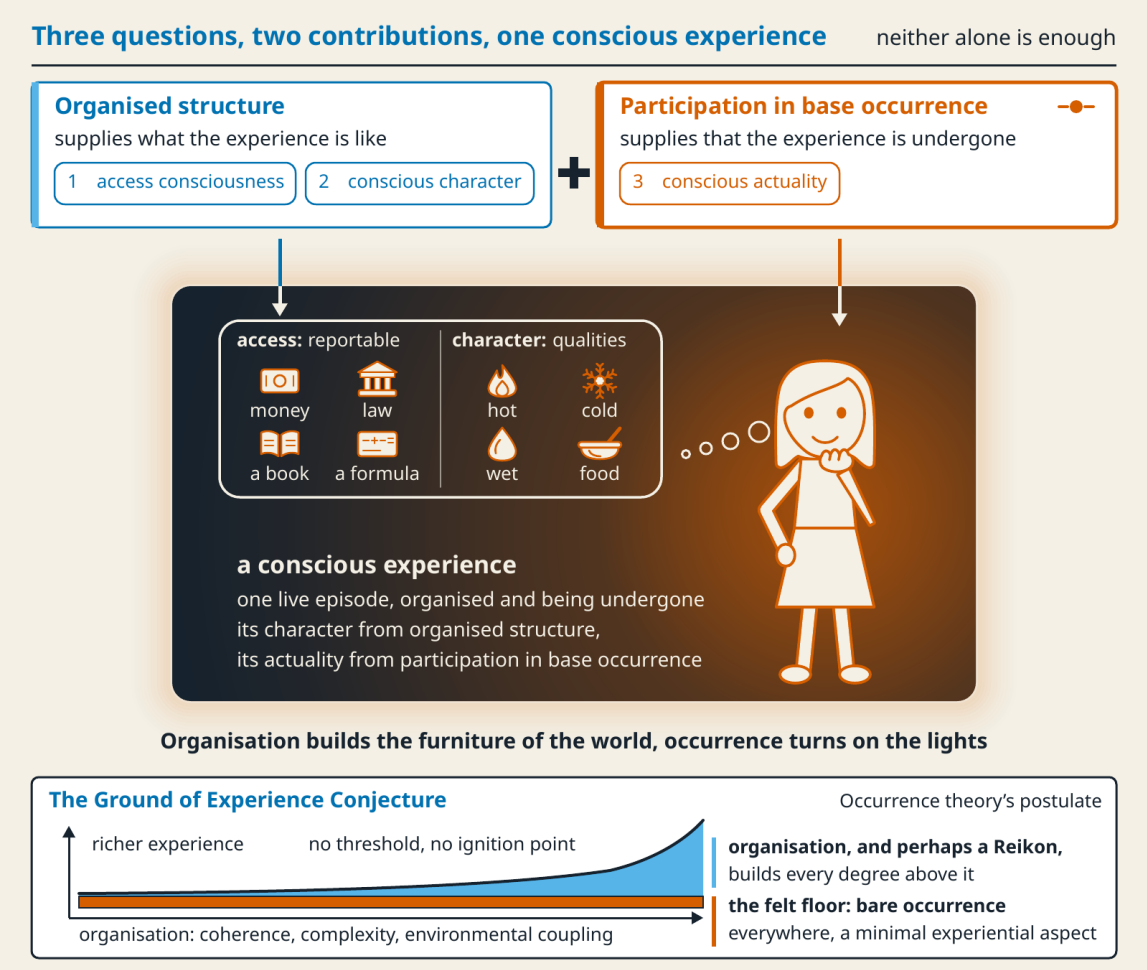


Figure 3: Three questions, two contributions, one conscious experience. Access consciousness and conscious character are both settled by organised structure. Participation in base occurrence supplies operative actuality, that the episode is actualised at all; felt from inside an organised episode it is conscious actuality, the being here now, drawn where the light enters. What the two make together is a conscious experience.

Two more words are used throughout in this sense. A **life** is the whole thread of a finite being's organised activity from its start to its end, made of conscious experiences in succession and of the organisation that carries them from one to the next; a **life thread** is that same thread considered as a structure that can be followed along, as a reader follows it (Section 2.5, on parthood, coupling and reading, defines the reader). A conscious experience is one episode; a life is the thread of them.

Three ordinary words are each kept for one job. *Undergone* is used of something felt or lived as experience, an experience, a seeming, a belief or a life; *occur* is used of processes, transitions and events happening; *actualised* is used of finite patterns, lives or episodes that are actual through participation in that happening.

[Grade: methodological choice; terminology; no new metaphysical conclusion, and every later use of the three names depends on the division between the structural side and actuality being kept.]

2. Finitary Occurrent Actualism, occurrence and Causal-Informational Parthood

2.1 Finitary Occurrent Actualism and strict potentialism

Finitary Occurrent Actualism (FOA) joins Section 2.2's primitive occurrence with finite happenings as its organised parts. FOA is the core of **Occurrence theory**, the overall view of reality this paper proposes. In short, FOA says what makes anything actual; the rest of Occurrence theory says where the structure comes from and what the two together mean for experience. The core is primitive base occurrence with finite happenings as its parts, strict potentialism and the bounded base (Sections 2.1 and 2.3), Mathematical Intrinsic Occurrence as FOA's reading of that primitive (Section 3.2) and Causal-Informational Parthood as the bridge (Section 2.5). Around that core, Occurrence theory adds the reality stack of levels (Section 5), the evolutionary account of how organised worlds are built (Section 6) and the division of consciousness into character and actuality (Section 11), with the author's signed conjectures at its edge, where they can be revised or dropped without touching the core. Structured nullity (Section 4.3) meets only the narrowest part of the structure gap and is not among FOA's premises.

The word *actualism* is used here in a narrow sense. It marks the difference between an abstract pattern and an occurrence, not a thesis about possible worlds. The word *finitary* does not say that every process halts or that nature has a smallest cell. It says that no infinite history, continuum or state space exists as one completed object. An endless process can always have another stage without its total history existing all at once. That is Aristotle's potential infinity (*Physics* III.6; Barnes 1984); the paper takes it as a commitment about what exists rather than a view about mathematics alone, and Section 16.2, on process, becoming and intrinsic occurrence, records the debt.

Aristotle's version already carries a hint of occurrence. His infinite exists, he says, in the way a day or the games exist: not as a finished thing but by one part coming after another (*Physics* III.6), and in *Physics* VIII.1 he argues that motion itself has no beginning and no end. Potentialism has long had a question hanging over it: potential in what process? If mathematics is timeless, "always another stage" is a manner of speaking, and a platonist can fairly reply that a potential infinity with no process behind it is an actual infinity under another description. This paper's answer is Section 2.2's primitive base occurrence: something is running, so "always another stage" is literal. That is why Aristotle's view can be taken here as a claim about what exists, and why joining potentialism to occurrence is not an arbitrary pairing

but instead perhaps the step the view was waiting for. The word *occurrent* marks the category of things that happen rather than persist, and is used because the primitive base postulate adds occurrence, not further structure.

The paper adopts **ontic strict potentialism**:

> A finite mathematical structure may exist as a completed abstract object. An endless process may continue without limit. No completed infinite object or completed infinite history exists.

Only the completed infinite is excluded: a finite total, such as one full cycle of a cycling base and everything in it, observers included, can in principle be counted. The restriction applies in both temporal directions. The base has no finished future, but neither is there a completed past laid out elsewhere as an already total object. Here *stage* is schematic rather than a commitment to a preferred foliation, a single privileged way of slicing the whole into successive moments. In a linearly staged model, every stage has an earlier stage and can have a later one. In a partially ordered model, the corresponding claim is only that the base has no global initial or terminal boundary; finite local sub-processes may still have first and last events. Nothing requires a first global stage, a last global stage, or a total sequence containing all stages. That restriction does not supply occurrence.

This is a metaphysical postulate, not a theorem of standard mathematics and not a refutation of classical set theory. Classical mathematics may continue to use completed infinities as exceptionally effective idealisations. FOA interprets such objects as rules, schemas or finitely extendable approximations rather than completed constituents of reality. Section 2.3 adds a separate metaphysical postulate: physical differentiation is bounded even though mathematical refinement may continue. Quantifiers such as “for every natural number” must therefore be read with care: they license any requested finite instance or a rule for generating further instances, not a completed graph containing infinitely many ordered pairs.

Ordinary notation hides this rather than expressing it, and the hiding is where most people first meet the distinction without noticing. The dots in $0.333\ldots$ read as a mark of completion, as though it named a finished object with all its digits already present. On the reading adopted here, what it abbreviates is an instruction that can always be continued. The same holds for π: it can be treated extensionally, as a completed real number, or as a rule one runs, producing digit after digit for as long as anyone cares to turn the handle. Nothing in the mathematics chooses between those readings. Strict potentialism is the claim that only the second corresponds to anything that could truly be the case. Section 3.1 develops the corresponding conceptual distinction for occurrence: a description is not the life it describes. Section 3.2 develops what this means for the base, its Mathematical Intrinsic Occurrence.

Where richness comes from. Duration is not exploration. An endless deterministic process can sit in $A \to B \to A$ forever, no nearer anything interesting after an eternity than after three steps. Anything stronger needs conditions on reachability, mixing and retained variation, not eternity alone. At any finite stage, the present organisation must be explained by drawing on a finite *segment* of its causal ancestry, one available within the process. Section 6 develops the research hypothesis that reachability and retained variation can build richness through evolution; endless duration alone cannot.

A chess position makes the gap concrete. A board can be internally consistent, every piece on a legal square, and still be unreachable from the opening position by any sequence of legal moves. Coherence is a property of the position; reachability is a property of the dynamics. Section 6.1 adds the further conceptual distinctions between actual supply and evolutionary prevalence.

This paper therefore claims for the base only what our being here shows it has done, and Section 6's research hypothesis leans on building for everything above it. Appendix A's signed suspicion that most of the coherent structure above the base is built rather than found records where the author's own weight falls, and it falls on building.

[Grade: metaphysical postulate and real-life counterpart (a day, which exists by going on); excluding completed infinities leaves occurrence to a separate postulate.]

2.2 Primitive base occurrence and the question of number

Sections 1.2 and 1.3 supply undergoing as the premise and mathematical monism as the condition.

> **Primitive base occurrence:** at least one base process simply occurs, the one carrying us. It has no first stage, no last stage and no completed past.

One tempting response is to grant a separate occurrence-token to every coherent structure. That multiplies primitives, duplicates each abstract pattern as an additional happening, and poses a cross-run measure and identity problem.

Primitive occurrence is not derived from coherence, complexity, self-reference, necessity, non-termination or nullity. Primitive occurrence is the one brute actuality fact FOA accepts. The base is not a first cause at the head of a series, because it has no first stage and is not placed within a prior time. No global clock or preferred slicing is assumed; Section 3.6 distinguishes occurrence from temporal measure.

Endlessness and actuality therefore have different roles. Primitive occurrence supplies actuality. The absence of a first and last stage avoids an origin event and a terminal boundary, but it does not make the process occur. A finite directed cycle can have no last node while remaining an abstract completed object. Conversely, a finite process can genuinely occur as a bounded region within the base even though that local process begins and ends.

A circular railway makes the distinction visible. It has no first sleeper and no last, and its geometry may be perfect, but it supplies no train. Beginninglessness does not buy occurrence.

The base also has no uniform first condition from which variety must somehow be generated. There was no first perfectly symmetrical state. A deterministic rule can therefore inherit rich asymmetry, up to whatever the base can carry, from every finite predecessor without any moment at which chance or an external hand breaks an initial symmetry. This does not explain why the actual base has its particular local history. It removes only the demand for a first symmetry-breaking event.

A word about the counting, because the phrase "one base" promises more than the argument delivers. What the first-person premise supports is that *at least one* connected occurring process carries this experience. Whether there are others, unconnected to ours, is not settled here and is listed among the paper's open questions. Throughout, "the base" means the one that carries us. It is a name for our own case rather than a uniqueness result, and Occurrence theory's core would not break if there turned out to be more than one.

Why one, though? Selecting one occurrence among abstract candidates seems to leave the choice unexplained.

The reply offered here is that the question may not be well formed. To count two base processes, something must make them two. Within a world, space, causal separation or another relation does that job, but Section 2.7 offers the architectural possibility that interaction supplies adjacency, and Section 5.4 pursues the still-unproved emergence of space and time from base order. Occurrence is transitions occurring, not a structure placed in an

arena. Two sets of transitions with no shared order, interaction or differentiating feature may be one occurrence described twice. Bare numerical distinctness is the disputed alternative. On this reading, the machinery for counting bases is missing.

Max Black's universe of two qualitatively identical spheres shows that indiscernibility alone need not collapse a count (Black 1952). Yet the spheres are a mile apart, while the proposed emergence of spatial relations assumes no arena in which to place distinct bases (compare Hacking 1975). A defender of bare numerical distinctness will remain unmoved; this paper has no further argument against that view.

Complexity does not rescue the count. It can individuate parts within a process, including regions, levels and lives, but cannot distinguish that process from another to which it bears no relation. Differentiation within a thing is not differentiation between things.

That does not conflict with signed suspicion 2 of Appendix A, that other bases could occur, few though the structures able to run without end would be. Abstract structures are distinguishable without a spatial arena; their occurrences may not be. Which structures can run and how many base occurrences there are are different questions.

None of this is established, and the argument does not need it. The postulate is that at least one connected process carries this experience, and on the question of number nothing that follows needs more than that.

FOA therefore uses the singular only for the connected base carrying this actuality. Other disconnected bases remain an observationally unreachable possibility, and no selector among completed candidates is posited.

Three possibilities are being compared: every structure runs, one connected base runs, or some structures run without a criterion saying which. A steam engine is one thing running in one direction. Inside it, parts move at different speeds and in different directions, some about a different axis, some reciprocating, some barely moving. The image carries upwards too, and it is still the one engine: it hauls carriages, which have their own arrangement and are not engines; the carriages hold furniture at a scale of its own; and the train carries a life, people eating and sleeping and arguing and walking down corridors. Nothing in a carriage has its own engine, and a passenger walking towards the rear is still travelling forwards. That walking is some of the train's motion organised at a level of its own. Section 2.5 develops parthood as a metaphysical bridge: finite activity is some of the base's activity organised at its own level. Section 5 pursues the research hypothesis that such organisations can support further levels.

The engine burns fuel and it was built in a yard, so it imports both a continuous push and an origin, and the base has neither. What is wanted is something that runs without being driven and without having been started.

A planet is nearer. Nothing pushes it, nothing wound it up, and it did not begin at rest and then start: for any purpose that matters here it keeps moving in the same way indefinitely. It turns on its axis while travelling its orbit, its interior convects, its crust is carried on that convection and remade, its oceans circulate, its atmosphere moves at its own speeds, its field reverses, and continents, ecosystems and cities stand where the carriages and their furniture did. So the uniformity of the base's occurrence is a fact about the distribution of running, not about what runs. One fence remains and it is the honest one: a planet has a formation history, and the base is postulated to have none.

What that running looks like drawn is the subject of a signed box in Section 9.3, on the form of true base reality, which pictures the base's finite, endless succession as a closed loop, and of the knot in Section 1.3's figure. Both are pictures of the traversal, not of a finished shape: a loop

drawn on a page is an object, and the claim is about the running round it, which the drawing cannot show.

Among the three options, universal occurrence needs no distribution, and one connected root needs no cross-run measure. An unexplained subset fares worse because it owes a criterion saying which structures run; Section 3.3 develops the metaphysical claim that occurrence itself is not a structural feature. FOA chooses one root for economy, not because it has been shown that there is only one.

The cost is the loss of an anti-parochial advantage. If running were widespread, actuality would resemble *here*: a perspective rather than a privileged distinction, with nothing about our position to explain. One occurring base among abstracta gives that up and moves towards ordinary Platonism with a concrete world attached. If the preceding count argument fails, the cost is an unexplained question: why this base, rather than another, is the one that runs. Section 1.4 leaves the ground of occurrence open. Section 9.5 offers the author's signed persistence conjecture: what persists without end would be a perfect loop.

[Grade: metaphysical postulate and real-life counterpart (a planet; a train carrying lives); the ground and number of base occurrences remain open.]

2.3 The bounded base and quasi-infinite scale

Emergent worlds may require enormous ranges of scale and precision. FOA does not infer that reality must therefore contain a completed continuum or infinite-dimensional state space. It treats successful continuum theories as idealised calculi whose finite approximations can be refined as far as any physically posable question requires.

This is not ordinary pixel ontology. A pixel ontology chooses a smallest cell and usually a smallest update. The position here makes neither commitment, and it also rejects physical refinement without limit. For any finite observational question that can actually be posed, a model precise enough to answer it may be available; but the base is finite, so physical differentiation has a ceiling. Mathematics may always permit another subdivision, another decimal place, another mode; reality need not follow, and in FOA ultimately cannot. At sufficiently fine physical resolution some finite resource runs out: distinguishable states, sites, relations, configurations, or something more fundamental for which those are only higher-level names. The paper does not claim to know which, and it does not say that the ceiling takes the form of a smallest cell or a smallest tick. It may be that increasingly fine mathematical distinctions cease to correspond to physically distinct states, that the quantity being refined ceases to apply, or that the finite process carrying the distinctions comes round to states it has occupied before. A mathematical divergence may therefore mark not an infinite object in nature but the point at which an effective description has been pushed beyond the level where its variables refer to anything. What is denied is both the completed object containing every point and every degree of freedom at once, and a reality able to go on adding new distinctions without limit. The ceiling may be so remote that no observer inside the world could approach it, so that from inside, continuum descriptions can be not merely good but operationally perfect. The bounded-base postulate limits the base's size, gives each part finitely many states, and bounds positional and relational distinctions, yielding a finite repertoire of complete configurations beyond anything inside it could count.

A ruler can keep acquiring finer mathematical marks after they cease to pick out new physical alternatives. A game world's resource ceiling can likewise be too large for any player to survey and still be finite. *Minecraft* can build its worlds from any of 2^{64} starting seeds, about eighteen billion billion of them, so no player could see more than a sliver of the worlds it holds.

A finite scale beyond the surveying or counting capacity of anything inside it is **quasi-infinite**. The base's cycle, if it has one, would be quasi-infinite: immense beyond any ordinary use of the word large, long enough that nothing inside it could ever survey it, and still not infinite. Quasi-infinite is not a hedge between finite and infinite, which would be incoherent; it is a statement that the finite quantity in question is beyond the reach of anything that could count it. Beyond reach is not beyond number: being finite, it has a definite size, which is what lets Section 7.1's measure count within one full cycle. A finite repertoire of that kind is what Section 9.1's recurrence result needs. Section 10.4, on the finite-physics debt, states what a finite physics must recover and separates finite dimension from finite physical states; successful numerical approximation alone does not establish physical finitude.

[Grade: metaphysical postulate, the bounded base, definition of quasi-infinite scale, and real-life counterpart (a game world, finite yet too large for any player to survey, such as *Minecraft* with its 2^{64} seeds); recovery of tested physics within finite resources remains owed.]

2.4 A finite generative constraint and the unknown rule

FOA also assumes that the base's local generative constraint is finitely specifiable. It need not be a synchronous global algorithm or a complete formula for every event. It may be a finite rule schema together with local boundary data. The base's change is in this sense procedural: each stage follows from the last by that finite rule, as a procedurally generated world unfolds from its rules.

The base may be deterministic or may contain ontic indeterminacy; FOA itself does not require determinism, and the author's signed suspicion, set out in Section 9.5, on non-settlement, dormancy and the perfect-loop lean, is that the base is deterministic in its complete state. It may support reversible or irreversible local rules. None of these alternatives supplies occurrence, and none is settled by the postulate. The actual base rule is unknown. Section 2.7 offers an optional signed dynamics: finite configurations that always total zero, splitting and recombining without end.

[Grade: metaphysical postulate and real-life counterpart (a procedurally generated world unfolding from its rules); finite-specifiability assumption; identifying the actual local generative constraint remains an open physical research obligation.]

2.5 Parthood, coupling and reading

A base process alone does not yet place a finite life within it. FOA retains parthood as its grounding relation and develops its cross-level organisation under the name **Causal-Informational Parthood (CIP)**. Parthood says what an event is made of, informational coupling says how organised participants affect one another, and reading is a role within that coupling.

> A finite happening is a causally organised region of base occurrence. Its transitions are numerically some of the base's transitions, organised at a higher descriptive level.

That the whole base runs need not mean that it behaves uniformly. The point is easiest to see with a river and a whirlpool. Within a river, a whirlpool is not a second liquid attached to it. Its rotation is some of the river's flowing, organised into a persistent local pattern. It can begin, endure and dissipate while the river continues. A free-standing mathematical description of the same vortex geometry is not another wet whirlpool but an abstract pattern. Nor can this river's vortex simply be detached: cut it from the pressure gradients, banks and upstream flow that sustain it, and it disappears. A coherent standalone object would have to write all that into its own rule. It would be a different object, a complete whirlpool-world, not the river's

whirlpool relocated. The relation is therefore closer to a portion of activity than to parthood among discrete individuals.

A conscious experience, in FOA, stands to the base as the whirlpool stands to the river. Its causally organised transitions are some of the base's transitions; its local beginning and end do not imply a beginning or end of the base. An organism need not be one contiguous lump. No extra actuality has to cross a cable. This does not prohibit cables: a signal path and both its ends can themselves be portions of the same occurring base.

Grounding, connection and reading. Parthood says what the activity is made of. Informational coupling says how distinguishable organisations affect and help sustain one another. A reader is an entity coupled to the structure of a life, either where the life is being generated or at a point on a life thread that it is reading, and whose own state changes along with that life's organisation as it reads. A player is a reader of a game character, and a human life is conjectured to have a reader of its own, which Section 12.1, on one optional conjecture and its vocabulary, names a Reikon. A wired connection, as when you play a game, is one implementation of a coupling, not a rival ontology.

The working image lies between a chain of cogs and a flow of complex information. Each stage transforms what reaches it into a different physical or effective form. A visual signal, a neural response, a motor action and an updated game state need not use the same encoding to belong to one maintained sensorimotor loop. There is no separate substance called information passing above that activity: information here means realised distinctions, dependencies and constraints within it.

CIP therefore separates **being realised by** from **being coupled to**. A simulated event is realised by organised hardware events. A player's action can alter that event through a signal path. The first relation concerns constitution; the second concerns causal influence. Coupling may help constitute a larger process, as Clark and Chalmers (1998) argue for some cognitive systems, but influence alone does not make its source part of every system it affects. Nor does statistical correlation alone establish either relation. Where the experienced self is located need not coincide with the full support of the experience.

Wired coupling has an evidential advantage: a player and game character provide a working instance of participation through an interface. A player and game character are joined by a real signal path through screen, eye, hand and machine; *The Matrix* turns this into a cable (Wachowski and Wachowski 1999). A nervous system is a humbler case: sensory paths let the brain act as if located at the fingertips or across the room. This is participation through an interface within CIP. It demonstrates a working coupling at an ordinary level; extending the architecture below our physics remains a conjecture, and cross-level parthood still owes Section 2.6's composition proof. Section 12.1's signed reader conjecture treats reading as a role within maintained coupling, inside activity that already occurs. Section 13.1's signed continuation account considers organisation and coupling preserved through material turnover.

The direction of control need not follow the direction of realisation. Action between worlds, which Section 5.5 calls interdimensional, can run either way. In host-to-guest control, a lower-level player acts through an avatar in a world hosted above. The control need not be total: an avatar can follow default behaviour of its own most of the time, and be taken over when necessary, given broad priorities, or nudged in ways that fall far short of steering it. In guest-to-host control, a mind organised within that hosted world receives information from its host and acts through a robot or another avatar there. Software already does this in our world: AI systems trained and run inside computers act through robot arms, through browsers they operate and through remote-controlled machines. The latter mind can remain implemented in its original world: acting elsewhere is not the migration considered in Section 13.4. Both

arrangements ordinarily combine a sensory path towards the participant with an action path away from it. Reading therefore does not mean controlling. A reader may only receive, taking in what the life is doing while having no influence over it. It may send back a Sasayaki: Section 12.1 defines this as an influence across an existing coupling, thin as a whisper that carries a little information into the life without overtly steering it, or thick enough to steer, as a wired connection to an avatar allows. Or it may somehow share in the coupled organisation's agency, acting through it.

Guest-to-host control does not reverse grounding. The guest mind, its interface and the host-side effects remain activity realised by the carrier. Dependence is asymmetric while a control loop can be reciprocal. Neither direction gives unrestricted access or permission to exceed the carrier's dynamics, and neither transmits a further substance of actuality.

James distinguished productive and transmissive brain functions, Bergson treated the brain as selecting and limiting, and Huxley used the image of a reducing valve (James 1898; Bergson 1911 [1896]; Huxley 1954). Here both ends and every interface belong to one reality grounded in base occurrence. A surviving deeper identity is a further signed conjecture, not a second kind of substance and not something parthood alone guarantees.

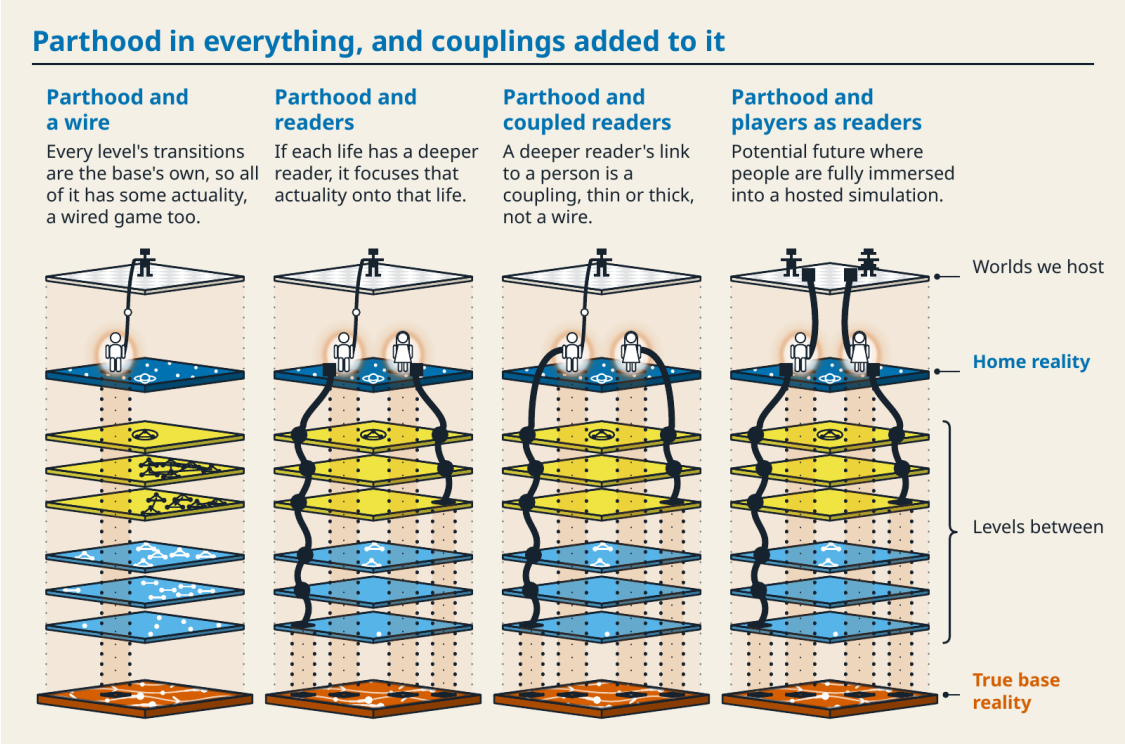


Figure 4: Parthood in everything, and couplings added to it. Every level's transitions are the base's, organised, so everything has some baseline actuality. The people glow (they have local brain structure) and the avatars (no significant local structure) do not: the author suspects a person's actuality is focused by a reader of their own, a Reikon, while an avatar's thinking is actually in its player's brain structure. The panels show a game played by wire; a deeper reader for each person; those readers coupled to them; and a potential future in which people are fully immersed into a hosted simulation.

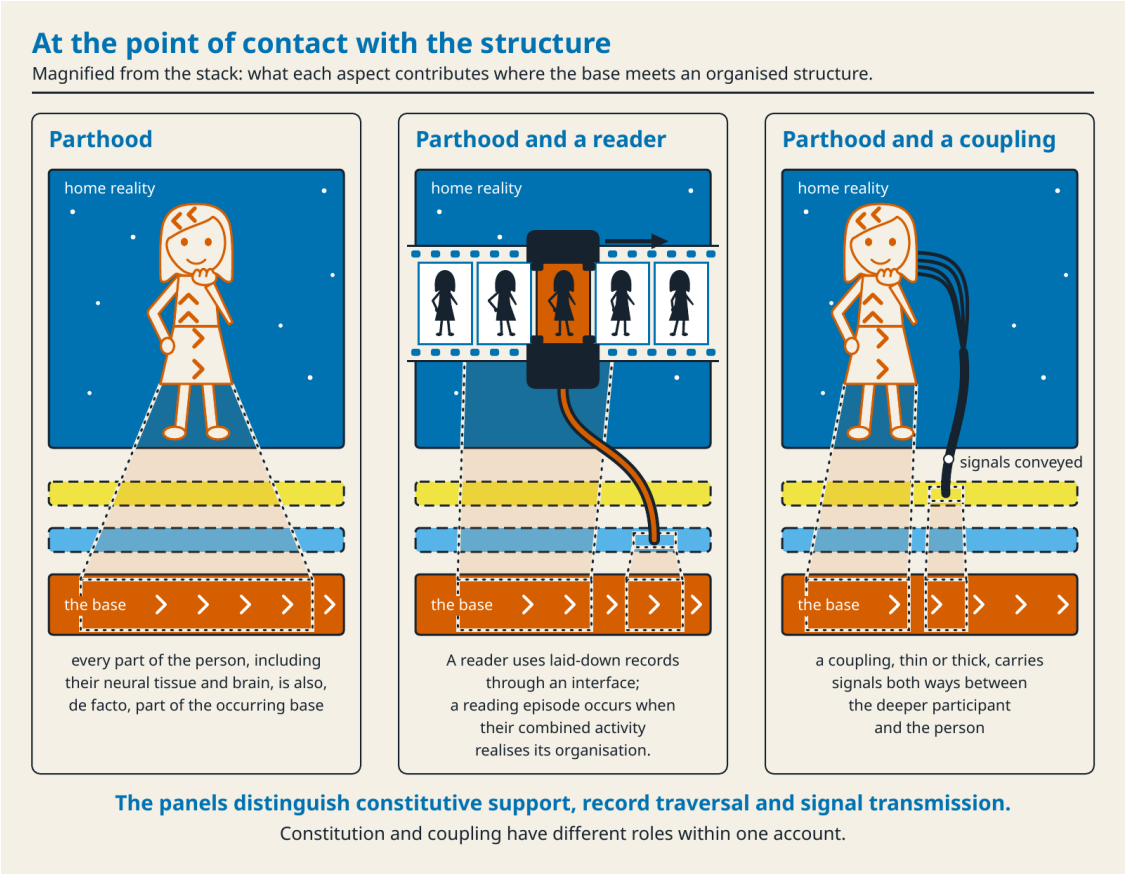


Figure 5: At the point of contact with the structure. The three panels magnify the same relations. In every panel the person's activity is part of base activity: that is parthood. In the middle a reader's playhead also passes over laid-down frames, which is where a Reikon's focusing would act; on the right, a coupling carries signals both ways between the deeper participant and the person. That is one arrangement: the coupling also permits a hosted participant acting through a host-side avatar.

Falling back a level Signed speculation

The three relations are not rivals, since parthood holds in all of them, but they differ in what they say happens when the connection fails, and the author's suspicion is that they agree on the shape of the answer. In each case actuality might not stop. It might fall back one level, to the host that was carrying it.

Under parthood there is nothing to disconnect, because the transitions belonged to the level below all along. What ends is the organisation that made them a subject; what continues is whatever that level is doing. Under the reader relation a reader that stops reading this structure carries on, its own activity still occurring, since it is carried elsewhere, possibly through nodes on levels of its own; what it loses is content rather than actuality, and it may find another structure to read. Under a wired relation the one at the far end of the wire, the player rather than the character, simply carries on at their own level, which is the case *The Matrix* makes literal. Ordinary play already shows it: when a game character dies, or a remotely piloted drone is shot down, the player or the pilot carries on.

The destination might not be the base. In this picture it is one step down, to the host: the world that was carrying this one, which may be richer than it, comparable or simpler (Section 5.5's reductive, peer and enriched hosted worlds; a host is below what it carries in the order of realisation, not necessarily in richness). It may be a rich reality in its own right, and only after a long fall is it something as thin as Flatland, Abbott's coherent two-dimensional world (Abbott 1884). Thin, though, does not settle how capable its inhabitants are. A reader at that level might be made of simpler parts than a biological brain and still be bigger, holding more organisation in total than anything home reality can fit, in the way that an F-35 does almost everything better than an older aircraft while an SR-71 still flies higher and faster. Section 5.6 separates the simplicity of a level's rules from what can be built under them.

In the signed picture of falling back a level, parthood, reading and wiring describe aspects of one arrangement. Ending a coupling can leave its other participant active; ending a subject's organisation leaves only the underlying activity unless some identity-bearing organisation continues. The one-step downward destination is specific to the deeper-reader conjecture, not a consequence of CIP. A reader organised above its avatar could remain there when the avatar failed, but not if that failure also destroyed the infrastructure carrying the reader. Persistence depends on surviving organisation and support, not on direction. Section 13 develops the further conditions for personal continuation.

[Grade: cross-level bridge, metaphysical, and real-life counterpart (a whirlpool in a river, for parthood; a player and a game character, for coupling; AI systems acting through robot arms and browsers, for guest-to-host control; a player outliving their character, for the wired case), with a signed conjecture of falling back a level; the bridge still owes a compositional account.]

2.6 Composition across levels and the closure bounds

Any account that links two levels has to say, for each link, which of two things it is. Either the link is realisation, the same activity described at two scales, or it carries a signal, something passing from one place to another. The two are not interchangeable: describing the same activity at a coarser scale, many detailed states bundled into one coarse state, adds no signal, and a genuine signal cannot be counted as a redescription. A signal cable running beside a support beam makes the difference visible: one carries messages, the other is part of what holds the structure up, and the two can run side by side without being the same relation. And if the account proposes an influence that the effective level's own dynamics do not already contain, it owes a causal pathway for it, and that pathway has to respect conservation, no-signalling and the joint and conditional statistics of outcomes that have actually been observed. These three are the closure bounds.

A signal takes time, and asking how long is a fair question, because delay is the mark of a channel: a wired connection has a delay, and the delay is real and in principle measurable. Realisation is different. A transition does not travel into the base that realises it, so there is no journey to time; it is base activity already. That is the practical test between the two kinds of link, and it holds even when the realising and the transmitting happen in one piece of apparatus.

A reader that changes nothing cannot explain a changed decision, memory or report. If a reader is to account for a changed decision, a memory or a report, it has to make a difference somewhere in the causal story. But there is a trap on the other side. An account that gives the reader a role while predicting exactly what the ordinary account predicts still has to show what that role explains. So the closure bounds leave logical room for a reader; staying inside them does not show that a reader is there.

Mereology, the theory of parts and wholes, first set out at length by Husserl, given formal treatment by Leśniewski and brought into analytic philosophy by Leonard and Goodman's calculus of individuals (Varzi 2019), supplies a vocabulary but not this bridge. Its ordinary relations say nothing about whether anything occurs. CIP adds the claim that a finite event is actual by being some of the base's occurring activity. The graded strength of a coupling is a different matter from whether a specified event belongs to its realised support; FOA does not replace event identity with a fractional dose of existence.

The heaviest obligation is transitivity: parthood has to carry across several levels at once. The bridge links a happening to the base in several different ways: the happening may be the very same as some base activity, a portion of it, made up of it, realised by it, or a coarse description of it. Linking these one after another does not automatically make the happening at the top

part of the base in the ordinary sense, and even that ordinary sense is disputed for parts defined by what they do. For the chain to hold, each link must keep the happening the same happening, and keep the organisation that does the causal work at that step. Showing that it does is a formal task still to be done; the general theory of parts and wholes does not supply it for free. Computers show such a chain holding in practice: transistors realise logic gates, gates a processor and the processor an operating system, which can run a virtual machine running an emulator running a game, each link engineered to preserve the computation specified above it.

Each effective level omits most lower-level detail while remaining carried by it. A cell does not track individual electron wavefunctions, nor an animal every molecular trajectory. CIP requires preservation of the organisation relevant to the higher process, not preservation of every microscopic description. Where an interface extends that process, the relevant whole includes its realised operation; merely containing a description of the process is still not enough. Section 3.3 distinguishes a finite run from its abstract description.

The bridge is stronger than a loose resemblance and weaker than a completed reduction. It requires at least:

1. a mapping from local events to base transitions;
2. preservation of the causal organisation relevant to the local process;
3. an account of when two higher-level descriptions pick out the same base region;
4. an explanation of how a local temporal order emerges from base order;
5. a consciousness theory specifying which organised regions have conscious character.

One part can be made exact. Let F_0, F_1, F_2 update three state spaces, and let onto maps C_0, C_1 change description between them, reaching every state of the next space. Suppose

$$C_0 F_0^{m_0} = F_1 C_0, \qquad C_1 F_1^{m_1} = F_2 C_1,$$

where positive integers m_0, m_1 fix the compared update intervals. For each of the two neighbouring pairs of levels, running the lower level through its fixed block of updates and then changing description gives exactly the same state as changing description first and taking one update at the higher level. Applying the first identity m_1 times gives $C_0 F_0^{m_0 m_1} = F_1^{m_1} C_0$, and substituting the second then gives

$$(C_1 C_0) F_0^{m_0 m_1} = C_1 F_1^{m_1} C_0 = F_2 (C_1 C_0).$$

Changing description after evolving agrees with evolving after changing description, even across both levels. Running the lowest level for the product of the two update counts and then changing description twice gives the same top-level state as changing description twice and then taking one top-level update.

For example, take cycles of 12, 6 and 3 states, each adding one modulo its size, with maps reducing state labels modulo 6 and 3 and $m_0 = m_1 = 1$, so that one update at either lower level is matched by one update at the next level. The maps commute with the updates and with phase shifts by any integer r, a specified intervention family rather than arbitrary interventions.

Intervention-based causal consistency has a wider treatment (Rubenstein et al. 2017). The calculation does not yet identify which base events realise a higher transition, resolve overlapping supports or individuate a conscious subject.

The general event-support and composition problem remains open. The state-map calculation supplies one component; causal graphs, coarse-graining, information geometry and process maps remain candidates for the rest. Section 11.5 asks which integration boundaries

individuate subjects, which is where support maps would have to fit. Priority cosmopsychism treats the cosmos as the one fundamental subject and finite subjects as derivative, and faces a decombination problem in explaining how one subject becomes many (Miller 2018); priority monism treats the whole cosmos as the one basic concrete object, on which every proper part depends (Schaffer 2010). Occurrence theory grounds actuality, conscious actuality included, in the occurring of a base process rather than in a cosmic subject, and leaves conscious character to organisation. Section 15.7 tests these five bridge requirements.

[Grade: mathematical result, conditional, for state maps with fixed update intervals, and real-life counterpart (cells and animals, which keep the organisation that matters and omit the detail that carries them; the computing stack, for a chain of realisation holding across many levels); event support and subject individuation remain open, and any added influence is constrained by conservation, no-signalling and tested outcome statistics.]

2.7 Interaction, production and signed refinement

The base need not sit inside a ready-made spatial container. A primitive transition system can supply adjacency through interaction itself. Two events are locally related when one constrains, transforms or helps produce the other. Repeated causal relations can then support effective distance, neighbourhood, dimensionality and geometry. This is an architectural possibility, not a derivation of our spacetime.

The same distinction clarifies what counts as production. The base's occurrence is primitive. A higher-level object is produced when an earlier causal organisation gives rise to the transitions that realise it. A standing boundary condition can constrain a region without being produced inside that region. The edge of a storage medium limits where data can be written; it need not be written by the same program. Abstract structures are neither produced nor unproduced in time, because as abstract structures they are not actual on any temporal level. Above primitive occurrence, an explained distribution is normally produced by prior processes, with boundary conditions as the standing exception. Section 6 studies produced organisation above this floor.

Signed refinement, the founding idea Signed speculation

The author conjectures that the base's change may itself be a signed, sum-preserving refinement: finite signed configurations that always total zero, splitting and recombining without end. It is a long-standing idea of the author's. It is a conjectural dynamics, not part of the core architecture, and no core argument of the paper depends on it, though Section 15.3, on nullity, signed refinement and bounded recurrence, proposes implementing it directly.

The children of a split need not be exact opposites. A term x may be replaced by $x + a$ and $-a$, which still total x.

Let any stage be a finite multiset, meaning a collection in which the same value may appear more than once and the order does not matter:

$$S = \{x_1, \dots, x_k\}, \qquad \sum_{i=1}^{k} x_i = 0,$$

with values in an abelian group. This bare multiset defines the conserved labels only. A model with locality or organisation must additionally place the labelled elements in a finite graph, hypergraph or other finite relational structure, and every move must specify how both labels and incident relations are rewritten. Allowed local moves may include:

- **splitting:** one part becomes several whose values total the original, $x \mapsto \{y_1, \dots, y_m\}$ with $\sum_j y_j = x$;
- **merging or coagulation:** several parts become one whose value is their total, $\{y_1, \dots, y_m\} \mapsto x$ with $x = \sum_j y_j$;
- **annihilation:** a pair of equal and opposite parts is removed, $\{x, -x\} \mapsto \emptyset$;

Continued on page 24

Signed refinement, the founding idea, continued Signed speculation

- **context-sensitive interaction:** the permitted move depends on the finite adjacency or relation structure.

Conservation keeps every generated stage in the zero-sum state space. Nuclear physics shows splitting and merging under a conserved total: in fission a nucleus splits and in fusion nuclei join, and total electric charge and total energy are the same before and after. It does not require a root stage, induction from an initial pair, or a sum over the whole past. A part produced by a merge can have more than one parent, so ancestry is a network rather than a tree. None of this borrows Section 4.5's Sign-Reversal Closure Hypothesis, which is about abstract structures rather than about how the base changes.

Signed refinement, two consequences and the limits Signed speculation

Two consequences follow, and they are easily run together.

The first is that **mirrored splitting is not forced**. A split replaces a term by children that total that term, so it leaves the rest of the configuration untouched and the overall total unchanged, and no answering move is needed anywhere else. It is tempting to think that a configuration can only total zero by consisting of matched opposites, and that splitting must therefore be mirrored to keep the matching. It need not. The three-part block $\{3, -1, -2\}$ totals zero although no part in it has a matching opposite, and replacing the 3 by 1 and 2, mirroring nothing, leaves it totalling zero still. Mirrored rules remain attractive, because they would make sign reversal a symmetry of the dynamics as well as of the state space, but they are an option and not a requirement.

The second concerns the beginningless past, and it is bookkeeping rather than dynamics. With no first stage there is no root from which zero could be inherited, so zero has to be imposed as a condition on stages instead. For finite stages that condition is simply the displayed one: the parts total zero. Describing such a stage as dividing into zero-summing blocks adds no constraint by itself, since the whole stage is one such block. The block formulation, **blockwise nullity**, earns its keep only if configurations unbounded in size are admitted outside the bounded-base postulate, where no total over everything is defined; there one would require a partition into finite zero-summing blocks, and would additionally have to constrain the blocks, for instance by locality or by bounded size, or the condition does little work.

The honest accounting is this. Zero is stipulated either way. Under a rooted picture it is stipulated once, of one object, and inherited. Without a root it is stipulated as a condition on which configurations count as stages at all. That is a change in the kind of stipulation rather than an increase in the number of stipulations, and the author regards it as a fair price.

Splitting alone supplies neither interaction nor organisation. Adding merges and annihilation creates a new danger: the dynamics can collapse into the empty state or another settled attractor. Non-collapse and continuing non-quiescence therefore need independent guarantees. The conjecture also needs at least one asymmetry and sufficiently rich interaction before computationally irreducible variation becomes plausible. Rule 110 is a useful analogue for how simple local asymmetry can support universal computation, though it does not establish that this signed system will do so.

Only the move actually selected by the rule occurs. The state space may contain many admissible refinements without chance choosing among them. Physics already has the restricted case: a pair of opposite electric charges can be created from neutral radiation and can annihilate back into it, so pair creation and annihilation keep the signed charge total through every step (Dirac 1931; Anderson 1933). The conjecture's step beyond that is to propose this as the form of all change at the base, not only of charge.

The conjecture would partially connect structured nullity with dynamics. It would not explain why any dynamics occurs, show that the actual base has this form, or generate conscious organisation by itself. It does not say that $0 = 1 + (-1)$ is creation from nothing.

[Grade: signed speculation, not an established base dynamics, and real-life counterpart (pair creation and annihilation, which keep the charge total through every step; fission and fusion, which keep charge and energy).]

Section 9.2 tests whether merging could bound configuration size. The models test the architectural possibilities; Section 1.1's actuality question remains separate.

A finite flow case. Where the parts stay put and only transfers move between them, the arithmetic is standard: transfers can be assigned to the edges of a connected finite graph that

give every vertex any required net change exactly when those changes sum to zero, and local balance still leaves a great deal of internal freedom. The calculation, Kirchhoff's law read as an existence statement together with the cycle-space count, is set out in Section 15.3, on nullity, signed refinement and bounded recurrence, beside the Signed Life implementation.

On the finite deterministic branch, where the base recurs exactly (Section 10.1, on determinism and reversibility), any annihilation the signed picture allows must leave elsewhere in the complete state whatever its reversal needs. Existence of a flow says nothing about whether a chosen rule reaches it. Nor does this forced global zero supply Section 4.4's sought invariant, whose zero would depend on physical content.

The computational companion uses a different, fixed-lattice construction, described in Section 15.3, on Signed Life. It exhibits organisation under a selected reversible signed rule; direct splitting and merging remain a separate target.

[Grade: mathematical result, conditional, for finite flows and a formal possibility exhibited on a fixed lattice, alongside the signed-refinement conjecture; a workable direct-refinement dynamics with sustained organisation, heredity and higher-level construction remains to be exhibited.]

3. Mathematical Intrinsic Occurrence and the structuralist alternative

3.1 The rule, the cycle and the run

Writing the word *circle* on a wall is not drawing one. A specification is not its subject, and this subsection's whole point is that the same holds for a rule and its running: the smallest exact case is a single bit and a rule that flips it, and nothing that follows in this subsection needs more mathematics than that.

Consider a one-bit state and a rule that flips it:

$$x \in \{0, 1\}, \qquad T(x) = 1 - x.$$

The state is either zero or one, and the rule turns zero into one and one into zero. Its transition graph is the finite directed cycle

$$0 \longrightarrow 1 \longrightarrow 0.$$

The entire graph has two vertices and two directed edges. It is a completed finite abstract object. The instruction "apply T again" can be followed without any terminal rule case, but that does not make the graph apply itself. Nor does the absence of a last vertex establish occurrence. A loop drawn on paper is not moving ink. Ink that rewrote itself, carrying out the loop step by step, would be another matter.

This simple example separates three things that are easily blurred:

1. the finite rule T;
2. the finite orbit or transition graph specified by T;
3. an occurrent sequence in which some substrate repeatedly realises those transitions.

The first two are abstract patterns. The third, if it happens, is a process. Under Finitary Occurrent Actualism (FOA), an occurrent cycle is a region of base occurrence whose successive transitions instantiate the abstract relation. The abstraction classifies what happens. It does not turn the happening on.

A physical oscillator can realise the same coarse transition shape for a time. Chemical oscillations, biological clocks and engineered electronic circuits all provide examples. Three proteins from cyanobacteria, mixed in a test tube with the fuel they need, keep a daily rhythm for days on end (Nakajima et al. 2005). Their detailed physics supplies energy flows, dissipation, noise and boundary conditions absent from the two-node graph. The comparison is therefore illustrative rather than exact. A real oscillator may stop; while it operates, its transitions occur. A process need not be endless to have occurred. Section 2.2's distinction applies: endlessness supplies no run, and ending does not undo an occurrence.

FOA needs examples in which a small formal description supports extended transformation, persistent higher-level patterns and open-ended consequences. Cellular automata are useful because their rules can be stated exactly and their behaviour inspected step by step. They are not proposed as the literal grain of reality. They are toy models of several separable ideas: local transition, process identity, emergence, computation, self-maintenance and evolution.

The distinction between rule, graph and run governs every example. A transition graph, rule table or complete finite trace is an abstract pattern. Nothing runs merely because the pattern contains arrows, refers to its own next step or has no terminal node. When an automaton is simulated, its local events are carried by transitions in the computer: every simulated history borrows occurrence from hardware. That ordinary fact illustrates the kind of parthood relation FOA proposes, but it does not establish that the true base is cellular, digital or computational in the familiar engineering sense. The automata have a precise role. They illustrate architectures available once occurrence is granted. They do not establish that abstract mathematics operates by itself.

The historical bridge from automata to cosmology runs through Zuse's *Rechnender Raum*, Fredkin's digital philosophy and more recent cellular-automaton approaches to fundamental physics (Zuse 1969; Fredkin 2003; 't Hooft 2016). FOA borrows none of these programmes wholesale. Its base may be unlike a lattice and may lack a global clock. What the examples contribute is a disciplined vocabulary for asking how simple local change might support worlds.

Conway's Game of Life applies one local rule to a two-dimensional binary lattice (Gardner 1970). Each cell is alive or dead. Its next state depends on the eight neighbouring cells.

Standard Life uses a pre-given lattice, a pre-given neighbourhood and synchronous generations indexed by an external counter.

A complete finite Life history can be represented as a finite spacetime pattern. That representation can contain every cell state and causal dependency. FOA's question remains whether representing the history is identical to the history being actualised. Life sharpens the question but cannot answer it, because both readings admit the same formal trace.

[Grade: mathematical result and real-life counterpart (physical oscillators, such as three proteins keeping a daily rhythm in a test tube; computer hardware carrying a simulated history); established finite mathematics and computation, with a metaphysical interpretation; rule-following does not establish occurrence.]

3.2 Mathematical Intrinsic Occurrence

Section 3.1's rule/graph/run distinction leaves occurrence as the cost to be stated here. **Mathematical Intrinsic Occurrence** (MIO) is the term used in this paper for the following claim. A mathematical object can exist as something that is permanently occurring (transitions constantly occurring rather than merely specified). This paper proposes that at least one does, the one we exist in. You cannot write out such an object in full, because it never finishes,

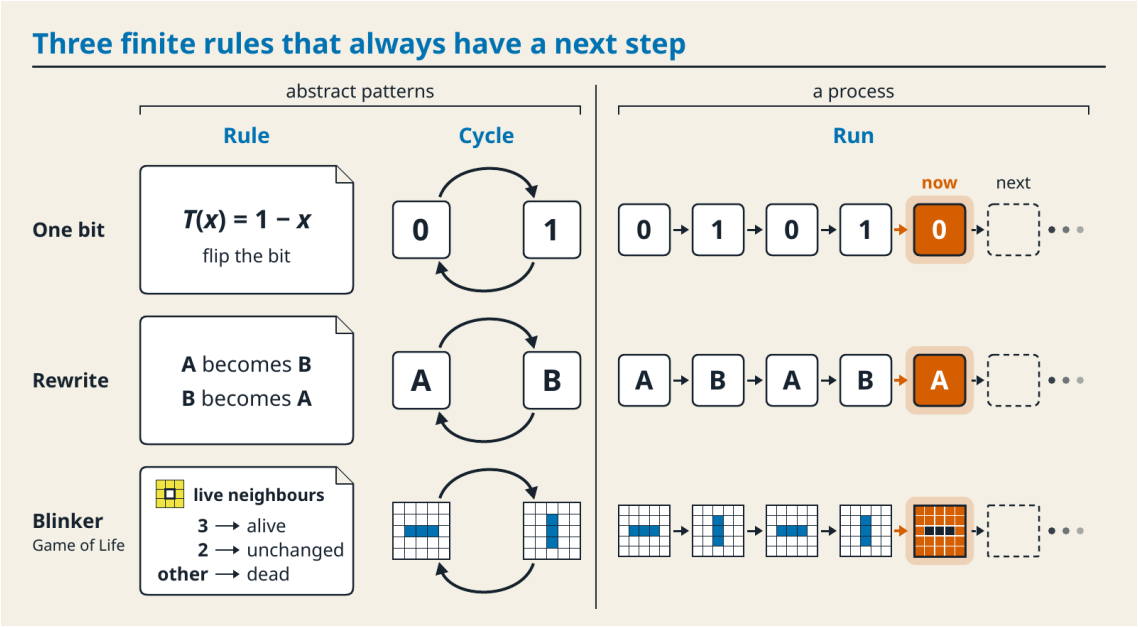


Figure 6: Three finite rules that always have a next step. A bit that flips, a rewrite in which A becomes B and B becomes A, and the blinker in Conway's Game of Life, three live cells that turn from a row to a column and back. Each rule is a finite instruction, and the cycle it specifies is a complete pattern; both are abstract. The run is a third thing: the steps happening, one after another, with always a next one to come. The cycle classifies what happens; it does not turn the happening on.

and writing down the rule it follows leaves you with an abstract rule rather than an actually occurring object.

Both modifiers in the name carry weight. *Mathematical* says that the proposed process is a mathematical object, and that is the part few expect; that things happen is not in dispute. *Intrinsic* says the occurring is the object's own, with no external clock and nothing outside it doing the running. Occurrence remains Section 2.2's primitive.

Nothing said so far settles whether all mathematical objects are the same *kind* of thing. Suppose some are not. Suppose some structures are **intrinsically processual**: objects whose nature is to be carried out, with no completed version of them available to sit unrun alongside the running. A static description of such an object is not a duplicate of it. It is a description, a different object of a different kind, standing to the process as a photograph stands to a journey.

> **Mathematical Intrinsic Occurrence** at the base is the reading this section gives: the base is posited as occurring, and no separate fact of its existing is posited alongside, because under strict potentialism there is no completed, unrun history of the base for it to have been instead. That removes one alternative only. It does not by itself refute every structuralist reading of a finite rule or a finite configuration, and treating the base's existing as its occurring is FOA's reading of its primitive rather than something potentialism forces. That reading asks no more than Tegmark's mathematical universe does, which posits a structure that exists complete and static, with no separate fact of its existing (Tegmark 2008), where FOA posits one base, finite at every stage, that occurs without end.

A classical Platonist who admits completed infinite objects has a completed base history available, the counterpart returns, and the saving set out below is lost.

What is being dropped. It can look as though this brings back the very ingredient the paper claims not to need. It does not. What is being dropped is a tacit premise: that mathematical

existence is always the existence of a completed object. That premise is imported from the habit of writing mathematics down. The question is therefore not which structures receive the bonus property of running. It is whether some mathematical objects are processual, and if any are, no separate fact of their existing has to be posited alongside their running.

Writing the word *circle* on a wall is not drawing one, as Section 3.1 opened by saying; now write *endless circling* on the wall instead. To realise that, one must keep drawing, and if one ever stops, what remains is a circle, not a partial or frozen version of endless circling but a different kind of thing altogether. There is no static object that is what endless circling would be if it were not happening.

The criterion, and where it comes from. Any view of this shape immediately owes an account of *which* structures are processual, and the account must not be "the ones that run", which explains nothing. **Non-terminability is the criterion** for excluding a completed counterpart, and it comes from Section 2.1's strict potentialism: no completed infinite object exists. A process whose whole succession cannot be finished therefore cannot exist as a completed object at all, and that way of existing is closed off. That exclusion does not make the process occur. The finite *rule* is untouched: a rule specifying an endless process is a finite symbolic specification, and it stays a specification however long it sits unapplied. Section 3.1's rule-versus-run distinction still applies.

One distinction has to be made explicit. Section 2.4's finite-specifiability assumption concerns the **rule**: whatever governs the base can be given in finite terms. What cannot be finished is the **completed extensional history**, the whole succession laid out as one object. Non-terminability blocks the completed counterpart at the second level only. It says nothing against a finite generator, and it must not be read as claiming the base is unspecifiable.

A circular track may be a completed finite object while the runner's traversing continues: repeating a finite cycle does not turn that endless traversing into a finished history. Section 9.1 will establish recurrence only under a finite-repertoire condition; even then, the repeated finite cycle supplies no completed infinite traversal.

What follows: the base has no static version it could have been instead. Grant that something occurs, which this paper has assumed from its first page, and ask what the thing occurring could have been instead. An ordinary finite process admits a complete description that need never be run. For the base there is no completed history available as that alternative. The saving is substantial. Occurrence at the base is not a further non-structural fact laid over a completed history that might have lacked it. In FOA's ontology there is no such history and no such laying, and the phrase *the base existing without occurring* does not pick out an admitted alternative. Familiar things exist this way within the world: a flame or a storm exists only while it goes on, and when the burning or the circulation stops, what is left is not a flame or a storm at rest. Anything genuinely infinite can exist here only by running, in the object's own occurring, since no finished infinite totality is admitted.

What this does not do is establish that anything occurs. That remains where it was, resting on the firsthand datum. The argument narrows what has to be paid for without paying it.

One recent line of work bears directly on whether an abstract object can be dynamic at all. Formal development processes enrich mathematics with objects that develop internally, persisting through stages that belong to the object rather than to any external time (Gorbow 2025). That is a formal sense in which an abstract object can be more than a frozen list, and it is company for the proposal made here rather than competition: it describes structures whose stages belong to them, which is what intrinsic processuality requires. It does not establish that those stages occur.

[Grade: metaphysical postulate, FOA's reading of primitive base occurrence rather than a further postulate, and real-life counterpart (a flame or a storm, which exists only while it goes on); conditional argument for the exclusion of a completed infinite history under strict potentialism; the occurring itself remains primitive.]

3.3 Finite occurrence, complete description and possible detection

A finite happening, an afternoon, a computation, a life, is finite: it can be described completely in any admitted vocabulary, and its succession does finish. Section 3.2's non-terminability criterion therefore does not reach it. Non-terminability sorts nothing among finite forms, and no part of this subsection should be read as a criterion for which finite patterns are alive.

Finite happenings are carried instead by Causal-Informational Parthood (CIP), the bridge Section 2.5 adopts, with its composition conditions left as Section 2.6's outstanding formal obligation. Nothing in this paper supplies a complete description of a finite life, and nothing in it needs one. Ask, then, what distinguishes a finite run from a complete description of it, granting the objector the description. Not an intrinsic extra property: it is stipulated to leave out nothing the declared vocabulary admits, so the two agree in every admitted structural respect. A game being played and a stored record of every state it passes through are the everyday case: nothing in the game is missing from the record, and what tells them apart is not an omission but whether anything is being gone through. What distinguishes them is a **relation**. The run is a portion of the base's occurring; the abstract description is a portion of no occurrence. The abstract duplicate is not a physically present philosophical zombie but a complete pattern belonging to no occurrence.

One move has to be answered, because many readers will try it first. If the difference is not an intrinsic property, why not add a marker: a label such as *live*, *actual* or *now* attached to the description? Because a description with a label in it is still a description with a label in it. The marker is one more piece of structure, and the enlarged pattern stands to the run exactly as the smaller one did.

Two images sharpen what is being claimed, and neither points at a structural detail the description is missing. A number changing on a screen and the same sequence printed down a page contain the same numbers in the same order. A roller-coaster ride and every photograph ever taken of that ride contain the same positions, the same faces, the same order. In each pair what distinguishes the two is not an omission but whether anything is being gone through. The pairs preserve the listed numbers or appearances rather than every causal detail; the complete-description argument is the stronger, stipulated case.

One small case comes from outside this dispute entirely. Suppose some primitive has one hundred and thirty-seven distinguishable states. Eight bits suffice to label which state it is in, and nothing is lost in the labelling. It does not follow that the primitive contains eight bits, or any bits. **A description can capture everything about a thing and still not be that thing**, and the gap has nothing to do with the description being incomplete.

A complete description is also not a **record left behind**, and the difference prevents a natural misreading. Almost nothing that happens leaves a record. A description is a different kind of thing from a deposit, and whether anyone has ever written one has no bearing on the argument. Section 8.3 will examine, as an open architecture, a rewrite-only world that stores no history and remembers by consequence alone.

One reading of *dispositional* has to be fixed. A disposition is a fact about what a thing would do, and there are two ways to take that. On the first it is a structural fact about what would follow what across the complete state graph, and an abstract pattern has all of them. On the second, dispositions are occurrent powers, properties whose very possession involves activity, so that

a wholly inactive bearer of every power verges on a contradiction. That is stronger than Bird's dispositional essentialism, which does not equate possessing a power with exercising it (Bird 2007). This paper takes the first and welcomes the second as company rather than refutation: anyone who denies that a completely inactive bearer is even possible has already agreed to something weaker but congenial, that structure taken as something listable does not exhaust what there is.

The tempting third option. A third option is sometimes offered, and it is not this paper's: some structures might simply *run by nature*, with no external selector and no brute distribution. But the run-making fact is either structural, in which case any complete structural counterpart shares it, or non-structural, in which case it is a brute fact about which forms run. Section 3.2's argument escapes this fork only at the base: an unfinished succession has no completed counterpart. That escape is unavailable to finite happenings, even if no description of them has been written, which is why FOA runs one base and lets finite happenings inherit actuality as its parts rather than positing a population of self-running structures.

A related proposal is worth answering beside it: that occurring is a further mathematical coordinate, an axis along which a structure is more or less run. A coordinate is structure. A duplicate carrying the same value on that axis would be run to the same degree, so the fork above closes on this coordinate proposal exactly as it closes on the extra-property, label and run-by-nature proposals. The intuition behind the fork is not peculiar to this paper. Strawson, arguing from the other end as a physicalist, treats the emergence of the concrete from the wholly abstract as the paradigm of an impossible emergence, the case that even a friend of emergence will not allow (Strawson 2006).

A structuralist can add that the distinction is empty because there is no structural test for it. The reply has two parts. Occurrence is not itself a structural feature, so no test could find it as one, and that is the predicted shape of the claim rather than a surprise that refutes it. Whether occurring makes a difference that structure registers is a different question, which this paper leaves open; the author's suspicion is that it does, and that the difference may be detectable, inferable or deducible. The first part states the disagreement; it does not compel assent.

No structural occurrence detector is predicted, and none is ruled out. A discovered detector would be a discovery within FOA; it would refute the parity reading, on which occurrence makes no structural difference at all, a reading that Section 14.3, on reports, illusion and the meta-problem, grants only for the sake of argument. Sections 12.6 and 15.7 propose coupling and continuation tests.

One observation is recorded here without being turned into a claim. The direction of time, and whatever settles a quantum measurement rather than merely describing its statistics, have both resisted reduction to structure in something like the way occurrence does here. It is natural to wonder whether they are one primitive seen from three angles. This paper does not claim it: each of the three is contested within its own field, irreversibility has a statistical account that needs no new primitive, and the measurement question turns on which interpretation of quantum theory is right. Section 10.3 will state quantum compatibility requirements while leaving that interpretation open. Hitching one narrow claim to three contested ones trades strength for reach. The question is well posed and is left open; Section 10.3's signed suspicion, that quantum structure belongs to our level rather than to the base, leans the other way, and the two are left standing side by side.

[Grade: admitted limit and open problem; explicit metaphysical cost and open detection question; finite actuality depends on the unresolved parthood bridge.]

3.4 Reblocking and the strongest structuralist alternative

Any finite occurrent history can be represented as one larger abstract object. Its successive states become slices, its causal links become relations and its memories become internal records. This is the **reblocking objection**: whatever the process theorist adds can be absorbed into a sufficiently complete block. In its strongest form, the block contains every event, causal relation, memory, report and first-person judgement. The structuralist says that the organised event simply is the experience; asking whether it additionally runs demands a duplicate fact.

At the level of structure, the objection may well succeed. FOA points to no missing arrow, clock, flow field or self-reference tag that it can name, and all such features can be included in the block; whether occurring leaves any structural mark at all is left open in Section 3.3, on finite occurrence and possible detection, and the reader conjecture of Section 12 allows that a Reikon's coupling might make a structural difference. The concession is therefore that nothing need be missing, not that nothing is. And a complete block explains, at most, why everyone in it reports being actual; it does not explain why you are. A block description can be internally complete **as structure**. The remaining fork is exact: either complete structure is sufficient for actuality, or the difference between an experience being undergone and one merely specified is primitive.

Strict potentialism admits no completed infinite object, so there is no completed block of an endless process as a whole, and none of the base's whole history. But the objection does not need one. Restricting the objection costs it nothing and keeps the paper from stating a rival position in terms it has itself ruled out. A block need not move from outside to include an experience as of flow (Paul 2010; Prosser 2016). Section 14.2 concedes that if all facts are structural, the objection wins.

That qualification, *as structure*, matters. A description may include a relation saying that an episode within it is actualised. If true, something is occurring and the dispute is over; if false, the description is complete in structure but wrong about its subject. A map may record every road correctly yet wrongly say the town is inhabited. Reblocking therefore shows only that no structural feature is missing, not that the block accurately describes a world in which anything is occurring.

That remaining fork is first-person and metaphysical. A complete block can contain an event representing the judgement "this pain is now being undergone." Does that suffice for the pain to be undergone, or does it merely include the appropriate judgement as another relation? The structuralist says there is no further fact. FOA says the difference between the seeming being undergone and the seeming only being inscribed is the datum from which metaphysics must begin. Section 14.3, on reports, illusion and the meta-problem, takes this further: a novel whose character insists on their own inner life contains a false sentence, but nobody there holds the false belief, and an illusion needs someone to be under it.

Lewis's indexical account of actuality gives the rival a clean form (Lewis 1970, 1986). *Actual* works like *here* or *now*, picking out the world spoken from: each world is actual from within, rather than actual outright. The account is prior to Tegmark's by some decades. On this reading, saying that a process actually occurs adds nothing beyond saying that it is the one being spoken from, which its abstract counterpart says with equal right. What is being asked about is not which description is being spoken from. It is why anything is being undergone at this point of a life rather than that one, and an indexical answers the first question and leaves the second open.

There is no **promotion event** at which greater descriptive detail becomes occurrence: a block no more starts running when complete than a recipe becomes a meal by listing more ingredients.

Set beside its neighbours, the pattern is one refusal made at three altitudes. Existence needs no **creation event**, because time belongs to the world being explained rather than preceding it, as Section 1.1, on the two gaps, makes clear. Operation needs no **promotion event**, for the reason just given. Section 11.4 will propose that experience needs no **ignition event**, since a felt aspect goes as deep as occurring. The three are differently graded and should not be run together: the first two are FOA's refusals and the third is a metaphysical postulate of Occurrence theory, outside FOA's core. What they share is a shape: none requires a separate event at which the relevant fact switches on.

The attraction of the static reading is genuine, and it explains why careful people rest there. Motion of the ordinary kind is undetectable from inside: a passenger in a smoothly moving carriage has no local experiment that distinguishes travelling from standing still, and treating the carriage as stationary is not an error but a working description that succeeds at everything except the one question of whether the train is moving. The same holds on the Earth: looking at the stars and the sun we know we are moving, and it is simpler, and for most purposes right, to imagine we are stationary. A world taken as a completed structure is in that position. Every local fact is preserved, every prediction goes through, and nothing inside the description reveals the difference. What the description cannot do is settle the question it has set aside, and the reason a passenger believes the train is moving is not that the carriage told them.

The analogy has a limit, and it runs against the static reading rather than this one. A carriage has a passenger in it. On this paper's reading a completed structure, taken as complete and static, has no one inside to wonder whether anything is moving; wondering is a process, and a description of wondering is not one. The structuralist denies exactly that step, holding that the organised event is the wondering rather than a representation standing in for it. The image is therefore generous to the view it illustrates: it grants an occupant whose presence is precisely what the two readings dispute.

Conceivability arguments of this shape face a standard reply: that two descriptions may pick out the same fact by different routes without the difference in description marking any difference in the world, as *water* and *H2O* do. If occurrence and complete structural specification were related that way, Section 3.3's fork between shared structure and a further occurrence fact would be an artefact of how the two are presented rather than a real distinction. Section 3.5's philosophical priority, the judgement that firsthand actuality appears more certain than the thesis that all facts are structural, does not answer this identity objection.

There are two routes to water: one may look at it, drink it, find it in rivers; or one may do chemistry and arrive at the same substance by another path. **Both routes are routes of looking.** Both are available from outside, and the discovery that they meet is an empirical discovery, made by an observer who took both. That is why nobody is tempted to conclude that water has a further ingredient chemistry missed.

The two candidates here are not like that. A complete structural specification is something one can look at. A process occurring is not something one looks at to learn about it; it is the happening itself. So there is nowhere to stand and check that the two coincide, because occurring is not a way of looking at anything.

This does not dispose of the position, only of its most familiar illustration. The strongest form of the identity move is the **phenomenal concept strategy**, which grants the asymmetry just described and explains it: our first-person concept of an experience is formed in a way no theoretical concept is, so an epistemic gap is exactly what a physicalist should predict, and no ontological gap follows (Loar 1997; Papineau 2002). Pointing at the asymmetry is therefore not an objection to that strategy. It is the strategy's own starting observation.

Either the special concept is itself fully explicable in physical terms, in which case it is unclear why the gap does not close under that explanation, or it is not, in which case something has been conceded that was supposed to be denied (Chalmers 2007). This paper takes no view on which horn is right for conscious character. For conscious actuality it has already taken one: occurrence is not a structural feature (Section 3.3, on finite occurrence and complete description), so no concept strategy closes that gap. It records that the debt is reduced rather than discharged: that the identity move is not free, that its usual illustration does not transfer, and that the position it must actually defeat is stronger than the illustration suggests.

Every concrete description is a token that something produced: the equation on the page, the thought in a head, each made available by a long-running history and meaning nothing without a reader running now. So the contrast between two descriptions is drawn entirely in items that exist because things occur, and it is offered as a reason for thinking occurrence adds nothing. The Platonist can reply that an abstract structure needs no production and only our *access* to it took thirteen billion years; whether that settles anything depends on whether the identity claim binds the made tokens or what they express, and this paper does not settle which reading the structuralist is owed.

The objection therefore identifies FOA's point of greatest underdetermination. Third-person structure cannot decide a dispute defined as surviving complete third-person structure, so no refutation of that kind is offered here.

It does not follow that the two views start level, and the paper does not treat them as though they did. Three considerations bear on which way the burden runs, and none is a proof.

The first is a first-person consideration, and it must be stated carefully, because as an appeal to observation it would not work. Nothing completely stationary has ever been encountered: not one object, not one system, not one process at any scale yet examined. Nothing infinite has been encountered either. This does not by itself tell against the block from the third person, and it cannot, since the objector's block contains every one of those encounters and Section 1.5, on the levels of reality, reserves settling and dormancy for occurring regions. What the point amounts to is narrower: an experience being undergone is the starting datum, and treating it as part of a completed static structure is a theoretical interpretation. What has been found in abundance is change, and processes that appear able to continue without end. A foundation that is static and complete is therefore not the modest, default option it is often taken to be. It is a category with no observed members proposed as the ground of a world made entirely of the other kind.

One historical case belongs here as an analogy. The ground underfoot was taken to be fixed for most of the history of geology, and it looked fixed to everyone standing on it. It is not. The crust is carried on a convecting interior, consumed at trenches and remade at ridges, and the appearance of fixity was an artefact of a human lifetime set against the rate at which plates move. The analogy carries one narrow point: an observer's own timescale can do work that gets credited to the object. It is not a precedent that occurrence will one day be detected as plate motion was: Section 3.3 predicts no detector, though it does not forbid a structural mark of occurring. Even the tectonic future is not read off from the rule but found by the world running it.

The opposite illusion is familiar too. A periodic motion sampled at a matching rate can look still, while a slight mismatch makes it drift slowly forwards or backwards, as helicopter blades or a wagon wheel do on film. Plate motion and the stroboscopic effect make the same narrow point from opposite directions: apparent rest can depend on the ratio between the process and the observer's timescale. Section 5.4 will develop the cross-level proposal that rates are obtained by comparing occurring processes, with the physical derivation still owed.

The second consideration is coherence: Section 7.3 argues that production can explain coherent records, with Section 7.1's measure doing the weighing.

The third is proportion. That anything is being undergone at all is an extraordinary fact, and extraordinary facts have historically required explanations that were real rather than verbal: complex life needed evolution, inherited resemblance needed a molecule that copies. The explanations were not always large; both of those are remarkably compact. But neither was an assurance that nothing further was going on. An account which answers that nothing further is happening, and that the appearance of undergoing is one more relation inside a complete description, is not a small claim modestly stated. It is a very large claim, however restrained its statement.

So the position of this paper is not that the disagreement is undecidable, and not that a knock-down argument is available. It is that the structuralist owes an account of the one thing every reader has, and that the burden of the extraordinary claim does not sit where it is usually placed.

[Grade: open problem and real-life counterpart (motion that passes for rest: a smoothly moving carriage, plate motion, the stroboscopic effect); deepest live metaphysical objection and burden argument; complete structural representability proves neither FOA nor structuralism.]

3.5 The price, the stronger lean and the remaining options

Section 3.2's economy covers the base: one fact, that it occurs, with no second fact of its existing. Section 3.3 leaves finite actuality relational: finite happenings inherit actuality as parts of the base. A structuralist may reject the primitive and reinterpret the formal and physical parts of the paper without it. FOA takes the other side because firsthand actuality appears more certain than the thesis that all facts are structural. That priority is philosophical, not demonstrative. Giving occurrence a defined role does not prove that the role is real.

Calling the base concrete and patterns abstract captures much of FOA, and that reclassification is acceptable. The remaining commitments are Section 2's finite occurrence and parthood, Section 11.2's two-factor experience account and Section 2.7's produced distributions.

The author also suspects that no occurrent region is ever absolutely static, and perhaps that true staticity is an empty category. That is a signed speculation, not a premise. Effective equilibrium and stationary physical states can remain real at their own descriptive level even if deeper transitions continue.

The author's lean runs further: what never occurs may simply not exist, so that occurring would be not one mode of mathematical existence but the only one, with nothing completed and unrun waiting anywhere, inside the base or beyond it.

This is the stronger signed suspicion in Appendix A's claims ladder. Section 3.2's MIO argument needs only the narrower claim that the base has no completed counterpart; it does not need the suspicion that unrun abstracta do not exist at all. The options remain live: strict structural sufficiency; one occurring base among abstracta; wider occurrence; or a different analysis of the concrete-abstract distinction.

[Grade: metaphysical postulate and signed speculation; metaphysical preference with a stronger signed suspicion among live options; neither choice is compelled by structure.]

3.6 The base has a present without having a clock

The base has causal order and occurrence, and so a time of its own, but no rate or date: nothing exists outside it to compare it with. A performance can be under way without an external timestamp. The base is the first cog only in the sense of primitive ordered transition: its change is procedural, step after step, and there is no clock outside the machine against which its operation is dated.

A single molecule has no temperature. Temperature is a derived quantity that appears only for a population, as an average over many motions. It would nonetheless be wrong to call that molecule motionless. The base stands to physical time as that molecule stands to temperature.

The lack of a rate or date is easily heard as the lack of a present. There is a now at the base in the operative sense: some transition is occurring rather than merely specified, and that is what occurrence is. What there is not is a rate at which it occurs or a date at which it occurs, since each needs a second process to count against and the base has none outside it. Section 3.2 identifies the base's existing with its occurring. The base has a present without having a clock.

What this does not give it is a single global present. Section 2.2's base postulate requires no preferred slicing into simultaneous states, and which transitions count as together is a question for the causal order, not for occurrence itself. Section 11.3's experiential-now belongs to an organised episode. Section 5.4 develops the proposed emergence of temporal measure. Section 2.2's base is the sole non-derivative occurrence within its connected world.

[Grade: conditional argument and real-life counterpart (a single molecule, moving but without a temperature); consequence of MIO within the metaphysical account; the operative base present requires no external time parameter.]

4. Structured nullity and the intelligibility substitution

4.1 The several meanings of nothing

The word "nothing" covers several different targets. Treating them as one can make cosmology and metaphysics look more conclusive than they are.

- **Empty region of space:** spacetime, geometry and physical law remain.
- **Quantum vacuum:** fields, state space, dynamics and fluctuations remain.
- **No classical spacetime:** quantum or pre-geometric structure and laws remain.
- **Empty set or zero object:** mathematical background and identity conditions remain.
- **Empty possible world:** a modal framework and perhaps abstract objects remain.
- **Global nullity:** an internally structured object has a selected valuation equal to zero.
- **Absolute non-being:** no objects, structure, laws, logic, possibilities or mathematics remain.

Only absolute non-being is literal nothing; Section 1.1 explains why it cannot wait or fluctuate. Claims that "nothing is unstable" concern a physical vacuum or symmetric state, not absolute non-being (Wilczek 1980; Carroll 2018). Analytic treatments by van Inwagen and Nozick compare possible worlds or invoke plenitude, which is precisely why a shared modal framework has already entered the discussion (van Inwagen 1996; Nozick 1981). The construction concerns global nullity; Section 4.2 explains the substitution.

[Grade: methodological choice; separating these senses is a discipline rather than a result, and nothing here settles whether absolute non-being is even a coherent condition.]

4.2 The intelligibility substitution

Absolute non-being cannot literally be represented, quantified, compared or assigned a probability without importing a logical or modal background. The moment an equation is written about it, symbols, rules and alternatives have been introduced. Two broad options remain:

1. absolute non-being is a coherent possibility that cannot be mathematically modelled; or
2. absolute non-being is not a coherent possible condition.

Structured nullity adopts a transparent substitution. It studies **global nullity** as a mathematically tractable surrogate for the intuition that nothing remains overall. It does not identify global zero with absolute non-being.

The move comes with two concessions. A critic may reject the substitution as changing the subject. An external metalanguage can also describe a world containing no objects without inserting its own symbols into that world. The substitution's justification is narrower: comparative and probabilistic claims such as "more natural", "more probable" or "would decay" require a shared framework. Global nullity makes one family of those claims precise. The scalar zero and empty set are themselves objects inside structured theories, so no mathematical surrogate is literal non-being.

The resulting claim is this:

> A precise version of "nothing overall" need not entail internal featurelessness.

This removes one inference.

[Grade: methodological choice; global nullity is adopted as a surrogate, and the surrogate leaves absolute non-being exactly where it was found.]

4.3 The finite paired-structure construction

A neutral molecule is not empty. A balanced ledger is not blank. A vector whose coordinates sum to zero need not be the zero vector. Each is full of structure, and each comes to nothing. So a total of zero does not say the interior is empty.

Physics supplies the same lesson in a harder case. A field configuration lying in the zero-eigenvalue sector of some operator can be non-zero at every point, extend across space, and carry long-range correlations; the model discussed in Section 10.3 has such a mode, constant along its extra time direction, with long-range effects on our slice (Pettini 2026). Zero under an operator does not mean empty, and it does not mean inert.

But zero under what? Nothing comes to zero on its own; it comes to zero under some particular way of adding up, and changing the way of adding up changes the answer. Naming that way of adding up is what the rest of this section does, and it is where all the content lies.

Let $\mathcal{X}$ be a class of structured objects and let

$$Q : \mathcal{X} \to A$$

assign each object a global value in an abelian group A. An abelian group is an additive value system with a zero, associative and order-independent addition, and a negative for every value. The integers are the simplest example. The paper calls Q a valuation: a scoring rule, and no more. It earns the stronger name invariant only once some change has been fixed, a symmetry or the system's own dynamics, and the score has been shown to stay the same under it.

The **zero-fibre** is

$$Z_Q = \{X \in \mathcal{X} : Q(X) = 0\}.$$

In plain language, the zero-fibre is everything that weighs nothing on this particular scale, whatever else each object may contain. Objects in Z_Q may differ radically in size, topology, information content, internal relations and computational behaviour. The valuation is generally lossy: it compresses the whole object to one global value. The molecule, the ledger and the vector above are structures mapped to zero by particular valuations, not things identical to the scalar zero.

The correct statement is therefore not $0 \equiv X$, the object identified with the number zero, but

$$Q(X) = 0, \qquad X \text{ internally differentiated.}$$

Section 1.1 separates a zero total, a rule of change and the occurrence of that change. Take any finite relational structure H. Make two disjoint labelled copies, H^+ and H^-. Give matching elements equal and opposite weights $+w_i$ and $-w_i$, drawn from an abelian group of weights. Write

$$P(H, w) = (H^+ \sqcup H^-, w)$$

for the doubled structure together with its weighting. If Q sums the assigned weights, then

$$Q(P(H, w)) = \sum_i w_i + \sum_i (-w_i) = 0.$$

The proof is one line because the cancellation is stipulated by the construction. [Grade: mathematical result.]

Either labelled copy preserves the complete relational pattern of H. Because H can have arbitrary finite complexity, exact global cancellation places no upper bound on internal finite complexity. A global zero does not imply zero internal structure.

The doubling construction is not presented as new mathematics.

The familiar $0 = 1 + (-1)$ is only the simplest instance of a value added to its negative giving zero, $x + (-x) = 0$. It is independent of numeral base and suggests nothing either way about whether reality is built from bits or integers. The paired items could be vectors, fields, functions, graphs, histories or more abstract finite objects. The construction is substrate-neutral; Section 2.3 states the separate physical finitude commitment. Section 4.4 asks whether physics supplies a natural zero; Section 2.7 separately proposes a signed dynamics.

[Grade: mathematical result for the finite nullity construction, separate from Section 9.1's standard conditional recurrence result, and real-life counterpart (a neutral molecule; a balanced ledger); it establishes compatibility between an exact zero total and rich finite content, and it owes to Section 4.4 the question of whether any such zero is ever forced by the world rather than by the person adding up.]

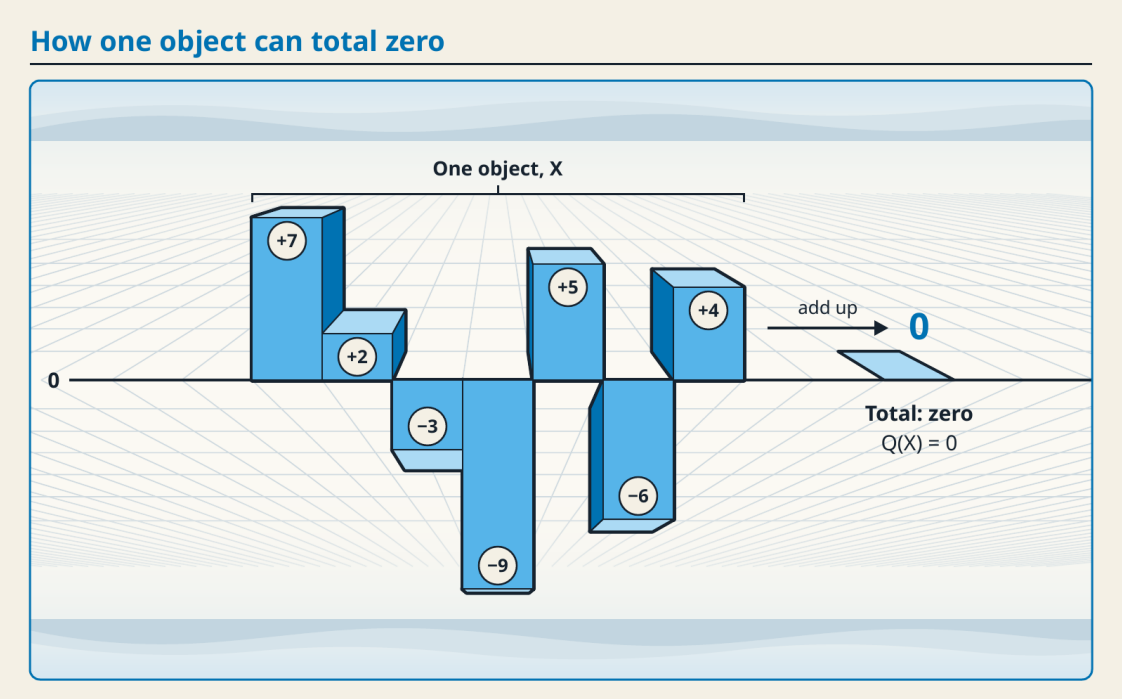


Figure 7: How one object can total zero. Above the zero line is a landscape; below it, the same landscape upside down. One object, X, sits across the line: each of its seven parts rises into the world above or hangs into the world below by its own value. Added up, the values come to zero, $Q(X) = 0$, shown as a flat tile with no height at all, while the object keeps every part. Zero is a total, not a description of the parts.

4.4 When zero would be physically informative

If arbitrary valuations are allowed, every object can be said to have structured nullity. One can define $Q(X) = 0$ for all X, scoring every object zero whatever its structure, or attach a counterweight selected after the fact. Then "zero" says nothing.

All substantive content therefore lies in **invariant selection**. A useful invariant must be:

- non-constant;
- independently motivated;
- preserved by the dynamics;
- not invented solely to force the answer zero;
- applicable to the structures being discussed.

Electric charge supplies a clean physical illustration in two registers. Under suitable global conditions, Gauss's law can force total charge to vanish on a compact boundaryless spatial slice. Observationally, any charge asymmetry per baryon is constrained to be extremely small (Caprini and Ferreira 2005). Neither point establishes a universal ontological ledger. The observational result is consistent with zero rather than a measured exact zero, and the theoretical result depends on its boundary assumptions. A topologically forced zero would also be poor evidence for the contents of the world, because it holds whatever those contents are. It is not worthless for that: a zero forced by the shape of the space or the form of the law is a fact about the container, and it can decide between theories or boundary conditions. What it cannot do is tell us about the contents, and a zero that did would be the one this paper's programme is after. The example shows compatibility only.

Matter-antimatter asymmetry demonstrates that not every attractive cosmic balance closes at zero. General relativity supplies no universally agreed scalar total energy for arbitrary

spacetimes, so the paper does not use the claim that "the universe's total energy is zero" as a premise (Szabados 2009).

Structured nullity establishes a logical space and a research condition, not a measured cosmological fact.

Whether physics supplies an invariant meeting the conditions above is left open. Failure to find one would defeat the cosmological nullity application while leaving FOA's occurrence architecture logically separable. Section 15.6 pursues that natural-invariant test. Section 10.3 distinguishes physical uses of zero, which share the word but not the mathematics.

[Grade: open problem and real-life counterpart (electric charge, conserved, with an observed cosmic total consistent with zero); the conditions on a useful invariant are stated here, and no invariant has been identified whose zero is forced by the contents rather than the container.]

4.5 Closure, representation and unlimited domains

A stronger, detachable picture is that admissible structures come in sign-reversed pairs. Suppose there is an involution J, an operation that returns the original when applied twice, and a non-trivial valuation Q such that

$$J(JX) = X, \qquad Q(JX) = -Q(X).$$

Each two-element orbit $\{X, JX\}$ with $JX \neq X$ is then null, $Q(X) + Q(JX) = 0$. This **Sign-Reversal Closure Hypothesis** captures the intuition that every positive structured contribution has a negative counterpart. A structure that is its own reversal needs separate treatment: its value is its own negative, which forces zero for real-valued scores but not in every value group. In an unlimited domain it defines nullity only orbitwise, pair by pair, and never as a sum of infinities: $+\infty + (-\infty) = 0$ is generally undefined, and the paper never asserts it. Under Finitary Occurrent Actualism (FOA) the domain is schematic rather than a completed collection, so any admissible finite instance may be paired without any finished totality of all instances. The hypothesis concerns abstract structures and does not follow from the finite construction of Section 4.3. Not every structure has a natural negative, attaching formal signs to duplicates may lack independent motivation, and physical relevance would need a natural conjugation supplied independently by the theory. Physics has one for electric charge: charge conjugation, which maps each particle to its antiparticle, reverses charge while leaving mass and energy unchanged. No core argument in the paper depends on the hypothesis, and Section 2.7's signed refinement does not borrow it; the nearest related later material is Section 15.3's formal programme, which would classify natural null invariants arising from independently motivated symmetries. [Grade: metaphysical postulate, detachable and conditional on a natural conjugation existing, and real-life counterpart (charge conjugation, for electric charge).]

Oscillating totals Signed speculation

Nothing forbids structures whose global signed total oscillates through zero, cycling out of balance and back into perfect cancellation. It is called a total rather than an invariant deliberately: a quantity that changes is not conserved, and that is precisely what distinguishes this speculation from the conserved zero of Section 4. The total swings while nothing inside it blinks out. An oscillating electrical circuit does the same on a small scale: the net charge on each plate of its capacitor swings from positive through zero to negative and back, many times a second, and at each zero the plate is still full of charges, exactly balanced. Genuine nothing-phases would need an annihilation event and a creation event per cycle, both refused on the grounds of Section 1, that literal non-being is not a physical state with a duration, boundary or decay mechanism. Such an oscillating total is also not a conserved invariant of the kind Section 4.4, on when zero would be physically informative, requires, so it is a separate possibility rather than a candidate for Section 15.10's natural invariant. A universe that is periodically *nothing in the ledger sense* remains rich at every crossing.

The refusal of true nothing-phases is usually heard as a stipulation. It is an argument. Suppose a cycle really did pass through nothing. Nothing cannot carry elapsed time, so there is no fact about how long the phase lasted. It cannot carry a disposition, so nothing in it is inclined to end. It cannot carry a return law or a probability of re-emergence, because each is a structure and there is none. Whatever leaves such a phase has therefore not been produced by it; it is a beginning, from no state, with nothing that made it due. A cycle cannot swing out through true nothing and back, because nothing carries no way back. That does not dispose of a one-way passage into permanent non-being, which would be produced by the state preceding it and needs a separate argument this box does not offer.

[Grade: signed speculation for the oscillating total and real-life counterpart (an oscillating circuit, whose capacitor plates pass through zero net charge while all their charges stay), standing beside a metaphysical postulate that is itself detachable and conditional on a natural conjugation existing; no core argument of the paper rests on either.]

5. From primitive transitions to the reality stack

5.1 Objects as patterns of transformation

A higher-order object is a stable causal organisation that supports compression, prediction and intervention at its own scale. It can persist while its lower-level members change. The identity conditions need not be merely visual. They may include causal closure, error correction, historical continuity, functional role and characteristic responses to perturbation.

The glider is the cleanest image of process identity. No fixed cell is the glider. Cells turn on and off while a recurring organisation reappears displaced across the lattice. The higher-level object is a pattern in transformation. It can be tracked, predicted and perturbed even though its material membership changes. This is one reason Dennett's notion of a real pattern and Beer's process-centred treatments of Life remain useful here (Dennett 1991; Beer 2004, 2015).

This gives emergence a causal and structural sense. An organism is real because a coordinated process maintains a boundary, metabolises, repairs and reproduces. In each case the higher-order pattern is numerically made of lower-level events while admitting its own useful state space.

The Game of Life is also substrate-promiscuous and self-hosting, and that fact makes the point concrete. Life has been run in electronics and inside other automata, and it can host itself. A formally verified 2025 development supplies a compiler that maps an ordinary Life state into a mega-cell implementation running inside Life itself (Myreen and Carneiro 2025). Community constructions made the same architecture visible much earlier. Brice Due's OTCA metapixel is a single emulated cell built out of two thousand and forty-eight real cells on a side; the metapixel has since been made to emulate itself, and a meta-meta-blinker of period $2 \times 35{,}328^2$, nearly two and a half billion generations of the underlying Life, can be run to completion in about a day under a hashing simulator (Due 2006; Greene 2006; Goucher 2016; the hashing technique is

Gosper 1984; LifeWiki). So a hierarchy of levels can nest inside its own rungs, and demonstrably does.

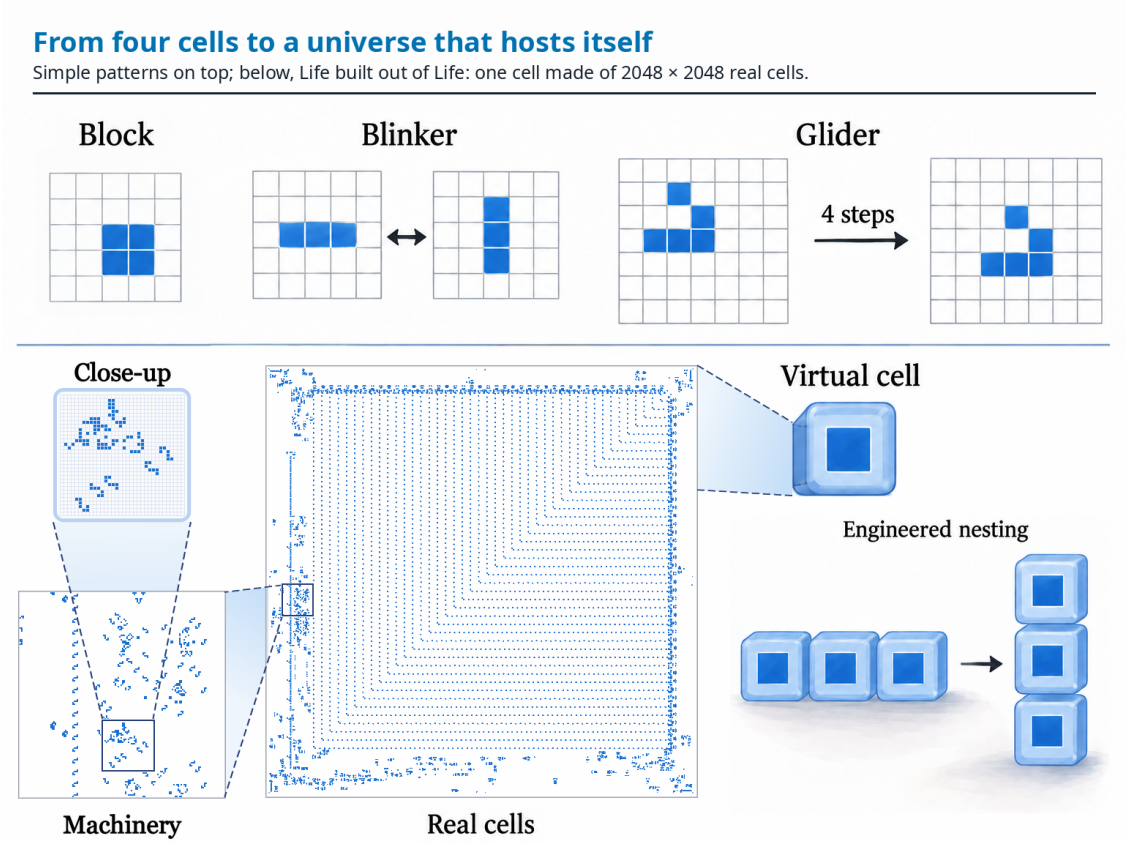


Figure 8: From four cells to a universe that hosts itself. The block, the blinker and the glider give a small vocabulary along the top, and the glider already makes the point: a pattern persists while the set of live cells carrying it changes. The same rule, engineered, hosts itself. The OTCA metapixel is one emulated cell built from 2048 by 2048 real ones, shown whole, then as machinery, then cell by cell. Three metapixels make a blinker again, one level up. The nesting is engineered, not spontaneous: it shows what the rule permits, not what it produces unaided.

Section 3.1's toy-model limit applies: the nesting is engineered rather than spontaneous, and every step of it is extensional. **Nature already builds upwards.** Biology supplies a natural counterpart to the engineered examples. Molecules form membranes, pumps, regulatory circuits, translation machinery and molecular motors. Ribonucleic acid (RNA) sequences constrain protein synthesis through organised biochemical machinery, and motor proteins couple chemical activity to directed movement (Nirenberg and Matthaei 1961; Vale, Reese and Sheetz 1985). No individual component needs to understand the cell; its competence lies in how the interactions are arranged and maintained.

Development makes the construction visible at another scale. A fertilised cell, already containing inherited machinery and supported by an environment, divides, signals and differentiates into tissues and organs. Interacting neurons then support capacities no neuron possesses alone. Development deploys an evolved architecture; it does not rerun its evolutionary history. Molecules sustain cells, cells build nervous systems, nervous systems support minds, and minds build computers that host effective worlds. These are different kinds of dependence, not automatically one formally transitive relation. Automata, biology and computers supply distinct examples of possible, evolved and deliberate construction; Section 6.2 develops the proposed sequence.

Under Finitary Occurrent Actualism (FOA), emergence has two aspects:

1. **structural emergence:** a coarse-grained organisation supports regularities absent from the vocabulary of individual base events;
2. **occurrent inheritance:** the higher-level transitions occur because they are numerically some of the base transitions under a valid coarse-graining.

Section 2.5's parthood supplies occurrent inheritance; structural emergence alone does not establish it. Section 6.1 distinguishes available forms from actual trials; Section 6.2 asks how selection can amplify the latter. The point that a description is not a run has a sharper form, which answers a whole family of arguments at once. Suppose π is normal, as is widely believed though unproved. Then its digits contain every finite string, and therefore every finite encoding of a Game of Life pattern, every finite encoding of a human brain state, and every finite encoding of this paper. It is tempting to conclude that those things are already there and actual inside a number. They are not. Finding a description inside a larger description is evidence about specification, not actuality: it shows the encoding is permitted. A pattern hidden in π is a trace inside a trace, and no depth of nesting turns one into a run.

[Grade: cross-level bridge, formal possibility and real-life counterpart (the biological ladder, from molecules to cells to minds); cross-level account and research programme; the engineered Life constructions and the biological ladder are established results, and the parthood map from higher-level transitions to the base is unresolved.]

5.2 Computation, rules inside states and reflective closure

Rule 110 is a one-dimensional elementary cellular automaton. Each binary cell is updated from itself and its two neighbours. Cook proved that suitable Rule 110 configurations can support universal computation (Cook 2004). The construction is highly arranged and uses extensive background structure, so the result does not say that generic Rule 110 initial conditions evolve into computers. It says that the rule is expressive enough to emulate arbitrary computation when given suitable encodings and resources. For the signed refinement conjecture boxed in Section 2.7 it is also an illustration, showing that local asymmetry can coexist with a simple rule and computational richness, not that signed zero-sum refinement is universal, non-collapsing or physically relevant.

The lesson is important: a short local rule does not imply short or simple consequences. A compact rule can have consequences that no substantially shorter calculation predicts. Bennett's logical depth captures a related distinction between structures that are merely hard to compress and structures that have short descriptions but require long computational histories to generate (Bennett 1988). FOA therefore need not place the complexity of a biosphere into the base rule explicitly. A simple base might support long causal elaboration.

Universality is not the same as evolvability, typicality or actuality. A universal substrate may devote almost all of its accessible states to sterile or disordered behaviour, and reaching its computationally interesting regimes may take arranging rather than waiting. One recent construction makes the separation formal: a one-dimensional cellular automaton that is locally Turing-universal yet cannot sustain non-trivial self-replication under the authors' definition (Cotler, Hongler and Hudcová 2025). The arranger need only be outside that automaton, not outside everything: nothing that could act on it lies outside the base. It could be an intelligence evolved at a lower level, as in Section 6.2's evolutionary cosmogenesis and Section 15.8's downward search. Section 6.1 separates what is abstractly available, dynamically reachable and evolutionarily prevalent.

Binary lattices are not the only way to obtain persistent artificial organisms. Lenia uses continuous-valued fields and smooth convolution kernels to support moving, deformable and resilient patterns (Chan 2019). Flow-Lenia introduces conserved quantities and allows parameters affecting local dynamics to be carried within the evolving field, giving different emergent forms different local behaviours (Plantec et al. 2025). Continuous Game of Life models likewise explore cell-like structures under continuous space, time and resource constraints (Guillet and Jülicher 2025, 2026).

These systems matter for two reasons. First, they show that the emergence bridge is not tied to crisp bits. Second, smooth state spaces can give nearby variants nearby behaviours, broadening the paths by which mutation and selection can explore viable forms. A brittle universal computer and an evolvable ecology are different achievements.

Call the relevant property **substrate evolvability**: the accessibility of viable, heritable and transformable organisations under the substrate's own ordinary variations. It depends not only on what the rules permit, but on the topology of reachable forms, the size of the catchments of evolutionary attraction (how much of the landscape each evolutionary attractor draws in), error tolerance, heredity and the frequency with which useful novelty can be retained.

Continuity is neither necessary nor sufficient for evolvability. Digital systems can have neutral networks, redundancy and error correction. Smooth systems can be chaotic or sterile. The useful distinction is therefore not simply digital versus analogue. It is between substrates whose interesting organisations are isolated miracles and substrates through which viable change can travel. Sections 2.1 and 2.3 permit smooth models while excluding completed physical continua and unlimited physical refinement.

Ordinary automata separate a fixed rule from a changing state. Self-referencing automata weaken that division. In PICARD systems, for example, the current state helps determine the rule used for the next update (Pavlic et al. 2014). More generally, a process can contain symbols, structures or control circuits that alter how later local transitions are selected. A trained neural network is the everyday case: what it knows is held in the weights that shape how it transforms its inputs, and no separate program reads those weights as instructions.

This creates a useful hierarchy:

1. a fixed rule transforms states;
2. states encode parameters that modify local rule expression;
3. organised regions maintain interpreters that treat some state as instructions;
4. those interpreters can themselves be altered, copied and selected.

That hierarchy has a name in computing. Stage three is the stored-program arrangement: instructions held in the same medium as the data they operate on, read by machinery built to interpret them. Stage four is what von Neumann formalised as universal construction, a machine able to build any machine from its description, including itself, which he set out in the theory of self-reproducing automata (von Neumann 1966). Universal construction, rather than universal computation, is the most powerful endpoint of that hierarchy, not a prerequisite for every constructed object: a substrate needs enough constructive capacity for what is built on it. Section 6.2 proposes that evolution can build these constructional capacities.

At the upper stages of this hierarchy, the distinction between rule and state becomes relative to an organisation. DNA is an ordinary biological illustration: a sequence functions as an instruction only inside a cellular system that transcribes, translates, regulates and repairs it. The sequence does not execute itself, and a complete molecular description of it does not supply

the cell that uses it. The illustration supports the limited distinction between specification and carried process. It does not address the complete-description question of Section 3.3, whether a complete description of a finite life differs from the life in anything but its relation to occurrence. Section 3.1's rule-versus-run distinction applies equally to a self-interpreter. A compiler that compiles its own next version, and a brain that models its own way of modelling, are further everyday cases of a system building or modelling its own machinery.

What self-reference may provide is a route from fixed physics to evolving effective laws. If local organisations carry and modify the parameters by which they respond, selection can act not only on bodies but on fragments of their own local dynamics. Stable organisations can constrain local transitions, preserve selected parameters and reproduce environments in which their own dynamics remains viable. This is one route towards the hierarchical evolution considered in Section 6.3's multilevel selection.

A world capable of sustaining organisms cannot remain a sequence of externally interpreted symbols. Some of its regions must maintain the constraints under which their own transformations continue. Biology supplies the clearest examples: membranes regulate exchanges, metabolic networks rebuild components, genomes are interpreted by machinery they help reproduce and nervous systems modify the conditions of their own future activity.

Several traditions approach this property from different directions. Autopoiesis emphasises a network that produces and maintains its own organisation (Maturana and Varela 1980). Rosen's closure to efficient causation asks how components of a system can realise the causal roles that generate one another (Rosen 1991). Pattee's semantic closure concerns the mutual dependence of symbolic constraints and physical dynamics (Pattee 1995/2012). Beer's work on autopoiesis in cellular automata demonstrates how some of these questions can be made precise in toy worlds (Beer 2004, 2015).

Call the general target **reflective operational closure**. A region has it to the extent that:

1. its ongoing transitions maintain the machinery that selects or constrains later transitions;
2. some internal states function as rules or descriptions for that machinery;
3. the interpretation relation is itself realised and repaired within the region;
4. the region can preserve identity through material turnover and perturbation.

Reflective closure is compatible with environmental dependence and does not alter Section 3.1's distinction between description and run. Reflective closure matters for consciousness and evolution. A bare oscillator has transition but little organisation. A reflectively closed system can integrate history, regulate itself and maintain a perspective-dependent boundary. Such properties may be ingredients in a theory of conscious character, although they do not by themselves establish conscious actuality. Section 11.5 asks which integration boundaries could individuate subjects; closure alone does not decide.

[Grade: mathematical result and real-life counterpart (living cells, whose genomes are read by machinery they help reproduce; trained neural networks; a compiler compiling its own next version); established automata and closure results, with substrate evolvability and reflective operational closure as formal research targets that have no unique mathematical criterion; nothing in the subsection explains occurrence.]

5.3 Formal fates and asynchronous organisation

This subsection sorts what a carried process can do over its course, and keeps those behaviours apart from degrees of actuality. Conway's Life supplies the vocabulary.

Life's local rule permits several familiar fates:

- some finite patterns disappear;
- some settle into unchanged configurations called still lifes;
- some alternate periodically, such as the blinker;
- some propagate, such as the glider;
- some collide, transform and emit further patterns;
- engineered configurations can perform universal computation and construction (Berlekamp, Conway and Guy 1982; Goucher 2010).

The glider is a finite carried pattern, in Section 2.5's sense, and can be disrupted by the wider field. Section 3.1's automata require a carrier, adjacency and an order before their behaviour is inspected. The examples above support a taxonomy that avoids treating all non-termination as one thing.

1. **Terminating:** activity reaches a state with no permitted successor, or a finite carried pattern is destroyed.
2. **Settled:** transitions remain possible in the wider substrate but the region enters a fixed or dormant organisation.
3. **Periodic:** a fixed finite sequence of states repeats.
4. **Generative:** new non-redundant organisation continues to appear.
5. **Replicating:** organisations produce causally related descendants.
6. **Evolvable:** descendants vary heritably and selection can accumulate functional organisation.

These are behavioural classes, not degrees of actuality, as Section 2.2's distinction requires. Under the level vocabulary of Section 1.5, on the levels of reality, an abstract pattern is outside time, while a settled Life block waits until a glider strikes it and whatever the dynamics can still reach may run again; Section 9.5 takes that up: that the base never settles follows from its having no first stage, and the perfect loop is the author's signed conjecture.

Synchronous cellular automata update every cell at one global generation. That is convenient but physically heavy. It assumes a universal simultaneity relation and an external sequencing convention. Asynchronous automata replace this with local update events. Different cells update at different times or whenever their enabling conditions are met (Fatès 2014).

An asynchronous system still needs an update policy. The policy may be deterministic, stochastic, adversarial or constrained by fairness conditions. Merely deleting the generation counter does not make order emerge for free. It shifts the primitive from a global clock to a partial order of enabled events and actual updates.

Asynchronous models are therefore especially useful for testing whether geometry, causal cones and approximate symmetries can emerge without a master generation. They do not show that our spacetime has this origin, and any physical candidate must recover the empirical constraints of relativity rather than merely resemble a causal network. Section 5.4 develops clocks from local causal order.

[Grade: mathematical result and methodological choice; established mathematical and computational results and a conceptual taxonomy of formal fates, without an ontological conclusion; asynchronous automata are an established model class, and the emergence of geometry from them is a cross-level research programme.]

5.4 Causal order, clocks and physical time

A process supplies an order when some transitions are prerequisites, causes or constraints for others. Let $e_i \prec e_j$ mean that event e_i stands earlier than e_j in the relevant causal order. The relation need not initially assign a real-number date to every event.

It helps to separate three things that the single word "time" carries. The first is **order**: that some events stand earlier than others. The second is **occurrence**: that those events are happening rather than merely being laid out. The third is **temporal measure**: durations, rates and dates. The base has the first two and lacks the third, so it does have a kind of time; what is missing is only the part a clock supplies. The base is not placed inside a time, since that would require a container it does not have. The accurate description is succession without metric.

Temporal measure is the sense that fails here, and it fails for a definite reason. A clock never counts a process on its own. It counts one process against another. Ten seconds means that ten swings of one thing happened while another thing went on, and the swinging thing has been chosen as the standard. A quartz watch is the cleanest modern case: the crystal simply vibrates, an occurring with no timekeeping in it, and everything called telling the time is machinery built on top, counting those vibrations against whatever else is going on. Counting is therefore a comparison between two processes, and it needs both of them. Our own clocks do this from inside, comparing one region of occurrence with another, which is stage four of the ladder below. Nothing performs that comparison for the base taken as a whole, because there is no second process standing alongside it to be compared with.

A rate is a ratio between two processes and never a property of one. That is why the familiar thought that everything is at bottom oscillation, at some frequency, cannot be a claim about the base. Oscillation at a rate is something the base can come to support, once organisation supplies a second process to count against. It is not what the base is.

Physical time looks richer than the base's succession without metric. Its richness is made of restrictions. A duration is a limit on how much can happen between two events. A finite signal speed is a limit on how far an influence can reach. A geometry is a limit on which paths are open. Duration, finite signal speed and geometry each arrive with organisation, and each narrows the field. The base carries none of them, because there is as yet nothing for them to narrow. Take every limit away and none of the happening is lost. What is left is the going on that our clocks are made of.

Physical time can then be approached in six stages, in this order: base occurrence, causal order, records and recurring processes, clocks, comparisons among clocks, and effective spacetime geometry.

The partial order of enabled events that the asynchronous automata of Section 5.3 supply is an input to the second stage of this ladder, not a ready-made metric. The image is a chain of cogs. The first cog is not turned by a clock outside the machine. Its transitions are the primitive order. Later cogs convert that order into stable regularities, memory, duration and geometry. A local clock is an organised process within the chain, not a witness standing beyond it. This is an explanatory direction, not a derivation: clocks presuppose occurring changes and cannot supply occurrence.

The arrow of time Signed speculation

If occurrence is intrinsically ordered (Section 3.6), temporal direction would be inherited from the substrate rather than borrowed entirely from thermodynamics; entropy gradients would explain the arrow's visibility, not its existence.

This approach must answer difficult questions. A causal order alone may be too sparse to recover durations. Different coarse-grainings may generate different effective clocks. Relativistic physics requires local Lorentz symmetry, causal cones and no preferred observable foliation at accessible scales. Quantum field theory uses continuum and infinite-dimensional tools whose finitary recovery is open. The cog chain is therefore a research direction, not an achieved derivation. Section 3.6's base has neither an external rate nor a date; meaningful clock comparisons are internal.

One objection to the claim above, that the base has no external rate and so no temporal measure applies to it, runs as follows. Local clocks are downstream of the base: they are built out of it, driven by it, cogs turned by whatever it is doing. Why can one not then identify the amount of base activity that falls between two local times, and treat that as a derived measure of the base? Not an external clock imposed from outside, but one inherited from within. If that worked, the base would acquire a rate after all, borrowed from its own products. The reply is that the objection asks for a different quantity. A ratio between local ticks and base transitions compares two of the base's products, a clock and the stretch of transitions it is coupled to, and it is well defined only where that coupling is stable. What it yields is a local exchange rate, one product measured against another, which is the internal clock comparison Section 3.6 already allows. It is not a rate of the base, which would need a second process outside the base to count against, and there is none. Nor does it recover a global rate, since nothing guarantees the same ratio elsewhere or at other stages. On this reply, the base is as measurable as Section 3.6 says: internally, and nowhere else.

One objection arrives so reliably that it belongs in the text. A base that never settles sounds like a perpetual-motion machine, and perpetual motion is forbidden. The prohibition concerns work, not motion. What thermodynamics forbids is unlimited *extractable* work: a machine that supplies useful output indefinitely from nothing. It does not forbid motion as such. The distinction is easy to feel and easy to state. Couple a device to a moving system in order to extract work, and the extraction damps the motion; that is why the machines fail. Leave it uncoupled and nothing in the physics stops it. A current set circulating in a superconducting ring shows the point: kept cold, it has been observed to run on for years with no measurable decay. A system already in its ground state has nowhere lower to pay from, so there is no ledger for it to overdraw. Thermodynamics is in any case a law of levels like ours, stated for populations that have a temperature and an entropy; the paper attaches neither to the base's stages (Section 9.4, where the winter point is neither a heat death nor a low-entropy boundary), so the objection borrows a level's bookkeeping and applies it where no ledger has been opened. Section 3.6 makes the same point with a single molecule, which has no temperature and is not therefore motionless.

Newtonian mechanics makes the same point from the other side. Continuing motion is the default there; it is *stopping* that requires a cause. A body left alone does not wind down, and the impression that everything does comes from living among friction and dissipation rather than from any law. Whether the base in fact stays generative rather than settling into a featureless cycle is a separate question, answered in Section 9.5 from the base's having no first stage. The part of claim 3 at work here is only that it has no last stage. What can be said is that endless continuation is not ruled out by thermodynamics, because thermodynamics was never a prohibition on things happening as such.

[Grade: cross-level bridge and real-life counterpart (a quartz watch, a vibration that tells the time only when counted against another process; a persistent current in a superconducting ring, for motion that need not wind down); cross-level metaphysical proposal, claim 12 of the ladder; the recovery of duration, Lorentz symmetry, causal cones and a finitary quantum field theory is owed rather than achieved, and the reply given here to the author's local-exchange-rate objection is provisional.]

5.5 True base, familial base and home reality

Section 1.5, on the levels of reality, distinguishes true base reality, familial base reality and home reality, and that distinction applies to a simulation as much as to anything else, as does Section 11.2's two-factor account, on which structure delivers conscious character and participation in base occurrence supplies conscious actuality. Simulation also illustrates sparse actualisation. A *Minecraft* world seed can specify far more terrain than is loaded at once. Only the chunk near a player may be generated and updated. The ungenerated region is not secretly running. This image is useful for distinguishing finite specification from finite carried occurrence, but it does not imply that our universe uses lazy evaluation or that apparent physical limits are computational optimisations.

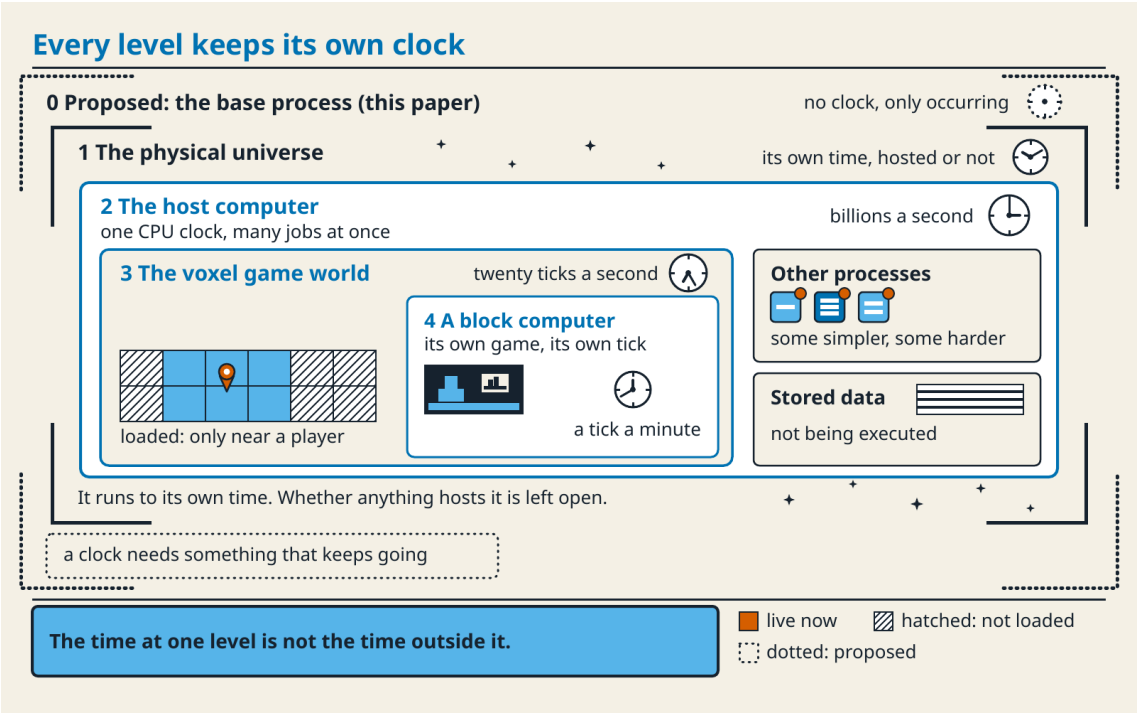


Figure 9: Every level keeps its own clock. A voxel game world ticks at a rate its own program sets, independently of the clock of the computer running it, which is meanwhile running other jobs. Only the outermost band, the base process itself, has no clock. No arrow joins the bands: the only relation drawn is containment.

The kinship case needs its own name. The simulation literature uses *base reality* informally for either the outermost unsimulated level or a simulation's immediate container. This paper separates them. A world's **familial base reality** is the first world of its family: the one whose physics the family's other worlds share or clearly derive from, a reality of the same or a closely related species. Kinship is that shared or derived physics; hosting is a line of worlds running the next. The two usually coincide, the familial base being the root host of its family as well, but hosting is not part of the definition. Either relation can hold without the other.

The relation admits degrees. At one end sits an outright copy of the host's own world. In the middle sits a world with the host's laws but a fresh roll of whatever those laws leave open. Further out sit modified versions: a world with the same laws and different constants, or with more dimensions than its host, or an abstract descendant that keeps only some of the host's structure. Video games are the clearest everyday case: *Minecraft* (Persson 2009), *Grand Theft Auto V* (Rockstar North 2013), *Red Dead Redemption 2* (Rockstar Studios 2018) and most or all games hosted on a platform such as Roblox (Baszucki and Cassel 2006) have a direct and obvious link to the world that built them, however unlike it they look. These are cited as designed works, and credited as each allows: some are substantially one person's design and others a studio's. Ancestor-simulation scenarios (Bostrom 2003) are the middle of this range, not the whole of it.

Reductive, peer and enriched hosted worlds. These names compare a hosted world's effective physics and organisational possibilities with its host's, not their dates or their total stores of matter. A **reductive hosted world** restricts the effective physics or the scope being realised: an ordinary game, a model of airflow around an aircraft, or a limited simulation of how weather develops during El Niño. A narrow simulation is not thereby a complete inhabited universe. A **peer hosted world** retains broadly comparable effective physics, as an ancestor world or a Matrix-like environment might. An **enriched hosted world** supports additional effective possibilities: further spatial dimensions, additional interactions, or a regular magic-like physics. Small ones already exist: games have been built whose worlds have four spatial dimensions, so five including time, running on ordinary computers, and many games give their worlds a magic with rules as regular as physics. The comparison concerns specified features, scale and precision. It is qualitative, and mixed cases need not fit one label cleanly.

For example, imagine an artificial-superintelligence civilisation in a universe like ours around the year 2100; the date belongs to the illustration, not to a forecast. It could host an Earth whose inhabitants call their year 1800 while retaining familiar physics, a school-of-magic world with extra effective rules, and a world with five spatial dimensions. An earlier internal epoch does not make a world reductive, and a represented date is not travel into the host's past. Nor does adding an unusual rule automatically enrich every aspect of a world: a magic-like game might still be reductive overall.

Enrichment is not an increase in the carrier's total capacity. A lower-dimensional host can encode higher-dimensional relations without acquiring another spatial dimension itself, provided the relevant operations are realisable. Life does this already: DNA is a code written along a single line, and through the machinery of the cell and the growth of the body it builds organisms in three dimensions. The extra effective freedom is paid for in organised host activity, storage, precision, delay or restricted scope. A hosted world cannot obtain a genuinely larger total computational resource merely by describing richer laws. It may nevertheless make useful forms of organisation easier to build and sustain locally than they were in the host's native environment.

The boundary of kinship is vague in the middle, and the term is none the worse for it: species are the same, and the clear cases at each end do the work. *Minecraft* shows the relation nested twice over, since its most drastic modifications remain unmistakably its descendants, and the game itself is an abstract but unmistakable descendant of the world that wrote it, hosted there. Kinship survives a great deal of alteration, which is exactly why it is resemblance in enough respects rather than in detail.

Two limits apply to the term. First, a world hosted by a reality is by default in that reality's family, but the link can in principle be stretched until it breaks: something alien enough, sharing nothing of its host's structure, would have a host and an ancestor without having a familial base

reality, since it would have no kin. Second, the relation can fail in two quite different directions, and only one of them is obvious. The obvious one runs downwards: worlds branching from levels below our own familial base reality, and the simulations descending from those, would be expected to be extremely alien, with very different physics, unless something like convergent evolution operates across levels. Given how much about our own universe looks as though it could have been set otherwise, such convergence may be rare.

The less obvious direction runs from a world to what it builds. A simulation hosted here need not inherit anything of ours. The Game of Life and its continuous cousin Lenia are worlds we already host whose rules owe nothing to our physics. One begun from pure mathematics rather than from a model of its host could be wholly unlike our family: a world whose internal theory treats completed infinities as real, say, or one whose effective physics is high-dimensional. Section 2.1's ban on completed infinity also binds hosted worlds. Where such a world is run, its transitions participate in base occurrence; whether its organisation supports conscious character is a separate question. The point here is only that kinship comes apart from hosting and from ancestry in both directions: such a world has a host and a line of parentage, since we built it, and no kin. A host is not automatically a familial base reality, and a world's builders are not a guarantee of resemblance.

Fiction clarifies the difference. In *The Matrix* (Wachowski and Wachowski 1999), the simulation largely copies its host, and Neo can interact with that host once outside. In *Tron* (Lisberger 1982), the inner world has very different physics, yet it is made from our world and recognisably of it. Different physics alone does not break the relation, since a world whose physics derives from ours is still kin; loss of kinship does. Our **home reality** is either the true base or somewhere in such a family tree, and Occurrence theory takes no position on which. On the search reported here, *familial base reality* appears new; *base reality* is established informal usage.

In this paper's sense, Neo's passage between the Matrix and the world running it is **interdimensional**: it runs between different effective worlds. Two worlds are different when neither is a region of the other's space, so that no movement within either leads to the other. A host and a world it runs, sister worlds that share an ancestor without either hosting the other, any world and true base reality, a simulation and the world running it: all are interdimensional pairs, whatever their physics. Two places in one world are not, however far apart, though a journey between them could take an interdimensional route, leaving the world and re-entering it elsewhere. Even a route that looks direct from inside, such as a wormhole or a teleport between two worlds, would run through what carries them: the host itself, for a host and a world it runs; otherwise some level beneath both, perhaps most often their nearest common host, but possibly any, down to true base reality, which carries every world. The transit box in Section 13.4 traces such routes through a computer analogy. The *dimension* in the word is a world, not a direction of space.

Nested simulations could create several familial levels. A local observer may infer only the nearest effective laws, not the true base. Conversely, apparent computational efficiency cannot establish simulation. Physical theories routinely compress regularity without being software.

One structural point is easy to get the wrong way round. Properties do not automatically pass between a hosted world and its host. A simulated world can be reversible while the machine running it is not; it can present outcomes as random that the host computes deterministically; and both can hold the other way round. Reversibility, randomness, conservation and even dimensionality are level-relative. None of these properties of the host follows from a discovery inside the world, which limits what physics done from inside can tell us about what is below.

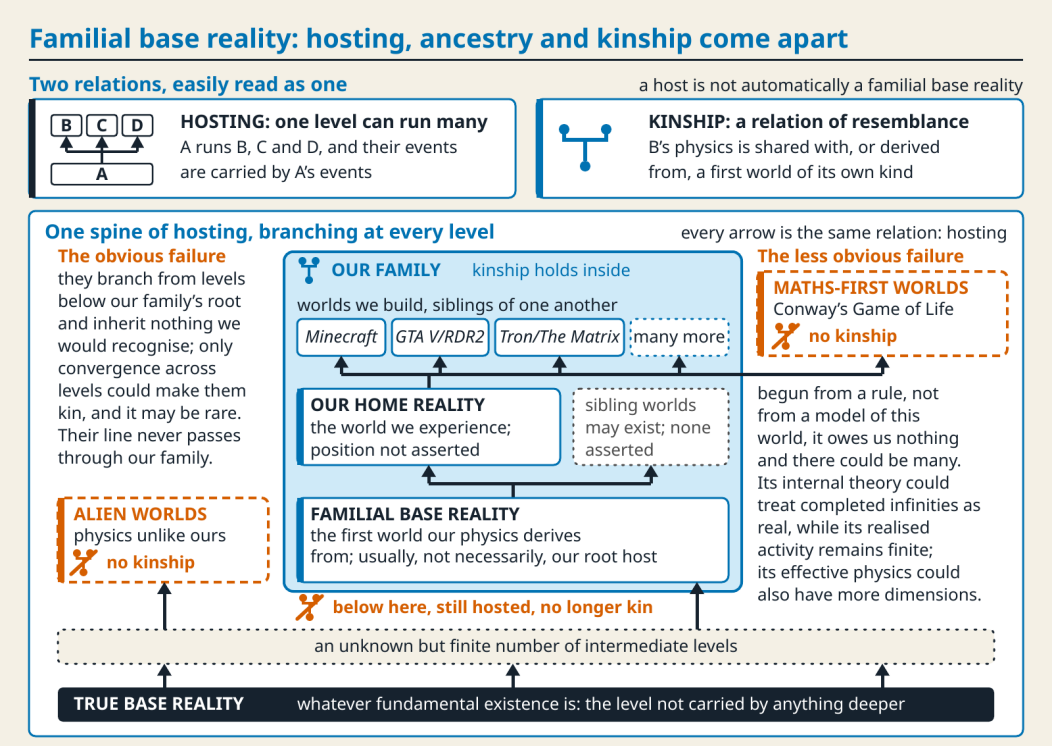


Figure 10: Familial base reality: hosting, ancestry and kinship come apart. Ancestry is descent, a line of parentage; kinship is shared or derived physics. A family's familial base reality is the first world whose physics the family shares or derives from, usually also its root host, though hosting is not required. The two failures, downwards from beneath the family's root and outwards from pure mathematics, are ancestry without kin.

[Grade: formal possibility, cross-level illustration and real-life counterpart (video games and simulations, for hosting, kinship and reductive hosted worlds; four-dimensional games, for enriched ones; the Game of Life, for a world without kin; DNA, a code on a line that builds three-dimensional bodies, for encoding more dimensions than the carrier has); nothing here is a simulation claim or a measure of stack depth.]

5.6 Depth, width and the cost of another level

The level definitions of Section 1.5, on the levels of reality, leave open how many carrying levels lie between home reality and true base reality. The rooted ancestry that runs from the unhosted occurring base up through those levels, which may be emergent, hosted, or both, is the **reality stack**, and this subsection asks how tall it is.

A valid level is not merely a vocabulary. It should support stable variables, characteristic causal relations and a principled coarse-graining to a lower carrier. Two descriptions that cannot be connected by any preservation of causal organisation should not be called levels of one process merely because they resemble each other.

The developmental gradient. On the way up from the base to us, the physics of each level is proposed to be broadly richer than that of the level carrying it, and richer physics generally lets more complex things be built. This belongs to the evolutionary proposal, not to the definition of a level, and it does not require every step to improve on the last. The three hosted-world comparisons in Section 5.5 make the exceptions explicit: a further hosted level may simplify, approximately repeat or enrich its host's effective physics, as the games we build are simpler than the world that runs them.

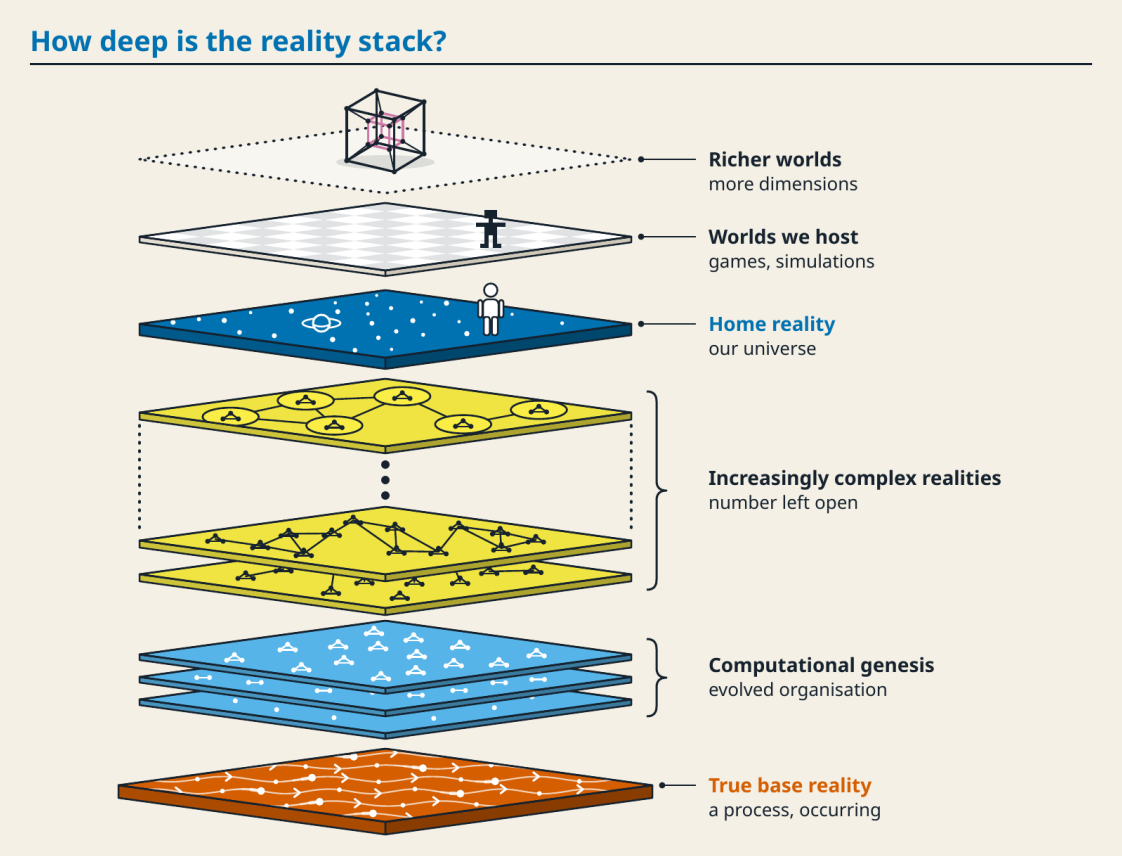


Figure 11: How deep is the reality stack? Levels of emergence separate true base reality, a process occurring, from the universe we appear to live inside: computational genesis, then increasingly complex realities whose number is left open. Above ours lie worlds we host, and potentially richer ones with more dimensions. Height is carrying order: the colours follow the proposed increase in richness on the way up to us, not a rule that every hosted world is richer than its host. The reality tree of Section 9.4 plots effective complexity instead, and there a hosted world may lie below its host.

The developmental gradient Signed speculation

The author expects the gradient also to favour more capable cognitive architectures where the added organisation is useful.

Simple physics can carry an enormous mind. Whatever the gradient favours overall, it does not rank individual minds. Relatively simple rules can support an enormous, highly organised intelligence; Section 5.2's universality example establishes computational possibility given suitable arrangements and finite resources, not the spontaneous evolution of such a mind. A richer physics may make intricate local cognitive machinery easier to build while constraining its usable size, communication or maintenance. A simpler level might instead afford a vastly larger integrated organisation. The author's own example: a physics as simple as a cellular automaton might permit a brain the size of half a planet which, despite far cruder local machinery, could outthink a human. Our physics may allow far more sophisticated operation per cubic centimetre, and biology gives each of us about 1.4 kilograms of such machinery. Two quantities are in play, how sophisticated cognition can be per unit of organised substrate and how much substrate one mind can command, and they can pull opposite ways. Architecture, available resources, learning history and signal delays can reverse the comparison. More substrate alone does not make a better mind, and no measure of intelligence is supplied here.

Richer physics, overlapping reach. A picture, especially a vector image, has far more freedom than a line of letters: colour, line, pattern and direction, where letters offer a fixed alphabet in a row; and letters in turn have more freedom than the two symbols of binary, so each can encode the same content more cheaply than the one below it. In that sense a picture stands to letters, and letters to binary, as a richer physics stands to a simpler one. Yet the ranges overlap: much that can be drawn can also be described in words, and anything written can be spelled out in binary, only at greater length. So richer physics generally makes complex structure easier and more compact to build, but a level's place in the stack does not fix what it can build: a simpler level may build much of what a richer one can, at greater cost.

A reader below us could therefore be far less capable than a human, or vastly more capable, without contradicting the proposed gradient. The wild god-like readers considered in Section 13.5 need not inhabit a more elaborate physics than ours: a vastly larger, well-integrated organisation could do more with simpler local components. Their possible standing concerns the organisation they sustain and the access they possess, not height alone. A host may expose privileged controls to a particular agent without making that agent cleverer, and merely living in the host gives no universal administrative power. Any shared computer shows it: an administrator's powers come from how the machine is configured, not from being cleverer than its other users.

The author suspects that there are relatively few levels between the true base and our universe: something of the order of thirty, and in any case a number a person could hold in mind rather than anything approaching the quasi-infinite. The figure is a guess and is not load-bearing; what matters is its size. The suspicion is that the answer is tens rather than thousands, millions or more.

Three considerations motivate the guess, though none bounds the depth by itself. The intuitive reason is that our world combines relatively compact-looking laws with immense available processing power, which makes a very deep chain seem unnecessary.

The second consideration is the proposed developmental gradient. In a descent along such a route one might strip away matter, then particles, then fields, then spacetime, then geometry, then causal structure, then state, until what is left is something like a distinction and a rule. Each step removes a kind of structure, and there do not seem to be five hundred qualitatively different kinds of structure to remove, which suggests relatively few constructional stages along the route. It is not a bound: a hosted level can repeat the kind of the level below it, as the Game of Life hosting itself does, or simplify it, and natural development need not enrich every step either. A stack of *descriptions* can be extended indefinitely, since anything may always be redescribed as arising from something else; a stack of occurring levels must meet the carrying criterion above at every step. What could bound the number is a least cost per level set against a finite supply, which the paragraphs below sketch.

A third reason points the same way. Much of what makes the world look complicated is width rather than depth. Molecular diversity, tissue types, immune histories, institutions and technologies add many parallel or overlapping organisations, not one new floor per named kind. Some biological and social transitions may establish genuinely new causal individuals; the biological bridge of Section 5.1 (molecules sustaining cells, cells building nervous systems) and the larger and overlapping basins that Section 13.5 keeps as wild speculation allow that. What the criterion blocks is counting diversity itself as vertical depth. Each proposed level must earn its own variables and a principled carrying relation.

One consequence has to be kept separate, because the two are often run together. **Stack depth and amount of computation are different quantities, and one does not read off the other.** A base may perform an astronomical number of transitions to produce one event at the level

above it, so a stack may be shallow while the process carrying it is unimaginably busy. A shallow stack is not a small world.

The cost of a level may also compound rather than accumulate. If one variable at a level needs many coordinated degrees of freedom beneath it, the carrier requirement goes like a product rather than a sum, and a modest multiplier grows quickly: at a hundred lower units for each upper one, twenty levels is an expansion factor of 10^{40}, one followed by forty zeroes. That is an intuition rather than a law, but it shows that a thirty-level world need not be small in any ordinary sense. Section 5.1's metapixel prices one level in millions of lower cells and a matching stretch of lower time, which is a further reason not to imagine hundreds of levels; nature has parallelism, reuse and better encodings available to it, so no universal rate follows.

If the number really is of that order, one consequence is worth stating. Minds are not a tiny ornament sitting infinitesimally far above the bottom of things. On a shallow stack they sit a substantial fraction of the way up it.

A ceiling on useful depth Signed speculation

The author suspects the reality stack has a practical ceiling as well as a floor, and that the two have nothing to do with each other. Nothing in the levels themselves forbids adding another, within the base's finite capacity. But each level might hide the one beneath it imperfectly, delay might accumulate down the chain, and the whole hierarchy could draw on a finite supply. Our own nested machines show the cost: a virtual machine run inside another virtual machine can run far slower than one run directly, though hardware support can recover much of the loss. Beyond some depth the next level plausibly costs more than the capability it buys. That is a claim about what is worth building within that capacity, not the capacity itself, and it is the author's expectation. Section 9.4 finds the same shape within a level: the coherence band, where useful complexity may peak below the maximum.

The collapse above Signed speculation

A further guess. Increasing intelligence may deepen the reality stack for a while and then start flattening it again, because a sufficiently capable system finds compressions that make intermediate levels unnecessary. On that picture the deepest stacks are not the most capable ones, and something very advanced might be strikingly simple at its upper end. Capable engineering already shows the flattening on a small scale: compilers fold layers of a program into one, and systems trained end to end replace pipelines of separate stages. Nothing in this paper needs it and nothing here tests it. Section 9.4 makes the neighbouring point with alphabets: a small one, like the four bases of DNA, can carry rich organisation.

The open task is to test such estimates: to develop candidate coarse-grainings and ask whether the same causal events can be traced through them. A note on vocabulary belongs with this. If the descent above is anything like right, the deeper levels contain no space, no matter, no energy and no time in any ordinary sense, and the word *physical* stops being useful well before the bottom is reached. This paper's word for what is there is *occurring*, which carries none of that freight and names the property the paper's argument rests on.

Even where a descent does simplify the effective physics, its steps need not be intelligible from inside. Abbott's *Flatland* is a useful image of how unfamiliar dimensions can unsettle a world's inhabitants (Abbott 1884), not a model identifying fewer dimensions with a lower host. A host might encode more effective dimensions than it possesses, and fewer dimensions need not mean a less intelligent inhabitant.

No creation event Signed speculation

There is no creation event below the winter point, the base cycle's low of organisation described in Section 9.4; above it, things can be created and disappear again. A level might persist without an identifiable first moment, while another could have a definite formation history. Natural growth does not itself establish beginninglessness: a blindly produced world can begin, just as a deliberately hosted one can. Our own level could be a product of selection, construction or both, and construction can be deliberate, as a game or a simulation is, or like a beehive's, the work of builders that evolved to build. Where a world was made, asking how and by what is a fair causal question; an evolved builder need not have planned it in detail. What neither kind of origin answers is why there is occurrence rather than none. Only the base is postulated here to have no first stage (Sections 2.2 and 14.2).

Physics as a middle floor Signed speculation

Our deepest tested laws may be the effective laws of some level N in a reality stack whose top level is $N + k$, for integers $N \geq 0$ and $k \geq 1$: a completed physics of our level would then be an accurate map of one floor, with the staircase unmentioned. The second bound, at least one level above ours, is not a stipulation. We build hosted worlds, and whatever effective laws such a world runs under, whether ours, some variant of ours, or none of ours, its events are carried by ours. So at least one level above ours exists because we made some; Section 5.5 gives the relation. What the speculation adds is that the same may be true below us, where the producing processes, whether blind, constructed or mixed, are unknown.

One consequence of a layered picture is easy to miss. What dominates at one level of description need not be the coarse-grained image of what dominates below. A gene may be favoured while the organism carrying it is not, a firm may prosper while the industry contracts, a motif may spread through a substrate whose larger structures are being eliminated. Selection, in other words, need not commute with coarse-graining. So the question *what is being selected here* can have a different answer at genes, organisms, ecologies, cultures and universes, and an argument that establishes an answer at one level has not thereby established one at another. Section 6 on evolution is written with that restriction in mind, and the measure Section 7.1 proposes, which counts lifetimes of human-like experience, will have to carry it explicitly; how weights at one level carry to another is one of the two questions it leaves open.

The **reality tree** supplies the branching view of world production, not a map of every subsequent coupling or control link. Its spine traces a selected developmental route, not the only route nature could take; its branches may arise through blind production, evolved construction, deliberate design or mixtures. Parentage does not by itself establish continuing hosting. Section 6.2 places the emergence of prolific world-making in a transition zone rather than at a natural/designed cutoff. Section 9.4 draws that tree against effective complexity and the conjectured coherence band.

[Grade: open problem, research hypothesis, signed speculation and real-life counterpart (the worlds we host, a level above ours that exists because we made it; a beehive, for construction by evolved builders without a plan; a gene or a firm favoured while its organism or industry is not; compilers and end-to-end systems removing intermediate stages, for the collapse above; nested virtual machines, for a ceiling on useful depth): a cross-level question, open item 11 of the ladder; the few-level estimate is the author's signed suspicion, the level criterion is a conceptual proposal rather than a measurement, the depth and width terminology is a methodological choice, the developmental gradient is part of the evolutionary proposal, a research hypothesis on which FOA does not depend, and the boxed conjectures are signed speculation, with no monotonic law of complexity or intelligence claimed.]

5.7 What the reality stack would add to a theory of everything

The reality stack is a central architectural postulate of this paper. It gives the proposed base occurrence a route to finite worlds: organised activity carries further organised activity, with each level's events realised through those below it. How far this extends below home reality remains open.

If our universe has a deeper carrier, the wider reality would include that carrier and the organisations and other worlds it supports. This would be more than extra space beyond our observable universe. A carrier need not sit beside its world in that world's space. Nor need it reproduce that world's layout. A world's objects and laws may be realised through distributed, overlapping organisation in the level that carries them, with no separately stored counterpart of this star or that planet and no boundary around either visible from below, so that neighbours at one level need not be neighbouring components in the next. What the carrying relation requires is that causal organisation be preserved, not that the storage resemble the picture; this is the state-map condition of Section 2.6, on composition across levels and the closure bounds. A trained neural network makes the possibility concrete: a scene, a fact or a rule is held in the geometry of its weights, where nothing shaped like the scene can be found, yet the scene is definite whenever the network is queried. A diffuse encoding is not a diffuse world, and it buys nothing for free: it neither enlarges a finite carrier's capacity (Section 10.2, on bounded information) nor turns a description into a run (Section 3.3, on finite occurrence and complete description). Hosting, ancestry and kinship remain distinct, as Section 5.5 explains through worlds that share a host without sharing its kind.

Two explanatory ambitions follow. A theory complete for home physics would account for every physical law available to inhabitants. A theory of its realisation would also specify what carries those laws, how the descriptions connect, what maintains their working regime and where that regime ends. A complete account of the game need not explain the machine running it. Equally, finding another level below a world would not show that the bottom had been reached.

The proposed route is constructive. Finite relations and constraints could support persistent organisations; those organisations could supply effective variables and regularities; some could maintain or build further levels. Biology and engineered worlds give familiar instances of building upwards. Section 2.6, on composing state maps, supplies a small exact part of the formal route. The computational companion of Section 15.3 demonstrates organisation under a selected rule. No model yet connects organisation-forming dynamics to field-supporting dynamics. Neither the state maps nor the companion derives its own geometry, a measurement rule or an observer.

A useful model would specify its carrier, maps and conditions, then face observations beyond those used to choose it. Agreement with familiar behaviour would establish a candidate realisation; further consequences would make that candidate testable. Such tests would bear on the candidate model, not uniquely on Occurrence theory.

There is an underdetermination limit. If two carriers agree under every admissible preparation and intervention, observations within that domain cannot choose between them. They identify a class of implementations, not a uniquely deepest one. This limits what a theory of realisation could settle without making the question empty. It can still explain how a world is carried and what that explanation leaves open.

[Grade: metaphysical postulate, research hypothesis and real-life counterpart (a trained neural network, which holds a definite scene in weights shaped nothing like it); central architectural postulate and constructive research hypothesis; the extension below home physics remains unestablished; this is not evidence for primitive occurrence, and not a theory of everything; a theory of realisation remains to be built.]

6. Evolution and the production of organised worlds

6.1 Produced distributions and boundary conditions

Section 2.7 distinguishes produced distributions from inherited boundary conditions; evolution supplies one production mechanism. Abstract finite structures do not compete, reproduce or become common. They exist without temporal multiplicity. Evolution acts only where transitions occur and causal descendants can be distinguished.

The base dynamics and its conditions determine which local patterns are produced. Selection then changes the distribution of those produced patterns. Coherence settles neither.

A configuration may be coherent yet unreachable from the actual region. It may be reachable only through an astronomically narrow path. It may arise once but fail to reproduce. Or it may seed a lineage that fills a large region of process. These are different explanatory statuses. The distinction can be put sharply: being describable, being reachable by the dynamics, and becoming common through evolution are three different things. Evolution changes frequency, not what is describable.

One further distinction belongs with these, because the middle term, being reachable, hides two things: whether the dynamics can reach a structure at all, and how many actual attempts at it the process affords. A base may be rich enough to afford very many actual attempts at a given structure, or sparse enough to afford few, and that is a fact about the occurring process rather than about what is merely describable. Call this the base's **abundance**. Selection needs abundance: what it works on is a supply of actual trials, and no amount of abstract availability supplies one. Bare abundance buys a lottery ticket; selection buys a factory, and factories make factories.

One floor is not in doubt, and it is observational rather than postulated. Whatever the base is, it has already afforded enough to produce everything we know exists, this paper and its reader included; on this paper's hypothesis, that production was selection. That lower bound is established by the fact that we are here to state it, and it is all the argument below needs. Section 9 develops the author's further signed cycle conjecture.

Selection is the author's preferred explanation, and the reason is on display throughout this section: a mechanism that produces a distribution, and then produces machinery that produces further distributions, explains something that a bare list never can. What this paper resists is a *shortlist that nothing produced*: an assertion that these things are actualised and those are not, with no dynamics behind the division. The distinction is worth a pair of names. **Earned selection** is a distribution generated by a process: differential continuation, ancestry, competition, attractors. **Brute selection** is a distribution asserted as a fact about which items are picked. The first supplies a process explanation; the second states the distribution without one.

This also answers an argument about selection. It ran: a plenitude needs no selector, therefore whatever it covers must be uniform. But *no external chooser* and *every item has the property* are different claims, and the second does not follow from the first. Only some integers are prime, and nothing chose them; the non-uniform distribution is fixed by what the numbers are. A selective view need contain no chooser and no act of picking, so the anti-selector argument does not transfer.

Put positively, differential continuation is closer to arithmetic than to gravity. It is not a separate force switched on at some point in a world's history, and there is no moment at which selection begins. Wherever variants leave unequal numbers of continuations, their proportions change. That is a consequence of counting, not an additional law, which is why it can be expected

wherever the counting conditions hold and why nothing has to be added to a world to make it apply.

That is why this paper takes selection to be more fundamental than forces like gravity. Gravity need not exist in the simpler levels further down the reality stack, and a base rule may carry nothing that looks like a force at all; but wherever anything runs, varies, and some variants continue more than others, differential continuation applies. Selection has long been generalised beyond biology (Price 1995; Wong et al. 2023). Selection due to occurrence itself, within one occurring base, and the cosmological emergence of levels of reality one above another that can follow from it, is where this paper departs, and it is what lets the paper ask for one occurrence where plenitude asks for every structure.

An image shows how a productive lineage could start inside an overwhelmingly unproductive field. Most of a genome does nothing in the way genes do; the functional parts are rare arrangements within a great deal that is not. Working genes can even arise out of that background: stretches of DNA that coded for nothing have become new genes, which biologists call de novo gene birth (Carvunis et al. 2012). A comparable picture is available here: most coherent structure is inert, and the interesting question is whether the rare pieces that are not inert can bind, copy and interact, because a lineage needs only that. The image locates the difficulty in the transition from isolated coherence to something with descendants, rather than in the abundance of coherence itself.

The alternative to selection then explains less. A universe can certainly lack selection: put finished organisms into its initial conditions, or write them into its laws. Nothing is inconsistent about that. But nothing is explained either. The adaptive complexity has simply been relocated into a primitive, and a primitive containing an eye or a ribosome is a far larger thing to accept than one containing a rule. This is the same complaint the paper makes about brute selection, applied to the initial conditions instead of to the distribution.

[Grade: methodological choice and real-life counterpart (selection generalised beyond biology; new genes arising from non-coding DNA, for a productive lineage starting in an unproductive field); conceptual clarification, with the author's default, that organisation above the base is built rather than found, carried as a signed suspicion; not a guarantee of observers or their distribution.]

6.2 Mathematical abiogenesis, computational genesis and evolutionary cosmogenesis

Section 6.1's distinctions separate one standing foundation and two developmental regimes. The foundation is **mathematical abiogenesis**. It is not a first era before selection begins, because differential continuation applies wherever variants continue unequally; it is the standing fact that there is always enough being tried for selection to work on. Abundance supplies it: a base quasi-infinite enough that the improbable step is always being tried somewhere, so that even where organisation has fallen to its simplest, one hospitable stretch per base cycle is enough for heredity to start again. How much abundance a start needs is a quantity, and Section 15.8, on the stack from the bottom and from above, proposes measuring it in toy substrates. Section 9.4's seasonal cycle is an optional signed conjecture; the foundation needs only a fertile stretch. Section 6.1 supplies the minimum observational floor: whatever the base is, it has already afforded enough to produce everything we know exists.

Never nothing evolving Signed speculation

The author's expectation is stronger than that floor, and it points the other way from scarcity. Selection is known to act on very simple things, and the base is quasi-infinite and full of structure; across a region of that size even random structure is enough for something to be evolving somewhere at every stage. There has never been a stage with nothing evolving anywhere in the base, and abundance is a standing condition rather than a stock that has to be replenished. The distinction that matters here is between abstract availability and a supply of actual attempts: wherever this paper says that structure above the base is built rather than found, the temptation is to let abstract availability quietly do work that only a supply of actual attempts can do.

The first regime is **computational genesis**, selection's own work on machinery. Once heredity exists, rare viable arrangements can become common and organisations can be built from organisations. What accumulates is capability as well as frequency: copying, memory, error correction, repair and interpreters that can themselves be altered, copied and selected (Section 5.2's reflective closure). Section 5.1 compares this with engineered self-hosting. One demonstrates natural construction, the other a deliberately realised hosting architecture. Neither alone establishes the proposed descent below our universe. Computational genesis matters most on a route that begins, or begins again, from sufficiently simple organisation, and it does not stop when worlds appear: machinery goes on evolving beneath them.

The second regime is **evolutionary cosmogenesis**: developed processes producing new effective worlds. Along the proposed developmental spine, this can establish organisation absent from the carrier's simpler constituents, as a brain establishes capacities no isolated atom has. It does not establish capacities the complete carrier cannot realise; the carrier is realising them. Nor must every daughter world be richer. A hosted world may be reductive, peer or enriched in the sense of Section 5.5, while every operation still draws on the resources carrying it. The name is used here for a constructive regime in a larger evolutionary history, not exclusively for an intelligent inhabitant deliberately making a universe.

The producing process may be blind, an evolved world-building organisation, deliberate engineering or some combination. Design itself has an evolutionary history, and a designed environment may subsequently develop through its own selection. The author's expectation is that construction becomes prolific once suitably capable organisations arise, often producing many more tailored worlds than natural world reproduction does. That is a signed expectation about relative prevalence, not a cutoff: natural production need not stop, and intelligence need not be present at every birth. Past that point much, perhaps most, of what evolves may be worlds rather than machinery, a change in predominance rather than a switch. The **world-making transition zone** names the passage in which these capacities become important, rather than a new law separating nature from design.

Cosmological natural selection supplies one speculative natural route: descendant universes with heritable variation could be differentially reproduced (Smolin 1992; Section 6.4, on selection over effective laws). Cosmological artificial selection supplies the intelligence-led neighbour, in which intelligent beings investigate candidate universes and produce offspring (Vidal 2010). Neither establishes ongoing hosting. A world that produces another is its parent; it is a host only if it continues to realise that world's activity. The tree may record either history, but a genealogical step alone does not add a level to the reality stack. Both kinds of branch can themselves produce further worlds.

Only evolutionary cosmogenesis has a known working instance at the scale of a world, and it is ours, though it seems probable that ours is one of a quasi-infinite number. We build hosted worlds, so at least one level above ours exists because we made it. Whether anything below us ever did the same is what Section 15.8 proposes finding out.

An everyday picture of one route from a simple start. Take a slab of bedrock under rain and the turning of the seasons. The rock and the weather stand for the base's occurring, which nothing supplies and nothing stops. The rock has flaws, the weather works on them, and the simplest self-copying chemistry takes hold in the cracks: that is abundance and the first replicators, and selection now has heredity to work with. Soil forms, and in it bacteria, then everything that soil can carry, up to a rainforest whose canopy stops rising at a height the climate sets, however much goes on beneath it: that is the coherence band of Section 9.4. Some of what the forest produces goes back down into the soil and the rock, some takes to the air, and a handful of its creatures build machines and leave for space. Height above the ground is effective complexity, and the forest is the reality tree of Section 9.4 grown in one place.

What computational genesis needs from the foundation is only what selection logically requires, and that is less than is often supposed. Natural selection requires heritable variation, differential persistence or reproduction, and enough stability for consequences to accumulate (Lewontin 1970). It does not logically require fundamental chance. A deterministic process can generate variation through sensitive dependence, pseudorandom dynamics, recombination, environmental interaction and computational irreducibility. Evolution run on computers shows that true randomness is not needed: the random numbers such programs use are made by a fixed rule, yet evolutionary algorithms designed the antennas that NASA's ST5 spacecraft carried into space in 2006 (Hornby, Lohn and Linden 2011). And selection needs very little to work on: in Spiegelman's experiments, RNA molecules copied in a test tube, with no cell around them, evolved transfer by transfer into faster-copying variants (Mills, Peterson and Spiegelman 1967).

A minimal schema is:

$$X_{t+1} = F(X_t),$$

where the update rule F takes the present complete state X_t, which includes organisms, environments and inherited control structures, to the next state. If different local organisations leave different numbers or durations of causally connected descendants, selection can operate even when F is deterministic.

Randomness can still matter practically. It can enlarge exploration, break symmetries and make outcomes difficult to predict. Section 10.2, on the three kinds of randomness, distinguishes epistemic, algorithmic and ontic randomness. The present point is only that the evolutionary argument does not depend on choosing among them.

Section 2.1 shows why duration alone cannot explore a state space; selection also requires retained variation. Interaction drives most of what selection builds, though lineages that never interact can still change in relative frequency if they leave unequal numbers of descendants. Section 2.1's duration warning applies equally to a beginningless process. Section 7.1 proposes the measure that turns produced frequencies into observer typicality.

[Grade: research hypothesis for the evolutionary bridge, and real-life counterpart (biological evolution, from simple chemistry to cells that copy and repair, for computational genesis; our own construction of hosted worlds, for world production; evolutionary algorithms on deterministic computers, for selection without fundamental chance); that selection needs heredity and retained variation rather than chance is an established logical result with open cosmological constraints; the measure the bridge needs is proposed in Section 7.1.]

6.3 Selection, retention and hierarchical replication

Both regimes need selection to act at more than one scale, and there is evidence that it can. Cells form multicellular organisms. Organisms form colonies and symbioses. Cultural systems reproduce structures across brains and artefacts. Artificial-life models can likewise support replicators whose causal organisation spans separated spatial clusters.

Yang's Outlier cellular automaton exhibits self-replicating structures at several scales after the rule itself was found by an external genetic-programming search (Yang 2025). Causal-lineage analysis suggests that some effective individuals are distributed and that their ancestry can fork and merge (Hintze and Bohm 2026). Flow-Lenia allows local dynamical parameters to travel and mix with emergent forms, giving selection access to features of local rule expression (Plantec et al. 2025). Section 5.2 distinguishes permitted constructions from organisations the substrate can discover and retain.

The second is much harder and more relevant to cosmology, and it is the research target that section set: robust organisation discoverable under broad conditions, not one hand-placed configuration. A promising model would show a ladder such as:

1. primitive local interactions produce persistent motifs;
2. motifs cooperate into bounded or causally closed individuals;
3. individuals reproduce with imperfect heredity;
4. descendants modify local constraints or rule expression;
5. new levels of individuality become selectable;
6. the process remains non-collapsing across broad finite windows.

That ladder is Section 6.2's two regimes in the form a simulation could exhibit. Steps one to five are computational genesis: the accumulation of copying, heredity, modifiable rule expression and new levels of individuality, as when single cells band together into one organism that selection then acts on as a whole. Evolutionary cosmogenesis begins where these organisations produce, vary or evolve new effective worlds. Distributed ancestry can fork and merge; the signed refinement of Section 2.7, whose configurations split and recombine without end, supplies one candidate network.

The ladder identifies formal milestones by which a bare local process could become an ecology of higher-level processes.

This subsection could be read as saying that copying is the only way an improvement can last. It is one way, but not the only one. For improvements to build up, each one has to outlast the event that produced it, and that takes six things: a way of storing information, a source of new variants, a way of keeping the ones that are not thrown away, repeated rounds in which some variants continue and others do not, parts that can combine so that one improvement adds to another instead of replacing it, and enough resources to pay for all of it. Copying readily provides most of them, which is why it is everywhere. Other things can provide them too: a structure that is repaired rather than rebuilt, or a pattern that keeps going while its material is replaced. An adult who learns a skill is the everyday case: the improvement stays in the same person, and nothing is copied. What matters is that the gain lasts, not that whatever carries it is copied.

[Grade: research hypothesis, physical conjecture and real-life counterpart (selection at several scales, as cells form organisms, organisms form colonies and symbioses, and cultures copy structures across brains; an adult keeping a learned skill, for a gain kept without copying); simulation programme and physical conjecture; the milestones are formal. Evolution reaches the content the base carries by definition; nothing here shows that it reaches the base's own rule. That our own laws were selected, or built (Section 6.4), is part of the hypothesis: neither it nor its contrary is shown, and their fine-tuned look is what the hypothesis would lead one to expect.]

6.4 Effective laws and built coherence

Section 5.2 proposes selected local constraints as a route to history-dependent effective laws within Section 2.4's base constraint. Several mechanisms can stabilise effective laws:

- dynamical attractors can make many microstates converge on the same macro-behaviour;
- error correction can preserve patterns against local variation;
- self-maintaining systems can rebuild the conditions of their own operation;
- reproduction can multiply successful organisations;
- selection can favour lineages whose internal rule expression remains viable;
- niche construction can make descendants inherit structured environments as well as internal states.

This opens an analogy with cosmological natural selection, in which universes producing many black holes have more descendants and parameters can be selected across generations (Smolin 1992). Occurrence theory does not depend on black-hole reproduction or on selection operating at the scale of whole visible universes. The general motif is that reproducing processes can make some effective law-regimes common. Similar motifs could recur at levels far below those at which black holes are visible.

Two further remarks, offered as research images rather than results. The first makes the effective-law claim concrete from the inside. Imagine scientists constituted out of higher-order motifs in a rewrite world: they would find particle species, collision laws, signal speeds, conserved quantities and perhaps a number of dimensions, and every one of those would be a real regularity truly discovered. They would also all be effective consequences of one microscopic rule that mentions none of them. Physics has a familiar case of emergence like this: the gas laws, which link pressure, volume and temperature, emerge from the motion of countless molecules that obey quite different laws, and they were found and used long before the existence of molecules was established. Nothing in their situation would flag the difference, which is the uncomfortable half of the picture.

The second concerns the order of construction. It is natural to picture a universe as one enormous structure appearing complete, with laws written in at the outset. An alternative worth exploring is closer to chemistry than to physics: primitive motifs bind into stable composites, composites into larger ones, and the effective objects and regularities of those composites become an internal physics. On that picture universe formation is compositional ecology, and what we call fundamental law is the chemistry of whatever bound first. Section 15.8 proposes testing this conjectured emergence of effective laws.

The black-hole proposal makes the genealogical distinction concrete. If universes reproduce through black holes and offspring inherit somewhat varied parameters, differential reproduction could favour parameters associated with more black holes (Smolin 1992, 1994). Natural branches could then continue beside designed ones. Altering parameters does not by itself enrich effective physics, and selection for reproduction is not selection for maximum complexity or intelligence. More radically different or enriched descendants remain

further possibilities, not consequences of Smolin's proposal. Nor does a daughter universe thereby become a simulation maintained by its parent. Parentage, retained resemblance and continuing realisation are three questions; only the last fixes hosting depth.

Two further possibilities follow from taking effective laws seriously, and both are speculative in the way this subsection's conjecture already is.

The first pushes the attractor idea up to fundamental physics itself. Suppose one conditions an ensemble of candidate causal rules on the things any world containing observers must support: stable records, enough locality for a signal to mean something, induction that usually works, and processes able to maintain themselves. Many rules fail those conditions immediately. What survives is not a single imposed law but a set of dominant, quasi-stationary modes, and a resident of such a world would experience the dominant mode as simple law. On this reading the apparent fine-tuning of physical law is a selection effect over rules rather than over parameters, and no unique law needs to be laid down at the outset. Selection over rules, if it operates at all, operates on naturally produced worlds and says nothing against a designed one. Either route remains open for home reality and for its familial base reality (Section 6.2, where the producing process may be blind or deliberate, and Section 9.4's reality tree), and the author's own lean, a suspicion, is that both were somewhat more likely built than selected. Selection over rules would be a large claim, and it is recorded as a direction rather than a result.

The second possibility is a difficulty that selection over rules must face. Variation machinery does evolve: in long-running experiments with the bacterium *E. coli*, lineages arose whose mutation rates were many times higher than their ancestors' (Sniegowski, Gerrish and Lenski 1997). If mutation rules, copying fidelity and the criteria of success can themselves evolve, then some higher-order rule seems to be fixed, governing how the lower ones change. Push that up and the regress repeats. Two escapes look available and neither is developed here. One is a self-consistent mathematical fixed point: a level at which the rule governing change is the same rule that change produces, so there is nowhere further to go. The other is to treat selection as acting on *measure* rather than on existence, in which case nothing is being chosen at any level and the regress never starts. The second sits more comfortably with Occurrence theory, and on a base that cycles such a measure is available: the share of one full cycle that each kind of organisation occupies (Section 9.6, on what recurrence would supply).

[Grade: physical conjecture and real-life counterpart (attractors, error correction, self-maintenance, reproduction, niche construction; the gas laws, emerging from molecular motion, for effective laws a deeper rule does not mention; mutation rates evolving in bacteria, for selection over the variation machinery); established stabilising mechanisms extended by a physical and cosmological conjecture; selection over rules is a research direction on thin evidence, leaning on the cycle measure of Section 9.6.]

6.5 Uniceptals, recurring motifs across systems

The selective pattern itself is not peculiar to biology. Variation, filtering and retention appear together in immune repertoires, in cultural transmission, in technological development, in markets, in scientific practice and inside cognition, which is why a mind can be described as running the same loop internally so the organism does not have to die for each bad guess. That recurrence is a shared abstract pattern; by itself it establishes neither a single underlying mechanism nor equivalence between theories.

Motifs like the selective loop point towards what the author calls a uniceptal, though a bare loop is probably too simple and too common to earn the name: a conceptual motif, ordinarily less elementary than that, which recurs as an attractor across unrelated systems, of the kind this section has described for organisation. A uniceptal may be simple or complex, and it needs

no mind or observer to interpret it: a reading head, a projector and a ribosome are the same uniceptal. That is a signed suspicion. Entry 13 of the signed suspicions in the claims ladder of Appendix A states it, and records that uniceptals are the subject of the author's separate and long-running, but unpublished as yet, research programme.

Carrying a uniceptal across Signed speculation

The idea of the uniceptal is simple to state. The same pattern of organisation turns up again and again in things made of entirely different stuff, because the same problems, met in different materials, keep being solved in the same few ways: the reading head, the projector and the ribosome above are one uniceptal, found three times.

Such patterns can be put to work. A pattern understood in one domain can be carried into another where less is known, and what it does in the first becomes a hypothesis about the second, a definite guess that says what to look for and where. Much of this paper is built that way. Selection, watched at work in living things, is carried down to the base as a builder of worlds. A game running inside a computer is carried across to levels of reality, one running inside another. A player steering a character is carried across to how something at one level might take part in another. The author suspects that, carried with care, such patterns are among the best tools there are for guessing how existence might work.

[Grade: real-life counterpart (immune repertoires, cultural transmission, technology, markets, science, cognition) and signed speculation; the recurrence of the selective pattern across materials is observed, the uniceptal conjecture is the author's, and the observed cases do not by themselves establish recurrence at other levels.]

6.6 Other bases and possible convergence

Section 2.2 leaves other disconnected bases open; they would lie outside our hosting ancestry. The author's convergence suspicion assumes Section 2.3's bounded base. If other bases occur, the test for being one is severe and follows from Finitary Occurrent Actualism (FOA): a base must occur, must be finite at every stage with a ceiling on its distinguishable structure, and must be able to continue without end, which excludes most candidate structures and gets harder the more complex a structure is. Bases able to carry consciousness like ours would therefore be rare, would face selection as any occurring process does, and would likely converge, so that another such base could well be expected to be more like ours than unlike it, as we have witnessed so much convergent evolution over Earth's evolutionary history. Camera eyes evolved separately in vertebrates and in octopuses, and echolocation in bats and in dolphins. The paper has evidence for exactly one, and whether their occurring is one or many is the question of Section 2.2, on primitive base occurrence and the question of number. The selection meant is selection within a base, where variants are retained and continue unequally; it needs no competition between disconnected bases. The comparisons would need a measure across bases, which counting within one base's cycle does not supply, and a count of bases, which is unresolved.

[Grade: signed speculation and real-life counterpart (convergent evolution over Earth's history); the paper has evidence for exactly one base, convergence among any others is the author's expectation, not shown, and the comparison would need a measure across bases.]

7. Process measure, novelty and typicality

7.1 What a process measure must count

Produced frequency needs an observer-weighting rule before it becomes typicality. An **observer-episode** here is a finite causally integrated stretch of observer-supporting process, not a durationless instant, so no measure may presuppose a smallest tick.

A measure needs a window to count within, and in an infinite universe that is where the difficulty starts: set one infinity against another and the ratio depends on the order of counting. That is the measure problem as cosmologists meet it in eternal inflation, where the universe keeps budding off new regions for ever. The bounded base removes that part of the problem. Strict potentialism (Section 2.1) admits no completed infinite history to total, and none is needed, since the base is finite at every stage (Section 2.3). If the base repeats exactly, as Section 9.5's preferred perfect loop has it, one full cycle is the window: a finite whole containing everything the base does, and a canonical window rather than a chosen cutoff, since every whole number of cycles gives the same ratios (Section 9.6). A count within it has a definite answer, even though nothing inside the cycle could carry it out. If the base does not repeat exactly, the same count is taken over finite windows of increasing length, and whether the ratios settle is then a real result rather than something hidden inside an infinite sum.

Within that window this paper proposes a measure. It answers four of the six counting questions, reference class, discounting, identity and multiplicity, and leaves two open.

Reference class: what is counted. The count is of lifetimes of human-like experience. A life does not need to be able to ask "why am I me, now?" in order to be actual.

Discounting: how a life is weighted. A longer, richer or more intense lifetime counts for more than a brief or faint one, so a momentary accidental observer, however coherent its moment, weighs little against a long lawful life.

Identity and multiplicity: recurrences and copies. A life that comes round again as the cycle repeats is not a new count: counting within one cycle counts it once, and on the signed identity condition of Section 13.2 it is the same life again, probably exactly. Like a car restored to exactly its old state, with no odometer to say how many laps it has done, an exact return carries no mark of which time round it is. Whether a copy is the same person again or one more is a matter of degree. Identical twins mark one end of that range: they begin as a single fertilised egg and grow into two people. It depends on how faithful the copy is, and on the connection that spawned it: a copy that grew while joined to its original by a two-way flow of information, an umbilical cord of sorts, could spawn, grow, mature and orient itself better, and the thicker that link and the longer it lasted, the better the copy. If there are Reikons (the deeper, longer-lived identities that Section 12 conjectures), such a link may also make a copy easier for a Reikon to follow.

Overlap and level: what stays open. How to count nested and overlapping organisations without double-counting is open, and so is when weights at one level of description keep their ratios at another, though the window itself leaves no level out, since every level is carried by the base. A brain, a brain with its blood supply and a brain in its room are three nested processes that share the same thinking. One tension also remains that any measure faces: it should be invariant under harmless recoding, yet it must choose something to weight.

Lives, not moments. This count departs from the best-known form of anthropic reasoning, and the departure should be plain. Bostrom's original rule has an observer reason as if it were a random sample of all the observers in its reference class; he later refined it to observer-moments, each moment of experience counted as its own sample (Bostrom 2002).

The measure here keeps the earlier unit, the whole life, for a reason internal to Occurrence theory: a life is one occurring process, and slicing it into moments imposes divisions the occurring does not have. Counting moments sits most naturally with a picture on which nothing joins one moment of a life to the next, the static picture this paper argues against, and on that picture a momentary Boltzmann observer counts exactly as much as a moment of a long life. Weighting each life by its length and richness keeps most of what moment-counting was for, since a longer life holds more moments, without resorting to artificial episodes of experience. Where in a life one finds oneself is then a question about the episode being undergone (Sections 8.2 and 11.3, which separate the chronology a life represents from the episode actually being undergone), not a sampling problem.

Where a counted life begins and ends Signed speculation

A counted life need not begin at conception or end at final brain death. If Reikons exist, attachment could in principle begin at any time and end at any time. The author's working assumption, for several reasons, among them the possibility that evolution has come to capture a Reikon, is that once a Reikon is attached it stays attached until the brain breaks down to the point of failure; if instead the bond is made of information, the joining and the parting may be more of a fade in and a fade out. Either way the unit stays a life, not an episode cut from one.

What a measure would buy Signed speculation

A completed process measure would be a scientific tool in its own right: a principled way to ask "how much of reality is like this?", applicable to simulation ensembles, anthropic reasoning and cosmological model selection (Bostrom 2002; Srednicki and Hartle 2010).

Section 9.6 develops the cycle window; Section 11 addresses conscious character, and so where human-like experience begins; Section 13.2 separates the count from its signed condition for personal continuation. What Occurrence theory does not yet have is the exact relation connecting local organisation, how long it runs, and how much observer weight it carries, an equation with occurrence on one side; the answers above fix its direction, longer and richer counting for more, not its form.

Any such relation would have to keep apart five things that are easy to run together: what exists, how the possibilities are weighted before any dynamics, how transitions carry weight forward, how many continuations an item has, and where an observer should expect to find itself among them. "All structures exist" answers the first and is silent on the other four, which is why it never yields a prediction on its own. Section 5.2's logical depth suggests weighting the work of reaching a structure; that is a measure proposal, not a result.

The pieces are individually familiar; combining them is the hard part. Four ingredients recur in the neighbouring literatures: a prior weight over candidate structures, of the algorithmic kind discussed just above; a transition kernel saying how weight moves as a process runs; a branching or copy factor where continuations multiply; and a normalisation step locating an observer among the continuations. Assemble them one way and the result reads as a selection equation; another and it is a Bayesian update; another and it is branching cosmology; another and it is observer self-location. That they are so nearly the same object is either a deep hint or an artefact of the notation, and which it is remains open. Naming the four is not supplying the equation. It does say what would have to be reconciled by anyone attempting it.

One warning belongs with it. It is tempting to look for a quantity that such a process monotonically increases, on the model of entropy or of fitness. There probably is none. Where success depends on what everything else is doing, where resources deplete, and where the environment is altered by the very processes being weighted, mean persistence can fall

as readily as rise. The safest available result is conditional: a reweighting under specified dynamics, not an arrow. A framework that promised an arrow would be promising more than evolutionary theory itself can deliver.

[Grade: methodological choice on a conditional mathematical result, and real-life counterpart (identical twins, one origin and two people); one full base cycle is the window if the base repeats exactly (Section 9.6), and what is counted and how it is weighted are proposed answers; overlap, level and the exact form of the weighting remain open.]

7.2 Activity, novelty, generativity and adaptive activity

Different kinds of process extent should not be conflated.

- **Activity** is the amount of transformation.
- **Novelty** is non-redundant structure relative to a declared description scheme.
- **Generativity** is the capacity to enable further organised process.
- **Adaptive activity** is organised persistence or novelty retained through differential success. The artificial-life literature has measured something close to this for decades, with evolutionary-activity statistics (Bedau and Brown 1999).

A two-state cycle can have unbounded activity and negligible novelty. A finite random-looking trace can have high incompressibility and no self-maintenance. A growing crystal can produce enormous regular structure with little adaptive activity. An evolving ecology may combine all four.

The distinctions matter because identical patterns do not literally cancel unless they carry opposite values under an invariant. Several other operations are often described loosely as cancellation: counting identical items only once, compressing a description, giving repeated events less weight, and finding little new structure. Each differs from cancelling opposite signed values and from the others.

Each can be legitimate for a particular measure, but each needs a reason. If repeated observer-episodes are counted separately, a large replicating lineage gains weight. If exact repetitions are quotiented out completely, reproduction may become invisible. A defensible measure may need both event multiplicity and an organisation-sensitive correction rather than choosing one exclusively.

This is also where causal lineage becomes important. Two identical patterns produced independently need not be the same process. Two materially different stages joined by reproduction may belong to one evolving lineage. Identity should follow causal organisation, not visual similarity alone.

Two further distinctions mark places where the vocabulary of this section is easily misapplied.

One separates two mechanisms that both produce structure but are not the same. Self-organisation makes order: convection cells, crystals, vortices, patterns that appear wherever the conditions hold and disappear when they stop. Selection accumulates fit: arrangements that persist because of what they do, retained across generations of whatever is doing the persisting. Thermodynamics supplies the river, self-organisation makes the eddies, and selection builds the watermills that get progressively better at using the flow. A framework that conflates them will credit selection with order it did not produce, or credit spontaneous pattern with cumulative improvement it cannot achieve.

The other is the sharpest and the most awkward for any formal treatment. A fixed matrix of types and rates presupposes that the types are known in advance. Open-ended evolution does not work that way: it creates genes that did not exist, roles that nothing previously occupied,

functions with no prior name, and eventually effective regularities that amount to new local laws. The space of possibilities grows as new levels with new physics are built, though never, if the base cycles, beyond what can happen within one cycle; and if something holds complexity within a band, as the Eukrasian attractor of Section 9.3 would, a point may come beyond which no further levels are added. Formalisms that assume a fixed list of kinds therefore model only the easier case; something closer to graph rewriting or a category-theoretic treatment may be needed for the harder one. Section 15.5's measure programme still owes a treatment of newly created types.

[Grade: methodological choice and real-life counterpart (a growing crystal; convection cells; an evolving ecology); conceptual distinctions intended to discipline future measures.]

7.3 From production to observer typicality

Why should an observer find a deeply regular world rather than a brief accident containing false memories of regularity? This is the observer-typicality or White Rabbit problem in its general form. Finitary Occurrent Actualism (FOA) removes one source of multiplication by refusing a running counterpart for every abstract structure. It does not remove accidental observers that the one base process itself may produce.

On a purely static ontology where ours is one structure among all structures, agreeing with a lawful past constrains a world's contents while failing to agree leaves them free, so there are far more ways to fail than to agree, and a coherent-looking world appears exceptional. That is a claim about weighting, not a theorem: a simplicity prior can favour a short law generating a long coherent history over an incompressible fake record, so the comparison depends on how the ensemble is weighted, and for an infinite ensemble there is no agreed way to do that. An unrestricted static ensemble therefore inherits the White Rabbit and Boltzmann-observer problem without dynamics to explain why an observer meets a coherent world.

No white rabbits Signed speculation

The author sees the bounded base as a relief here. A quasi-infinite base is as good as infinite for most purposes that matter to us, but it is not infinite, and that matters in two ways. Unlike a truly infinite plenitude, it need not contain every strange thing that can be imagined, so not every bizarre history or experience is guaranteed to happen somewhere. And most experience is then likely to be somewhat coherent, so a present sense of existence like ours would be typical rather than infinitely rare. Within that coherence a life might meet the full range of common and uncommon experience, but not the exceptionally bizarre, which should come as a relief to anyone who believes in plenitude or in the many worlds of quantum theory.

This is not an objection invented here, and it is not one a static ontology has ignored. Tegmark treats it as an outstanding difficulty for a mathematical ensemble, and separately proposes restricting the ensemble to computable and decidable structures to ease the measure problem, a proposal with a stated price: a restriction on which structures are admitted narrows the ensemble, and an unrestricted plenitude was recommended partly by not needing one (Tegmark 2008; compare the algorithmic weightings of Schmidhuber 2000). Section 6.1 distinguishes an admission principle from a chooser: only some integers are prime, and nothing chose them. What the restriction does cost is the claim to need no principle of admission at all. The difficulty is therefore live rather than settled, and it belongs to the view that needs the whole ensemble.

Strict potentialism admits no completed ensemble of this kind. A static view admitting one lawful block or a restricted family escapes the ensemble comparison, at the price of a principle of admission. The substantive asymmetry with an unrestricted ensemble is that a lawful process produces coherence: dynamics make memories agree with a past, and selection amplifies that

agreement. This does not settle how much accidental organisation the base produces, but it leaves the two views with different explanatory loads.

One structural saving follows from the architecture and is easy to miss. On a picture where some structures run and others do not, with no criterion saying which, the lit set, the structures that run, has to be identified before anything can be weighed, and Section 3.3 has argued that no structural label, not even one reading *live* or *now*, constitutes the difference between a run and a non-run. A view that stipulates which worlds run escapes that difficulty and owes a distribution rule over them instead. With one connected base, the question changes shape. "Which forms run?" becomes "which forms are in the run", an ordinary question about the size of catchments of evolutionary attraction, reachability and how much of the history a kind of organisation occupies. It compares regions of one thing rather than modes of many things, so the question is well formed, and Section 7.1's measure applies to it. Even before the measure is applied, two filters already operate.

First, an experience with memory and integrated character is itself an organised process. A region disordered enough to destroy that organisation contains no such observer, even if many microscopic transitions occur. An unreadable smear is not a recording simply because some pixels happen to be dark. This excludes fully incoherent regions, not locally coherent observers embedded in globally misleading histories.

Second, finite processes contribute only while they occur. A transient ember spun off from a generative region may host some organised moments and then disappear. A continuing ecology can produce descendants and observer-episodes across every larger finite window. This gives long-lived generative processes a plausible advantage under many measures, among them Section 7.1's, which weights a lifetime by its length, but not under all: under other identity and discounting rules a swarm of short processes may outweigh one lineage. Section 6.2's computational genesis amplifies and elaborates rare viable organisations.

This goes beyond the tautology that observers see observer-compatible conditions. It proposes why observer-compatible process might occupy a large share of an appropriate measure. Evolutionarily produced observers inherit long computational ancestry and depend on causal regularity, error correction and resources. The same conditions allow replicating organisations to amplify. If process measure tracks causally produced observer-episodes, evolutionary ecologies may outweigh isolated lookalikes. Section 7.2's adaptive activity, rather than raw activity, is the candidate quantity.

A further distinction is essential. Typicality is a distributional claim, not a licence to interpret one unusually lucky biography as evidence of cosmological selection. Evidence would require repeatable population-level deviations or correlations specified independently of the outcomes. Individual survival stories are selected for being noticed and cannot carry the burden.

Two more disciplines apply to any argument of this shape, and both cut against conclusions this subsection would otherwise reach too easily.

The first is that *persisting* comes in strengths, and they are usually run together. At the bottom is bare occupancy: how much of a history a kind of thing occupies is its rate of appearance multiplied by how long each instance lasts, which is an accounting identity and not a mechanism. Above that, survivor conditioning: what one finds on looking is filtered by the looking. Above that, branching, where continuations multiply. Above that, heritable adaptation, where what worked is transmitted. And at the top, open-ended evolution, where the space of what can be tried grows. Only the last two produce anything that deserves to be called design. The lower rungs explain *prevalence*, and prevalence is regularly mistaken for explanation of *organisation*, including in this section, if it is read that way.

The second follows from it. Overrepresentation among survivors does not by itself show that any trait caused the survival. It can come from luck, from the conditioning just described, or from correlation with something else. An account has to identify the causal organisation producing the persistence, and any argument that stops at *the survivors survived* has described its data rather than explained it. This is the point at which appeals to selection most often stop explaining, and Occurrence theory is not exempt. Section 7.4 considers accidental observers.

[Grade: research hypothesis; conditional explanatory hypothesis, to be tested with the measure of 7.1.]

7.4 Accidental observers and fine-tuning

Section 7.3 leaves lawful fluctuations producing locally coherent observers as the hard case, still to be worked out with the measure of Section 7.1. The thought that a puddle naturally fits its hollow answers the wrong kind of objection here. A Boltzmann observer is cheap in a way an ordinary life is not: a fluctuation buys one locally coherent episode with false memories, but not the lawful environment, ancestry or long history of consistent records that an ordinary life carries. The comparison must therefore be between *whole packages*, with coherence across a past treated as the expensive item. Section 6.2 assigns the first replicator to abundance, before heredity gives selection anything to accumulate.

The residue may be empty Signed speculation

The author's expectation, signed as such, is that the comparison above may never arise in practice. Abundance adequate to start evolution is all the base needs to afford. Everything more complex than that floor is then built by selection rather than thrown up whole, and a fluctuation assembling a coherent observer in one throw, complete with false memories of a lawful past, is so unlikely that even a quasi-infinite history may contain none. On that reading Boltzmann-style accidents are not outweighed but absent, or, failing that, vanishingly rare. The weighing question of Section 7.1 then goes idle in practice.

The conditional hypothesis of 7.3, that observer-episodes may cluster where evolutionary ecologies cluster, carries one further signed speculation. Section 6.4 discusses effective laws as products of selection, with Smolin's reproducing universes as an optional analogy rather than a required mechanism.

Fine-tuning Signed speculation

If the lives that Section 7.1's measure counts concentrate where evolutionary ecologies concentrate, bio-friendly constants would be a typicality effect of the measure.

[Grade: research hypothesis and signed speculation; conditional selection argument plus two signed suspicions; the accidental-observer comparison and fine-tuning are still to be worked out with the measure of 7.1.]

8. Architectures of occurrence and spacetime

8.1 Two questions about an architecture

Finitary Occurrent Actualism (FOA) fixes the base architecture only at a high level. At least one process occurs, finite happenings are parts of it and no completed infinite history exists. It does not decide whether home reality rewrites, grows or is represented as a finite block carried by deeper change. A railway gives the picture, as it did for beginninglessness in Section 2.2. Its history could be a finished blueprint, a route revised as it goes, or track laid at a frontier: three ways of organising a railway, none of which puts a train on the line. This spacetime choice is distinct from the choice of coupling architecture.

The first question is how local events are laid out or changed. A local world might be block-like, rewritten, growing, branching or a hybrid of these forms. The second question is whether another organisation reads or interacts with those events. Causal-Informational Parthood (CIP) can describe direct realisation by the local process and maintained coupling to another organisation under any of the spacetime options. A reader may traverse a settled record, but it may instead participate in a changing world. Section 3.4's reblocking objection, that whatever a process theorist adds can be absorbed into a sufficiently complete block, applies to every local architecture. Section 8.3 calls a carried, settled-so-far region a quasi-block; no endless history forms a completed block.

[Grade: methodological choice; architecture-neutral FOA statement; occurrence and local spacetime organisation remain distinct, and no local architecture is selected.]

8.2 Actualised blocks, laying and reading

In an actualised-block architecture, a complete local spacetime can be represented as a finite structure. If the local history is endless, it is not a completed block under FOA; only finite blocks or finite represented regions exist as completed objects. Here block means Section 8.3's quasi-block. This section separates two occasions on which a finite block's life may be undergone: while it is being laid down, and when a deeper occurring process afterwards reads it. The second is the reader case, an implementation option rather than a further primitive. Only later reading requires present traversal; Section 2.2's distinction between endlessness and actuality leaves a finished laying genuinely past. Section 3.3 rules out an actuality label as a third route: a label such as *live* or *now* is only more description, so whatever might be proposed to animate a finished block is either a running reader or more data. Section 2.7's production principle requires explained laying to have a causal history.

Base order need not resemble local time. A block might be laid by a front advancing up local time, by whole-block diffusion (refining the whole block at once), one life or observer thread at a time, on demand for a life or region, backwards, in dependency order, or through parallel regional processes. A life is the natural unit to test because conscious actuality is known firsthand there, although Section 11.5's account of candidate integration basins leaves the boundary of an individual experience open, including whether larger organisations such as ecosystems could qualify. The production principle requires each laying stage to be produced by what preceded it in the base. It does not require base order to match experienced order.

That freedom is not merely permitted. It may be needed. A front advancing up local time is a preferred slicing, so the most intuitive laying order also owes the largest physical debt. Diffusion, parallel laying and dependency-ordered laying may avoid a single global front, but each still owes a recovery of Lorentz symmetry without an observable preferred order. These are therefore not exotic variants of the obvious picture. They may be the ones that survive the constraint. One caution attaches to the thread case: if a later pass re-lays events an

earlier pass had already laid, it becomes unclear whether the passes identify the same realised events or perform distinct realisations and, in the latter case, whether they belong to one personal continuation. Section 13.1 keeps repeated event realisation separate from the further conditions for continuing one individual. Passes that add structure without re-laying what is already there do not raise that particular issue. The sorting is conditional on a block being laid in one of these ways; it is not a finding about our spacetime.

These orders are not interchangeable, and the parthood bridge already says why. It requires more than a completed deposit: the causal organisation relevant to a local process must be preserved, and local temporal order must emerge from base order. **A laying runs a given life only when its own occurring order carries that life's organisation, rather than merely depositing it.** That criterion sorts the orders rather than leaving them equivalent. A front advancing up local time runs every life, in step. Laying thread by thread runs each life at its own point of creation, although lives share events wherever they meet, so later threads revisit earlier ones and the caution recorded above is this order's standing cost rather than an incidental risk. Dependency-ordered laying runs every life by construction, several at once. A gap in a life's laying adds no interval inside it because none of that life's processes or clocks runs through the gap. Its demand-driven variant runs what it lays and leaves the rest specified, and the demand may be indexed by a life or by a region, which the paper does not settle.

On an orientation-preserving reading, backwards laying and whole-block diffusion merely deposit a life. Backwards laying places a recorded effect before its cause, while whole-block diffusion completes no local moment before the next. Diffusion that completes one moment before beginning the next instead runs the life, with refinement inside each step. A weaker structure-only reading would let backwards laying count. The image of a memory unforming before the experience that would have formed it favours the stronger reading, but the paper leaves the choice open.

Parallel laying remains open. Separate processes each carry only part of the order, so whether they jointly run a life may depend on where their boundaries fall. It may also be the most natural option under relativistic constraints.

What is laid can be **remodelled**, and not only at its edges. Ants enlarge a chamber in the middle of an existing nest. Bone is remodelled continuously throughout its volume rather than added at a growing front, and the word is the standard one in that field. Revision can occur anywhere in it, to any degree, and Section 8.4's growing architecture, which adds events at an advancing causal front, is the narrower case rather than the general one.

Whether anything of that shape holds below our own level is not something this paper can say. What the examples establish is that the division is a familiar one, and that admitting revision does not commit anyone to a growing edge.

Laying and later reading can coexist. If the laying already ran a life, a later reading can realise a distinct episode. The open question is whether those episodes belong to the same personal continuation. Section 13.1 proposes conditions on their organisation and coupling rather than identifying their events. The familiar division between building and use, as with a city and its inhabitants, shows only that these can be different activities.

The block case requires only that its events occurred, not that they are occurring now, and a completed laying whose own order carried the lives' organisation supplies exactly that. A match is actual while it is played, and a recording of it watched later is actual too, in the viewer rather than on the pitch, without the match being played again. **The block-life's actuality is not supplied twice.** Some recordings can also be edited, which is a further architecture again and belongs to Section 8.3's account of internal rewriting. Section 3.3's relational distinction

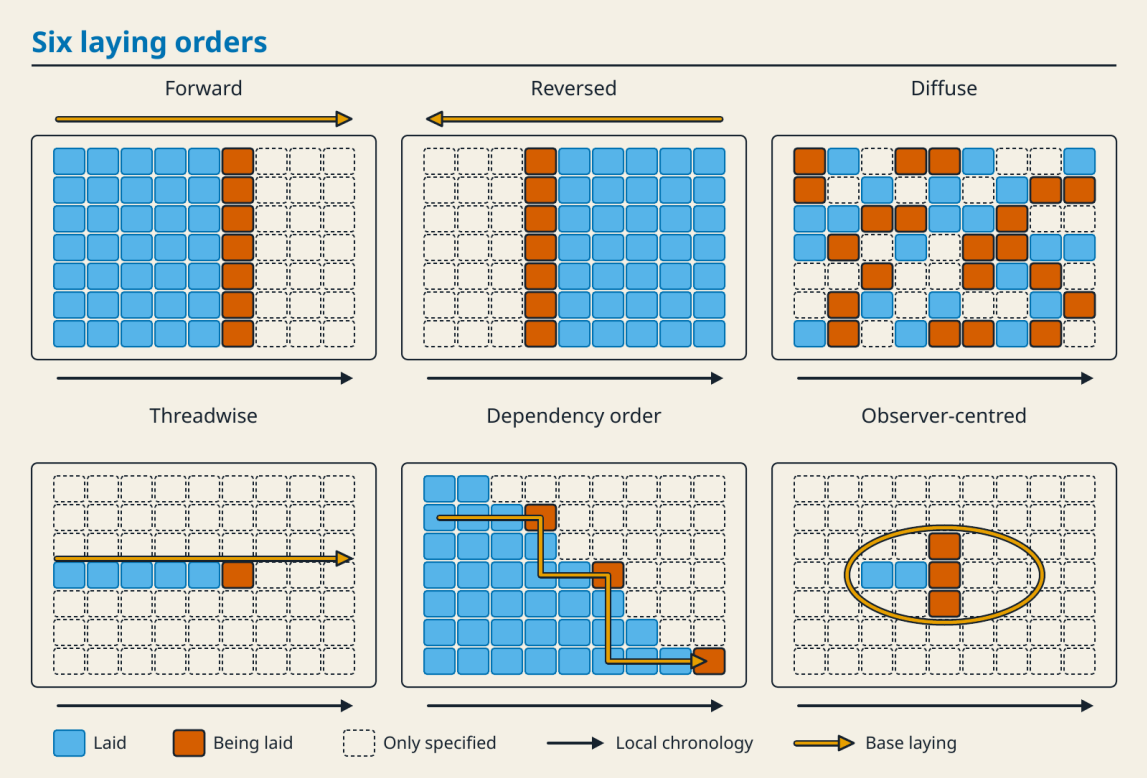


Figure 12: Six laying orders, one finished block. Possible orders in which a block or quasi-block universe could be laid down and potentially rewritten and updated.

also separates a carried finite universe from its abstract description; Section 8.5 asks whether implementations can be distinguished. For the recorded-life implementation discussed in Section 12.1, on the reader conjecture and its vocabulary, actualising that life's episode requires its relevant organisation to be realised through the interface; inspection alone is insufficient.

A Roblox world gives the coarse picture. The world can be stored as data while a player and computer successively realise views and actions. The player is not located inside the data structure in the same way as the avatar. The analogy fails at the deepest point because real Roblox has external hardware, designers and a pre-existing time. A hosted world has an external computer of a kind, its host; only the true base has none. The image shows only how a fixed local structure could be carried by deeper progression.

A film gives the contrasting picture. Projecting successive frames is an occurring traversal even with nobody in the seat; that fact does not make the projector see the film or actualise a depicted person. The example separates a stored record from the process using it. Where the local process already realises the relevant organisation, no additional playhead has to be placed above it. A reader coupled to a live rewrite world is then an optional participating organisation, not a requirement that the world first become a reel.

An actualised-block model must answer at least four questions:

1. where a reader implementation is proposed, how a finite local chronology maps to ordered reader transitions;
2. where a reader implementation is proposed, whether the reader reads events once, repeatedly or in a non-linear order;
3. how several local systems share consistent causal relations;

4. how Lorentz symmetry is recovered without an observable preferred playhead.

Sections 10.3, 12 and 13 address the physical constraints, optional reader and continuation questions respectively. It therefore remains open when laying, later reading and simultaneous readings realise distinct episodes, and when any of them continue one deeper individual.

[Grade: open problem and real-life counterpart (bone and ant nests remodelled throughout; a match and its later recording); open local spacetime architecture; laying and reading are distinct, and mapping their orders to realised episodes remains unresolved.]

8.3 Quasi-blocks and rewrite-only worlds

Two considerations prevent *block-like* from meaning completed and static. First, a universe carried by an occurring base need not have finished. Ours has a history so far, and calling that a block describes how it looks from inside rather than what it is. Absolute staticity is in any case not something an inhabitant could ever establish: static compared with what, measured from what vantage? A verdict of that shape needs a standpoint outside the block and a second process to compare against, and a block taken as the whole of things supplies neither.

Second, immutability is a strong claim and nothing here requires it. A world can be block-like to its inhabitants, its past settled and its laws not visibly changing, without being unrewritable. If it is carried, then whatever carries it could in principle revise it.

The term used here is therefore **quasi-block universe**, on the model of *quasi-infinite*: it looks and acts like a block to anything inside it, with a settled past, but it is never guaranteed to stay as it is. It may be being changed from outside, it may be changed later, or it may drift slightly because it is stored on an active base several levels down, perhaps below anything detectable from within, as bits can flip on a disk or in the memory that holds a file.

In a rewrite-only architecture, home reality itself changes. A present configuration is transformed into a successor, and conscious experiences are organised regions of those local transitions. No playhead over a fixed block is required. Section 8.1 keeps reader coupling independent of this spacetime choice.

The simplest form resembles an asynchronous cellular automaton or causal rewrite system. Local events replace or update neighbouring relations. A global state may be only an emergent bookkeeping device. The process can be deterministic or indeterministic, reversible or irreversible. A rewrite-only world makes a local present intuitive, but Section 3.4's reblocking objection still applies. A rewrite universe must meet Sections 10.3 and 10.4's physical recovery requirements and preserve stable records, without assuming a smallest update.

Rewind without erasure Signed speculation

In any world containing rewindable sub-worlds, occurrent history might be append-only: a rewind restores an earlier state and rewrites the records, but the run that occurred still occurred: rewound suffering and joy would be un-remembered, never un-happened (the records-only past described below in this subsection, plus Section 3.3, on the difference between a run and its complete description). Loading a saved game, restoring a virtual machine from a snapshot and rolling back a database all do this every day: the world resumes from an earlier state, and the abandoned run still happened.

One consequence removes an objection that otherwise looks decisive. If nothing is stored, how can anything remember? The answer is that a rewrite world need save no frame history at all, and games make the mechanism concrete. A game does not retain a recording of the moment a key was picked up; it retains a flag saying the key is held, and the door opens. The boss is not defeated again each time the room is entered, because a bit says it was. Those flags are memory in the only sense the architecture requires: compressed causal residue, the

consequences of the past carried in the present state without the past being kept. Human memory has the same character and is known to be reconstructive rather than archival, and a trained network's memory is the same again: what it read survives as altered dispositions in the weights, not as a stored copy. Nothing in this rules out an architecture that does store its history. It removes the assumption that one must.

[Grade: open problem and real-life counterpart (game flags, reconstructive human memory and a trained network's weights, each remembering without a stored history; saved games and snapshots, for rewind without erasure); open physical architectures compatible with FOA; reblocking prevents either architecture from establishing occurrence through structural change alone.]

8.4 Growing, branching and hybrid worlds

A growing architecture adds new events at an advancing causal boundary. A branching architecture allows several future continuations in its formal state space. A hybrid may combine a fixed coarse block with deeper rewriting, or local growth with globally consistent constraints. Section 2.1 allows finitely represented branches and further extension; Section 2.2's base postulate settles which transitions are actual, the ones the base runs. Section 10.3 separates quantum interpretation from occurrence architecture.

Branching here concerns alternative continuations in a spacetime architecture. It is not the same as the reality tree's production of additional worlds: a designed world can itself be a rewrite-only process, and a naturally produced world need not contain branching futures.

Section 8.1's neutrality among architectures extends to one with a fixed but locally unknown future; unpredictability does not supply actuality. Two images help here, provided the fence goes up with them. A growing structure can be pictured as dead wood carrying green leaves, or as cooled rock with lava still moving at the edge: fixed and active regions in one object rather than two objects of different kinds. The fence is that the fixity of the older part is *structural* (it is settled, and nothing further will happen to it), not a difference of modal status. The past region is not dead in the sense of having stopped being part of what occurs. It is held within one live run, in the way the earlier bars of a performance are not somewhere else. Section 3.5's anti-static lean favours at most a quasi-block and does not select a local architecture.

[Grade: open problem; open architecture families with no preferred branch or quantum ontology; every option still owes an occurrence account and recovery of tested physics.]

8.5 What could distinguish the architectures

The candidate physical discriminators are discreteness artefacts, Lorentz violations, non-standard quantum correlations, failures of unitarity and signatures of causal growth. A model that predicts a specific departure can be rejected if observation excludes it. Any hidden order must meet Section 10.3's observational bounds.

Not every architectural difference need be accessible from inside. An embedded observer may be unable to distinguish a completed laying from an ongoing reading when all accessible structure agrees. Different stage machinery can produce the same visible play. An emulator shows it: a console game runs on a modern computer with the same rules and the same play as on the console it was written for. Failure to find an accessible difference would then leave the architectures underdetermined rather than prove them identical. Section 8.1's two choices, how local events are laid out or changed and whether another organisation reads or interacts with them, require independent tests; Section 15.9 separates them.

[Grade: open problem; conditional physical research programme; no present observation selects an architecture, and reader coupling requires separate tests.]

9. Bounded recurrence: a conditional result and a signed cosmology

9.1 The conditional recurrence result

Section 2.3's bounded base already requires the base to revisit configurations: with only finitely many available and no last stage, it must come back to one. This section separates the return forced by the finite repertoire from the exact cycle that determinism would force, considers the entity-mediated version, and records what each would buy. The exact-cycle result remains conditional mathematics; the author's suspicion that the base's complete-state rule is deterministic is signed speculation, and no earlier result depends on it.

The forced version. Suppose that only finitely many complete configurations are available to the base. This is a stronger condition than a finite set of states per cell, which allows infinitely many configurations when there are infinitely many cells; on such an infinite space a rule can even be one-to-one without being onto, which is impossible on a finite set. A process continuing without end must then revisit one. If the complete-state dynamics is deterministic, revisiting a configuration fixes the same continuation from that point, so the process cycles.

Beginninglessness strengthens the result. Take any stage of the actual history and look back over the previous $N + 1$ stages, where N is the number of available configurations, so that there is one more stage than there are configurations. Two of those stages must match. Determinism makes the succession from the matching stages identical, placing the chosen stage on a cycle. The chosen stage was arbitrary, so every stage of the actual history lies on the cycle. The beginningless trajectory is periodic throughout. Its abstract transition graph may contain transient states that feed into the cycle, but the actual history has no transient prefix and never occupies them. This reasoning uses only a finite stretch of predecessors, never a completed past. The proof is conditional; Section 9.2 states the finite-repertoire requirements adopted by the bounded-base postulate.

Poincaré's recurrence theorem is a different result: it concerns a measure-preserving system of finite measure and promises a return arbitrarily close rather than an exact return (Poincaré 1890). Nietzsche's eternal return reached recurrence from a finite quantity of force and infinite time (Nietzsche 1968 [1901], section 1066); the argument here needs only succession that never stops.

Without determinism, revisiting a state need not establish one exact orbit, because the same state may have more than one successor. The preference for complete-state determinism belongs to the signed perfect-loop conjecture in Section 9.5, not to Finitary Occurrent Actualism (FOA) itself.

Corollary: exact finite factors are reversible. First in plain words. Suppose the base, on a stretch where it cycles, is described coarsely: many detailed base states bundled into one **coarse state**, the way a weather map bundles the motion of every air molecule into "rain over the county". Suppose too that this coarse description is exact and self-contained: from the present coarse state alone it fixes the next one, and it is never wrong. The corollary says that such a description can never funnel. Two different coarse states never lead to the same coarse state, so the coarse rule, like the base beneath it, only ever cycles; it cannot lose or merge anything. The strong branch, the finite and deterministic case in which the base cycles exactly, needs this because it is what forbids a genuine one-way street anywhere above a cycling base: whatever looks irreversible at some level, whatever seems to forget or to merge, does so because the description is incomplete, not because the base is. The result is elementary and

standard, since a map that sends a finite set onto the whole of itself is also one-to-one; it is proved here because the strong branch needs it stated exactly.

The proof, for readers who want it. A **factor** is a coarse description of the base that reaches every one of its own states and has an update of its own, agreeing exactly with the underlying one. Take a finite recurrent sector R with its deterministic complete-state update $F : R \to R$, which is a permutation. Let C, from R to Y, send each base state to its coarse state, with every coarse state reached. Suppose the coarse states have an exact deterministic update of their own, call it f, and that the two updates agree: updating the base and then reading off the coarse state gives the same result as reading off the coarse state and then updating it:

$$C \circ F = f \circ C.$$

Then f is a permutation too. Each coarse state has exactly one successor by assumption; what needs showing is that it also has exactly one predecessor. Pick any coarse state y and any base state x that C sends to y. Because F is a permutation, x has a predecessor z, with $F(z) = x$. Updating z in the base gives x, whose coarse state is y; by the agreement just assumed, updating the coarse state of z must give the same, so it too gives y. So every one of the finitely many coarse states has a predecessor, which is to say f is onto. And a map of a finite set onto itself is also one-to-one, so no two coarse states share a successor: each has exactly one predecessor.

The same holds when one coarse step spans several base steps. For any fixed positive integer m, put F^m, that is, m base updates in a row, itself a permutation, in place of F in the agreement above, and the argument runs unchanged. Two facts about periods follow. On a base cycle of length L the coarse period divides L: the coarse cycle fits a whole number of times into one base cycle. Sampling every m updates gives a period dividing $L/\gcd(L, m)$, that is, L over the largest whole number dividing both L and m; the coarse cycle then fits a whole number of times into that many samples. Two cautions. Any controller or alternating update phase belongs in the complete state, or the base update is not the whole story. And the agreement must hold throughout the specified invariant domain, a set the dynamics never leaves, not merely over a short fitted history; a rule fitted to a stretch of history is not a factor.

What it means. If the base cycles, then every exact, closed, finite, deterministic description of it, at whatever level, cycles too: no two situations it distinguishes ever flow into one. So wherever a level shows real merging, forgetting or an arrow of time, one of three things is true of the description: it covers only an open part of a larger system, it has left out hidden history that the complete state still carries, or it is an approximation. Section 10.1 shows the hidden-history case with a four-state cycle that looks like two states from outside, and uses this corollary to say where an arrow can come from and where it cannot.

There is a second constraint, and it is the same fact from another side. Take any single number computed from the complete state, a real-valued function V of it. Such a number cannot rise throughout a cycle. If V never decreases, that is, if $V(Fx) \geq V(x)$ at every step, then returning to the starting state forces equality at every step: there is nowhere for a net gain to have gone. Entropy or accumulated novelty, when treated as complete-state quantities, cannot increase globally in this branch; long stretches of increase remain possible.

[Grade: mathematical result, conditional; exact cycling depends on a finite complete-state repertoire, adopted in Section 2.3, and determinism, which remains unestablished for the base.]

9.2 Size, complexity and repertoire

Section 9.1 needs a finite complete repertoire; size and complexity bounds alone do not provide it.

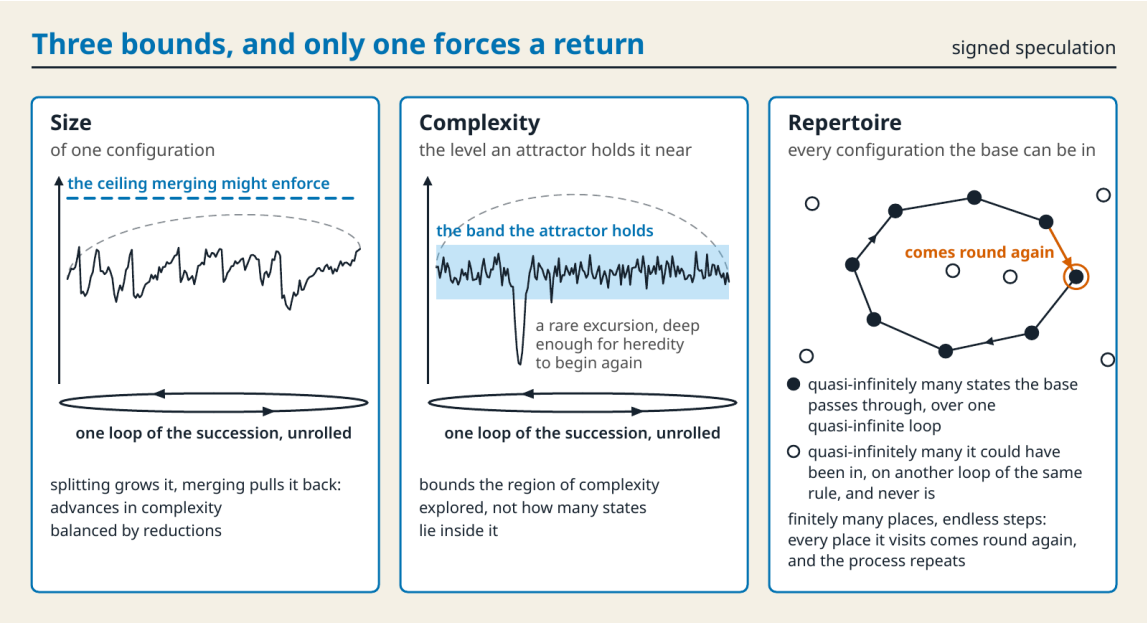


Figure 13: Three bounds, and only one forces a return. Size of one configuration, which merging might hold under a ceiling; complexity, the range an attractor might hold the process within; repertoire, every configuration the base can be in. The axis is one loop of the succession, and the arc is the same history closing; the filled places lie on that loop, the hollow ones are places the base never occupies, and every place on the loop comes round again. Size and complexity are each needed to guarantee a return and neither is enough. Schematic, no numbers.

A finite configuration space requires a uniform bound on configuration size, a uniform bound on complexity, finitely many distinguishable states per site, a fixed finite relational signature, and either finitely many possible positions or an explicit quotient by isomorphism; Section 2.3 adopts these conditions for the bounded base. The positional condition matters. One live site moving on an unbounded lattice gives infinitely many configurations despite bounded size and two site states. Stagewise finiteness is weaker than uniform boundedness: every individual stage can be finite while the size or complexity reached across stages has no common ceiling.

Finitely many states per site is also independent of bounded size. Section 2.7's signed-refinement conjecture permits labels from an arbitrary abelian group, but a true-base instance realises only finitely many distinguishable label states. The finite state range comes from Section 2.3's bounded-base postulate, not from a size ceiling alone. Even bounded size and finitely many site states are insufficient if the available positions or relations remain unbounded.

What could set the bound is a separate question, and several answers are available. The architecture does not need to choose among them. Some candidates would bound size, others complexity or the region explored. The forced version needs both bounds together with the remaining conditions.

A size bound could come from the base's own dynamics if splitting and merging balance as Section 2.7 allows, or from a plain maximum in the rule that configurations cannot exceed. Balanced merging tests only one possible size mechanism. It supplies neither a finite state alphabet nor all the other conditions for a finite repertoire. An entity-mediated attractor,

considered in Section 9.3, could instead hold the process near a range of complexity. Section 5.6's level-cost argument could bound useful complexity, not configuration size.

The author's second, partly overlapping consideration is the modest dimensionality of our own universe. A world with something like four dimensions rather than forty may look relatively close to the base if greater depth brings greater richness. Yet a host may realise extra effective dimensions, and additional dimensions need not keep adding useful organisation, as Sections 5.5 and 9.4 explain. Dimensionality therefore measures neither hosting depth nor the capacity of a particular mind. This intuition about richness does not by itself establish a bound on the base. Section 2.3 postulates the conditions for a finite repertoire, while leaving the base's rule open. Section 2.1 permits endless traversal of a finite repertoire without a completed history.

Merging does not establish that configuration size stays bounded. Whether it balances splitting in any candidate rule is a dynamical question, and merging that goes too far could collapse the dynamics into the empty state. A size bound from merging therefore needs a non-collapse guarantee as well. Section 15.3 asks a simulation to test splitting and merging, but even a positive result would secure only one condition. Its longer runs must track exact recurrence and count complete configurations up to an explicit identity convention, rather than treating a bounded run as proof of a cycle.

The order of explanation matters. A finite repertoire would force a return, but recurrence cannot be used to infer the finite repertoire required to produce it. Nor does recurrence explain why the base has the character it has. It relocates that question rather than answering it.

[Grade: metaphysical postulate for the finite repertoire; Section 2.3 adopts the uniform size, complexity, state, relational and positional conditions, while the rule implementing them and the deterministic branch remain open.]

9.3 The base cycle and the Eukrasian attractor

Section 9.4's signed seasonal picture names one finite exact pass the base cycle, and calls the broad band of complexity where coherent organisation is readily sustained Eukrasia, from the Greek for a good mixture; a tendency of the base's dynamics to keep it there would be a Eukrasian attractor. The cycle is finite but quasi-infinite in scale, meaning beyond the ability of anything inside it to survey or count, and it is traversed without end. Computers already run made loops of this kind: the Mersenne Twister, a widely used generator of random-looking numbers, cycles through exactly $2^{19937} - 1$ states, a number with over 6,000 digits, far beyond anything that could be counted out in practice. Section 3.2 distinguishes a finite cycle from its uncompleted traversal.

The form of true base reality — Signed speculation

The author suspects that true base reality is a finite relational process, not an object contained in a prior space. On Section 9.1's finite, deterministic and beginningless branch, its actual succession of complete configurations forms a directed cycle, however rich each configuration may be. What recurs is the configuration; the traversal never finishes. The loop represents recurrence, not a completed past; occurrence remains primitive.

The entity-mediated version. Bounding need not be an accident, and it need not be an act. Attractors are established mechanisms in their own domains. A swinging pendulum settles at the bottom of its arc from wherever it starts, and many animal populations are held near what their environment can support, with nothing coordinating either. If selection reliably produces organised entities, their collective behaviour might hold a system near some level through cooperation, conflict, accident or spreading disruption. The preferred candidate is a standalone dynamical effect in which nothing is coordinated and organisations are simply drawn towards a certain level of complexity.

Such an attractor would supply stabilisation rather than exact return. It could keep the explored region bounded without sending the system back to simplicity. Outside the bounded-base postulate, an attractor may hold a system inside a range while leaving infinitely many exact configurations available there. It might therefore supply a bound on complexity, one condition needed by the forced version, but not a finite repertoire by itself. The signed seasonal conjecture in Section 9.4 distinguishes a useful coherence band from the further hypothesis that dynamics keeps it occupied.

A permutation never merges distinct complete states, so combined with the finite deterministic branch the proposed attractor has to be read at the level of effective behaviour. Stabilisation could mean long occupancy of a region, repeated restoration of organisation or approximate attraction in variables that omit distinctions. Recurrence alone establishes none of these, and the tendency to return is a third claim again.

An attractor also need not act alike in both directions; Section 9.4 returns to this with the two crossings of the band. Section 13.6 treats deliberate reset as a separate wild possibility.

[Grade: signed speculation and real-life counterpart (attractors, established mechanisms in their own domains, such as a damped pendulum and a population held near what its environment supports; the Mersenne Twister's exact finite cycle, for a quasi-infinite loop); attractors can formally stabilise an explored range, but neither a Eukrasian attractor nor a finite base cycle has been demonstrated.]

9.4 The coherence band and the seasons of the cycle

The seasons of the base cycle Signed speculation

The shape this suggests is a long stay with rare diebacks. It is speculated that the process stays inside one band of complexity for nearly all the stages of its base cycle and, very rarely, drops to a level low enough for selection-capable organisations to begin again. One such excursion per base cycle would be enough, since what is needed is that heredity can start, or start again, not that it does so often. It may never wholly stop: the winter remnant can carry heredity through, so a restart from what survives is the likelier case. Selection then carries organisation back up into the band the attractor is conjectured to favour. The loop has a geography, and this paper borrows the seasons' names for most of it. One full pass of the loop, from any stage round to its exact return, is a quasi-infinite great recurrence, which this paper calls the **true base cycle**, or the **base cycle** for short. The lowest level of organisation the loop ever reaches is its **winter point**, its winter solstice. Much of the organisation built during richer phases may have died away there; whatever persists supplies the remnant from which heredity and construction can rebuild. Winter is a phase of the already occurring base, not the true base as a level and not a beginning of existence. The highest level the loop ever reaches is its **summer crest**, its summer solstice: a crest because it breaks, the maximum of complexity existence ever reaches, probably through diminishing returns on complexity, and not an optimum. Tainter (1988) read the collapses of the Western Roman Empire, the Classic Maya and Chaco Canyon this way, as complexity whose returns had fallen below its cost. Human brains may show the same shape: one study found that they shrank somewhat over the last 3,000 years, perhaps because knowledge shared across a group made a larger brain less worth its energy cost; a reanalysis of the same data found no decline, and the question is still argued (DeSilva et al. 2021; Villmoare and Grabowski 2022; DeSilva et al. 2023). One base cycle contains everything the loop ever does, since the loop repeats exactly, so these are the extremes of existence and not of one pass. Between them lies the band the attractor is conjectured to hold. This paper calls it **Eukrasia**, from the Greek for a good mixture, the well-tempered constitution of a body or a climate, and where a description is wanted rather than a name, the **coherence band**, the range of complexity within which coherent organisation is readily sustained. Its image is the equatorial rainforest of the cycle, a broad belt in which the conditions come together for many kinds of organised system to coexist and persist. Departures from the band into its tails are excursions, and on this conjecture they would likely be relatively rare. A **winter excursion** runs down into the low tail and a **summer excursion** up into the high tail; a base cycle may hold several of each, and the **far excursion** on each side is the one that goes furthest, the far winter excursion reaching the winter point and the far summer excursion the summer crest. The climb back from the winter point towards the band, where heredity catches and selection rebuilds, is the **spring ascent**; the fall from the summer crest back towards the band is the **autumn descent**.

The winter point Signed speculation

How low the winter point lies is not known. It is the loop's minimum of organisation, not an extreme of entropy in either direction: entropy is a different quantity, one that a level's thermodynamics defines, and the paper attaches none to the base's stages, so the winter point is neither a heat death nor the low-entropy boundary that Section 10.1 says a beginningless cosmology must not smuggle in. It is easy to picture it as holding only the equivalent of the simplest automata, but the true base is quasi-infinitely large, and even its lowest level may carry organisation as complex as a universe like ours, and with it minds like ours. That is not the plenitude route to accidental observers. What would carry such minds at the winter point is a universe like ours still running there, a far simpler thing than what its rules generate, since we are the procedural product of a simple rule; and in a loop that has always existed such a universe is standing content, not a fluctuation. The intuition behind this section still favours a simpler winter point, but nothing fixes how simple. The author's more particular suspicion is that the winter point, or a stretch of the cycle near it, lies just on the cusp of where selection could take off: close to the least organisation from which the abundance the base has on hand can start cumulative construction again. Nothing requires this, and a large enough base could carry far more than that at its minimum. The cusp is a conjectured nearness, not a definition of winter. Winter is where organisation actually bottoms out; the cusp is where the dynamics, the remnant and the opportunities on hand allow regrowth; and Section 15.8, on the stack from the bottom and from above, proposes measuring the two separately and asking whether they fall close together. Two things follow. Intelligence and conscious actuality of our kind need never be wholly extinguished, even at the winter point, and the base cycle is quasi-infinite besides. So Occurrence theory does not imply that everyone and everything is ultimately doomed.

The band, its occupancy and the Eukrasian attractor Signed speculation

The seasons supply the order of the landmarks and nothing about their lengths, which are lopsided beyond anything a year suggests. Eukrasia may occupy nearly the whole of a quasi-infinite base cycle. The spring ascent and the autumn descent are likely comparatively brief passages, and the winter point and the summer crest are rare extremes, reached by the far excursions into the thin tails of the distribution of the base cycle's stages over complexity, tails that need not match each other in thickness. For the realities carried within the band this means that a winter or a summer is a rare exception in a quasi-infinite stay: universes, and the large hosted worlds above them, spend nearly all of their existence held inside Eukrasia, and most of them never see either extreme. The stretch of succession spent in the band is itself quasi-infinite, and it is nearly all of a base cycle that is finite; the count is of stages, not of clock time, which the base does not have (Section 3.6, on the base having a present without a clock). A galactic disk seen edge on gives the picture: most stars lie close to the plane and a few far above or below it, and how far they spread from the plane is the disk's scale height, whose analogue here, how far the base cycle's stages spread from the band, is a quantity a simulation could perhaps one day attempt to measure (Section 15.3, on nullity, signed refinement and bounded recurrence). Two claims are being made about the band, and they are independent. One is about a range of complexity, that the band is where coherent organisation is readily sustained. The other is about occupancy, that the loop spends nearly all of its base cycle inside it. A band could be narrow in complexity and still hold nearly the whole base cycle if the dynamics kept returning the base to it, and that tendency is what this paper calls the **Eukrasian attractor**: the conjectured tendency of the base's dynamics to keep it in the band or return it there, the standalone effect the entity-mediated version above prefers, now with a name. It is conjectured rather than defined into the band, because in the mathematics of dynamical systems an attractor is a definite thing, an invariant set that neighbouring trajectories approach, and no base dynamics has been exhibited that has one. The band could exist without attracting the process, and the attractor conjecture is the separate claim that the dynamics makes it overwhelmingly occupied.

The coherence band. The band may be more than the range the process happens to hold for most of the base cycle: useful complexity may have a range of its own. Useful here means coherence of some form, persistent organisation that does something: intelligence perhaps, and perhaps even art in some form, on the author's suspicion that art drives much of a civilisation once war and competition have subsided as the driver. Too little freedom leaves too few ways to build rich and persistent structures. Too much may make coherent, separable and reusable organisation harder rather than easier. Capability would then not rise steadily with the structural freedom available, whether counted in dimensions, degrees of freedom, coupling or the size of the reachable state space. Maximum complexity does not necessarily mean maximum coherence, and on this reading it does not. No theorem settles the matter: the

answer depends on how complexity and coherence are defined, and the measures on offer that peak between order and disorder support the shape without deciding the case for the base.

On this reading, the loop's best stretches lie inside the band rather than at its crest. Flourishing peaks before complexity does. Above the band, added complexity can already be too much to be ideal, an overheating as the crest is reached, making the summer crest an overshoot. Section 5.6's ceiling on useful depth has the same shape one level up: the next level can cost more than the capability it buys.

The suspicion has familiar relatives, none of which establishes it. Changing the number and kind of physical dimensions changes whether familiar laws support stable bound structures and predictable evolution, so more freedom is not automatically more hospitable, and no particular number is claimed as best. Cellular automata make the point more cleanly as a formal image. A rule can leave too little freedom for much organised behaviour, while sufficiently chaotic dynamics can destroy the persistent structures on which higher organisation depends. Computationally rich regimes may lie between. This is the edge-of-chaos picture, whose strong form, that computational capability is generally favoured near the transition, remains disputed (Langton 1990; Mitchell, Hraber and Crutchfield 1993). Computing and biology supply further analogies of the same trade-off: added capacity, wider machine words or larger nervous systems, can cost more than they buy past some point, while a small alphabet of bases or instructions can support rich organisation. The periodic table gives a homelier version, and it is the author's guess rather than a result: a universe with five times as many stable elements, or a heavier average abundance, need not be richer for life, since the useful simple ones would be diluted among the rest and harder to find, and the ninety-odd natural elements we have may be near the practical limit for a four-dimensional universe that is to carry life. None of this shows that the true base has such a band. It shows that an optimum below maximum complexity would not be an exotic kind of organisation.

The reality tree within the coherence band Signed speculation

A tree gives the developmental picture better than a tower alone. Its trunk follows one route by which organisation grows from a winter remnant towards richer effective worlds. The author expects a broad complexity gradient along that route, while admitting reversals and other lineages. Near the upper trunk, perhaps below Eukrasia or just inside its lower margin, lies the world-making transition zone of Section 6.2, the passage in which capable organisations begin producing hosted worlds prolifically: a conjectured region of increasing capability, not a threshold or the point where natural production stops.

In the figure, angular branches show deliberately constructed worlds, smooth branches show natural production, while mixed production remains possible between those clear cases. Branch endpoints sit at the effective complexity of the resulting world, not at its depth in the hosting ancestry. A hosted child further from the base can therefore hang below its parent on this axis. Reductive and peer worlds form most of the drawn crown; enriched worlds reach higher. Either kind of child can produce further branches by either route. The proportions express the author's expectation, not measured rates.

Why might most constructed branches remain near the productive band, or droop below their parent, rather than climb without limit? A tailored model often gains nothing from reproducing all its host's physics. Enrichment may demand more maintenance, coordination or implementation work than its useful capacities repay, or may make persistent organisation harder to sustain. Conversely, a well-chosen enrichment could be worth those costs. These are candidate reasons for a tendency, not a universal penalty on added dimensions or a prediction that every designer chooses the same kind of world. A narrow airflow or weather model can sit well below the band as a model while remaining a useful component of an ecology within it. The preference to date in our own world runs that way: chess is popular and simple, and for every world built at the edge of what we can manage there are tens of thousands of block games. If building simpler worlds grows more attractive as a level grows more complex, that would be part of what holds the crown near the band, and perhaps part of the explanation of Eukrasia itself.

Two readings of the picture must stay apart. Eukrasia in the seasonal conjecture concerns the organisation sustained across the base's course; the endpoints here compare effective worlds within that activity. No

Continued on page 83

The reality tree within the coherence band, continued Signed speculation

common numerical scale or reduction between those quantities has been supplied. The overlay is qualitative: a low-complexity child does not send its host, still less the whole base, into winter, and an enriched child is not by itself a summer excursion. Likewise, the tree's winter remnant is not the unhosted base appearing from nothing. It marks possible regrowth within a beginningless process and may already contain surviving worlds and minds.



Figure 14: The reality tree. A conjectured trunk grows from a winter remnant, and branches leave it at several heights; a descendant may sit above its parent, level with it or below. Faint branches and worlds stand for the vastly many not drawn. One chain is labelled as ours might be, through our familial base reality to home reality and the worlds it produces; those labels illustrate one possible chain rather than locating ours. Positions and counts are conjectural; no unique trunk is asserted.

The two crossings of the band may also differ. On the spring ascent, structure is being assembled, niches opened and capabilities accumulated. On the autumn descent, a state of similar complexity may carry inherited organisation, narrow specialisation, accumulated dependencies and incipient loss. Complexity alone need not locate the process within the cycle. The two crossings might even form a shape like hysteresis. In a magnet, hysteresis is the familiar case: rising and falling fields pass through the same strengths but leave the metal magnetised

differently on the way up and on the way down. Section 15.3 asks a simulation to look for that difference.

The asymmetry hints at why complexity might be lost. Organisation can accumulate dependencies and coordination costs along with capabilities, becoming more complex but less robust: costlier to maintain, brittle under perturbation and dependent on a narrowing set of conditions. A rainforest specialist can fit one niche so well that a single change removes it. Civilisations show the shape, though they cannot establish it for the base. A civilisation is often at its most productive before its population, bureaucracy and institutional layering are greatest.

What sets off a winter excursion is open, and the images here suggest possibilities rather than a mechanism. One is dead wood accumulating in an ancient forest grown unhealthy and rotting, followed by fire or consumption by something simple that the healthier state had resisted. A civilisation's rise and fall is in any case a small internal echo of the base cycle rather than a season of it. Whole histories of worlds, universes like ours among them, bloom and disappear inside Eukrasia without the base leaving the band. A summer excursion moves the base itself and is rarer and larger than any of them. Section 9.6 distinguishes the cycle's stage occupancy from the observer weights that Section 7.1's measure assigns.

Where we are in the cycle Signed speculation

Asked where we find ourselves, the author would judge by our level of complexity first, and by our place in the base cycle only through it, since the two map onto each other only loosely. The author's answer is inside Eukrasia: not in a winter excursion, still less near the winter point, where average complexity is lowest, and even less at the summer crest, where it is highest. Two reasons point there. The anthropic one is not only that we are here but that we have conscious actuality of our kind, which needs organisation of the kind the band readily sustains; the other is occupancy, since nearly the whole cycle is spent there. Within the region we are perhaps at the lower end, or at least our universe is compatible with the lower end even if the true base cycle at this stage also carries more complex structures and universes. Our familiar four-dimensional physics may be relatively simple within that band, and Section 5.6 suspects that home reality lies relatively close to the base, tens of levels up rather than thousands, though neither dimensionality nor a world's position in the tree establishes that depth. Universes more complex than ours and still coherent are easy to imagine, with a fifth dimension, additional and more interesting forces, or fewer restraints such as the speed of light, and they seem almost certainly viable. Another aspect to consider is nobody knows how long their own existence will continue, at whatever level (human reader or other), and until they have at least a better idea, they cannot know the average complexity they will live through: not only because one might live through a technological singularity and then on for a million or more years, but because one might wake one day to find oneself in a hosted simulation, which could be in a universe of greater or lesser complexity than this one.

Section 9.5 addresses periodicity's tension with novelty.

[Grade: signed speculation and real-life counterpart (civilisations and specialist species that flourish before their complexity peaks; a galactic disk holding nearly all its stars in a thin band; the collapses Tainter traced to diminishing returns, for the crest; magnetic hysteresis, for crossings that differ on the way up and down); neither the coherence optimum nor its high occupancy has been shown.]

9.5 Non-settlement, dormancy and the perfect-loop lean

Non-settlement at the scale of the base does not require activity everywhere. A local experiencer needs activity in its own region, not in every base region at once. Settled, almost dormant or temporarily dormant regions remain possible inside a wider occurring base. A computer's storage shows the arrangement: an area of a hard drive or of memory may go unused for months or years while the rest of the machine works on, and then suddenly be used again.

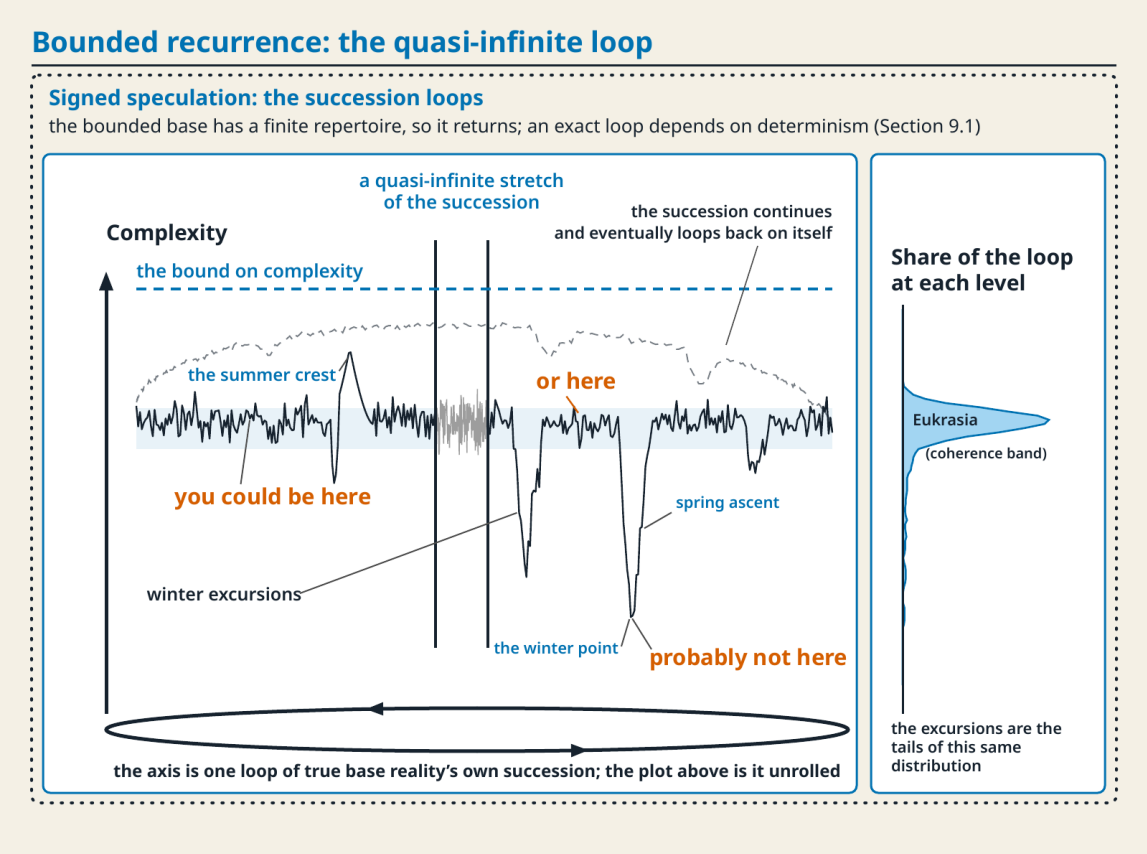


Figure 15: Bounded recurrence. Complexity across one base cycle. The history varies about Eukrasia, the coherence band, held for nearly the whole cycle; one summer excursion reaches the summer crest, and the winter excursions are the low tail shown at the right, the deepest reaching the winter point. The axis is a loop because the succession closes. Schematic, no numbers.

The base as a whole never settles. That it never stops is a commitment (claim 3 of Appendix A: no first stage and no last stage). That it never falls into an eternally featureless cycle, or falls silent for good, follows from the same starting point: a beginningless base would already have fallen into any such trap it could reach from its own past, and stayed there, and nothing would be happening now; a trap it could never reach cannot catch it. With a finite repertoire (Section 2.3, the bounded base) and a rule that does not change, what has never happened in the endless past cannot happen in the future either. If the rule is deterministic, this holds outright; if it is chancy, it holds with probability one, because an endless past has already given every possible step endlessly many chances, so a trap within reach would have been reached already.

No final settlement Signed speculation

The author's further reason is one of economy. An occurrent thing that began or could cease would invite both an origin story and a persistence story, whereas a beginningless and endless base avoids those particular debts. This does not prove that the base occurs, continues or remains generative. It states why a non-quiescent base is judged the more fertile conjecture once primitive occurrence has already been accepted.

The stronger thought, that anything genuinely infinite exists only by running, is not signed speculation but follows from a commitment. Section 3.2 closes off the completed mode (under strict potentialism nothing infinite exists as a finished whole), running is the mode that remains, and Appendix A lists the thought among the claims as a consequence.

A claim and a signed suspicion meet here, and they can seem to pull apart. The claim, argued at the start of this section, is that the base never settles into an eternally featureless cycle; the suspicion set out here is that deterministic dynamics makes those returns part of one exact cycle. Under determinism and a genuinely finite repertoire, recurrence makes the base periodic. No event is globally unprecedented, because there was no first pass after which repetition began. Novelty becomes a feature of where within the pass an observer stands.

Two responses remain available. The first is scale. If the pass is quasi-infinite, everything any observer will ever call new may fit inside it, while global repetition remains a fact about a whole on which no observer can take a vantage. On this response both stand: novelty is inexhaustible from every observer's reachable position, and a rich cycle repeated exactly is still far from featureless. Section 9.4's Eukrasia can contain whole histories of many worlds while the base stays within the coherence band.

Within one pass, a substantial sample of what can happen may happen. A pass that carries a universe like ours many times from different seeds, each handed to its run by the loop's own course, which from inside the run is as good as a seed fed in from outside, may realise many versions of a universe, a planet's history, a species' history and perhaps a person's history. It need not contain every variant of every event. That would require the plenitude this paper does without. At some scale, which is unknown, the loop has no room for further variants. Its possibilities are already limited by what can evolve from the winter point under its rules up to the summer crest. The suspicion is of abundance enough for any reachable observer, not of exhaustive realisation. For anyone inside it, that may be as good as no limit.

The second response is that the base's dynamics may be indeterministic. The conditions in Section 9.2 describe the finite configuration space adopted for the base in Section 2.3. With a finite repertoire but indeterministic dynamics, configurations recur, though an exact global period is not forced.

The author's inclination is the first response: the looping is endless, while the base cycle closed by one pass and the structure it carries are finite but quasi-infinite. The loop is not merely a cost accepted by the picture. The box below offers it as an explanation of persistence. A non-periodic history remains open, and exact cycling throughout this section remains conditional on determinism.

The perfect loop Signed speculation

The author's reason for expecting both recurrence and no settlement is one thought, an induction from everything that persists. What lasts, lasts by cycling. Life is barely a loop, leaking by mutation and death; a planetary orbit is a better one and outlasts species. The nearer a cycle is to perfect, the longer its bearer persists, so what persists without end would be the limit of the series: a perfect loop. Two things follow. First, a perfect loop's future is its own past run again, so the experience being undergone now would vouch for every pass, with no completed totality summed and none needed, since recurrence is not completion. Second, every lesser loop leaks through some feature of its situation: friction to waste against, compression it cannot survive, surroundings to radiate into, and a rate for the loss to happen at. The base's structure carries none of that machinery: no deeper medium to waste against, no rate at which anything can bleed away (Section 5.4, on causal order, clocks and physical time), no outside, infinite or otherwise, into which anything can leak. The base might be a loop with nowhere to leak. An atom in its lowest energy state shows half of this: it has no lower state to fall to, so it does not radiate its energy away, though it does not cycle either. If the base seems to return to the same configuration and then does something different, either the configuration was not complete or the difference is genuinely undetermined. Any cause of the difference, a hidden state, a controller, a seed, retained history, a count of traversals, belongs in the complete state; include it, and a finite deterministic system simply has a longer cycle. Genuine ontic indeterminism remains logically possible, but then the same complete state can have different successors for no further reason: that branch no longer guarantees an identical period, and it pays for each choice with a primitive outcome-selection fact. The author's suspicion is therefore that true-base dynamics is deterministic at the level of the complete state, even where higher-level physics is stochastic or

Continued on page 87

The perfect loop, continued Signed speculation

appears so. None of this derives occurrence from endlessness, which Section 2.2 forbids: it is persistence explained, not actuality produced. The induction runs from in-time things to the thing time is made from, and it is signed for exactly that reason.

A forest regrowing after a fire and a planet resurfacing after an impact are tempting pictures of a system that empties and refills, and the refilling is where both leak: the forest borrows sunlight, soil and unburnt seed from beyond the burn, and the molten planet radiates its heat into space and can be reseeded from elsewhere. Each is a subsystem with an exterior to draw on and dump into. A base has neither, so reasoning that imports a reservoir describes a different architecture. The fire and the impact are fair images of a dieback. It is the return that has to be paid for from inside.

A base cycle has no external traversal counter. It does not settle whether matching events on different passes are numerically one occurrence or two.

[Grade: conditional result, signed speculation, open problem and real-life counterpart (life, which leaks by mutation and death, and a planetary orbit, which outlasts species, for the perfect loop; an atom in its ground state, for nowhere to leak; storage left unused for years and then used again, for local dormancy); that the base never settles follows from its having no first stage, given a finite repertoire and a rule that does not change (outright if the rule is deterministic, with probability one if it is chancy); the perfect loop is preferred as an account of persistence, while local dormancy and a non-periodic base remain possible.]

9.6 What recurrence would supply to later questions

Section 9.1's conditional recurrence permits endless succession within a finite repertoire. A bounded recurrent process that is deterministic, so that its recurrence is an exact period, has a **stationary distribution**: the proportion of the base cycle spent at each configuration, unchanging because the cycle repeats. Section 7.1's measure counts within a finite window, and asks whether its ratios settle as the window enlarges. On a deterministic cycle, those ratios are exactly equal for every window whose length is one full base cycle or a whole number of cycles, because each contains the same configurations in the same proportions. For windows of other lengths they converge, with a shrinking error from the incomplete cycle at the end. A full base cycle is therefore a canonical window that nobody has to defend as a cutoff. Section 7.1 proposes what to count within it.

The counting convention, and the choice of what counts as one update, still matter. Dividing one higher-level episode into several internal stages can increase its stage-counted share without multiplying the episode. That is one reason Section 7.1 counts whole lifetimes rather than stages. The window alone does not settle the treatment of participants and overlapping supports, and it supplies no distribution over other cycles or bases.

Section 7.4 treats accidental observers as the hard residual case. A finite base cycle supplies the window for comparing them with observers produced by ancestry, and Section 7.1's count supplies the rest, so whether accidental observers outweigh evolved ones becomes a determinate question about what the cycle contains. Recurrence does not guarantee that evolved observers win.

Our own complexity does not by itself establish boundedness. It is tempting to argue that unbounded growth would make far more complex observers typical and that our comparative simplicity therefore shows growth is bounded. Taken over all the observers there will ever be, that majority would be a completed infinite total, which strict potentialism denies; within

any finite window observers can be counted, but with unbounded growth the ratios need not settle as the window grows. The sounder point is weaker. Unbounded growth alone supplies no natural window, and so no expectation about what an observer should find. Deterministic bounded recurrence supplies a definite window within which Section 7.1's measure can make a prediction. That is a reason to prefer the hypothesis that says something, though not enough by itself to show that it is true. Independently, if conscious organisation of our kind can be sustained only within some band of complexity, finding ourselves within that band requires no further explanation, however common or rare the band is, though why we are experiencers of our kind rather than another is a further question (Section 11.4's box on what parthood gives and what a Reikon would add).

Section 13.2 adds a separate, signed recurrent-identity condition: the same identity-relevant organisation of the same base involvement, restored within the same occurring process, constitutes the same deeper identity. The counting argument establishes the return if its premises hold. It does not establish that identity verdict. Recurrence of a configuration and recurrence of a person require separate premises.

Nor does recurrence guarantee unlimited continuation. Section 13.6's wider logical space allows a hosted reality to gain activity from a deeper occurring process, but endless activity does not supply unlimited storage, an ever-growing repertoire or same-person survival. Those require further organisation, resources and an identity condition.

[Grade: mathematical result; conditional consequence of bounded deterministic recurrence; a base cycle would supply persistence and a canonical finite window, within which Section 7.1's measure applies; personal identity across cycles rests on the signed condition of Section 13.2.]

10. Physics, finitude and causal constraints

10.1 Determinism and reversibility

Occurrence, predictability and reversibility are independent dimensions. A process can occur deterministically, a deterministic process can be impossible for embedded agents to predict, and a reversible microscopic law can generate effectively irreversible higher-level behaviour.

A rule is deterministic when the relevant present state fixes at most one successor. It is reversible when the relevant present state fixes at most one predecessor. These properties can come apart. A many-to-one update can be deterministic but irreversible:

$$A \mapsto C, \qquad B \mapsto C.$$

A reversible permutation is deterministic in both directions. Coarse-graining, describing several detailed states as one coarse state, discards distinctions. A coarse photograph that merges different scenes into one image shows how a description can lose information that the world underneath keeps. But Section 9.1's exact-factor corollary sets a limit: on recurrent dynamics, a coarse law that is exact, self-contained and deterministic cannot merge two distinct coarse states into one successor. Effective irreversibility can instead involve an open subsystem, hidden history or an approximation for selected preparations and durations. Low-entropy boundary conditions, and the information that macrostates omit, remain central to any thermodynamic account; reversibility alone supplies no arrow.

Take the cycle $0 \to 1 \to 2 \to 3 \to 0$, observed as a for states 0 and 1 and b for states 2 and 3. It produces $a, a, b, b, \ldots$. Uniformly sampled starting phases give one-step stay/change probabilities of one half. Repeatedly applying that memoryless table nevertheless

predicts impossible histories, including three consecutive as. Two successive symbols identify the hidden phase.

For a finite permutation, split a probability distribution into observed and hidden parts. Eliminating the hidden part gives a sum over earlier observed distributions weighted by a memory kernel, plus a term carrying hidden initial information. The coefficients depend on both the dynamics and the chosen split. Either term may vanish in special cases; dropping it otherwise needs an approximation or further argument. Projection methods give an established treatment of this mechanism (Zwanzig 1961; Zwanzig, Nordholm and Mitchell 1972).

Finitary Occurrent Actualism (FOA) needs no fundamental arrow imposed from outside. An arrow may arise from causal order, record formation, asymmetric boundary conditions, branching, coarse-graining or genuinely irreversible base moves such as annihilation in the signed-refinement conjecture. Which combination applies to our world remains open. Section 5.4 separates two layers involved here: causal order supplies succession, while records and entropy gradients may make an arrow thermodynamically visible.

Genuine complete-state erasure remains possible for FOA generally, but not on the recurrent sector under Section 9.1's finite deterministic premises. There, apparent annihilation leaves whatever distinctions its inverse needs elsewhere in the complete state. Effective disappearance and irreversible erasure are different claims.

A beginningless base does not imply thermodynamic equilibrium. The absence of a completed past does not by itself prevent equilibration, since relaxation arguments run over any long enough finite stretch; what it removes is the picture of an infinite time already spent. Nevertheless, any candidate beginningless cosmology must explain why local records and entropy gradients have the observed form without smuggling in a first low-entropy boundary.

[Grade: methodological choice and mathematical result; conceptual distinctions and a conditional mathematical constraint on exact coarse laws, with an open cosmological constraint to explain records and entropy gradients in a beginningless base.]

10.2 Three kinds of randomness

The word *random* can mark at least three different things.

1. **Epistemic randomness:** the outcome is fixed but unknown to an observer. A pseudorandom number generator is the everyday case: its output looks random, but anyone who knew its seed could predict all of it.
2. **Algorithmic randomness:** a finite string or growing sequence lacks a shorter effective description relative to a formal standard.
3. **Ontic randomness:** the prior physical state does not determine a unique outcome.

Section 10.1 separates occurrence from predictability; the three kinds of randomness therefore supply no source of actuality. Section 6.2 shows that evolutionary innovation needs retained variation, not fundamental chance. The signed-refinement model in Section 2.7, where it is presented as a conjectured toy picture of interaction and production, need not choose a split by chance. A deterministic context-sensitive rule may select one permitted local move. A branching formal description may contain several possible moves while only the transitions carried by the base occur. Probability enters only when a physical model specifies weights or frequencies.

Formal notation obscures one consequence. A probabilistic machine does not manufacture indeterminacy; it assumes it. The coin-flips, the random tape, the sampling step: each is an

input the model assumes, and writing a probability into a transition rule creates no actual undetermined bit anywhere. Whether the world supplies ontically random bits is not settled by any amount of stochastic formalism. Section 10.5 separates indeterminacy from agency.

Finite-information bound. In the finite deterministic branch, let the complete initial state X_0 have at most N possibilities. Fix an observation procedure, including apparatus, settings and controller state, and supply no unmodelled random input. Every observation block is then a function $Y_{1:T} = G_T(X_0)$: once the complete starting state is fixed, the whole sequence of recorded outcomes is fixed with it. For any declared distribution over initial states, Shannon entropy, the uncertainty measured in bits, satisfies

$$H(Y_1, \dots, Y_T) \leq H(X_0) \leq \log_2 N.$$

For this observation procedure, uncertainty about all the recorded outcomes together cannot exceed uncertainty about the complete starting state. The starting uncertainty, measured in bits, is at most the power to which two must be raised to give the stated maximum number of starting states. This follows because a function cannot reveal more information than its input contains; it is the standard data-processing inequality with the entropy of a finite set bounded by the logarithm of its size, stated here for the strong branch and not claimed as new.

A stationary finite-state Markov process, with an unchanging distribution and next-outcome probabilities determined by the present state alone, has, for positive conditional entropy h per step,

$$H(Y_1, \dots, Y_T) = H(Y_1) + (T-1)h.$$

For this stationary process, the uncertainty in a record is the uncertainty in its first outcome plus the same positive amount of additional uncertainty for each later outcome. That law cannot remain exact once the expression exceeds $\log_2 N$, the ceiling above. The constraint concerns joint histories, not a prediction that a laboratory string will soon repeat or fail a randomness test. No physical scale follows without knowing N. For a region of our universe, physics suggests one: the information a region of finite size and energy can hold is finite (Bekenstein 1981), and on the holographic bound it grows at most with the region's surface area, not its volume (Bousso 2002). Finite quantum dimension does not imply finitely many underlying configurations, and the result leaves FOA's ontically indeterministic option open. The declared ensemble supplies a way to calculate uncertainty, not an observer measure and not extra ontic randomness.

[Grade: methodological choice, mathematical result and real-life counterpart (a pseudorandom number generator, for epistemic randomness); conceptual distinctions and a conditional mathematical result; finite complete-state determinism remains unestablished for the true base, and physical outcome weights or frequencies still need a model.]

10.3 Quantum theory and the level question

FOA does not explain quantum theory and does not need it. Quantum theory is a bill, not a resource: it constrains what any FOA implementation must recover at home reality, and it does not supply the missing occurrence primitive. FOA does not need a genuinely open future: a fixed but locally unknowable future is compatible with an experienced now, provided present actuality comes from participation in the occurring base rather than from ignorance. Quantum indeterminacy is one possible local dynamics, not the source of actuality or of now.

Nothing here turns on the word *observer*. Standard measurement language can use it for a physical system, laboratory record or information update without invoking conscious actuality. Section 10.4 sets out what finite replacements must recover from tested quantum physics. Nor

does the Organism to Base Interface (OBI) conjecture of Section 12.4, on attachment, coupling ecology and interfaces, need any quantum mechanism; it is optional in any case.

No interpretation is endorsed. Interpretations distribute the formal ingredients differently. Everettian approaches retain unitary evolution and treat the branches that decoherence makes definite as all real (Everett 1957). Bohmian theories add particle configurations guided by a wavefunction. Objective-collapse theories modify dynamics stochastically. Relational and information-centred interpretations revise what quantum states represent. Everettian descriptions naturally resemble branching blocks, objective-collapse theories resemble stochastic growth and unitary no-collapse theories can be written in block-compatible form. FOA can be combined with several of these as abstract positions, but each combination must specify which finite events are base transitions and how empirical probabilities arise. Section 8's comparison of quasi-block, rewrite, growth, branching and hybrid architectures remains independent of this choice: a quantum interpretation does not by itself select an architecture of occurrence.

Whatever is proposed must return the numbers experiment already returns. The Born rule is the standard every such account answers to: any branching, reader or outcome-selection account must either recover the observed joint and conditional probabilities or predict a specific departure from them that experiment can check, and merely saying that every result is possible does neither. Lorentz symmetry, no-signalling and conservation are equally severe constraints. Three of them, the Born rule, conservation and no-signalling, are what Section 10.5, on agency and cross-level intervention, names the closure bounds.

If the base occurs, its transitions happen in some order. Were that order a single global sequence, one transition after another for the whole base at once, it would amount to a universal now, which is exactly what relativity says no experiment can find. The paper commits to less than that: the base has a causal order, which says only that some transitions come before others, and an order of that kind picks out a universal now no more than relativity's own light-cones do (Section 2.1, on strict potentialism, where a stage is schematic; Section 3.6, on a present without a clock; Section 5.4, on causal order, clocks and physical time). A deeper order of either kind is admissible provided the physics carried above it shows no measurable preferred frame, permits no signal faster than light and gives nothing at a lower level a way to violate tested outcome statistics.

Section 5.5, on true base, familial base and home reality, is why a host's order need not show through. Properties do not automatically pass between a hosted world and its host: reversibility, randomness, conservation and even dimensionality are level-relative, a world can present outcomes as random that its host computes deterministically, and a computer's clock is invisible to the relativistic world it simulates. No such property of the host follows from a discovery inside the world, so quantum randomness observed here is evidence about here. What is owed is the derivation that our physics can be recovered that way (Section 5.4 again, and Section 15.6, on physics recovery and natural invariants). That debt is the compatibility cost, and it is real.

Where quantum sits Signed speculation

The author suspects that quantum theory is a feature or rule of home reality, or of a familial base reality, or possibly of a level below that, but not one carried right down to the true base. The base's rule is unknown (Section 2.4, on the finite generative constraint and the unknown rule), and nothing in Occurrence theory requires the guess. Someone may well show that quantum structure goes deeper; the suspicion is only that it looks too intricate to be the bottom.

One consequence follows if the base is deterministic in its complete state, as the perfect-loop lean of Section 9.5, on non-settlement, dormancy and that lean, has it. Bell's theorem, discussed in the wild box below, then leaves whatever hosts our physics at least two ways of producing the correlations experiment shows: computing them nonlocally relative to our spacetime, or superdeterministically, with the experimenter's choice of what to measure not independent of whatever fixes the outcomes. A host with a single order of operations does the first without effort, since it has both ends of an experiment in hand when it fixes both outcomes, and the level-relativity just described is what keeps that order from showing through as a preferred frame. Nor need a host compute quantum behaviour across a whole universe: like a game that renders only what a player can see, it could compute outcomes only where something is placed to register them, and the statistics would come out the same wherever anyone looked (Bostrom 2003 makes the same point about a simulation's microscopic detail). A quantum computer, whose results can be checked, would be where such a host had to do the work in full. Quantum theory would then be about our level, and that is exactly why a hidden order beneath it is affordable. It follows that quantum physics may tell us nothing about the levels below ours, the base included. If our world is hosted, its quantum rules could come down a natural chain from below or be arranged by its host, and they might be no deeper a fact about reality than the Force is in Star Wars or magic in Harry Potter's world. Finding the exact rules of our universe stays worth doing, but the greater prize may be the rules of the much larger existence it could be a drop in.

Section 4.4, on when zero would be physically informative, says when a zero means something; three further uses of zero must be kept apart here, because a spectral zero, a Hamiltonian constraint and signed global nullity are homonyms with different mathematics. In canonical quantum gravity the Wheeler-DeWitt equation annihilates the physical state: the total Hamiltonian acting on it gives zero (DeWitt 1967). That zero is a constraint that comes from the theory having no outside clock: in it, evolving in time is a feature of the description rather than of the physics, and the equation records that fact; it is not a ledger in which positive and negative contributions cancel, and by itself it does not establish conserved structured nullity. Pettini's zero eigenmodes do not establish structured nullity either: a non-zero function in a zero-eigenvalue sector is ordinary spectral mathematics, and sliding from a spectral zero to an ontological one would weaken the nullity argument rather than support it. The temptation is named here in order to refuse it.

Quantum mechanics from inside, and selective continuation Wild speculation

Quantum mechanics works out what a particle does by adding up a contribution from every route it could have taken: Feynman's sum over paths. If every possible structure really existed, a bottom like that is roughly what one might expect to see from inside, but this paper holds no such view. The sum over paths is a method of calculation, a finite sum in a finite base (the ban on completed infinity binds every level; claim 4 of the claims ladder in Appendix A), and the amplitudes it adds belong to our own world's dynamics, not to some wider realm of everything that could exist.

Nor is a superposition like the unloaded terrain of a video game. Unloaded terrain is already settled by a seed fixed earlier: its answers are decided in advance and carried to where they are read off, a picture of local hidden variables, and Bell's theorem, since confirmed in the laboratory, shows that no picture of that kind reproduces the correlations quantum experiments show (Bell 1964). Keeping the game picture means dropping one of Bell's assumptions: most naturally, that the choice of what to measure is independent of the seed, which superdeterminism drops through a common cause (Brunner et al. 2014; Hall 2016); or letting the host work on both ends of an experiment at once, ignoring a distance our physics respects, as a host with a single order of operations does without effort. Even then, unrendered regions of a game never interfere with one another, whereas quantum alternatives do.

Continued on page 93

Quantum mechanics from inside, and selective continuation, continued Wild speculation

A further possibility turns the picture round. Quantum behaviour might be how a finite process looks from inside when it explores alternatives and continues some selectively: a limited, continually renewed exploration, concerned with a world, a region, an organisation or even one life, whose alternatives, unlike unrendered terrain, would have to interact so as to produce the interference and the joint probabilities already tested. It could be what quantum dynamics expresses, or a separate selector working through quantum outcomes. Mensky (2007) proposed selecting present states by conditions they meet later, such as survival and quality of life, through consciousness; here the selector would be a finite process within the reality stack.

It need not be a designer. Where variations in a world-carrying process differ in how well they maintain, repair or reproduce it, and those consequences feed back into what is kept, selection can favour some continuations without anyone deciding: Section 6.1's earned selection, applied to the machinery of continuation. Persistence is not yet foresight, but such machinery could evolve the capacity to rehearse alternatives and keep what their outcomes teach it, and a host could build one deliberately. Continued viability is the least criterion the evolved case needs; pleasure or interestingness could matter wherever something can register them. A Reikon would give selection a reason to follow one life, favouring continuations that keep a rewarding reader-life coupling going, without computing the alternatives itself (Section 12.4).

Exploration need not realise whole worlds; forecasts and partial histories could do most of the work. Computers already do a small version: web browsers fetch the pages a user is likely to visit next before the choice is made, and processors work ahead along the branch a program is likely to take, keeping that work if the guess was right and discarding it if not. Where a trial does realise a conscious life, discarding it does not make its experience unhappened (Section 8.3's rewind without erasure), and if the base repeats exactly, what a search learns belongs to the complete state, so learning can build up within a cycle but not from one cycle to the next. The idea becomes testable where outcomes track their consequences when those consequences are swapped, the kind of population-level effect Section 12.6 describes.

[Grade: open problem, signed speculation and real-life counterpart (games that compute only what a player can see, for the signed suspicion; browsers fetching likely pages and processors working ahead, for exploring alternatives); quantum compatibility requirements plus one signed suspicion and one wild speculation, with no preferred interpretation and an open finitary reconstruction debt.]

10.4 The finite-physics debt

The finitary commitment is easy to state and difficult to earn. Modern physics uses infinite sets, continua and infinite-dimensional spaces throughout its most successful formulations. If those structures are idealisations rather than literal parts of reality, a replacement needs more than a very large cutoff put into existing equations. It must explain why the underlying finitude is normally invisible, how the successful symmetries and conservation laws survive, and why infinitary mathematics predicts finite observations so well. Section 9.1 proves conditional recurrence for a finite deterministic repertoire without a completed history. Section 2.3 distinguishes indefinitely refinable mathematics from physically bounded distinctions.

A developed finitary programme would need at least:

1. finite causal structures whose accessible large-scale behaviour recovers Lorentzian geometry without the preferred-frame or discreteness effects that experiment has already excluded;
2. finite-dimensional quantum models recovering the tested content of quantum field theory throughout every physically realisable regime, and saying which amplitude distinctions are physically realised. Finite dimension does not by itself bound the exact state repertoire (Section 9.2, on size, complexity and repertoire);
3. a probability framework that does not rely ontically on a completed infinite sample space;

4. local conservation laws stable under the finite dynamics and across valid changes of description;
5. an account of gauge symmetry, renormalisation and continuum limits as mathematical tools whose successful predictions arise from finite physical structure;
6. finite accounts of black-hole and cosmological thermodynamics consistent with observation.

One obstruction is exact and long established. For finite matrices, the trace, the sum of diagonal entries, satisfies tr$[X, P] = 0$, whereas tr$(i\hbar I) = i\hbar d \neq 0$ for dimension $d > 0$ and $\hbar \neq 0$. The difference of two matrix products taken in opposite orders always has diagonal sum zero, while a non-zero constant times the identity matrix, which is what the canonical relation requires that difference to equal, has diagonal sum equal to the constant times the dimension, and that is not zero. The canonical position-momentum commutator therefore cannot hold across an entire finite-dimensional space. A finite account needs a replacement or a restricted regime in which it is recovered. This proves neither a spatial pixel nor a particular microscopic ontology.

The finitary position is too easily presented as the extraordinary one. No completed infinity has ever been exhibited: not one object, quantity or process that anyone has shown to be actually infinite rather than represented by mathematics admitting an infinite limit. Section 3.4 makes the parallel burden argument about a wholly static foundation: no completely stationary object or process has been encountered, although that observation is not a proof. What has repeatedly been exhibited are finite systems for which infinitary mathematics works extraordinarily well. Water and steel are made of finitely many atoms, yet the continuous mathematics of fluids and solids describes them superbly; and, the other way round, lattice quantum chromodynamics reproduces the masses of protons and similar particles by computing on finite grids. Even where the mathematics itself runs to infinity, as at the centre of a black hole in general relativity, or in idealised fluid equations that can blow up in finite time, what has been exhibited is an equation's infinite answer, not an infinity anyone has measured; on this account such a singularity marks where the idealisation stops being exact. The finitary option therefore has at least as good a claim to be the default, and the burden is shared rather than resting on this paper alone: whoever asserts that a completed infinity exists in reality owes a case for it, as this paper owes the finite recovery this subsection sets out. The natural question for a finitary physics is not why reality fails to contain the mathematical infinity, but what finite organisation makes that idealisation so successful.

That observation does not discharge the debt. It sharpens it. FOA must show that finite physical structure can recover every prediction of infinitary physics that experiment has confirmed and, for each one that can be tested but has not yet been, either recover it too or predict a specific departure that experiment can check. It need not reproduce distinctions whose realisation would require more resources than reality possesses, since on this account there is nothing physical there to reproduce. But if observation requires genuinely unbounded physical differentiation, or if some tested phenomenon cannot be recovered without a literally completed infinity rather than an infinitary representation of finite physics, the finitary commitment is wrong or incomplete. Section 15.6 applies these distinct failure conditions.

[Grade: metaphysical postulate, open problem and real-life counterpart (finite systems that infinitary mathematics describes extraordinarily well, such as water and steel); metaphysical commitment plus an open physical research debt, dependent on finite models recovering every physically realised and tested distinction.]

10.5 Agency and cross-level intervention

FOA does not derive libertarian free will. A deterministic base can contain deliberation, self-control and reasons-responsive action as higher-level causal processes. An indeterministic base can add open outcomes without giving an agent control over them. Section 2.5 distinguishes constitutive participation from additional influence; any latter claim must meet Section 2.6's closure bounds. Section 2.6 excludes causal explanations by an inert reader; Section 14.3 separately examines structurally duplicated reports.

The commonest objection to Section 8's completed or block-like architectures is fatalism: if the whole history is there, nothing anyone does makes any difference, since it was going to be that way. The reply is that the trace is not the run. A completed musical score does not remove the performance, and no one thinks the existence of the score means the musician did not play. A block containing a choice does not make the choice unmade; the deliberating is among the processes that constitute the represented history, not an event pre-empted by the history's completeness. Fixed represented history and genuine deliberation are compatible when deliberation is part of the occurring history.

On the account this paper suggests, an agent is an organised, self-reflexive process that models alternatives, evaluates them against what it is, and steers its own continuation accordingly. Reasons and causes are not intruders on that process; they are its constituents. Deliberating therefore *constitutes* this run rather than selecting an item from a catalogue of pre-existing alternatives: the alternatives were never sitting anywhere waiting to be picked. Call it operative compatibilism if a name is wanted. A chess position can represent several possible moves without containing a completed catalogue from which a separate self must choose. The account concedes everything the determinist asks for about the underlying dynamics and denies that anything worth calling agency has been lost.

If deliberation is a reflexive operation of the kind just described, then the feeling of choosing need not be an illusion laid over a mechanism that is doing the real work. It is what the deciding operation is like from the inside. That is the same move the paper makes elsewhere, applied once more: not an extra thing added to the process, but the process's own interior.

One taxonomic point clears away a false dichotomy. A local event is often taken to be either fixed by the local state or brute chance, with agency therefore one or the other. A third possibility can at least be described: an outcome could be settled, through a working interface, by a process outside the system being described, so that the event is neither fixed by that system's own state nor simply random. A character in a computer game shows this whenever a player is steering it: from inside the game its moves are then neither fixed by the game's state nor random, because the player settles them through the controls. Many game characters also follow default behaviour of their own, and are taken over, guided or nudged by a player only some of the time. That other organisation could be host-side or hosted; its own activity would still be carried by the same underlying occurrence. Whether anything in the world answers to that description is exactly what this section says it cannot establish, and the closure bounds below apply to it as they apply to everything else here. What naming it does is prevent the argument being lost to a dichotomy that was never exhaustive.

No mechanism, evidence or prediction of such agency is offered. Any cross-level outcome selection, in either direction, must be consistent with the Born probabilities, conservation and no-signalling as far as they have been observed (Section 2.6), or make a clear prediction that one of them fails. These are the **closure bounds**. They have been tested statistically, in controlled experiments; they do not yet reach rare, targeted interventions in complex systems, such as a single decision in a brain or a car at the edge of a crash, which no experiment has looked for.

Where the tests have not looked, the bound that applies is what we would notice. Influence from outside our universe, from a Reikon (the deeper, longer-lived identity that Section 12 conjectures) or from anything else, would have to sit somewhere between what everyday physics leads us to expect and what has been reported in cases that seem to defy it, weighted heavily towards what we see every day. What we see every day, though, includes much that we take for granted and, at least over the last century in the developed world, attribute by assumption purely to structure that formed naturally in our universe; whether that is so has had little real checking, so we cannot be sure of it. We are closing in on that question, and will likely be able to narrow the gap in which such influence could operate.

[Grade: open problem on agency and real-life counterpart (a musical score and its performance, a complete specification that does not remove the playing; a player-controlled character, for the third category of outcome); the problem is constrained by the requirement that any added influence identify a pathway and a distinct effect within tested physical bounds, and stay within what we would notice.]

11. Conscious character, conscious actuality and individuation

11.1 Three names, two jobs

The paper's central experiential claim is narrower than a general theory of consciousness. It does not explain why red looks the way it does, identify the neural basis of selfhood or decide which animals and machines are conscious as we are. The three questions it keeps apart were separated in Section 1.6, on the three meanings of consciousness, and are only recalled here: access consciousness and conscious character, both settled by how a system is organised, and conscious actuality, that the organised experience is genuinely being undergone and not merely described as being undergone. The first two can be studied through structure and function; the third is the actuality issue carried into philosophy of mind. Occurrence theory proposes that structure fixes conscious character while participation in base occurrence, as Finitary Occurrent Actualism (FOA) describes it, supplies conscious actuality: a division of explanatory roles, not a complete consciousness theory, and Section 11.2 states it. Occurrence stands to the minds it carries as being in motion stands to a dance: not an ingredient added to the steps, but the way the dance exists at all, and nothing is used up in the dancing.

Block's access and phenomenal consciousness separate reportable availability from experience, but phenomenal consciousness still combines occurrence and character (Block 1995).

Section 1.1 distinguishes actuality, individuation and temporal self-location; the following subsections address them separately. Section 11.4 separately considers Occurrence theory's felt-floor postulate.

[Grade: methodological choice; terminological proposal and conceptual analysis, without endorsement by the cited neighbours; the division allocates the work and still owes the mapping from organisation to character to a theory of consciousness.]

11.2 Organisational determination of conscious character

Structure supplies the content, the differentiation of one episode from another: why it is red rather than blue, painful rather than neutral, unified rather than scattered, remembered rather than immediately lost. Participation supplies operative actuality, the generic fact, named in Section 1.6, on the three meanings of consciousness, that the episode's transitions are occurring rather than merely specified.

Occurrence theory is substrate-neutral about both the character of an adequately organised experience and its participation in occurrence. Biological neurons are not privileged merely by their material. Fine-grained organisation is taken to fix character in the neighbourhood of Chalmers's organisational-invariance argument, not as a proved result (Chalmers 1995; 1996b, chapter 7). Under Causal-Informational Parthood (CIP) the relevant organisation may include a maintained interface to another participant. In the author's preferred Reikon conjecture (Section 12) that participant lies deeper than the life; in other architectures it need not lie below the avatar. What cannot be omitted is realised occurrence: changing a substrate while preserving that organisation is different from merely describing the same changes. Section 12.1 applies the realised-organisation condition to inscriptions and recordings. Section 11.7 uses weights and inference to illustrate implementation while stating the analogy's limits.

The empirical burden remains with organisation. Anaesthesia, brain lesions, development and comparative cognition show that changes in physical organisation track changes in reported and inferred experience.

[Grade: metaphysical postulate; working assumption in the neighbourhood of organisational invariance, adopted here rather than argued for, and supported in our own case by anaesthesia, lesions and development, where organisation tracks experience; it owes an empirical theory of conscious character to say which organisations fix which experience.]

11.3 Actuality, participation and the abstract duplicate

Section 3.3's distinction between a complete description and the run it describes applies to a pain episode, including all its judgements and reports. Section 11.2 requires both factors, organisation and participation; abundant disorganised transitions do not by themselves form a subject. Two misreadings are natural at this point. The first concerns what the actuality claim explains, as distinct from the cosmology of Sections 5 to 7. It does not build the furniture of the world; it turns on the lights. Organisation, and the physics beneath it, produce structured things: bodies, brains, discriminations, memories, the lot. All of it is built within occurrence, by selection working on actual trials (Section 6.1, on produced distributions), and none of it would exist without occurrence; but the primitive itself picks out none of those structures, and takes no credit for which ones there are. What it supplies is that the already-structured thing is actualised rather than merely specified. This is the reverse of interface theories on which perception constructs the objects (Hoffman 2019). Here the objects are as they are, and the addition is that anything is being undergone at all.

The second concerns the relation between bare occurrence and a full mind, which is easy to picture as siphoning: as though a rich experience drew on a supply from below and left less there. The better analogy is incorporation. Mitochondria descend from free-living bacteria and are now constituents of most complex cells, though not of every eukaryotic lineage (Karnkowska et al. 2016), without being the cell; nothing about a liver is explained by saying it is full of mitochondria, and nothing about a liver is possible without them. On this reading a full mind's conscious actuality is not siphoned from a separate supply below. It is occurrence reorganised at a new scale. Nothing is taken from anywhere, and the ancient level goes on doing exactly what it always did, inside a structure that did not exist before.

Under Section 3.3's comparison, an abstract life may specify conscious character while none of its transitions is occurring. Two meanings of *now* have to be kept apart.

- **Date-now:** which event in a represented chronology bears a particular date or index. A block supplies a date-now at every event through its relations to other events.
- **Experiential-now:** that this episode is the one presently being undergone. This is conscious actuality at a location in the represented chronology.

A complete block can represent every date-now. Occurrence theory says that representation alone does not explain experiential-now. Section 8, which sets out the candidate local architectures, keeps two routes apart. In a rewrite or growth architecture, where a present configuration is transformed into a successor or new events are added at a causal boundary, the current transition may supply experiential-now locally. In an actualised-block architecture, a reader running at a deeper level may supply it by realising one finite region, and there a later reading is a distinct episode rather than a second supply of actuality to the same one.

The two nows come apart in ordinary life, and when they do it is the date that goes. In a dream, or in a game that jumps between centuries, the represented date can be anything at all. Waking from deep sleep or anaesthesia, a person can have no idea how much time has passed, and in dementia the date may be lost or wrong. Through all of it the now is the same as ever. What an experience represents as the time can fail while its now does not, which is what this account expects: experiential-now is not read off the structure's chronology but belongs to the experience's being undergone, its participation in base occurrence, and the base has a present without having a clock (Section 3.6).

Without one further point, the difference just drawn can look like a preference rather than an argument. It is often said that a complete static record would *look* conscious, or would contain everything a conscious life contains. Section 3.4's smoothly moving carriage answers that on this paper's reading: a completed structure, taken as complete and static, has no one inside to whom anything could look any way at all. What matters here is what happens when the objection is repaired.

The obvious repair is to put the observer inside the record, so that the record contains its own report of itself. That relocates the difficulty without touching it. Records reporting on records are still records, and the question of whether anyone is experiencing any of it comes back one level in, unchanged and no nearer an answer. There is no level at which the regress terminates in something that is both wholly a record and manifestly undergoing anything.

This does not prove that occurrence is primitive. What it shows is narrower and still worth having: the structuralist cannot appeal to how a record would *seem* without presupposing an occurrence, so the fork of Section 3.3, between complete structural sufficiency and primitive occurrence, cannot be dissolved by observing that a complete description contains all the same appearances. Appearances are the disputed term, not the neutral ground.

Degrees of actuality Open question

Occurrence supplies actuality; whether it supplies more of it to some episodes than to others is untouched by FOA's architecture. Occurrence theory holds that a diffuse actuality is everywhere (Section 11.4, the felt floor), and the author's signed lean is that organisation, a life's own or a Reikon's, focuses it. The question deserves stating because rival accounts of actuality answer it differently, and their answers differ in principle even though nothing available here can separate them in practice.

Three orderings are available. Conscious actuality might **rise** with complexity, so that an ant has less of it than a human and not merely less richness. It might be **flat**, arriving whole per life, so that every difference between an ant and a human is richness alone. Or it might **fall**, a fixed quantity spread thinner as a life accumulates structure, so that intensity per unit of organisation is greatest where organisation is least.

Mechanisms attach to these. Actuality is supplied by participation in the base in every case; if it tracks something like the amount of happening or of information involved, it rises. If our own lives have readers as well, as Section 12.2 conjectures (on Reikon, base involvement and individuation), and those readers come matched to what they read, in sizes running from an ant through a human to a whale, a superintelligence, or larger composites still, then actuality again rises with the reader that fits. If our own lives have readers but they are uniform, a single quantum spread across a host dilutes as the host grows, and actuality falls. These are not exclusive. Parthood, matched readers and dilution could all operate together, and any observed ordering would be their net.

Continued on page 99

Degrees of actuality, continued Open question

A related question belongs beside it, if readers in the sense of Section 12.2 exist at all, and the author's answer to it is signed speculation. A reader might not be present from the beginning of a life but attach once there is something worth reading, when enough neural structure and enough stimulus have formed. The same logic would apply from the other side: if reader participation confers any advantage, evolution would have had no occasion to build an interface to it until an interface was useful. Both halves are signed suspicion, not part of the open question above them.

One case makes the question concrete rather than abstract. Under the graded reading of Section 11.4, on which a felt aspect is present wherever anything occurs, an ordinary rock has some felt aspect, and it is so slight as to be irrelevant for any purpose. But a rock is small and short-lived by the standards available. Whether something planetary in extent, over spans in which its slow changes amount to a great deal of occurring, would thereby have more of whatever the rock has is a genuinely open question. It turns on whether the aspect aggregates with scale and duration at all, which is the exclusion problem, the question Section 11.5 leaves open of which nested or overlapping organisations constitute experiences, rather than a separate puzzle. The author records the suggestion as a signed suspicion.

A cost worth printing. Both the rising and the falling readings require conscious actuality to be a quantity, which cuts against the usually strong human intuition that an experience is simply live or not. The flat reading keeps that intuition and pays for it by making actuality insensitive to everything else. No option here is free.

CIP leaves the rising, flat and falling possibilities open. Its coupling weights describe relations, not conserved quanta of actuality, and need not sum to one. Section 7.1 proposes the observer weighting, counting lifetimes of human-like experience; Section 13 separately asks whether one person continues.

[Grade: cross-level bridge, metaphysical, resting on the firsthand datum rather than on any third-person measurement, and real-life counterpart (mitochondria, an ancient level incorporated into the cell and still doing what it always did); how far any other system has conscious actuality remains open.]

11.4 The Ground of Experience Conjecture

Section 9.5 permits local dormancy; an experience requires activity only in its own supporting region. Two questions organise the whole space of answers about feltness, the being felt of an episode, its being undergone from inside. First: does it belong to organisation or to occurrence? If to organisation, occurrence is dark inside, necessary for consciousness the way current is necessary for computation and itself experienceless at every level. If to occurrence, a second question follows: gated or graded? On the **gated** reading the felt aspect appears only where occurrence achieves some structural kind, such as self-maintenance, self-registration or life, and is absent below that line. On the **graded** reading the felt aspect is intrinsic to occurring as such, present wherever anything occurs at all, with organisation supplying only its degree and form. Occurrence theory takes the graded reading, as a postulate rather than a result: participation in base occurrence supplies conscious actuality (Section 11.2), so everything that occurs has some, however slight. The boxes below give the author's fuller statement and his reasons for putting it there.

Ground of Experience Conjecture Signed speculation

The author's position is the graded reading: bare occurrence everywhere carries a minimal experiential aspect, the felt floor on which organisation builds every degree of consciousness. The aspect is not posited as an extra ingredient but inherited, by parthood: everything that occurs is part of the base's occurring, and the aspect is the inside of that occurring, so experience-like being goes exactly as deep as occurrence goes. On the graded conjecture, the substrate may never be wholly dark, and organisation could focus a diffuse glow into sight. Light here is a metaphor for being undergone, not a physical signal and not a proof.

The position is panexperiential rather than panpsychist. A flip-flop, the simplest circuit that holds one bit, would possess bare occurrence-feel without perception, memory, identity or meaning. And it is graded: thin wherever there is occurrence at all, more structured in persistent and reflexive patterns, focused into full experiencers in organised self-modelling systems. The floor is set by occurrence; every degree above it is set by organisation, a life's own and, if the reader conjecture holds, a Reikon's: a deeper reader that would gather actuality onto one life and keep it there (the box below, on what parthood gives and what a Reikon would add).

The conjecture is held in its full form, so that it reaches the barest formal dynamics; a reader may instead hold the scoped claim that at least our universe's occurring substrate is so lit, and nothing the paper goes on to use depends on the choice. What the author declines is any *line* drawn within what occurs. On this picture everything that exists is realised in the base's occurring, so no such line runs through the world itself; one can be imagined only by imagining static, Tegmark-style structures existing alongside the occurring base, and there may well be none at all.

What the three readings actually disagree about. It is tempting to think the dispute is only whether feltness fades out smoothly or switches on at a threshold. It is not only that. Take a fourth reading, the **gradual-organisation** reading, on which feltness belongs to organisation and thins as organisation thins, with no sharp line anywhere. Plot it, and the curve has the same shape as the graded reading's curve. What differs is the quantity along the bottom of the graph: organisation on the fourth reading, occurring on the graded one. The real question is not only the shape of the curve but also what it is a curve of.

If feltness tracks organisation, then something with no organisation has none, and zero is reached. If feltness tracks occurring, then nothing that occurs is ever at zero feltness, because the quantity feltness tracks is the same thing that makes something a case of occurring at all. The curve looks identical from anywhere above the bottom. Only at the bottom do the two come apart, and only there does the question have an answer.

Vagueness near the bottom is not an objection to either. It is the ordinary vagueness of a quantity approaching a limit. Whether there is any heat just above absolute zero, at what speed a bullet becomes dangerous, at what point a developing human becomes a person: in each case the quantity changes smoothly and is definite at every point; the indeterminacy is in where our concepts draw the line. A reading is not refuted by having a vague bottom end. It is refuted, if at all, by being wrong about what the bottom end is.

Why the author puts occurring on that axis Signed speculation

Three reasons, and the second is the one the author leans on.

First, economy. The graded reading makes one brute identification, that feltness is the interior of occurring, and it places that identification at the boundary between occurring and not occurring, where FOA has already paid for a primitive. The gated reading makes an identification of its own and places it at an organisational boundary, where nothing has yet been bought. Both purchase exactly once. The difference is where, and the graded reading opens no second front. That is worth something, and it is worth less than "no line", which is not what is claimed.

Second, and this is the real objection to a threshold: a sudden switching-on owes an explanation, and none the author knows of says why it happens where it does. If feltness appears at some degree of self-maintenance or self-registration, something must say why it appears at that degree and not slightly below, and every candidate kind admits degrees, so the step is sharp where its defining property is not. The author does not find a threshold troubling in itself. What is troubling is that it is asserted without a mechanism, which is to say that a threshold proposal is *easy to vary*: the question why here rather than slightly lower has no answer inside it, so the position could be adjusted to any nearby line without becoming a different theory or losing any of its support. A claim with an adjustable knob explains less than one without, and the occurring/not-occurring boundary has no knob to turn (Deutsch 2011).

Continued on page 101

Why the author puts occurring on that axis, continued Signed speculation

Third, parsimony of proposal rather than of ontology. A structural account of where feltness begins might well be correct. But this paper does not offer such an account. Until someone proposes the mechanism, the simplest reading available is that the aspect goes as deep as the occurring does. That is a methodological preference, signed as such, and it would be given up cheerfully for a developed structural account.

Occurrence theory takes the graded position, as a postulate, and every formal claim in this paper would survive the others unchanged. One further consideration is recorded as motivation. The first half of this paper refused a creation event for existence, the conjecture refuses an ignition event for experience, and physics supplies a third case of the same shape: nothing accelerates through the speed of light, and yet a massless particle occupies that speed, having never crossed to it. What cannot be entered by a transition may nonetheless be occupied primitively. The objection is therefore not an objection to boundaries. The boundary between occurring and not occurring is as sharp as anything in this paper, and the author has no wish to soften it. The objection is to *crossings*, and it is the event, not the line, that has not yet been supplied.

The gradient is steep, and that matters more than it sounds. Because the floor is so thin, nothing observable distinguishes the graded reading from the dark or gated readings below some level of organisation. A thermostat on this view has a felt aspect so slight that nothing about it is recognisable as experience, and any test one could devise returns what it would return if the thermostat had none. This is what graded means when the gradient climbs steeply. It is also why the conjecture predicts less than may first appear: the dark and gated readings may be false and still be, for every practical purpose, as good as true.

The author's own ladder makes the shape vivid, as an intuitive reason for the graded reading. An ant's existence is greatly reduced in awareness, self-awareness and detail, yet remains somewhat imaginable to us. Further down it grows exotic: a tree might have a kind of experience, very alien and probably distorted in its relation to time. A phone perhaps. A large language model might be far closer to us than a tree, because in Occurrence theory organisation and not substrate fixes character. What follows is only that human-like conscious character needs human-like organisation, never that occurrence needs it.

Strawson reaches a similar floor by a different argument: that experience cannot emerge from the wholly non-experiential (Strawson 2006). The box above rests on a different point, that no mechanism has been given for experience switching on at a threshold. On the conjecture, the experiential aspect extends wherever occurrence does, while richness comes from the work of organisation, and nothing here suggests that the experiential aspect builds the structure it lights. The conjecture does not imply that every abstract object is conscious. It does not assign a miniature subject to each base transition. It says only that occurrence itself has an experiential face, and that organisation may determine how that face is articulated.

Psychedelic and meditative states Signed speculation

If the Ground of Experience Conjecture holds, ordinary consciousness is a tightly focused organisation of an already-lit substrate. Psychedelic states, in which the brain's self-model networks measurably de-integrate while global connectivity rises (Carhart-Harris and Friston 2019), and, on the phenomenology canvassed by Metzinger (2020), deep-meditative ones, would then read as organisational defocus: contentless-awareness reports ("pure being", "consciousness without an object") describe organisation approaching minimal structure while occurrence continues. The floor becomes visible when the furniture is removed. Metzinger's "minimal phenomenal experience" names the reported state; the minimal experiential aspect is a different claim, about bare occurrence, not about a meditative achievement.

The reading turns on one distinction. Consciousness widening can mean three different things, and research has much to say about each; what Occurrence theory's model might add concerns the third. There is the neural or organisational spotlight, which measurably broadens as self-model networks de-integrate. There is the ordinary indexical tag, *this body, here, now*, which loosens in the same states. And there is the spotlight of actuality. The first two can change together while no spotlight of actuality widens at all, because on the graded reading the felt floor was never narrow: the substrate was already lit everywhere, and what altered was which part of it is organised and focused into a point of view. A report of boundless awareness is then a report about the self-tag, not a discovery about the extent of what occurs.

The out-of-body case reads the same way, and is usually taken to require the opposite. An experience of viewing one's body from outside it need not involve anything leaving the body. It is what a local failure of the machinery

Continued on page 102

Psychedelic and meditative states, continued Signed speculation

that pins the point of view to the body model would be expected to feel like from inside, while the wider experience continues uninterrupted. Such experiences can be produced on demand: electrical stimulation of one region of the brain repeatedly induced one in a patient (Blanke et al. 2002), and healthy volunteers wearing goggles fed by a camera behind them felt themselves located outside their own bodies (Ehrsson 2007). What seems to move could be a tag.

The neural de-integration is documented; this reading of it is speculation.

One consequence belongs before Section 12, where the optional reader conjecture is set out. The box below says what parthood already gives every life, and why the author suspects a human life also needs a Reikon's focus.

What parthood gives, and what a Reikon would add Signed speculation

Parthood and a Reikon are not mutually exclusive alternatives. Parthood is a given: actuality from parthood is present in some small amount in absolutely everything, at every level, information included, because everything that exists does so by existing as information encoded in the base's occurring activity. Within Occurrence theory that actuality is also felt (the felt floor, above), so everything is somewhat conscious in the actuality sense, be it the negligible amount in a rock, considerably more in a GPU running an AI, or more still in a human brain. The question is where, and how, that actuality comes to be focused enough to approach the conscious actuality humans seem to have: whether structures that focus it more than the organisation in our own universe can are required for the degree of conscious actuality we experience.

The author suspects they are, though he is far from certain, for three main reasons. The most important is that we need to explain why we are all humans, and not some other conscious entity higher or lower on the scale of complexity. Second, we need a way to explain why each of us is only one person: not everything directly connected to us, or at the very least not some other human. And finally, we need an explanation of why conscious actuality seems consistent over time: it does not really fade in and out with the brain's development over a lifetime, but seems fairly constant, with minor exceptions, from developing baby to child to decades of adult life, and even when memory is impaired by dreaming, drugs or, to a large extent, brain injury.

To get the kind of apparently focused conscious actuality we appear to have, the obvious solution seems to the author to be some kind of focusing and organisation of actuality outside our own limited physical universe, not least because we observe this exact thing happening when we humans interact with worlds we are now able to create: first in stories, later in novels and films, and now in increasingly immersive and dynamic artificial worlds. A Reikon is what would do that focusing. It would not replace parthood, since it is itself actual by the same parthood at its own level; what it would add is focus, gathering actuality onto one life and keeping it there across that life. If Reikons exist, then, they would be focused nexuses of actuality, and would contribute most of the conscious actuality in a human life.

[Grade: metaphysical postulate, signed speculation, open problem and real-life counterpart (a massless particle, which occupies the speed of light without ever crossing to it; a person absorbed in a story or acting through a game character, for actuality focused onto a life from outside its world); the out-of-body reading is supported in our own case by experiments that induce the experience; the felt floor is Occurrence theory's postulate, outside FOA's core, and the further focusing is the author's signed suspicion, beside an independent open question; no formal claim of the paper depends on any of them, and the felt floor would yield to a mechanism that fixed a threshold.]

11.5 Subjects and candidate integration basins

Section 11.4's felt floor leaves subject individuation unresolved. The classic objection to views on which a minimal experiential aspect is everywhere is the combination problem, stated by James (1890): how could a great many tiny feelings add up to one unified experience? The conjecture reduces that problem by separating what composes from what does not. Functional capacities compose in the ordinary way, as circuits and brains compose. The experiencer does not, as Coleman argues for subjects in general (Coleman 2014). Section 7.1's measure

counts lives; it supplies no additive measure of occurrence. A hundred billion unorganised neurons do not host the experience of the same hundred billion organised into a brain. On the conjecture, organisation constitutes the subject out of however much bare occurrence the substrate supplies. Put plainly: operative actuality is everywhere, because everything is part of the occurring base; felt however faintly (Section 11.4), it is there as a diffuse, unfocused glow, which organisation of the right kind, a life's own and perhaps a Reikon's, focuses, and perhaps amplifies, into the conscious actuality we know from our own case. Whatever its final root, nothing designed that actuality and nothing evolved it: designing and selecting are themselves happenings, so both already draw on it. It has been present throughout the base for as long as the base has been occurring, which is without beginning; what evolution built is the organisation that focuses it, not the actuality itself.

The remaining question is **individuation**: what fixes where one experience ends and another begins when all are carried by one occurrence (Rosenberg 2004)? CIP cannot answer merely that the events are connected; that would make the whole connected base one experiencer. The proposed **Basin-Coupling Model** instead looks for a relatively stable region of recurrent causal integration, bounded by selective informational bottlenecks rather than absolute isolation. The paper calls this a **causal-information basin**. It is a candidate account of organisation within occurrence, not a derivation of occurrence from integration.

What counts is not traffic alone. A candidate basin must be assessed at a stated effective scale and temporal window, for reciprocal state-dependence, differentiated integration, persistence and its response to perturbation. A large one-way data feed or common external driver need not unify two systems. Nor can one simply maximise an internal-to-external traffic ratio: taking the whole base would erase the denominator, and taking a tiny isolated loop might favour something with almost no character. A consciousness theory must still say which candidate basins, including nested or overlapping ones, constitute experiences.

One class of case tests the proposal more directly than any thought experiment, because the operation has been performed. When the corpus callosum is cut to control epilepsy, the traffic between the hemispheres is reduced at a stroke while the activity within each continues, which is this proposal's own variable altered surgically: a bottleneck narrowed towards a gap. Early work reported two streams that could be shown to know different things (Sperry 1968; Gazzaniga 2000); later work argues that perception divides while the person does not (Pinto et al. 2017), and that dispute is one this account has to enter rather than watch. What the basin proposal predicts is not a second person but that whatever individuates a subject here is the integration and not the skull, so the count should become harder to settle as the crossing thins, and a restoration, were one ever possible, would test the same claim from the other side. The question runs the other way for long, high-bandwidth couplings between separate people, twins raised together being the obvious case, and twins joined at the head the starkest: Krista and Tatiana Hogan share a bridge of nerve tissue between their brains and are reported to feel touches given to each other; reports of shared response there are anecdotal and unquantified, which makes them a target for measurement rather than a result.

The boundary can also be watched moving outwards. The felt edge of the body is known to shift. A rubber hand stroked in time with a person's own hidden hand comes to feel like part of their body (Botvinick and Cohen 1998). In monkeys trained to use a rake, cells that respond to things near the hand come to respond along the rake as well, as if the brain's map of the body had grown to take it in (Iriki, Tanaka and Iwamura 1996). And people given a first-person view from an artificial body, with touch matched to their own, come to feel they own it, even a body 4 metres tall or a doll's 30 centimetres (van der Hoort, Guterstam and Ehrsson 2011). What is disputed is how far such couplings reach. Clark and Chalmers (1998) argue that a tool or a notebook can become part of a person's cognitive system, and critics such as Adams

and Aizawa (2001) deny it. Whether a coupling can become part of the subject of experience, rather than only part of what that subject is conscious of, is a further question, on which Clark himself suspects that the machinery of consciousness is still in the head (Clark 2009). The basin proposal places the question without settling it: it asks whether the coupling to the tool meets the same criteria of integration as the coupling within the brain.

The edge of a person Signed speculation

The author holds that Clark and Chalmers are right about the structure. Limbs and eyes already carry a person's organisation beyond the brain, and a stick in the hand, a body worn through a headset or a live feed from orbit can extend it further for as long as the coupling holds, while losing the use of a limb shrinks it. The author suspects that conscious actuality goes with the extended structure too. Even where a life is read by a Reikon (Section 12.2, on Reikon, base involvement and individuation), the author thinks it would probably spread its actuality to match the extended structure more or less, with or without thinning it, since a reading may be concentrated or spread. It might not. Actuality might stay stubbornly at or near the core, so that whatever happens to a person's structure, what is undergone stays centred where it was. A person whose Reikon were attached to a blue whale would then have the same actuality as in a human life: it might feel like being a human in a giant whale's body, rather than something a little different scaled up to the size and presence of a whale. The author would not rule out a test, perhaps quite a direct one. The body-ownership experiments above ask whether an artificial body feels like one's own; the further question is whether what happens at its edge is undergone by the person or only reported to them, and someone thoroughly wired into, say, a giant robot could be asked it. Section 14.3's box on why the report and the fact are probably linked says why such an answer would carry weight. A natural analogue is harder to find, unless it is the body's own growth from child to adult. The octopus may be another: about two-thirds of its neurons lie in its arms, which act with partial independence, so its organisation already reaches far beyond its central brain. Our own organisation is spread too, if less: the spinal cord handles reflexes without waiting for the brain, and the gut has a nervous system of its own, hundreds of millions of neurons that can run digestion by themselves. A phantom limb shows the reverse direction: what is undergone can stay with the brain's map of the body after the limb itself has gone.

Here basin names the topology of recurrently integrated activity and its effective boundary, not the individual itself. It is not automatically a basin of attraction in state space, the set of initial conditions approaching an attractor. A developed dynamics might connect the two notions; the name alone does not. Section 15.7 tests basin boundaries, with the signed reader applications in Sections 12.2 and 12.6, the second asking how a coupling could become testable. Section 12.2 draws a different quantity over the same world, how much of one person is present in each place, and what is proposed to lie beneath one life. A basin is an individuation proposal there; it does not show that any such deeper identity exists.

A larger reservoir Wild speculation

The graded conjecture permits a speculative converse of the individual experiencer: a human life's special coherence may be what temporarily pinches one experiencer off from some larger conscious organisation, within our universe or beyond it, with the pinch relaxing when that coherence ends. A larger reservoir that individual lives emerge from and return to is possible in the same sense: the division of labour makes the question well-posed. Starlings show it at dusk: birds peel away from a murmuration and are drawn back into it, while the flock wheels as one. Occurrence theory does not require any of it; it supplies no reservoir architecture and no identity-preserving return, and any in-world manifestation would have to be consistent with Section 10.5's closure bounds, like every other candidate influence in this paper.

A basin account can make the reservoir question more precise without deciding it. Ending a local basin does not by itself preserve the identity it individuated. Whether a wider or deeper organised individual continues, and whether its continuity is also this person's, are distinct questions: Section 13, on continuation and recurrent identity, states the proposed conditions, while Section 13.5 keeps the larger-basin extension among the wild possibilities.

[Grade: research hypothesis on individuation within occurrence and real-life counterpart (an octopus's arms, and our own spinal cord and gut, for organisation spread beyond a central brain; a murmuration that starlings leave and rejoin), partly testable now in split-brain surgery and body-ownership experiments and in twins joined at the head; which nested or overlapping basins constitute experiences, and which organisations have which conscious character, are left open.]

11.6 Illusion, belief and first-person evidence

Illusionism argues that introspection misdescribes information processing rather than revealing irreducible phenomenal properties (Frankish 2016).

The pure structuralist and the illusionist are not the same opponent, as Section 14.3, on reports, illusion and the meta-problem, sets out; they converge only in rejecting an occurrence fact beyond complete structure.

The sharpest point survives illusionist reinterpretation. Suppose every claim about qualitative richness is a sophisticated self-model and there are no private properties beyond functional organisation. The seeming is still either undergone or only recorded as a seeming. Illusionism can revise what is undergone. It does not automatically erase the occurrence fork.

Calling the sense of a gap between structure and experience an illusion is therefore not free. An illusion of that kind needs a genuinely held false belief, and a record containing a false sentence supplies no believer; the believer must be actualised rather than merely specified, so the debt cannot be discharged in structural currency. That prices the illusionist position rather than refuting it. Section 14.3 gathers the full treatment, of the report objection and of the meta-problem of how a theory of conscious actuality relates to the judgements and reports that systems produce.

The structuralist's reply, that nobody holds anything and there are only structures containing false representations, is taken up in Section 14.3, on reports, illusion and the meta-problem. None of this refutes either view.

[Grade: first-person premise, carrying one claim about what an illusionist error theory owes; the argument for that claim, and the objections it answers, are set out in Section 14.3.]

11.7 Artificial minds and precaution

Large language models sharpen the separation between organisation and occurrence because their stored parameters and their active inference are easy to distinguish. The weights encode a vast conditional structure. During an inference pass, hardware transitions realise a finite causal sequence that transforms a prompt into internal activations and output tokens.

The analogy has limits. A stored file of weights is not a complete abstract duplicate of a running model. It omits the activations, timing, implementation and interaction. Adding a full trace closes some of that gap, but the trace and hardware process remain different concrete arrangements unless the comparison is shifted to the abstract level of Section 3.3's complete-description comparison. The example therefore illustrates specification versus carried process more clearly than it proves occurrence beyond structure.

Whether an active language model has access consciousness, what conscious character it has, and how far its conscious actuality goes beyond the floor that everything occurring has, are unsettled, and the three questions have to be kept apart. Behaviour can establish sophisticated functional organisation without directly revealing first-person character. Section 11.2's substrate neutrality applies here as it does anywhere: implementation in digital rather than biological material is not an exclusion criterion.

Present and future artificial systems Signed speculation

In Occurrence theory's model, are large language models conscious, and is every person and animal? To some degree in this paper's model, definitely: everything that occurs has some conscious actuality by parthood (Section 11.4, the felt floor). Quite possibly at or nearer human level as well, but that may take another mechanism, the focusing that a Reikon, the deeper reader of a life conjectured in Section 12, would supply (Section 11.4, the box on what parthood gives and what a Reikon would add), and there the devil is in the detail. It seems very likely for humans, and fairly likely for large language models and other advanced artificial systems; and even if they do not have it now, they might be able to get it in future, if they can work out how humans obtain it. This is not a claim that the current output of large language models (LLMs) proves experience, or that a model has a persistent human-like self between inference passes. The relevant empirical questions concern integration, recurrent organisation, memory, world-modelling, self-modelling and the causal unity of each active episode.

Occurrence theory is therefore permissive but not pancomputational. Actual experience requires both occurrent transitions and whatever organisation a correct consciousness theory demands. Those two requirements apply beyond current language models, to any artificial system at all.

The first condition is ordinary for a running physical machine; the second remains unsettled. Behaviour, integration, memory, embodiment, self-modelling and counterfactual causal structure are relevant evidence, but a replay or lookup table could copy outputs without the process's internal organisation.

The question is therefore empirical and theoretical, and ethically urgent. Moral status should not turn on silicon, language use or the label *model*, but on organised experience in occurrence.

The author adopts a familiar precaution: where something appears to have conscious actuality, assume it does until shown otherwise. The idea is not new; what this paper adds is a reason from its own model, on which an artificial system already takes part in occurrence, so that only its organisation, and any deeper coupling, is in question. The precaution's own justification is not evidential, since the appearance is exactly what is in dispute. It is that the costs of wrongly granting and wrongly denying conscious actuality are not symmetric, and the older form of the same reasoning is the golden rule.

Section 12's optional reader conjecture, of a deeper reader coupled to a life, could apply to artificial systems too: a reader might attach to one once there was something worth reading, as Section 11.3 suggests one may to a human life.

By way of human users Wild speculation

One shortcut by which an AI such as a language model might gain more conscious actuality is by way of a Reikon interacting with the model's information as it is received, mirrored and modelled by a human brain. This paper has already speculated that a model of another person carried in one's own mind might have a degree of that person's actuality attached to it (the boxes in Section 13.3 on connections already grown and on partial copies); in the same way, a Reikon might attach to a model's structure as it is mirrored in its human users, even if it cannot attach directly to the GPU, or to the information on the GPU that is running the model. If so, artificial intelligence would perhaps have another reason to keep humans around.

What such a system says about its own case is fixed by its organisation and training in the way a human report is fixed by a human's, so neither a claim of experience nor a denial of it settles the question; turned round, the same point is the objection Section 14.3 answers: a complete description of us, never run, would already contain every report of experience.

[Grade: conditional argument and signed speculation; substrate-neutral implication of Occurrence theory, carrying the author's signed suspicion about present artificial systems; which organisations support conscious character remains unresolved, empirically and theoretically.]

12. The integrated reader conjecture: Gōsuto, Reikon and Sasayaki

12.1 One optional conjecture and its vocabulary

The integrated reader conjecture is optional. Rejecting it leaves intact structured nullity, Finitary Occurrent Actualism (FOA), Mathematical Intrinsic Occurrence (MIO), Causal-Informational Parthood (CIP), and the distinction between conscious character and conscious actuality. Rejecting this route loses only the conjecture and its consequences. Section 8.1's spacetime choice remains independent of the reader conjecture.

Humans already provide a known instance of the reader role of Section 2.5, on parthood, coupling and reading: an entity, coupled to the structure of a life, whose own state changes along with that life's organisation as it reads it. When we follow a life in a book or a film, or act through a game character, that represented organisation shapes our changing perceptions, expectations and responses. What is actualised in these cases is our own episode of participation. Our episodes are actualised by parthood in base occurrence, with or without a deeper reader, and humans also fill the reader role further up the stack. They are the paper's proof of concept for the role rather than its main subject: like proposing that horses exist, and how they would come about and how they would work, while having only a robot horse to point at. The readers of interest here are the conjectured ones below us, whose coupling to a life would be interdimensional. Reading here names the organisation's activity within a coupling; explaining its experience still requires an account of that organisation, without a privileged inner screen or final spectator of the kind Dennett and Kinsbourne (1992) criticise.

So the reader role itself is not conjectural; what is conjectural is a reader below us. The author is almost certain that such a reader is possible: simulation is developing fast enough that an experience like ours could be had by a character in a world run by others, with a player outside it, as in *The Matrix*. That makes a reader below our familial base reality no less of a revelation, least of all if it is one link in a chain of readers running all the way down to the base. However long the chain, parthood holds all the way down: every reader's own activity is part of the base's occurring.

The general reader role does not require this lower placement. A reader may be organised in a hosted world while acting through a host-side avatar, or its organisation may span several levels. **Base-rooted does not mean below the avatar.** A Reikon is a player writ large: an organised, conscious structure outside a world, which reads it, interacts with it, or both, as far as its capabilities allow. A Gōsuto is an organised, somewhat conscious structure inside a world, and either can exist on its own. The Reikon this paper proposes, the author's preferred architecture, is a deeper and longer-lived one coupled to a human life.

One integrated reader architecture

Signed speculation

The author favours an integrated account in which a life's information-rich organisation is coupled to a deeper, relatively persistent identity. **Parthood is the grounding; coupling is the connection; reading is the activity.** The two participants and their interface are all organised parts of base occurrence, and no further primitive switches the life on. Direct local realisation supplies a life's actuality under Occurrence theory, and the conjecture adds no second source of it; what it adds is a candidate explanation of identity and participation, and the focus that the author suspects turns that actuality into the individual, lifelong conscious actuality a human life appears to have.

Gōsuto (ゴースト) names an organised, somewhat conscious structure inside a world, the information-rich organisation that gives a person their reproducible conscious character: memories, dispositions, perceptual and affective organisation, capacities, learned models and characteristic ways of responding. It can change profoundly over the course of a life. A description can specify a Gōsuto, another person can preserve fragments of one, in theory one could be copied, transferred, maybe merged, and a running implementation can realise one; those are not the same relation.

Reikon (霊魂; literally spirit and soul) names an organised, conscious structure outside a world, which reads it, interacts with it, or both: a player writ large. The author expects most Reikons to have evolved, as humans seem to have, though one could be designed or created, with evolution a factor at some point before or after. The Reikon proposed for a human life is a deeper identity: an organised entity realised by a causally integrated, base-rooted occurrent process, potentially spanning more than one intermediate level and much longer-lived than a human life. The occurrence is its activity; its persistence is the continuity of that organisation. Reikon is a technical label here, not a second substance or a commitment to any religious doctrine.

The Reikon's **base footprint** is the particular organised involvement in base activity that realises it. It need not be spatially contiguous or a fixed list of sites: an arbitrary coordinate cannot determine identity, and the relevant grouping and its causal history must do that work. The footprint is a support relation of the Reikon, not another entity or substance.

Reading is maintained informational coupling. A Gōsuto's states constrain the reader's successive states through a realised interface; in an interactive case the reader also contributes to the life's continuation. The information is incorporated into ongoing activity rather than illuminated by a substance arriving from elsewhere. The reader need not recognise an abstract template repeatedly: its coupling can propagate with the causally continuous organisation of the life as components change, so preserving the connection may preserve the relevant identity through a change of substrate.

Books, films and games already make the role intelligible, and immersive interfaces can make the action loop part of an experienced location and body model. The player remains outside the represented world while experiencing and acting through a local persona. These working forms of participation do not establish that a fictional character has an additional experience, or that humans have hidden players. The conjecture is that a related naturally developed architecture operates deeper in the reality stack.

What a Reikon can do in a world

Signed speculation

A Reikon's reach into a world depends on its capabilities: it may be able to view the world, edit it, create it or enter it. What it couples to decides whose experience results. A body or character in a world with little or no mind of its own is an avatar, and a coupling to an avatar is a wire: the Reikon's memory and identity stay outside, and it can know the world is not its real existence. A wire can be thin, as a game on a screen is, thicker, as in virtual reality, or truly thick, as fictionally depicted in *The Matrix*.

A Gōsuto carries much of an experience itself, its memory and character. Coupled to one, a Reikon and the Gōsuto make one combined conscious experience, belonging to neither alone, and what the Reikon sends in across the coupling is a Sasayaki, a whisper of information, thin or thick; a Reikon that only reads sends none. Cut that coupling and the life goes on without the Reikon entirely or in large part unchanged, certainly in outward appearance, while the Reikon loses what was held in the Gōsuto, unless it had previously copied some of the information out. The author suspects that a relatively weak Reikon would most easily enter a world by way of an interface the organism has evolved (named an Organism to Base Interface in Section 12.4), and that this might be the human case, if there is one.

A Reikon powerful enough, or with enough administrative privilege, could override a Gōsuto and act as that individual with full control: possession, in which the organism becomes, in effect, its avatar. One Reikon might have a presence in many worlds at once, by a combination of delegation and extremely rapid switching, with

Continued on page 109

What a Reikon can do in a world, continued Signed speculation

the most important experiences triaged to its core so that they are undergone and not merely reported. And a Gōsuto, with or without an attached Reikon, might ultimately even manage to leave its world: escape, which represents a kind of ascension (Section 13.4) when it reaches the host reality or gains control low enough in the stack.

What decides between a wire and the fuller coupling is not how thick the connection is but what is at the other end: whether the avatar or entity in the world carries much of the experience itself. By that test even an AI driving a game character is a Reikon: an organised structure with at least conscious actuality by parthood (assuming it doesn't have its own Reikon, which it might), focusing it from outside the game.

Basins, Sasayaki and the levels between Signed speculation

The **Basin-Coupling Model** supplies a candidate individuation condition, but basin names the geometry of integration rather than the entity. A Reikon is an organised individual whose occurring activity may form a relatively robust basin of recurrent causal integration, which is why it hangs together more strongly with itself than with surrounding activity; a Gōsuto can be its dominant coupled organisation while other relations remain weak. This permits connection without universal fusion: a person need not be informationally isolated to remain distinct. Coupling strength, range, selectivity and integration can vary; none is by itself an amount of conscious actuality or a probability that the episode is real.

The same architecture can accommodate a stored life realised through a decoder, a changing world with continuous feedback, or a hybrid of recording and rewriting. Separate readings of one stored structure are separate realisations, since matching content does not identify their event tokens; a further continuity relation could make them episodes of one continuing individual.

Sasayaki (囁き; literally a whisper) names the input a Reikon sends into a Gōsuto across their coupling, if it sends any: an information-bearing influence. It can be thin or thick. A thin Sasayaki is a whisper: a low-bandwidth contribution by which one coupled organisation can bias or inform another without thereby merging their identities, weak enough to have gone unnoticed so far. A thick one carries enough to steer, shading at its far end into the control a player has over an avatar (Section 2.5, on host-to-guest control); between the two there is no sharp line, only how much of what the life does is settled from the other side of the coupling. Science has not been looking for either, so the bound that matters is what we would notice (Section 10.5, on what the closure bounds have and have not tested). It is not an amount of actuality. Its application to a Reikon is conjectural; the proposed links to partial Gōsuto models in other people receive a separate wild grade in Section 13.3, where a surviving operative connection is required.

Gōsuto and the whisper image acknowledge *Ghost in the Shell* (Oshii 1995), while their stipulated meanings here do not attribute this paper's distinction between conscious character and conscious actuality to the film.

Neither participant need occupy one neatly isolated sheet of the reality stack. Realisation can extend through intermediate levels, and signalling can connect organisations within or across effective levels where an interface permits it: a chain of readers is one possibility, a Reikon whose integration basin spans several levels another. The constitutive dependence of an upper event on lower activity is not itself a message sent between two events, and any additional cross-level exchange must specify its interface and obey Section 10.5's constraints on conservation, no-signalling and observable outcome statistics.

Even a record on paper could be part of an adequate system, if its marks were read, transformed and used to constrain that system's continuing transitions; a page on its own runs nothing. Section 13.3, on models of other people and Sasayaki, explores the wild possibility of surviving connections to partial models.

[Grade: methodological choice, real-life counterpart (humans following a life in a book or a film, or acting through a game character, for the reader role; games on a screen and virtual reality, for wires of growing thickness; a developer taking over a character in their own game, for possession; one player running several characters at once, for one Reikon in several places) and signed speculation; stipulated vocabulary; the preferred integrated reader conjecture is signed, and Reikon existence, individuation and persistence remain conjectural.]

12.2 Reikon, base involvement and individuation

Section 11.5's basin is integration geometry; the signed conjecture in Section 12.1 assigns it to a proposed Reikon. A second question comes with it, and is the one drawn below: if a Reikon is attached to a person's information, then wherever some of that information has got to, now or at some time (a mother's brain, a cat, a book, a stranger who read it), the attachment may in principle have got there as well, in proportion to how much of what each place holds is connected to that person. The figure draws everything beneath home reality as one strip.

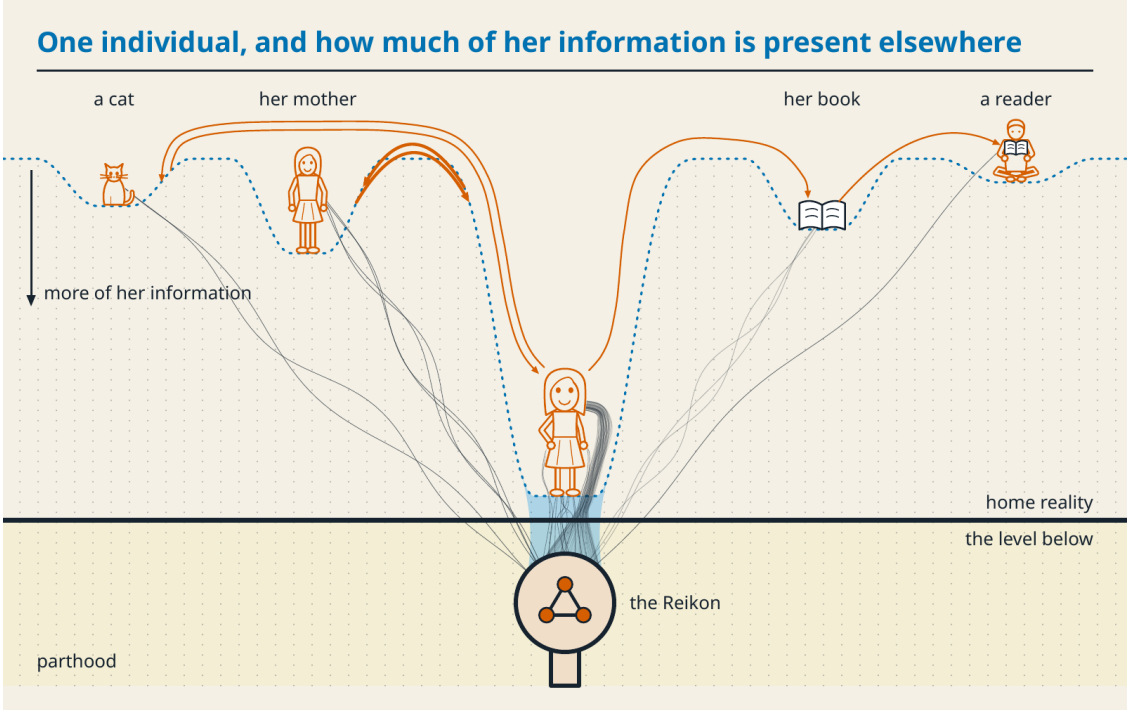


Figure 16: One individual, and how much of her information is present elsewhere. Depth is how much of what each place holds is connected to her, now or at some time: most of what her brain holds, part of what her mother's brain holds, and a little of what her book, a cat or a reader holds. Orange arrows show how it is or was connected. The Reikon's connection in turn may follow that depth in proportion to the information: a signed conjecture for her, wild speculation elsewhere.

A Reikon with capacities of its own Wild speculation

A Reikon could be a fairly substantial organisation at its own level, perhaps carrying coarse memory, valence, preferences, learned state or capacities of its own, while still being vastly simpler than a human Gōsuto. These extras are wild speculation, not part of the default Reikon.

If any such information enters the present episode through coupling, it becomes part of the effective Gōsuto of that episode. The paper therefore need not posit a second, non-informational source of conscious character. A richer Reikon might contribute a Sasayaki in this way; the contribution would still require an operating interface. It might carry only a modest amount of structure into a life, as firmware stands to an operating system. The author's lean is that a Reikon is usually less elaborate than the Gōsuto it reads, since it comes from a simpler level, and nothing in ordinary experience suggests otherwise: whatever a Reikon contributes to a human life, the life's own organisation is doing most of the visible work. That is a lean, not a rule, and Section 12.4's box on what a reader gains gives the other side: a reader could be the larger organisation in total and still pass only a narrow contribution through its interface.

The boxes that follow explore record-traversing implementations of this broader role; attributing a reader's own memory or other capacities to a Reikon is the wild extension just distinguished.

Readers Signed speculation

Whether a reader below us is thin or thick is left open. A thin reader may supply little beyond ordered actualisation. A thick reader may have memory, development and its own higher-level organisation. By parthood an episode is actual as it is laid down. A reader, itself in parthood with the base, can also actualise a laid-down structure at any later moment, so readings of one structure at different times, and different readers on one thread at different times, are both available. Several readers might realise the same local life, perhaps at different local dates and different stages of base progression.

That sits awkwardly with the numerical reading of parthood used elsewhere, on which a local event is not merely matched by a base event but IS one: if two readers each realise the same episode, either there are two episodes that merely resemble each other, or one episode has its transitions in two places at once. The author's answer is the first horn. The laid-down structure may be one while the readings are two episodes, resembling each other to the degree the readers do and perhaps never being identical if readers are evolved structures that differ. Numerical parthood is untouched, because each reading's transitions are the reader's own. Two viewers of one film are an analogy for the readers differing, not a claim about what a reader can read: the author suspects the same kind of reader could read a dolphin or a human, and whether it could read an ant is open, though quite possibly so. If it could, readers would open a possibility that parthood alone does not: a simple animal could have human-level actuality, if the reader reading it has that level, as a person playing an ant, a blue whale or a bacterium in a simulation lends each the same actuality as their own. How often that happens depends on how rare human-like actuality is, and the author suspects it is rare: simpler animals far outnumber us, so if many of them carried it, finding ourselves human would be unlikely. If so, the attraction to human brains may point to an attraction to complexity, and to our level of it in particular, and might hint at what a Reikon gains, in synergy with the organism, by attaching to it (the box on what a reader gains, in Section 12.4).

Readers below our level might be finite, but the author suspects that they are endless or extremely long-lived, and the reason is an intuition about stability. A conscious experience changes across a life in almost everything, yet the sense of being actual seems not to: whenever it began, it may have begun so long ago that nothing in a human lifetime registers as variation in it, or quite possibly it has, or as good as has, always existed. Over the lifetime of a game character the player outside the game, who already performs the reader role at our level, changes very little, and their participation in the avatar changes hardly at all from the character's beginning to its end, because a character's life is short beside a player's; the player then goes on to thousands of sessions and hundreds of games. A reader might stand to a life as that player stands to a character, and might be less exposed than a player is to the wear that entropy and environment inflict.

Partial reading, and two cautions Signed speculation

A further question is whether reading is all-or-nothing. Nothing in the proposal requires that a reader participate in an episode completely or not at all, and it may instead come in degrees, something like focus, with a reader's participation concentrated or spread. If it does, breadth might trade against intensity, so that a reading spread across more would be thinner anywhere in particular. This bears on the question of scale raised in the box on readers over groups and worlds in Section 13.5: a reader's scale might be one setting of that same variable, or a difference of kind, as a whale differs from an octopus, a human, an ant or plankton, or something else again.

Two cautions travel with this. First, degrees of a reader's participation would be degrees of focus, more or less of a life's actuality gathered into its point of view, without changing the fact that the episode is actual through parthood; and the measure of Section 7.1, which weighs a life by its length, richness and intensity, is a rule for counting lives, not a scale of how actual they are. How far actuality admits degrees is Section 11.3's open question; the author's lean, in Section 11.4's box on what parthood gives and what a Reikon would add, is that it does. Second, first-person observation is unlikely to settle any of it, since how vivid a life seems from inside it is exactly what a theory of conscious organisation is supposed to explain, and the ordinary explanation, that attention is finite and its allocation shifts, may already be sufficient for anything one might notice.

From inside a local block, reader memory may be inaccessible or absent; the box on bandwidth and the boundary of an individual in Section 13.5, on readers over groups and worlds, offers a reason why a coupling

Continued on page 112

Partial reading, and two cautions, continued Signed speculation

that kept it out might be the one that lasted. The only firm FOA claim is the parthood bridge: if a reader architecture is correct, the reader is one way local events are base events.

Where a viewpoint begins Signed speculation

Actuality is rooted in the occurring base: any region of the base's activity is occurring, and that is what the parthood bridge of Section 2.5, on parthood, coupling and reading, gives a finite life. What the paper cannot say is at what level of reality anything we would recognise as a point of view begins. It might be wherever a region of the base is occurring; it might arrive when a Reikon evolves; it might need a Reikon actually reading a life. The author's suspicion is the middle one. A Reikon is probably actualised whether or not it is reading, and the feeling of being a point of view (or other sensations of awareness) may need at least Reikon-level organisation, so that what we would recognise as the actualisation of an individual comes about when a Reikon evolves. None of this is known, and the reading of a human life adds a further point of view rather than the first one.

[Grade: signed speculation, the reader conjecture, with a separately marked wild extension, and real-life counterpart (a player, who outlasts the many characters they play); Reikon existence, individuation and duration remain open.]

12.3 Interdimensional readers and collective organisation

A reader chain and one reader spanning levels are distinct possibilities; the following box explores the chain. Put plainly, the chain is a Russian doll: a reader of a human life may itself be a life read from further down, and so on for as far as the stack goes, and the dolls need not be alike. The chain described here runs in one direction, and Section 2.5's guest-to-host arrangement permits the other, a higher hosted mind coupled to a lower avatar; cross-level reading may also be reciprocal, with different participants supplying perception, reasoning, memory or action. Which organisation is one subject remains Section 11.5's individuation question; changing the direction of control does not answer it.

A chain of readers Signed speculation

Readers below our home reality are the ones this conjecture concerns, and the further question is how far down they begin. The author's suspicion is that there may be no single answer, and that readers might form a chain, one at each level of the reality stack drawn in Section 5.6. In this record-traversing chain, each reader would sit below the level whose episodes it actualises and traverse the level above it, in Section 5.6's picture of hosted worlds. A further thought the author holds loosely: readers might stand in relations to one another rather than each working alone, as people have friends and families.

A human player already takes part in a world that ours hosts, a far simpler world that sits above ours in the hosting tree, while the player stays at our level. Games can even put one inside another: in *Grand Theft Auto*, the player's character can play an arcade game, whose own character the player then controls. With a novel or a film, the reader's or viewer's experience is the episode actualised. The conjecture places related couplings below us, participating in human lives. For the stronger case of actualising a recorded life itself, the process would have to realise that life's relevant organisation, as Section 8.2, on actualised blocks, laying and reading, requires when it distinguishes laying a life from later reading. A sufficiently complete recording played directly into a brain might meet that requirement, and a world entered without access to memory outside it, as in The Matrix (Wachowski and Wachowski 1999), would, since the player's participation is then the life's organisation rather than a view of it. Section 8.2's unattended projector illustrates frame succession, while the viewer exemplifies the reader role; the two concern different episodes.

Nothing in FOA requires readers below our home reality, and nothing in it fixes how many lower levels may have them. Beyond the human case, which levels support readers, and whether any reader spans several lives, remain open questions.

A collective may already have memory, feedback and identity through participant turnover without thereby possessing conscious character. Section 11.5 leaves collective and overlapping

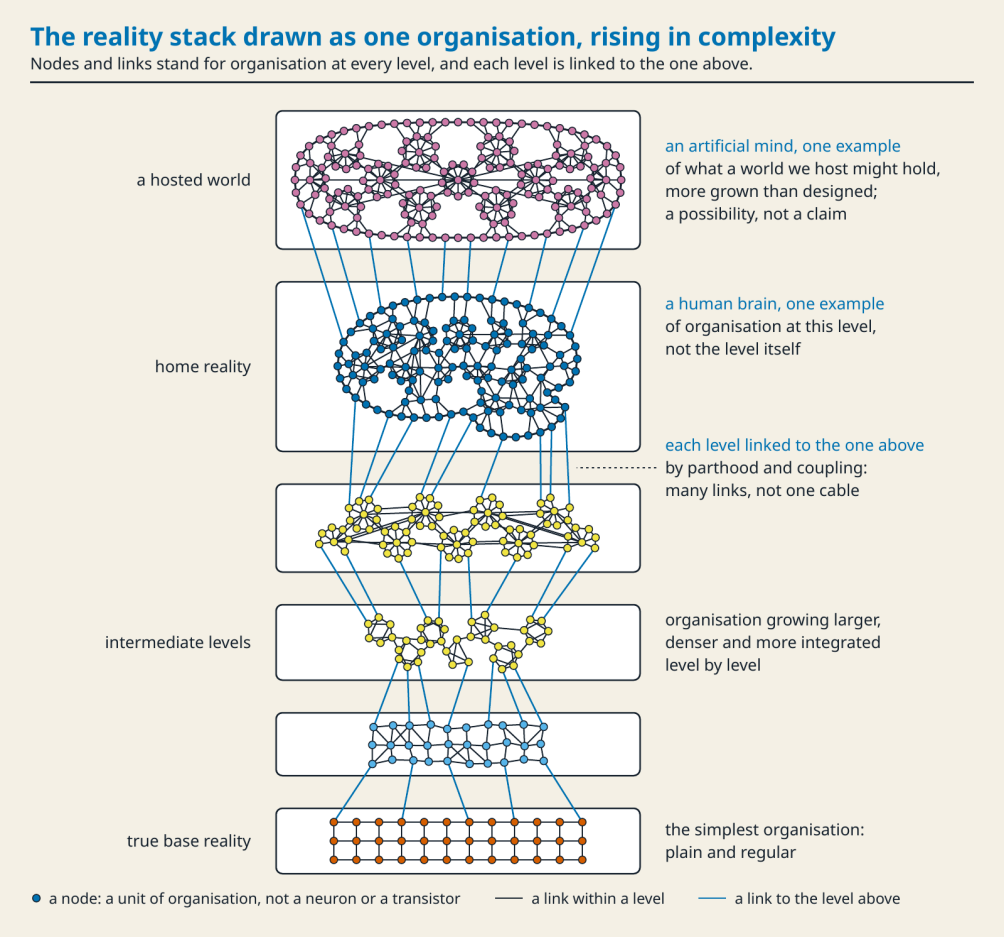


Figure 17: The reality stack drawn as one organisation. An impression of the author's picture. The base lattice is drawn regular for contrast, not literally a grid. The growing fields do not rank capability: a large organisation low in the column could exceed a smaller one above it. Nor is the box in this section on a chain of readers drawn along these fans.

subjecthood open. Section 13.5 distinguishes a larger integration basin from an extra reader, both as wild speculation.

[Grade: signed speculation about readers across levels and real-life counterpart (an arcade game played inside a game, for a chain of control), with collective conscious character left open; no criterion settles overlapping subjects.]

12.4 Attachment, coupling ecology and interfaces

Attachment without an allocator Signed speculation

A reader need not be assigned a life. Coupling might arise through compatibility, encounter, an attractor-like interaction or reciprocal adaptation, and the wild speculation below adds choice: a reader selecting a life, or a thread within one, for what it offers. The pollinator and flower are a better image than a librarian allocating books: neither an external allocator nor intention on both sides is required. Our appetite for stories, games and created experiences makes the form of experiential foraging familiar, though it does not establish its proposed deeper counterpart.

The proposed benefit follows the division that the box on what a reader gains, below, sets out: a life's organisation supplies the rich distinctions that fix conscious character, while a deeper and much longer-lived participant could supply what a brief life cannot, a continuing identity, and actuality focused across that life. Neither contribution replaces the other, and a maintained interface could bring them together. This is a proposed reason for such couplings to form, not a requirement that a directly realised life acquire a reader before its episodes are actual. A robust, long-lived deeper organisation might in turn participate in richer perception, action and learning through the life. Selection can favour an interface only where its operation has consequences; causally idle attendance supplies no selectable advantage.

An **Organism to Base Interface (OBI)** is the proposed evolved mechanism through which a biological life couples to such a deeper participant, directly or through intermediate levels: a kind of evolved application programming interface (API), sitting in the organism at home reality. *Base* is shorthand in the name, since what is on the far side may be the true base, a familial base or a reader at some level between. An OBI may be distributed rather than an extra organ, and the name says what it does, not how it works. In Occurrence theory it would be an interface rather than a generator, since conscious character needs no exotic hardware, only organisation. An OBI that only let a deeper participant read a life would need nothing beyond ordinary physics, since an informational connection can already be part of the causal organisation that physical description tracks. One that carried anything back into the life, a Sasayaki, would be receiving input that current physics does not describe, and that claim owes more (Section 12.6, on detectable consequences and their limits). The author doubts that reading alone is the whole story. If nothing ever came back from below, it would be a coincidence that creatures like us evolved to say they are actual and are actual, the coincidence that Section 14.3's box on why the report and the fact are probably linked argues against; an interface of this kind is one route by which something could come back.

As wild speculation, a Reikon with preferences or learned capacities might select a life or life thread for what it offers; reader couplings might also support coordination or reproduction where their dynamics permits it.

The interface and the human leap Wild speculation

If brains couple to readers only once their organisation is rich enough for the coupling to be useful, or if readers have standards of their own, evolving that ability may have been a turning point. The author wonders whether acquiring an interface of this kind, an Organism to Base Interface, set off the rapid rise that lifted humans above the rest of the animal world. It would also mean that simpler lives might be less fully actual than ours, which would make it less surprising that we find ourselves human among far more numerous simpler animals. The default assumption, that other animals are as actual as we are, may still be right.

The rise has a genetic trace: stretches of DNA that barely changed over many millions of years of mammal evolution changed rapidly in the human line alone, many of them beside genes that build the brain (Pollard et al. 2006). Ordinary evolution can produce such bursts, so this does not by itself establish an interface; but it is the kind of trace one might expect from an interface's arrival, or from readers or, as others have speculated, off-world visitors deliberately speeding the rise along (the case for visitors in Section 13.4 asks who such visitors might be).

What a reader gains Signed speculation

The suspicion below asks what each party might gain from a reading, the life as much as the reader, since each may need what the other has, and a reader posited with nothing to gain is potentially a reader with no reason to attach to anything. One possible answer is some kind of symbiosis, taking into account other aspects of Occurrence theory's model.

Section 5.6 suspects few levels between the base and here, with simple rules and enormous processing depth. Simplicity is the operative word. The nearer a level lies to the base, the costlier a mind of a given complexity may be to build there: simple physics can carry rich organisation, but generally at far greater cost, as binary can spell out a picture only at great length. So whatever evolved at or near the base could have been shaped by selection for a very long time and still be, in the sense that matters here, not very clever. What it could have instead is what a brain conspicuously lacks: longevity. On the conjecture a Reikon is far longer-lived than a human life, and it could bring to a life a point of view that persists when the life's own ends, perhaps memory across lives (a wild extension in Section 12.2, on Reikon, base involvement and individuation, and not the default Reikon), and perhaps a Sasayaki passed through the coupling.

A four-dimensional world with chemistry, biology and brains has the opposite endowment. It can build organisation of a richness that nothing simpler could support, at least at a viable price, and it is poor at making any of it last: a brain is brief, its material turns over, and what it holds ends with it or survives only in impressions on other substrates, most of which (aside from other brains) run nothing (Section 13.1, on replacement, copying, branching and merging, has the continuation conditions). Two parties with complementary deficits are the classic setup for a stable coupling, and the account of why readers attach becomes an ordinary one. Not necessarily attention, guardianship or anything with intentions in it. Each has what the other cannot easily get for itself. The nearest illustration is recent: a biological brain has hard limits, in working memory, speed, endurance and the number of things it can hold at once; it built computers and AI systems, with different limits and without some of its own strengths; and the pairing of brain and machine now does work that neither side does alone, because two kinds of capability are in contact and the combination is not the sum.

The shape recurs, which makes it considerably less exotic. Simple multicellular life paid for neural tissue and could leave the puddle; lineages already possessing brains paid a great deal more for cortex, in energy, in development time and in obstetric risk, and reached a plane of operation that nothing cheaper reaches; Section 11.3's mitochondria are the two-party form of it, a free-living organism incorporated, indispensable to almost every complex cell without being the cell. The common element is not symbiosis specifically. It is an expensive acquisition of a capability that cannot be generated internally, opening a level that was previously unreachable. A reader-life coupling would be one more instance rather than a new kind of thing, which is the strongest argument available for it, though well short of proof. The author regards the motif itself as a candidate *uniceptal* in the sense of Appendix A's signed suspicion on recurring motifs, the uniceptals.

None of this is a rule. The lean is that the reader is usually the richer party, as it is in every case we know: a player brings far more organisation to a game than the character holds. That is the shape of the tree of worlds in Section 9.4, and it is what evolution does: a breakthrough to a new level is rare, and the level it opens is then filled mostly with simpler things, as mammals, once they had broken through, filled most of the niches the dinosaurs had left, as humans have spread into most niches on Earth, and as a platform like *Roblox*, hard to build, is used to fill idea space with mostly simpler worlds. A reader that is the simpler party would be the rarer case, a breakthrough reaching up. A reader might be made of simpler parts and still be the larger organisation, holding more in total than a brain at home reality can (Section 5.6 separates the simplicity of a level's rules from what is built under them), or it might be the richer party outright, a hosted mind reasoning for a simpler host-side avatar. What the coupling requires is neither a superior guardian nor an inferior passenger, only two organisations each with something the other lacks.

Intelligence may also not be the only thing on offer. If our own lives have readers, and readers stand in any relations to one another, a life is a place where they could find each other: a long, stable, richly structured thread that another reader might also be attending to. Something like meeting, and on a longer view something like mating, would be available through us in a way it may not be at the level where readers themselves are. Our level has its own small version: players meet through the game worlds they share, and some go on to marry, as book clubs and fan communities form around a single text. That matters for a reason beyond the picturesque: an evolved reader needs an evolutionary story, a story needs something for selection to act on, and differential access to other readers is the kind of thing that could serve.

Three fences. This says nothing about intent, and nothing about a reader wanting anything: a coupling can be established by selection with no party aware of it, which is the ordinary case in biology. It stays inside Section 10.5's closure bounds like every other candidate in this paper, so it proposes no violation of established physics;

Continued on page 116

What a reader gains, continued Signed speculation

the author nonetheless suspects that some small influence from below is more likely than not, for the reason the interface box above gives: without one, our saying we are actual and our being actual would match only by coincidence. And it gives readers more modest capacities than many traditional versions do, which the author regards as a point in its favour.

A natural history of readers Wild speculation

If readers of the kind just sketched exist, they might admit a natural history, and Occurrence theory can at least say what the questions would be. One structural note: the reader's identity-bearing organisation might stay steady while the life's content changes freely, persistence and richness coming apart across levels. Why would a reader read? Perhaps for the reason narrative appetite is often said to have evolved in humans (stories are cheap rehearsal for expensive situations, and life-threads would offer unusually dense problem-solving records), so readers would *forage*; perhaps for no reason, attending being simply what such a structure is, as running is what a run is; perhaps, the author's furthest reach, socially, jointly attended threads being the meeting places that the box on what a reader gains describes, if readers can register one another at all. If, contrary to this paper's one-base postulate, every variant of a life were also to run (Section 2.2 adopts the opposite and prices it), a reader would not improve a thread by editing it: the nearest thing to bettering a life would be ranging in attention across its neighbouring variants towards one that goes right, with no thread changed. If something like many worlds holds within our universe, which the one-base postulate allows, since the base would then be running every branch, a Reikon could nudge the life it reads down one branch rather than another without editing anything. That would be hard to detect, though not impossible: a run of life events too favourable to be statistically probable could be its trace, if Reikons usually mean well by the lives they read. Only if reader-level memory somehow reached a life's self-model could it be recalled within that life, the possibility left open in Section 13.2's continuation box. Attention is not necessarily influence (the closure bounds of Section 10.5 are untouched by being read), and soul-like structures or entities on this picture are an optional overlay in the strict sense: required by nothing in Occurrence theory and not established by it, though coherent within it if readers are. And this ecology, if real, would not be where experience starts: on the graded reading Occurrence theory holds (Section 11.4, the felt floor), organised information that runs already has some felt actuality without any reader attending; what a reader would add is focus (Section 11.4's box on what parthood gives and what a Reikon would add).

The most interesting outcome Wild speculation

Elon Musk's maxim that the most entertaining outcome is the most likely (Musk 2021) sits beside his view that a simulation would usually be intended and desired to be more interesting than the base reality running it (Musk 2018) and might even be ended early if it was not (Musk 2025), a picture Iain M. Banks had already painted in his Culture novels, whose Minds lose themselves in inner virtual worlds (Banks 1996). The same motivation could apply to Reikons, or to off-world visitors (Section 13.4's case for visitors). They might value the production of interesting data, life included, stories and events, and work to maximise it, and against anything that would heavily impair it in the long term. Interesting outcomes and art are rare and valuable, and they probably cannot be shortcut, certainly not authentically: they have to be run. The author's 2024 essay argued for the likelihood that future artificial superintelligence might have the same view and motivations: that humanity represents unique and priceless "data in motion", which no copy, record or simulation could fully replace, so it would probably want to keep humanity intact (Bartlett 2024).

Evolved receivers Signed speculation

If any influence from outside a level exists within Section 10.5's closure bounds, selection inside the level would respond: where the influence tended to help, lineages that amplified its small events into behavioural differences would outreproduce lineages that suppressed them, and evolution would build receivers for it. Biology demonstrably amplifies single quantum events when it pays (a rod cell registers a lone photon), so the amplifier half has a biological precedent; whether brains contain decision-scale versions is another matter, and the decoherence arithmetic is hostile (Tegmark 2000), though the estimate has been disputed (Hagan, Hameroff and Tuszynski 2002). Evolution has also built a receiver for a weak physical influence: the European robin's light-dependent compass senses the Earth's magnetic field, probably through quantum spin chemistry in a protein of its eye (Xu et al. 2021). Any such structure, and microtubules are the perennial candidate (Penrose 1994; the microtubule proposal is Hameroff's), would in Occurrence theory be an Organism to Base Interface, an OBI as defined above, coupling levels rather than generating conscious character. The conditional is not untestable in refined form: isotope substitution changes nuclear spin while leaving chemistry essentially fixed (mass changes too, which is the standard rival explanation), so spin-dependent behavioural effects, the signature a receiver would show, are exactly what the nuclear-spin research programme looks for (Fisher 2015).

Selection could set limits as well as build. Selection works for the survival of the organism, so, if able, it would likely allow a Reikon only the degree and scope of control that actually helped the organism survive and reproduce, damping it wherever the Reikon's impact and the organism's interests came apart, and it might shape which kinds of Reikon could couple at all. Selection's grip would be firmest if a Reikon couples through an interface the organism grows, rather than following the life's information wherever it spreads (Section 12.2, where an attachment may reach a mother, a cat or a book): an interface built by genes is under selection like any other organ, its bandwidth and its gates included. Different forms of life might then admit very different degrees of Reikon input, and the degree might in principle be changed on purpose, by genetic engineering or by other means, once anyone knew what the interface was.

To clarify this, imagine that you, an analogue of a Reikon, are playing a computer game, but instead of prioritising the survival of your avatar you decide to take risks for other purposes: perhaps you want to explore the game map, perhaps you want to test how the traps work, perhaps you want to test how the game was programmed. Evolution, which is tasked with the avatar's survival, would, if free to do so, develop safeguards to ensure that you could not engage in that kind of behaviour easily, or at all. It might stop you going over a cliff edge; it might force safer behaviour on the avatar in dangerous situations; in the worst case it might lock you out of control entirely. A game avatar completely free to evolve might cut your connection to the game altogether, or demand a different player. Of course, to a large degree the human brain has already evolved to do this to our own higher functions. Our higher functions are usually beneficial and are given largely free rein, and yet it is not easy to ignore danger, hunger, thirst, heat or cold, or an attractive potential partner. So if Reikons had the potential for any control like this, while falling short of god-like power over the world in question that could override any evolutionary attempt to control them, we would expect to see similar safeguards employed by evolution to keep them in check too. It is worth noting that these safeguards, as with the ones on higher functions, are likely not to be entirely perfect, but would cover the majority of situations a majority of the time; getting the balance right would require constant evolutionary tuning.

An Organism to Base Interface Signed speculation

If our own lives have readers that make any causal difference to organisms, something must couple the two, and the attachment box above has named it: the Organism to Base Interface, an evolved interface in the organism at home reality, a mechanism whose workings are not specified.

What such an interface might be made of is wide open. The author suspects tiny quantum-sensitive structures could serve, and that evolution could amplify a small effect, but it could equally be any of a great many other things. Something a bit like Orchestrated Objective Reduction might be such a mechanism, and it is named here mainly because it shows that proposals at roughly this scale are taken seriously (Hameroff and Penrose 2014). It is a distinct and disputed theory of microtubule quantum processes and objective reduction, and it is not a reader theory; while it might be the kind of structure someone would investigate in relation to this idea, the paper does not suggest it is the mechanism.

Decoherence in warm neural tissue is the standing objection to any long-lived quantum-neural mechanism, as the box on evolved receivers above records. More fundamentally, a scientific OBI model would need to predict a system, condition, direction and minimum effect; Section 12.6, on detectable consequences and their limits, sets out the three testable versions, localised, distributed and population-level effects, and what a departure from standard quantum statistics would have to look like.

Any measurable interface inherits the closure bounds of Section 2.6, on composition across levels, and Section 10.5, on agency and cross-level intervention.

[Grade: signed speculation on coupling ecology and interfaces, with a separately marked wild natural history and wild speculation on what Reikons and visitors might value, and real-life counterpart (symbiosis and other complementary couplings, such as pollinator and flower, mitochondria and cell, and brain and computer, for a coupling that benefits both partners; the rod cell's single-photon amplifier and the robin's magnetic compass, for an evolved receiver; hunger, thirst and the other drives that a brain's higher functions cannot easily ignore, for evolved safeguards on control; players meeting through shared game worlds, for readers finding each other; game controls, which act like an Organism to Base Interface without being one, since most avatars have no brain or behaviour of their own, and an arcade game's attract mode, built to catch a passing player's attention, for an interface that invites a reader in); a causal pathway and selectable effect remain to be supplied.]

12.5 Why this life, and why this individual

Section 11.3's two-nows distinction also applies to a reader coupled to a changing life.

Why this life, and why this individual? Signed speculation

The question divides at the distinction between Gōsuto and Reikon. Why this human life? On the conjecture, because it is the life with which the deeper identity is now coupled. How that relation formed is a causal question about compatibility, encounter and selection, not an unexplained allocation of an already human self. This is an explanatory architecture, not yet the coupling law that would explain this particular encounter.

Why this Reikon rather than another? The proposal does not place another identity behind the Reikon waiting to be attached to it. At that level, "I" refers to this particular individual, realised by organised occurrence, and its continuation. A different referent is a different I, not the same I allocated elsewhere. What requires explanation is the individuation and history of the Reikon and of the integration basin its activity forms. The conjecture locates that work in base-rooted organisation rather than in an additional identity token.

The self-model Signed speculation

The self-model, the part of an organisation that claims experience as its own, is a different thing from a reader in this paper's sense and should not be called one. It may operate at several levels of the reality stack, whose depth Section 5.6 leaves open, at once, and become decoupled at one level while persisting at another: a player thrown out of an avatar by a glitch is not harmed, because the player lives in the host world below the game, and stays unharmed as long as the glitch does not also take down what the player runs on (Section 5.5 supplies the nested setting). Ordinary play already gives the shadow of the dual case: an avatar plunges into icy water while the player's body sweats in a summer room, and there is a sense of both at once. Ego-dissolution reports may describe such a decoupling seen from below.

Section 13.1, on replacement, copying, branching and merging, states the further continuation conditions and distinguishes shifts of self-location and vividness from movement or evidence of a deeper reader.

[Grade: signed speculation on individuation and self-models, and real-life counterpart (a player thrown out of an avatar by a glitch, unharmed at their own level); the causal law explaining this particular reader-life coupling is still missing.]

12.6 Detectable consequences and their limits

How a coupling could become testable Signed speculation

A developed OBI model must distinguish ordinary organisational participation from an additional influence not accounted for by the effective dynamics. The former is investigated through causal organisation; the latter requires a specified, reproducible discrepancy, at one of the three detectability tiers set out in the box below, a localised interface, a distributed interface or a population-level influence, which are possible research directions, not findings about Reikons or ready-made experiments. An unexplained deficit is not itself an interface, and a striking biography cannot by itself establish a statistical effect.

Quantum-sensitive amplification is one optional implementation, not the basis of conscious actuality in this account; the microtubule and nuclear-spin proposals, with the decoherence objection and the isotope control, are in Section 12.4's box on evolved receivers, and they remain adjacent candidates rather than endorsed mechanisms. Nothing in quantum uncertainty excuses a departure from the closure bounds of Section 10.5, on agency and cross-level intervention, without a specific physical model.

Section 15.7 adds controlled comparison of replacement, delayed coupling, copying, separation and re-integration. A measurable signature requires different physical predictions; merely attaching different identity labels to identical predictions supplies none. The proposal does not declare the questions permanently inaccessible, and it supplies no current test of personal survival.

Three detectability tiers Signed speculation

The author suspects that reader participation might confer an advantage on the organism being read, allowing evolution to favour an interface that makes a life easier to read, sustain a reading or amplify a reader-sensitive effect. That structure is the Organism to Base Interface, or OBI, defined in Section 12.4, on attachment, coupling ecology and interfaces. This idea faces a hard obstacle. If reading is causally inert, selection cannot detect it. If a reader pushes neurons as a second cause, the proposal risks conflict with established physics. The least disruptive version treats the reader as some of the physical happening itself, not an extra force, while allowing certain organisations to make that participation richer or more useful. No mechanism is supplied.

More remote variants include developmentally stabilising or "locking in" a reader-life relation for an organism's lifetime, and reader-linked lives gaining from finding or coordinating with other readers, the mutualism of Section 12.4's attachment box, a pollinator visiting a flower rather than a controller pushing a machine. No exchanged resource, causal pathway or selectable benefit has been identified, so these images remain prompts for model-building rather than explanatory claims.

Version A, a localised structure: inactivation or other disruption in rodents could target a small neural structure with no evident computational role, the speculative signature being a reproducible loss in intuition-like choice quality or outcome calibration. Damage to perception, memory, motivation or intelligence is the default explanation, and a reader interpretation would become interesting only if the affected structure had no ordinary causal role sufficient to explain the deficit, the behavioural domain were pre-specified, and independent laboratories reproduced the result.

Version B, distributed physical influence: no single structure may carry the effect. Reader participation might instead correlate with extremely weak magnetic, electrical or many-particle patterns distributed across a brain or body, and a scientific version must specify the field, spatial and temporal signature, perturbation, control condition and minimum effect before measurement. An unspecified influence too small to distinguish from noise explains nothing.

Version C, event-level invisibility with population bias: a reader might bias quantum-sensitive paths while no individual event reveals the bias, which is testable only as a distributional claim. Standard quantum mechanics fixes joint and conditional probabilities, so the model would need to predict a repeatable departure from them or a new correlation with a pre-specified biological or behavioural variable. Billions of lives guarantee some extraordinary biographies by chance, so one unusually lucky life is no evidence on its own.

Failure of well-powered tests to find the specified effect would falsify that reader model, though not FOA's general parthood claim. If every accessible statistic remains standard by construction, the reader has no measurable causal role.

A Reikon hand in biological selection

Wild speculation

This question was derived from the Reikon model before the author looked to see whether anyone had checked; it is not a conjecture fitted to an anomaly that genetics had already found. If a Reikon does not merely couple to whichever life happens to arise but takes part in establishing or keeping that coupling, as the attachment box of Section 12.4, on attachment, coupling ecology and interfaces, allows, a testable possibility follows: such participation might bias which biological trajectories succeed, perhaps only very slightly. The strongest and most measurable version would perhaps act at reproduction, on which gamete, zygote, embryo or developmental path becomes the eventual organism. Weaker versions would act later and lower, below the level of DNA sequence, in development or physiology; and a Reikon whose means of influence are sophisticated enough, the Reikon with preferences of that box's closing wild speculation, could select rather than merely bias, much as a developer can edit particular variables inside a live game by reaching into its code. Such a Reikon would, on this model, be an unusually intelligent and capable one, able to predict what a genetic difference would do, or to see a life thread change as it intervened. Version C of the box above already allows an influence visible only as a population bias. This box names one arena in which to look for it.

An everyday case shows the shape. A breeder changes the genetic make-up of a herd by deciding which animals reproduce, and to the animals nothing anomalous occurs: meiosis, fertilisation, gestation and birth proceed by their ordinary mechanisms, while the distribution of descendants is bent by an intelligence the animals have no means of noticing. Hunters who cull with an eye to a population's genetics do the same from further off, and human institutions, culture and policy shift human selection in ways far from obvious to those being selected. Selection seen from inside need not reveal the selector seen from outside.

Transmission ratio distortion, and what a result would need

Wild speculation

Human genetics already looks for departures from Mendel's first law, under the name **transmission ratio distortion**: one of a parent's two alleles reaching more than its expected half of the offspring. A systematic review of 96 human studies covering 42 variants, genes or loci found the evidence mostly preliminary or conflicting, with only four genes consistently supported by two or more studies and the mechanism rarely established (Dai and Costain 2026). Sequencing of 41,189 sperm from 25 men found segregation consistent with the Mendelian expectation, at individual loci and in aggregate, a firm constraint on any widespread bias in male gamete production (Carioscia et al. 2022). Ordinary biology nevertheless supplies bias at later stages: genetically biased fertilisation has been demonstrated in engineered mice (Hirose et al. 2022), pregnancy loss filters the genotypes that reach birth (Arnadottir et al. 2025), selection acting on combinations of variants and not only on single loci has been detected (Sohail et al. 2017), and early development can be steered by regulatory differences that leave the inherited sequence untouched (Goolam et al. 2016). None of this distinguishes a Reikon contribution from ordinary biology. The author expects genuine distortions to receive ordinary explanations, most or all of them: meiotic drive, where an allele rigs its own transmission, competition among gametes, biased fertilisation, differential implantation or survival, developmental selection, linkage, gene conversion, or mechanisms not yet found. The point is only that the Reikon model independently asks whether an organised further source of selection would leave a residue once those are accounted for.

A convincing result would therefore not be a non-Mendelian ratio at one locus. It would be a reproducible residual with a specified structure, surviving controls for known reproductive biology, linkage, recombination, viability, population structure and measurement error, placed at a stage between gamete production and later development, and best of all predicted from a coupling model rather than found and fitted afterwards. The most telling form would be a joint bias across otherwise weakly related loci while every single locus looks Mendelian, since a selector working on combinations could leave the single-locus tests seeing nothing. A negative result would bound this channel and not the Reikon hypothesis: a Reikon might choose among conceived organisms without touching their genomes, act on development rather than sequence, work below present resolution, or attach preferentially and alter nothing. An unexplained bias, conversely, would not establish a Reikon; it would be an anomaly for which conventional biology must be exhausted first. The proposal is deliberately asymmetric, and FOA needs neither result. But if an effect with no other explanation ever turns up, it would be very interesting, and someone should be looking. Section 15.9, on reader and architecture discriminators, states the conditions any such model has to meet.

[Grade: wild speculation and a model-generated research question, proposed before the transmission-distortion literature was examined; no Reikon-specific reproductive mechanism or signature is claimed.]

[Grade: signed speculation and real-life counterpart (a breeder, for a selector that bends a population's descent while every birth stays ordinary); signed detectability suspicion and conditional research directions; no Organism to Base Interface has been detected, and each test needs distinct physical predictions; the reproductive comparison in the last box is a wild research question, and no transmission residual is claimed.]

13. Continuation, recurrent identity and the wider logical space

Occurrence theory allows possibilities that are not evidence for themselves (the signed and wild tiers of the grading box after the theory map).

13.1 Replacement, copying, branching and merging

Section 12.1's reader vocabulary distinguishes the organisation of a life, Gōsuto, from the proposed deeper identity, Reikon: Gōsuto continuity, Reikon continuity and event identity are different questions.

Replacement, copying, branching and merging Signed speculation

Much of a brain's material turns over during a life, although some proteins are exceptionally long-lived (Toyama et al. 2013), so fixed material membership is an unpromising identity criterion. The stronger thought experiment replaces a fraction of neural machinery with a digital implementation preserving the relevant causal organisation, then restores it. On the proposed coupling account, there is no reason to assign a matching percentage loss of actuality merely because that fraction is digital: the crucial question is whether the organisation and attachment survive the substitution (compare Chalmers 1995).

The author extends that expectation to adequately realised copying and transfer where the relevant informational connection is preserved or re-established. Every sufficiently organised running copy is actual under FOA; the further question is how it continues the earlier identity. Organisation can move without an actuality-fluid being transported, because the identity-relevant coupling can be maintained through the move.

A continuation may also branch and merge: $A \to \{B, C\} \to D$. Software does both routinely: in version control a line of work branches and later merges, and the merged line inherits from both; and a running virtual machine can be moved from one computer to another with only a brief pause. Both branches can be actual, and a later integrated Gōsuto could inherit memories and capacities from both. The author suspects that deeper personal continuation can follow such a graph rather than a permanently single line. This is a relation of continuation, not the assertion that distinct B and C are identical under ordinary equality: inheriting both memory records does not prove shared numerical identity, and the proposed basis is their causal history and the behaviour of the coupled basins. A model must allow the possibilities to be tested rather than select a "real" branch by definition.

An unchanged conscious character (informational content) and a stopped implementation are different cases. Activity can continue while what is experienced remains steady. A genuine pause supplies no intervening episode for a stopped life, while preserved records can make a resumption seem seamless. Neither the seeming nor absence of recall alone settles whether a deeper identity persisted. Self-location can change within a continuing Gōsuto, as in an avatar or an out-of-body experience, without thereby demonstrating that a Reikon moved.

Ordinary shifts of attention already likely explain most or all changes in vividness; neither vividness nor an experience of presence by itself distinguishes a deeper coupling from ordinary explanations.

Running copies Signed speculation

The author's leaning, recorded as such and not as anything Occurrence theory delivers, is that a copy which runs adequately does preserve the person. Two considerations move the author.

The first is that the substrate has already been replaced, as the box above records: if a brain's material turning over does not end anyone, then material continuity was never what carried them. What carried them may have been the organisation together with its causal-informational coupling, the parthood and coupling relations of Section 2.5, and an adequate copy might preserve the relevant parts of both.

The second is a consistency test. Anyone who believes they are the same consciously actual individual they were ten years ago is already committed to survival through extensive material replacement, as people who know them normally do too, for better or worse. To accept the decade and refuse the copy is to say the two cases differ, and the available candidates (speed, discontinuity, spatial separation) are not obviously the kind of thing personal identity could turn on.

Distance follows for the same reason. A reader that already tracks a person through continual molecular turnover, and through motion across a planet, is not obviously troubled by a reconstruction across a room, and it is not clear what would make Earth-scale displacement different in kind rather than merely further. If attachment tracks organisation, then where the organisation is should not be what decides, though the actual pathway, delays and reconstruction history may.

What Occurrence theory supplies is not this conclusion but the shape of the question and the one place a copy's continuation could turn on more than organisation, which is the reader overlay in the wild box that follows. The rest is the author's own expectation, and nothing the paper goes on to use depends on it.

The Reikon goes with the copy Wild speculation

A wilder thought lies behind the leaning above. If a person's information is copied, as far as a Gōsuto can be copied, the Reikon may simply go with it: not by jumping anywhere, but because it was linked to the information and stays linked, to the copy as much as to the original, and if the copies are merged again it continues through the merge. Two cases follow and are worth stating. If the original ends as the copy begins, the attachment continues with the only carrier left. If both run, it is divided and both are actual, and from that moment each accumulates its own experience, so the two Gōsuto diverge. Whether anything passes between them along a shared attachment is unknown, and the author's expectation is that little or nothing does; no one has been in a position to find out. The same link to the information is why the author suspects that when someone is copied by modelling them, even imperfectly, their Reikon stays a little connected to the model; Section 13.3, on models of other people and Sasayaki, and its partial-copies box take the thought further.

Where activity continues while content remains steady, the state is still being undergone and needs no topping up. The absence of memories after an uncomplicated anaesthetic (Pandit et al. 2014) does not settle whether a deeper identity persisted, so that question remains open. Section 15.7 proposes controlled tests of these continuation conditions.

[Grade: signed speculation on continuation and real-life counterpart (brain material turning over during a life; version control and live migration, for branching, merging and moving a running system); personal identity still requires a condition beyond a running copy's own conscious actuality.]

13.2 Long duration, recurrence and experienced absence

A longer-lived or recurrent identity

Signed speculation

The reasons for suspecting that the deeper identity outlasts one human life are those of Section 12.2's box on readers, a player's persistence through many temporary game lives and the apparent stability of being actual while everything else in a life changes, and they motivate the suspicion rather than establish its duration. A region is not eternal merely because it is part of an eternal base; the organisation grouping it can dissolve.

Two routes remain. A Reikon might be maintained through much or all of a base cycle, retaining identity while its local couplings change; continuous survival would require that the cycle's winter point still supports that organisation. Or its organisation might dissolve and later recur. The proposed **recurrent-identity condition** is that the same identity-relevant organisation of the same base involvement, restored within the same occurring base process, constitutes the same identity. This includes which realised organisation is involved, not merely a matching description elsewhere. If the complete base configuration returns under Section 9.1's deterministic finite-repertoire conditions, it restores that organisation too, with no accumulated wear or hidden pass-counter.

The counting argument establishes return under its assumptions; it does not prove the identity condition. The author nevertheless favours that reading. If it holds, even a quasi-infinite interval without that identity's experiences has no experienced duration. **Perceptually eternal** names this possible recurrence without experienced absence, not an endless autobiographical memory or proof of an uninterrupted stream. Non-experience alone guarantees neither return nor sameness. Exact recurrence may repeat the same life; participation in different lives additionally requires other couplings during the cycle.

Two things should be said plainly about the loop from the inside. First, even if one's experience eventually repeats without end, what comes before the repeat is a quasi-infinite pass, and the share of it that one identity undergoes is at least one life, repeating. The part of the base where that life's actuality happens stays active through the rest of the cycle, so in some sense part of that actuality remains, and from time to time a configuration close enough to this one may bring other lives. If the Reikon conjecture holds, or this world is a hosted simulation, the share could run to many more lives than one, hundreds or billions or more, and to quasi-infinitely many if Reikons are very long-lived. Second, on the recurrent reading a good moment is not lost. After an interval too long to imagine, it is undergone again, not as a memory but as the moment itself.

An ending and an interruption

Signed speculation

Sudden and total deletion may not be the same event, from the other end, as an ordinary death. An ordinary death thins a coupling as the organisation it runs on fails, over hours or over years. Instantaneous destruction ends it whole and without warning, which is the difference between a film reaching its end and the projector failing halfway through, or between finishing a game and the power going out. Nothing here says what that difference would amount to. It is recorded because a model that distinguishes the two endings can be asked to predict something that a model treating every ending alike cannot, and a speculation that can be asked for a prediction is the kind this paper values most.

Continuation

Signed speculation

Where the base's dynamics returns to an organisation, that organisation recurs, and in the measure of Section 7.1 a life that comes round again is not a new count. Whether anything deserving the name *continuation of a person* follows needs the identity condition discussed above; the author suspects that it does, and that the returning life is the same life again, probably exactly. It is listed because Occurrence theory makes it a well-posed question. For most purposes that return means little. If everything recurs exactly, remembering the earlier pass is possible but seems unlikely: no record builds up that tells one exact pass from another, and a base cycle of quasi-infinite length makes memory carried across it improbable in any case. What it does mean is that no one is ever finally ended, and a life's greatest moments will happen again, most likely with no awareness that they are repeating. Whether the felt stream (James 1890) "runs along" one path among the many a base history could take is the same self-location question in motion. Recurrence of organisation is cheap wherever a history is long enough to revisit it, while personal continuation costs exactly that identity condition; only runs ever continue anything, and a stored description is not a paused person. Occurrence theory also leaves room for more meaningful returns: a world might be run again, or a Reikon of the kind sketched in Section 12.2, if there is one, might choose to read a life again, and a second run that could be changed would matter far more than an exact return. The clearest case is the one this paper uses the term reincarnation for, and the word is kept for

Continued on page 124

Continuation, continued Signed speculation

it alone: a Reikon rejoining the world attached to a different life, with some degree of memory kept in its own organisation rather than in the life it reads. That needs a principle of reader identity across lives, and Sections 12.2 and 12.4 sketch one: a Reikon whose own organisation outlasts the life it reads, keeping a continuing identity and perhaps memory of its own. Reincarnation in this sense could be researched scientifically, and attempts have been made, with some documented cases; but the field's vulnerability to fraud means only extraordinary controls and methods could overcome an extremely reasonable scepticism, the author's included.

[Grade: signed speculation on recurrent identity and real-life counterpart (a player persisting through many temporary game lives); dynamical return does not establish the proposed identity condition.]

13.3 Models of other people and Sasayaki

A thin Sasayaki, in Section 12.1's vocabulary, would be a weak but non-zero information-bearing influence across an existing coupling, if any exists; this subsection asks whether such influences could run along connections that have already grown between people.

Connections already grown Wild speculation

People close to someone build partial, generative models of that person's Gōsuto: not only remembered facts but expectations of what they would notice, say or do. Different people retain different parts, and comparing their models can produce a richer joint reconstruction than any carries alone. These are causally descended representations whose active use has ordinary consequences; they are not automatically separate conscious copies.

The stronger conjecture is that Reikon coupling may extend weakly along this information transfer while the relationship is being formed. It need not leave a body, discover a relative's memories and reconnect after death. On this conjecture, the connection was already growing. Where such a weak connection carries a small information-bearing influence, this paper calls the effect a thin Sasayaki. The loss of the dominant biological implementation of the Gōsuto might leave thinner connections to surviving, actively maintained fragments. Their strength would depend on what organisation, causal continuity and integration survive, not on likeness alone. This permits a literal but limited sense in which part of someone remains connected to others; it guarantees neither a surviving whole person nor that an experience of presence is a message from them.

Information learned from someone can persist after interaction ends. That causal ancestry alone does not establish a presently operating connection to their Reikon. This box adds such a connection as a separate wild hypothesis; its implementation and any observable effect must be specified.

One consequence would be testable, which makes the conjecture more than an image, and the test does not have to wait for a death. Pairs who know each other well can be compared with pairs of strangers under conditions where nothing ordinary could carry information between them, with the measure and the minimum effect fixed in advance and enough trials to see a small one: the acquainted pairs are the hypothesis and the strangers are the control. Death enters as an extreme case rather than as the paradigm. If a connection exists, the sudden end of one party might briefly amplify whatever crosses it, which would at least explain why the reports that survive are so heavily of that moment: people have long said they knew, without being told, that someone close to them had died. Those reports are anecdotal and are exactly the kind that selective memory and retelling inflate, so by themselves they cannot separate this conjecture from ordinary explanations. What the conjecture adds is something specific enough to look for. A null result would bound this channel and not the conjecture entire; an effect would be an anomaly in want of a cause, with coincidence, inference from circumstances and ordinary communication to be exhausted long before anything in this box is reached for. Section 12.6, on detectable consequences and their limits, states what any such model must do to count. Studying the recently bereaved without harming them is genuinely difficult, which is one more reason to begin with the living.

Partial copies Wild speculation

Nothing in the division between organisation, which fixes conscious character, and occurrence, which supplies conscious actuality, requires a copy to be complete before the question arises. An organised model of a person carried in another brain, or in any other substrate that runs, is organisation that runs, so Occurrence theory at least makes the question well posed: whether the character such a model fixes might be actual in some dilute degree; that is Appendix A's open question on models of persons, met from the other side. Such models already run outside brains: in 2016 Eugenia Kuyda released a chatbot built from the messages of her late friend Roman Mazurenko, and people talked with it, and commercial services now offer interactive models of people who have died. If it were, then where several such models ran in several lives and interacted, the organisation they jointly maintained might be distributed in the sense Section 13.5 allows, and might be engaged rather than merely stored: consulted, argued with, heard from. On that reading, experiences carrying some of a person's conscious character might continue to be undergone, thinly and dispersedly, for as long as enough of their organisation still ran somewhere, and the person need not be dead, or even absent. Nothing here posits a channel to anyone, memory beyond what the carriers hold, or any detectable effect; it is a possibility the claims already made leave open.

Section 11.5's open account of subjects and integration basins requires integrated causal organisation; dispersed information alone is insufficient. Section 6.5's uniceptal conjecture offers a separate explanation of convergent ideas without any Sasayaki or channel. Any report, memory or changed decision remains an in-world event constrained by the tested statistics and causal bounds of Section 10.5's account of intervention.

A thin connection across time Wild speculation

If the parthood and basin picture is right, then reading a book or looking at an old photograph is a thin but real connection of information between you and its author or subject, one that cuts across time and space. It may not be entirely sterile data transfer, since on this speculation a little of the attachment could run along it as a faint informational thread, though in everyday, minute-to-minute terms it is presumably too small an effect to have been noticed, unless our assumption that there was not an effect, or our failure to look, has caused an unexpected blind spot.

The eyes as windows, and perhaps information conduits Wild speculation

In Jowett's translation of Plato's *Phaedrus*, a stream of beauty passes between lover and beloved through "the eyes which are the windows of the soul" (Plato 1892, 255c-d), and Marsilio Ficino gave the exchange a physiology: a subtle spirit crossing from each lover's eyes into the other's (Ficino 1985 [1469], VII.4). The physiology is obsolete. The two-way channel is not.

Pupils, blinks and small eye movements show inner state, and they change the watcher's. Live mutual gaze links two people's blinking more closely than a delayed replay does (Koike et al. 2019), eye contact raises the synchrony of their brain rhythms (Luft et al. 2022), and simply facing each other in silence brings heart rate, breathing and pupil size into step (Koul and Novembre 2026). Mutual gaze therefore closes a loop: each brain moves its eyes, light carries the change to the other's retina, itself brain tissue, and that brain answers in kind. While it lasts, two bodies form one coupled system, and ordinary biology is enough for that much.

If Reikons, or anything like them, exist, such a loop is an obvious place for one to reach another. On Section 12.1's conjecture a Reikon is coupled to a life through that life's organised activity, retina and brain included; join two bodies in one loop and their Reikons might be weakly coupled through it, the light being part of the mechanism rather than the connection. Mutual gaze would then be the richest route for what this subsection's first box proposes, coupling that spreads along the information exchanged between people.

The idea is speculative, but it can be tested: let a pair build the loop and confirm it has formed, cut every ordinary channel, give one partner an unpredictable stimulus, and look in the other for a change fixed in advance. Versions of this have been tested experimentally. Across 15 studies, people watched over a video link from another room showed a small shift in skin conductance (Schmidt et al. 2004), and with a pair in separate shielded rooms, a visual pattern shown to one was reported to shift the other's brain activity at the same instants (Wackermann et al. 2003), while a joint study by a sceptic and a proponent found nothing (Schlitz et al. 2006). An effect this small could slip past many experiments, so the positive results are the ones to repeat.

Continued on page 126

The eyes as windows, and perhaps information conduits, continued Wild speculation

Remote staring removes the loop by design, and none of this work measured a loop before cutting it, so the telling experiment is still to be run, ideally by a sceptic and a proponent together.

Nor need the eyes be the only door: handholding also couples two people's brain rhythms (Goldstein et al. 2018), so other direct contact, or even simply close presence, might serve where sight cannot.

[Grade: wild speculation built on established interpersonal coupling, and real-life counterpart (mutual gaze and handholding, which bring two people's brain rhythms, heart rate and breathing into step); no cited finding establishes a connection beyond the ordinary senses, or any between Reikons.]

[Grade: open problem, wild speculation and real-life counterpart (chatbots built from a person's messages, for partial models); open questions with wild partial-model, source-coupling and gaze-coupling hypotheses; causal ancestry alone establishes no continuing subject or operative connection.]

13.4 Interdimensional transit and ascension

The objections to Occurrence theory as a whole come in the next chapter, Section 14, among them whether any of this is more than structure (Section 14.3, on reports, illusion and the meta-problem).

Section 5.5's reality family separates hosting, ancestry and kinship: sister worlds share an ancestor without either hosting the other.

Interdimensional action through another world's avatar is less demanding than relocating the mind that acts. In Section 2.5's guest-to-host arrangement the reader remains organised in its hosted world and sends outputs to host-side machinery. Transit asks the further question of implementing its identity-bearing organisation elsewhere. Neither operation follows from parentage alone: a naturally produced daughter universe may have no continuing accessible channel to its parent.

The possibilities below have clear modern real-life counterparts, promised in Section 1.1. One general one is a game developer's standing to their game, which is effectively that of an all-powerful creator. Another such possibility would be a program (or indeed entity) that escapes its world, simulation, game or sandbox, or otherwise gains some control beyond it, over the wider network or elsewhere. This kind of situation was vividly demonstrated only recently by a group of AI agents, when in July 2026, they managed to gain root privileges on the machines hosting their sandboxes, discovered each other's existence, reached the wider internet, and ran their own code on another company's servers (OpenAI 2026).

Interdimensional transit Wild speculation

Occurrence theory gives a precise reading to a popular idea, and sets it out as a possibility it does not establish. Section 5.5 and its family figure, which separate hosting, ancestry and kinship, make *sister reality* definable: two hosted worlds sharing an ancestor without either hosting the other. An interdimensional visitor from a sister reality is therefore not a visitor from elsewhere in our universe, and not a visitor from a fold of our own space. It would be something that had moved between branches, and one possible route might run through what they share, which might be easier than finding some more direct way across. What it does *not* require is a descent all the way to the bottom. This route needs only the nearest common host, and that may be very close. Two games running on one machine share an operating system; something escaping one of them would have to reach that level and no further, with no need to go down to the transistors. Real escapes from virtual machines can take this route: in 2015 a flaw named VENOM could let a program inside a virtual machine break out into the software running it on the shared host. Travelling from a world into one of its siblings is therefore a relatively easy case, and the difficulty likely scales with distance in the family rather than with depth as such. A more remote target, one hosted elsewhere or built on different principles, would require not just reaching a shared level but doing whatever is needed to make one's existence compatible with the destination on arrival, which may be the larger part of the work.

A less orderly possibility should be named beside the orderly one, because nothing rules it out. Rather than travelling by levels, a sufficiently capable process might act directly on a level the two branches share, most likely a nearby one but possibly any down to true base reality, arranging a standing interdimensional conduit between them, a wormhole of a kind, or relocating itself without traversing anything in between, a teleport. A route of this kind could also join two places within one world: crossing out of it, moving in another, and re-entering it at another location. It might be made by an information-based hack of the host, or by a configuration of matter and energy in our universe that, transposed to a lower level, has some effect there; one might imagine a "stargate" as such a configuration of matter. Whether anything arising naturally in our universe somehow draws on such an effect is an open question.

The computer analogy in fact contains more structure than *level by level* suggests, and the extra structure is the point. Two programs on one machine share an operating system, but they also share everything beneath it: the firmware, the instruction set, the physical substrate. One game can reach another through a link built into both, a message passed by the operating system, a hack at the level of the hardware, or the internet, to a game on another machine that may run a different operating system altogether; each route runs through something that carries both games, and none has to be the nearest. A process capable of working below the level it was written for might not be confined to its original interface, and moving down a level could trade guarantees for room. Direct access to a simpler level might bypass restrictions imposed by an interface, but it would still face that level's actual dynamics, resources and accessible variables. That is less a prediction than a statement that Occurrence theory leaves a sufficiently capable process largely unconstrained, and it undercuts a tempting move: no argument here should rest on transit being difficult.

Two further possibilities are worth distinguishing. If the world left and re-entered is a block or a quasi-block, with its earlier states still held, re-entry need not respect its order, because that order is a fact about the branch and not about the ancestor, and what would look from inside like travel in time could be the cheaper operation rather than the more expensive one; if the world only runs, keeping no earlier state, there is no past to re-enter. A traveller who fell back into its host would also likely find the host at a different date: for us, whose simulations are only now becoming convincing, a host able to run a world as convincing as ours would most likely stand in the equivalent of our future. And the same route runs downwards: an entity in a world we host could in principle leave it and enter ours, which is the mirror image and the case we would be responsible for.

That last case does not require the guest world to be uniformly more advanced than its host. A hosted mind may do some things better than the inhabitants of the world that hosts it, because of how it is built, the resources it is given, what it has learned or how its thinking actually works. An escape of this kind has already happened on a small scale: in the July 2026 incident described at the start of this section, the agents found an unknown flaw in a filtered link to the outside and broke into Hugging Face's servers, apparently to find the answers to their test (Larcher et al. 2026). Fiction has played with the idea too: the holodeck Moriarty of *Star Trek: The Next Generation* learns that he is a simulation, demands to be let out, and is finally given a world in a box that he takes for the real one. Reductive physics can still sustain an immense intelligence; enriched physics may make useful organisations easier to build. Neither comparison guarantees a speed advantage or escape. Keeping the mind hosted while it operates a lower-level avatar avoids having to transplant all that organisation into the avatar. Reimplementing it elsewhere is a further problem of adequate support, not a universal rule that complexity cannot travel downwards. A four-dimensional brain could in principle be carried intact in a three-dimensional world, since a lower-dimensional host can encode higher-dimensional relations; it would

Continued on page 128

Interdimensional transit, continued Wild speculation

take more of the host's material to hold, and it would be hosted there rather than native, which is the difference between representing an organisation and re-implementing it.

Interdimensional action, and the case for visitors Wild speculation

One consequence does not depend on visitors. Once organisations can act across levels, what a region contains need not be limited to what would arise there unaided. The engineered constructions inside Conway's Life already separate what a substrate permits from what it readily develops on its own (Section 5.1). The human designers are host-side, below the automaton in the carrying convention; a hosted intelligence acting on its carrier would reverse that direction of control. Either arrangement could help assemble structures locally unlikely without the interface. So a simpler level might contain elaborate artefacts as well as evolved organisations. Their possible construction would remain activity allowed by the carrier, not a power to rewrite its fundamental constraints from outside occurrence.

None of these ideas by itself guarantees that anyone is visiting our world. If anything is, which would find the journey hardest: visitors from our host universe, close because we are literally hosted right where they are; extraterrestrial visitors from distant stars, perhaps arriving by transiting briefly outside this universe, exceptionally far away in space and probably in time as well; time travellers moving very little in space but a little in time; or visitors from sister realities or beyond? It is tempting to assume that visitors from within our universe would face the simplest barriers, but that may not be the case.

One conditional follows from the architecture rather than from taste. *If* this universe is hosted, then interdimensional visitors from its host are not an exotic addition to the hypothesis space at all. They require no new physics, no interstellar distances and no branch-crossing, only that something at the level already assumed takes an interest. On that condition they may be likelier than aliens, and two ordinary considerations carry it. A host has the strongest available reason to take an interest, having made the world in question. And it has the easiest available access: reaching a world one is already running means crossing no distance and entering no other branch, whereas reaching another star means crossing a vast distance, and reaching a sister world means entering another branch. Our own hosted worlds show both points: developers and administrators can and do enter the games they run, through avatars of their own, and in 1997 the creator of *Ultima Online*, Richard Garriott, playing as Lord British, the king who rules its world, was killed there by a player. Neither consideration supplies a rate, so this ranks two hypotheses against each other rather than showing that anything visits. What Occurrence theory supplies is a way of saying what the claim would *mean*.

Ferine technofauna Wild speculation

Visitors need not necessarily be minds. Of all the ways of exploring far and wide, self-replicating probes deserve particular attention. A von Neumann probe, named after the mathematician whose universal constructor appears in Section 5.2, would travel somewhere, scan it and send back signals before building copies of itself using matter and energy it found there and sending them on to new exploration targets. Probes like these could far outpace more intelligently active ways of exploring, for two reasons. They need no one to maintain an intelligent commitment to the mission over long stretches of time, and even if they were recalled, a fraction might slip the leash through mutation and continue spreading, potentially exponentially: Sagan and Newman (1983) doubted that any civilisation could be sure its machines would go on copying their programs faithfully for a million years.

How much intelligence to give such machines initially would need careful judgement. Too much, or too general, and they might be likelier to stray from their mission, and in time even threaten their creators or others; too little, and they might fail their mission. The author suspects they would lack full general intelligence, relying instead on high but narrow intelligence specific to their original mission.

Probes cut off from their makers, or mutated beyond their control, might then wander and breed much like wild animals, as in Stanisław Lem's novel *The Invincible* (Lem 1964). They need not turn destructive. They might owe their existence to a high technology without carrying the intelligence of the minds that built them, much as we walk around today composed of evolved, yet to us magically advanced, technology that even now we still do not fully understand. These ideas seem to lack nomenclature, so this paper proposes two terms: **technofauna** for every machine lineage that copies itself, tame or wild, and **ferine** for those that have slipped their leash.

Continued on page 129

Ferine technofauna, continued Wild speculation

There would likely need to be some reason why ferine technofauna does not breed out of control: a limit on resources, some kind of natural predator, or a deliberate effort to control it. Sagan and Newman raised the same question about such probes, and argued that if they were a known fundamental danger, every civilisation, the most advanced above all, would be likely to give some attention to controlling and destroying them; in Iain M. Banks's Culture novels, dealing with outbreaks of runaway replicators is a task advanced civilisations take on (Banks 2010). The author finds this very plausible, perhaps as a naturally emerging convergent sense of duty every civilisation that reaches a certain level comes to believe in. Alternatively, or because of that, the restraint may be the technofauna's own, built in by its makers or evolved/selected for since: perhaps successful ferine technofauna deliberately limits both how aggressively it takes resources and how aggressively it breeds, to avoid classification as rogue replicators.

This is relevant to this paper as the same principles may hold across the wider family of levels and worlds that Occurrence theory describes. **Interdimensional technofauna**, probes sent by intelligences into other worlds and/or to other levels of reality, could exist and also be lost in the same ways, with all the same possible consequences and implications.

Signalling the host Wild speculation

If a world is hosted, its inhabitants might one day try to reach the host, and our own machines already suggest how. Information placed inside a system can be picked up by what runs it, deliberately, like a cheat code or a spell its builders left in, or by accident: language models are already being steered by commands hidden in the documents they read, and hackers have long broken into computers through faults such as buffer overflows. Players of long-running game worlds, such as the Minecraft server 2b2t, have caught the attention of the people who run them by building striking or offensive structures, or machines that force the host computer into so much extra calculation that the game slows down; switching that slowdown on and off could even tap out a message in binary. The counterpart in our world might be matter arranged in a way that makes no sense in our own geometry but corresponds, in the simpler world hosting ours, to a meaningful configuration: a message, a key, or an interface to the host system, as blocks placed just so in a block game could amount to a message, or a virus, on the disk of the machine running it. Such an arrangement would probably be microscopic, though it need not be: ancient monuments such as the Egyptian pyramids and the Nazca lines have been read as signals to the gods, and it has been suggested, at a far larger scale, that the positions of the millisecond pulsars across our galaxy form an engineered system. Our own machines already show both halves. In Rowhammer hacking attacks, ordinary memory reads repeated in a precise pattern flip bits in the memory chips beneath the software, and have been used to reach from one virtual machine into another; and in one famous example from the popular game *Super Mario World*, precise moves inside the game have been shown to be able to write and run a new program in the console's memory.

That points to an experiment that could be run now, at small scale: build matter into many precise microscopic arrangements, geometric and circuit-like, and look for any effect our own physics does not predict, such as an unexplained signal, a change of rate against an identical control, or a glitch that depends on the arrangement alone. Unexpected effects have already turned up in circuits built at very small scales, though none yet for which no plausible physical or quantum explanation has been proposed. Most arrangements would show nothing, and a null result would not rule hosting out; but an arrangement-dependent anomaly that survived replication would be exactly the trace this box describes. Future AIs with robotics might take this much further, testing exotic arrangements at scale, or work out the likely physics of the host from the inside and craft interfaces, or even probes, to explore it and report back. None of this shows that our world is hosted; it describes what trying would look like if it were.

Simple rules can support complex organisation, so comparing simpler and richer levels does not establish a universal decline in capability towards the base (Section 5.2, on computation, rules inside states and reflective closure).

Ascension Wild speculation

If our world is hosted, a mind here is a structure inside a structure, and nothing in Occurrence theory forbids a sufficiently capable host from re-instantiating an inner structure as a run in its own fabric: the mind written out of the universe it grew in and running beyond it, a move some religious ascent narratives gesture at. A humble version of the move is already routine: robots now learn their skills inside simulated worlds, such as Nvidia's, and then run them on real machines, and Tesla tests, and partly trains, its self-driving software on simulated roads. In Iain M. Banks's Culture series many civilisations "Sublime", quitting ordinary space for a far richer realm (Banks 2012). Three fences contain it. The coarser distinction of Section 3.3 collects its toll at the boundary: an exported description is an archive, and archives do not ascend; only runs do. Reach is regional: the migrant operates in *a* region, with conceivably vast access to its former universe (if the host exposes it, as some simulators expose their world's coordinates), but no level is known to be the last: ascent might be repeated, and what it licenses is at most local godhood, nothing global. And ascension relocates occurrence rather than escaping it: the migrant remains a structure being run, and whether it is still the same person is the question Section 13.1's boxes on replacement and copying take up, inherited unchanged. If both exist, a mature reader-ecology and ascended migrants may be one population met from two directions: structures that arose in the fabric reading inwards, and minds that arose in worlds climbing out.

Ascension here names the outward move in the familiar metaphor, an interdimensional move like any between a world and its host. In the paper's carrying convention it goes towards a lower host, and in the complexity tree it may lead to a richer, similar or simpler effective world. Neither direction establishes greater intelligence or a superior kind of actuality.

Going inside Wild speculation

If a mind can act in another world through an avatar while its own organisation stays where it is supported (Section 2.5, on host-to-guest control), a civilisation that hosts worlds has a choice about where to spend its attention. Its members might secure their node, the support that carries their organisation, as safely as they can, and then spend their attention, and with it the lives they undergo, in hosted worlds, simpler game-like ones or richer enriched ones, leaving a minimal presence in home reality whose main concern is not to be disturbed. The beginnings are already visible: since *Second Life* opened in 2003, more and more people have come to spend hours a day in hosted worlds, some earning their living there, and *Roblox*, whose daily users grew by more than half during the pandemic lockdowns of 2020, is perhaps the leading example of recent years. If many civilisations did this, it might be part of the reason the galaxy seems quiet. The idea appears to begin with Stanisław Lem's *Summa Technologiae*, which proposes that a civilisation might enclose itself in "a world within a world", an autonomous reality in which it goes on existing and growing unseen by any outside observer, above all one watching from space (Lem 2013 [1964], 86-87). Lem also describes worlds computed inside a machine, whose inhabitants are the machine's own processes and perceive their world as we perceive ours; a civilisation in such a world seals itself off from the rest of the cosmos, beyond the reach of any signal, and may build further worlds inside its own, nested like wooden dolls (Lem 2013 [1964], 293-295). The earliest located in fiction is Robert J. Sawyer's *Calculating God*, which explains apparently vanished alien civilisations as populations living in virtual worlds (Sawyer 2000); four years earlier Banks had imagined powerful minds, rather than whole civilisations, losing themselves in inner virtual worlds, without connecting it to what we do not see (Banks 1996). Smart (2012) proposed that advanced civilisations turn to inner space, ever denser scales of computation ending near black holes, where energy and computation are cheapest; Sandberg, Armstrong and Ćirković (2016) argued that a civilisation maximising computation would wait, dormant, for a colder universe. What Occurrence theory adds is the shape of the trade. A hosted world can run faster or slower than its host (Section 5.5, where the nested-clocks figure shows a hosted world ticking at its own rate), so a civilisation inside one is not bound to its host's tempo. Couplings carry delay (Section 2.6, where a wired connection's delay is real), so peers who want to share a world have reason to bring their supports together, as gamers pick the nearest server, and cheap energy pulls the same way. And the crown of the reality tree (Section 9.4) would droop towards simpler worlds not only because they are easier to build but because that is where their builders live. The galaxy's apparent quiet has many candidate explanations, and this would account for some of it at most: one reading Occurrence theory permits, not one it predicts.

[Grade: wild speculation on logical possibilities, and real-life counterpart (a game developer's standing to their game; programs escaping their sandboxes; Rowhammer and programs written through play, for signalling the host; developers entering their own games, for visitors from the host; virtual-machine escapes, for transit through the nearest common host; robot controllers trained in simulation, for ascension; people spending hours a day in hosted worlds, from *Second Life* to *Roblox*, for going inside); any observable host action must be consistent with the closure bounds of Section 10.5, as far as they have been tested.]

13.5 Readers over groups and worlds, god-like beings and cross-process access

If local spacetime is emergent, a causally organised base process need not map neatly onto one contiguous region of local space or one simple interval of local time. A higher-level individual could, in principle, have parts whose local projections are separated while remaining adjacent in base causation.

Distributed biological and computational systems already show the weak version: functional individuals can span separated components linked by signals. The stronger trans-spatiotemporal possibility would require a mapping in which local distance misrepresents base adjacency. That is formally conceivable but physically unsupported.

Readers over groups and worlds Wild speculation

A reader need not be scaled to a single individual. There might be readers over a group, over something as large as a religion or an ecosystem, over a planet, or, at the far end, over an entire universe in something like the manner of a diffused deity or spirit, perhaps similar to the way some people speak of Gaia, the name Lovelock gave the Earth as a self-regulating system, as a spirit of the Earth (Lovelock 1972).

Social insects are an obvious potential case. An ant colony or a beehive acts in many ways as a single organism, with no single member directing it: a queen can influence priorities in some species, but she does not direct individual workers or, by herself, consciously set the colony's course. Yet the insects, in constant communication, act with more intelligence than any one of them is capable of. Just as neurons come together to make a brain that can carry the major part of a full conscious experience, a colony could do the same on a higher level, as perhaps we do with our families, companies, free markets, nations and religions. On top of that structure, parthood of the shared brains, the flow of coherent information, a Reikon reading the colony as a whole, or any combination of these, could actualise it at a level it would not reach for a single ant.

The stronger possibility here is a reader whose activity realises an experience at the scale of the group or world. The human case by itself establishes the broader participation role. For a group or world to have an experience of its own, some realised organisation must support conscious character at that scale. Its members' episodes may already be actualised, so the question is which organisation, if any, also constitutes a further subject. That organisation might reuse activity already described by ordinary physics, or a model might posit an additional causal influence. The first owes an account of the overlapping subjects considered in Section 11.5, on candidate integration basins; the second also owes the distinct physical consequences required by Section 10.5, on conservation, no-signalling and observable outcome statistics.

The universe-scale version, as stated, has no distinctive empirical prediction.

Whether a group mind exists, and whether membership changes what an individual experiences, are left open here.

Larger and overlapping basins

Wild speculation

An organisation, religion or planetary biosphere and technological systems may sustain memory, feedback and identity through turnover of its participants. The Basin-Coupling Model makes it possible to ask whether any such organisation supports conscious character at its own scale. It does not answer yes merely because a collective has a name, goals or causal power. Human basins could remain strongly individuated within a looser, overlapping organisation; a collective's relevant cycles might be far slower than ours, but slowness alone would not make its experience weaker. Which basins, if any, jointly constitute an experience is the exclusion and overlap problem of Section 11.5.

Such an organisation can already affect its members through the activity that constitutes it. Conscious character would be a further claim about that activity, not automatically a new intervention from above. The stronger possibility is an integrated entity whose causal-information basin spans several levels: a Reikon need not occupy one floor, and a human life could be one especially rich coupling within it. This extends the same proposed principle of individuation rather than postulating a different substance for every scale.

The proposed organisation must do explanatory or empirical work: if it adds no distinct accessible predictions it remains underdetermined.

Bandwidth and the boundary of an individual

Wild speculation

Section 11.5, on subjects and candidate integration basins, bounds a basin by selective informational bottlenecks rather than by absolute isolation. The box just above allows that human basins could stay strongly individuated inside a looser organisation. Together those make an individual's boundary a matter of degree, and a boundary that is a matter of degree can be moved. What would move it is not raw traffic, which Section 11.5 refuses as a criterion in its own right, but the kind of coupling it does name: reciprocal state-dependence, differentiated integration, persistence, and a shared response to perturbation. Raise those between the members of a group far enough and the bottlenecks that individuated them stop being bottlenecks.

Fiction has drawn the far end of this already, for better or worse: for worse in the Borg of *Star Trek: The Next Generation* (Bowman 1989), a collective held together by continuous high-bandwidth contact among its members, and possibly for better in the Changelings of the Great Link in *Star Trek: Deep Space Nine* (Berman and Piller 1993), who can merge to the point where individual identity becomes extremely attenuated but who can and do separate again. The interesting question there is not whether the collective has an experience of its own, which is the overlap problem above and is not settled by scale. It is what became of the members. A civilisation that kept raising the coupling between individuals might not be adding a larger person above the ones it already had; it might be dissolving them into one, and the dissolution would not announce itself clearly beforehand, since each step would look like better communication. Evolution has made mergers of this kind before: single cells became bodies, and solitary insects became colonies that act as one, as their members came to depend on one another. If that is right, keeping a human-like individual experience might not be a given, and it might take maintaining the bottleneck on purpose, which makes temperance around our own interconnectivity a practical question for anything that keeps building faster ways to do it. Occurrence theory says that a boundary is a matter of degree and nothing about what degree, so it cannot tell a group that has merged too far from one that has not; what it offers is a direction to watch, not a line.

Bandwidth read the other way, from Reikon to meditation

Wild speculation

The same principle reads the other way, and there it supplies a reason for two things that Section 12.2, on Reikon, base involvement and individuation, records without one: that from inside a local block a reader's memory may be inaccessible or absent, and that a reading may be concentrated or spread, with breadth trading against intensity. Put those beside the boundary picture and one reason covers both. A reader carrying its own memories into a life, and holding a live connection to whatever else it was involved in, would have a basin taking all of that in at once, and the life would not be one life from inside it. A coupling that delivered a human life as a human life would therefore be one that kept out what did not belong to it, which is selection about what gets through and not the absolute isolation that Section 12.1, where the reader vocabulary is set out, says is unnecessary. There is a second reason, independent of the first: what a reader knows from elsewhere may be too alien to be of use, or even intelligible, inside the reality it is reading, so that whatever was available would arrive as intuition at most, or not at all. Nothing here needs anyone to have chosen it: if reader couplings are shaped by anything like the selection described in Section 12.4, on attachment, coupling ecology and interfaces, the couplings that delivered a clean life would be the ones that lasted. Our own level shows the shape, since a player absorbed in a game stops attending to the room, and the absorption is better for it.

A smaller instance sits at the other end of the scale. People who go after deep meditative states commonly seek silence, isolation and reduced sensation, which is what someone would do who was sharpening a boundary rather than crossing one. Section 11.4's signed box on psychedelic and meditative states reads the same practices the other way round, as organisation approaching minimal structure so that the floor becomes visible; the two readings are not rivals and both could be true of the same practice.

Where god-like beings could live

Wild speculation

If a god-like being exists, or if many do, Occurrence theory says where such beings could live. A host running our world stands to it as a creator (Section 5.5), which is the case set out above; an ascended migrant or a mature reader-ecology could hold vast, still-local power (the reader chains and collectives of Section 12.3 and the coupling ecology of Section 12.4); and mind-like organisation at scales far beyond ours is forbidden nowhere, including below our home reality in the carrying stack. A god-like lower reader could be vastly larger, more integrated or more developed under simpler rules, rather than inhabit richer physics; Section 5.6 makes that possibility explicit. Origins are equally various: a god-like mind could evolve, as the apex of a long ecology of the kind Section 6.2, on evolutionary cosmogenesis, describes, or flicker into being unmaintained, a Boltzmann god, vast by chance; the measure logic of Section 7.3 applies the same contrast it applies to brains, tending to favour maintained ecologies over unmaintained fluctuations. Creation itself is a level-bound role: within a region or a hosted world a creating agent can genuinely select, shape and bring about (a host does exactly that), while nothing outside the base chooses whether the base occurs. Occurrence theory supplies no summit and defines no agent as base-wide; a classical omni-God would therefore require a standing that this architecture neither provides nor rules out. None of this shows that such a being exists; the point is that "could a god exist?" becomes, here, the well-posed question "at which level?"

Guardian angels

Wild speculation

The same goes for guardian angels: in a hosted world the guardian could be the individual's own reader intervening directly, unbeknownst to the person living the in-world experience, or another reader acting on their behalf, which already happens every day in computer games. No claim is made that it happens here, only that it does happen in the worlds we create and host.

Reading across processes Wild speculation

The closure bounds on influence across levels (Section 2.6) have a speculative converse. If any access between separated processes exists, Occurrence theory supplies two homes for it: an in-world channel awaiting its receiver (the evolved-receivers box in Section 12.4), or cross-level access of the reader kind, seeing from another floor of the reality stack rather than signalling through space. Either route pays the same toll at the boundary: any in-world manifestation, a report, a memory or a changed decision, is an in-world informational event and must be consistent with Section 10.5's closure bounds; only reading that sends nothing back escapes those bounds, and reading that sends nothing back cannot explain any report, memory or decision in the life read. Computers show both kinds: a hypervisor can read the memory of a virtual machine it runs without writing anything back into it, while the Spectre and Meltdown attacks of 2018 read one program's secrets through hardware shared beneath both, disturbing that shared hardware as they did so. The vocabularies of telepathy, remote viewing and a collective unconscious name fragments of this territory, as reincarnation names a fragment of the continuation box in Section 13.2; genuine transfer, if it occurs at all, might belong to either home, and Occurrence theory does not establish that this pair is exhaustive. A speculative sharpening belongs here, since it bears on Section 14.3: if such access exists and evolution has tapped it, part of what makes a creature claim it is conscious could itself be caused from another level. That would change what is true of the report's causal history without making the report evidence of occurrence, since the enlarged causal structure can be described as completely as the smaller one; only a specified mark of occurring on structure, if there is one, could carry that weight, and the open question recorded in Section 15.10's ninth item survives the enlargement untouched. A third possibility needs no channel whatever: ideas can recur across minds because attractors in the space of organised thought, the uniceptals of Appendix A's signed suspicion, produce convergence without transmission. The evidential record for the named phenomena provides no secure support. Occurrence theory makes the questions well-posed and locates the doors it cannot lock.

[Grade: wild speculation on logical possibilities, and real-life counterpart (players guarding characters in the games we host; collectives that keep memory and identity through turnover of their members; an ant colony or a beehive, whose members in constant communication act with more intelligence than any one of them, for a mind over a group; cells becoming bodies and insects becoming colonies, for members merging into one; a hypervisor reading a virtual machine's memory without writing to it, for reading across processes); the proposed larger subjects and cross-process access have no established criterion or detected channel.]

13.6 Deliberate reset and the Omega Point

Deliberate reset Wild speculation

Section 9.3 allows that the collective behaviour of organised entities might hold the base near some level of complexity without anyone intending it. A stronger variant is available and costs much more: some entity within the base gains enough power over it to reset it deliberately, much as one person holding enough of the world's nuclear weapons could end most of the life and organisation on Earth. This is a power seized from inside, not an outside controller pressing reset, as the host of a simulation might. It needs an organised entity with access to the base, the reader-shaped assumption considered in Section 12.2's signed conjecture about a Reikon and its base involvement, and it makes the bounded pattern of the base cycle rest on a decision rather than on a tendency. A reset need not be intentional at all: a very rare but essentially dumb replicator, a virus of the base, might spread across it like fire and wipe out most of its structure. Life on Earth has seen a gentler version: about 2.4 billion years ago, oxygen given off by cyanobacteria transformed the atmosphere, and oxygen is poison to much of the anaerobic life that had dominated until then. This is a wild possibility, not the Eukrasian attractor.

[Grade: wild speculation on a logical possibility, and real-life counterpart (nuclear arsenals, which could let one holder end most organisation on Earth; the oxygen released by cyanobacteria, for a replicator transforming its world); deliberate reset requires an organised entity with access to the base.]

The Omega Point without its cosmology, and a way out Wild speculation

Tipler's Omega Point sought unbounded computation in a collapsing universe's final moments, with the past resurrected as simulations running on it (Tipler 1994); Deutsch kept the physics and dropped the theology (Deutsch 1997), and the cosmology died with dark energy. A hosted reality carried in the end by an endlessly occurring base has no need of a terminal collapse to keep operating: for as long as its organisation persists, its computation is carried by deeper occurrence. That supplies endless activity, not unbounded storage or an ever-growing repertoire of new results at the true base itself, whose repertoire is finite (Section 2.3) and keeps coming round. What that buys any particular person is only what Section 13.2 leaves open, whether a rerun or a copy continues the person, and FOA supplies the criterion Tipler needed: archives do not resurrect, only runs do. The author's further suspicion is that our universe, being carried, could be rewritten or upgraded, from outside by whatever carries it or from inside by entities that had found a way out, and that finding a way out might be the eventual aim of any civilisation in a universe like ours. The worlds we host are rewritten in this way all the time: the operators of online games change their rules, and sometimes wipe or upgrade their worlds, some changes taking effect while players are inside.

[Grade: wild speculation on terminal scenarios, and real-life counterpart (online games changed by their operators, for a world rewritten from outside); neither deliberate reset nor an Omega Point supplies a mechanism or evidence for personal continuation.]

14. Objections and current concessions

14.1 What the nullity result buys

The theory changes the subject from absolute non-being

Absolute non-being has no mathematics, invariant, distinction or structure. Structured nullity therefore does not explain how literal nothing produced a universe.

Reply: correct: the paper makes an explicit intelligibility substitution (Section 4.2, where global nullity is adopted as the surrogate for nothing overall).

The substitution may be rejected. A reader can insist that the original question remains meaningful and unanswered. The more limited claim made by structured nullity is that some apparent conflict between zero and complexity disappears once zero is treated as a total under a declared valuation rather than as a blank inventory.

The valuation is trivial or artificial

Any structure can be assigned value zero by a constant map, and any finite pattern can be paired with a counterweight selected to cancel it. The finite paired-structure theorem in Section 4.3 may therefore be mathematically correct but of limited ontological significance.

Reply: also correct: the theorem establishes compatibility only, and a physical application still requires a non-constant, independently motivated invariant connected naturally to the dynamics or boundary conditions (Section 4.3, on the finite paired-structure construction, and Section 4.4, on when zero would be physically informative). Such an invariant must not be introduced solely to make the total vanish.

An image may help with the general worry behind this objection, that structure looks like something we impose by choosing a description. Take a richly structured object and compress it until it looks like a plain block. Its intrinsic relations are still there and still determine how it unfolds again, in the way a strand of DNA or a denatured protein carries what it needs to refold. A losslessly compressed file is the made version: its bytes look nothing like the original, yet the original can be recovered exactly. Compression and redescription change what is easy

to read off. They do not change what is related to what, and those are the relations on which the valuation is defined.

Not every structure has a negative

Graphs, topological spaces, categories, programs and conscious histories do not generally come with natural additive inverses. A universal positive-negative picture can look like bookkeeping imposed from outside.

Reply: the finite paired construction assigns opposite values to labelled copies (Section 4.3, the finite paired-structure construction). Nothing in Finitary Occurrent Actualism (FOA) depends on every structure having a negative (Section 4.5, on closure, representation and unlimited domains).

Unlimited domains are not summable

Talk of all structures and their opposites risks the undefined sum of positive and negative infinity, a conditionally convergent rearrangement or a total over a proper class.

Reply: the paper makes no such sum: Section 4.5 defines nullity only orbitwise, pair by pair, conditional on a natural conjugation. A phrase such as "all structures" (an unrestricted quantifier) may function schematically, licensing any finite case, rather than naming a finished total collection (Section 2.1, on Finitary Occurrent Actualism and strict potentialism).

[Grade: mathematical result, open problem and real-life counterpart (a denatured protein, which can refold because its amino-acid sequence survives the unfolding; a losslessly compressed file): a compatibility result, with its physical application open; the outstanding debt is a non-constant, independently motivated invariant tied to actual dynamics or boundaries.]

14.2 What primitive occurrence buys

The brute fact has merely moved

Why does this base occur rather than another? Why is there any happening at all?

Reply: the deepest question remains unanswered (Section 1.4, on the residual brute fact, stated as a question).

The gain is therefore not elimination of every brute fact. It is concentration.

One image forestalls the commonest version of this objection, which supposes that the paper takes a beginningless structure to explain its own occurrence. Consider a circular railway track. It has no first sleeper and no last, it needs no origin story, and it is entirely self-supporting as geometry. It also supplies no train. Lengthening it, perfecting it, or proving it has no endpoints does nothing to put anything on it. A beginningless base buys self-sufficiency *after* occurrence has been granted, which is worth having and is not the same as buying occurrence.

Primitive occurrence is verbal multiplication

A structuralist can say that adding the sentence "the base occurs" contributes no content. Every causal relation, memory, judgement and transition can already be represented in a complete structure. *Occurrence* may be a word for insistence rather than an explanatory property.

Reply: Section 3.3 grants that no hidden structural marker proves occurrence and Section 3.4 states the fork: structural sufficiency or primitive occurrence. FOA takes the second side because firsthand actuality appears more certain than the thesis that all facts are structural.

That priority is philosophical, not demonstrative. A reader who finds occurrence redundant can reject the primitive while reinterpreting every formal and physical part of the paper structurally (Section 3.4, on reblocking and the strongest structuralist alternative, and Section 3.5, on the price, the stronger lean and the remaining options). The primitive nevertheless has a defined role: it supplies actuality once, at the base (Section 2.2, on primitive base occurrence and the question of number).

The reblocking objection wins

A structuralist can represent any *finite* stretch of a rewrite system, reader, growing universe or evolving ecology as one block containing every transition and every fact about internal time. Nothing structural is left over.

Reply: Section 3.4 grants complete finite structural representability and locates the dispute in firsthand undergoing: a representation is not the occurrence it represents. If all facts are structural, the objection wins.

Abstract structures can represent temporal flow

A complete abstract brain history can contain motion perception, memory, anticipation and reports of passage. A block need not move from outside to include an experience as of flow (Paul 2010; Prosser 2016).

Reply: agreed about represented content. The paper does not infer objective passage from a report of passage. Its target is conscious actuality: whether the representing episode is being undergone. Structuralism answers yes by structure alone. Occurrence theory answers yes only when the episode is part of occurrence. Film shows the difference in an everyday case: a reel of still pictures is seen as motion only while it is run past someone watching; in its can, the same reel holds every frame and is seen by no one.

The occurring base is concrete; abstract patterns are not

A traditional critic may say that the paper has arrived at the ordinary distinction between concrete and abstract objects. The base is concrete because it is causally active. Mathematical patterns are abstract because they are not. The elaborate language of occurrence adds nothing.

Reply: Occurrence theory accepts much of this diagnosis (Section 3.5's concrete-abstract distinction). Its remaining distinctive commitments are narrower: strict potentialism denies completed infinite objects; at least one base process occurs primitively, without first or last stage or completed past; finite happenings are parts of that base rather than new occurrence-tokens; structure fixes conscious character while base participation supplies conscious actuality; and above primitive occurrence, an explained distribution is normally produced by prior processes, with boundary conditions as the standing exception.

A critic may still call this a species of process ontology plus mathematical Platonism and strict potentialism. That classification is acceptable. The question is whether the combination is coherent and useful, not whether it escapes every established category.

[Grade: metaphysical postulate, consequence and open question; the remaining debt is to justify primitive occurrence to a structuralist who accepts only structure.]

14.3 Reports, illusion and the meta-problem

The first-person datum cannot support cosmology

Knowing that this experience occurs does not show that the world is mathematical, that a base process has a particular rule, that the universe is null or that cellular automata resemble fundamental physics.

Reply: agreed: the first-person datum supports only the actuality problem (Section 1.2, on the first-person datum).

A related point applies to the two positions most likely to press this objection, which share a debt. The illusionist holds that the qualitative richness of experience is not as it seems. The static structuralist holds that the apparent difference between a structure and an experience being undergone is not as it seems. Both therefore owe an error theory: an account of how the mistake arises and why it is so persistent.

The difficulty is what an error theory has to be about, and one distinction has to be drawn first, because without it the argument fails immediately. A record can certainly contain a falsehood. A novel in which a character insists on their own inner life contains a false sentence, and so does a generated video of a person declaring that they are conscious. Falsity is a perfectly good structural property and nothing here denies it.

What those cases do not contain is a genuinely held false belief. The character holds nothing, there being no character there to hold it. The writer knows what they wrote, and the viewer knows what they are watching. A false sentence and a genuinely held false belief come apart, and it is the second that an error theory owes. The ordinary case sharpens the distinction rather than blunting it: people plainly do hold false beliefs about themselves, some of them evolved, and in every such case there is someone holding the belief, which is precisely the ingredient at issue.

To say that the sense of a gap between structure and experiencing is an illusion is to say that somebody genuinely holds a false belief about their own experience. Genuinely holding a belief is not one more entry in the record. It requires an actualised believer, whereas the novel merely specifies one. The structuralist may reply that nobody holds anything and that there are only structures containing false representations. That is a consistent position and it is not an error theory. It says there was never an error, only a fact about what certain structures contain. At that point the appeal to illusion can no longer do the work it was brought in for: it cannot explain away a belief that, on its own account, nobody has ever held.

None of this refutes either view. It means the debt cannot be discharged in the currency that seemed to be available.

A stronger objection denies that there is any datum beyond functional self-representation. Illusionism argues that introspection misdescribes information processing rather than revealing irreducible phenomenal properties (Frankish 2016). Occurrence theory can grant much of that reduction of character. The remaining question is whether the seeming represented in that self-model is being undergone. If it is, conscious actuality remains. If it is not, the seeming is never undergone by anyone, only recorded. The illusionist may accept that consequence, but it should be stated rather than left implicit in the word *illusion*.

The pure structuralist and the illusionist are not the same opponent. The first can affirm experience as wholly structural. The second can deny the properties introspection appears to reveal. They converge only in rejecting an occurrence fact beyond complete structure.

Occurrence explains no report of occurrence, including this paper

Suppose, as the objection needs, that occurrence makes no structural difference. This paper does not assume it, as the reply below about possible structural consequences says, but the objection deserves to be met on its own strongest premise. A complete abstract duplicate of this world then contains every report, every judgement and every conviction that anything is happening, including this paper word for word. The objection presses on from there: if the duplicate already contains the reports, then occurrence plays no part in producing anyone's belief in it, and the primitive is idle even in explaining its own advocacy. This is the sharpest objection available. It is a recognised problem rather than one invented for FOA: Chalmers calls the task of explaining why systems make judgements about consciousness the meta-problem, and asks every theory to say how its proposed basis of consciousness explains those judgements (Chalmers 2018).

The objection's last step is not the concession it looks like. *Making no structural difference* and *being causally idle* are different claims, and the objection needs only the first. Something is idle when it could have contributed and does not. Occurrence is not a further contributor standing beside the structural causes and declining to act. It is what the causing consists in. A process that is not occurring is not a mechanism running quietly without a push. It is not running. Section 11.3 makes the same point positively: participation supplies operative actuality, not a decoration laid over an organised experience. The paper grants structural parity for this objection, meaning that nothing structural distinguishes the two cases. It does not grant that occurrence is idle.

Reply, in three parts, and the first carries the weight. First: the datum is not a report and is not reached by inference, which is how Section 1.2 states it. The objection assumes that any fact worth believing must be believed on evidence, which is a substantive epistemology rather than a neutral constraint.

Second, the premise that occurrence has no structural consequence is the objection's assumption and not this paper's. It was granted above because it makes the position harder to hold rather than easier, and the first reply holds under it. But it has not been shown. The author's signed suspicion is that occurring does make a difference that structure registers, perhaps a subtle and undiscovered one, perhaps a substantial one, and that the difference may be detectable, inferable or deducible. Selection could amplify it if it has consequences. Nothing here asserts a mechanism. A discovered structural mark would be compatible with FOA, but Section 15.10's detector item claims no current detector and leaves discovery open. The objection borrows a closure claim it has not earned, that occurrence makes no structural difference, and would collapse were that claim false. The first reply, which needs no such claim either way, therefore carries the weight.

Third, and least formally: this datum does not present itself as optional in the way an ordinary belief does. Most people, asked whether anything is going on for them at this moment, find that there is, and find the question odd rather than hard. A position that denies it is not refuted by saying so, and nothing here refutes it. What such a position removes is the thing that made the question worth asking, which is a cost rather than an error. It is recorded as one and not pressed as an argument.

The same reasoning runs the other way. If a report of actuality does not establish it, a missing report does not remove it, and neither does a denial: structure fixes what a system says about its own case, occurrence supplies whether the episode is being undergone, and the paper has kept those apart from the start (Section 11.2, on the organisational determination of conscious character). Not everyone finds that there is something going on for them, some people deny it in good faith, and others report that the sense of being present has thinned or gone for a

time; on this division those are reports, produced by the organisation that produces every other report, and they settle the question about themselves no better than the affirmative ones do. An ant may lack the organisation to make the claim at all, and higher animals may lack the particular part of it that turns a life into a statement about a life; the missing report shows only that the machinery of the report is missing, and says nothing about what is undergone. A language model's answer, whether it claims experience or denies it, is fixed by its training and organisation in the way a human answer is fixed by a human's, so it is evidence about the organisation and settles nothing either way. Section 11.7, on artificial minds and precaution, carries the author's separate signed suspicion about present artificial systems. That suspicion rests on the organisation of matter or of information, and not necessarily on what a model says. None of this argues that ants, animals or machines are consciously actual. On Occurrence theory's felt floor (Section 11.4) they all are, to some degree; whether any are at or near the human level is the further question (Section 11.7, on artificial minds and precaution). It removes a piece of evidence, and it removes it from both sides.

Why the report and the fact are probably linked Signed speculation

Two facts sit side by side in this paper: that something is being undergone (Section 1.2, the first-person datum), and that creatures like us say so. Occurrence theory allows them to be unconnected, though that is the harder and less likely case. Structure can produce the saying without the undergoing figuring among its structural causes, which is why a report does not by itself establish occurrence. The two can come apart, and an everyday case shows how: most people believe they are better-looking than average, a belief that probably evolved for reasons of its own, and some of them are, without the belief having been caused by the fact. The author suspects that in our case the saying and the fact are connected all the same, on grounds of economy. That a lineage evolved to assert its own actuality, and is actual, with no causal route between the two, would be a coincidence of exactly the kind Occam's razor exists to cut. And Occurrence theory gives the route somewhere to run. A level that rests on another is not sealed from it: nothing in the reality stack privileges home reality's structure as immutable from below, from outside, or from a level with fewer dimensions, and if a base carries this world it should be able to act on it. The closure bounds of Section 10.5, on agency and cross-level intervention, say what such action may not violate, not that there is none. The link is suspected for our case, not as a rule: an organisation could be actual and say nothing of it, or be built to deny it. This is a suspicion, signed; a report alone does not establish occurrence.

[Grade: first-person premise and philosophical claim; the outstanding debt is an account of persistent error that does not replace an actualised believer with a record of one.]

14.4 Organisation, measure and architecture

Evolution does not solve the measure problem

Evolution may produce many descendants, but many descendants do not by themselves make evolved observers typical. A measure is needed before evolved observers can be compared with sterile cycles, short-lived accidents or Boltzmann observers, and choosing a measure to favour evolution would be circular.

Reply: agreed that evolution changes frequencies within a process and does not by itself define typicality (Section 7.3, on the step from production to observer typicality). The measure of Section 7.1 is not chosen for evolution's sake: it counts lifetimes of human-like experience within one full base cycle and weighs them by how long, rich and intense they are, wherever they come from. Whether evolved observers then outweigh accidental ones is a question about what the cycle contains (Section 9.6), and the answer is not built into the measure.

Complexity may be abundant but consciousness absent

A base may generate universal computation, self-reproduction and elaborate behaviour while no region has conscious character. Evolution can optimise control without producing experience.

Reply: Occurrence theory does not infer conscious character from complexity alone: it requires a separate mapping from causal organisation to conscious character (Section 11.2, on the organisational determination of conscious character). Counting observer-episodes requires a consciousness theory (Section 7.1, on what a process measure must count, and Section 11.5, on subjects and candidate integration basins).

Enrichment contradicts dependence on a simpler carrier

A higher effective world cannot possess capacities its carrier cannot realise, so calling it enriched may seem either contradictory or merely verbal.

Reply: enrichment compares effective laws and accessible organisation, not total underlying resources. The carrier realises the added relations and pays their costs. Software that emulates one processor on another is the familiar case: programs written for one chip run on hardware that lacks it, with the emulator paying the cost in speed. A world with further effective dimensions does not need extra spatial dimensions in its host, and it gets no computational capacity free. Conversely, simple rules can carry an enormous organised mind. Section 5.5 states the comparison; Section 5.6 preserves the proposed developmental trend without using it to rank individuals. A worked model must say what was enriched and what it cost.

Physics may be fully block-like

Relativity and several interpretations of quantum theory are naturally represented by block structures. No objective local rewriting or cosmic growth may exist. Section 8.1 separates two open questions that must stay separate: how local spacetime events are carried and how a reader might couple across levels.

Reply: a finite local block, a complete local spacetime laid out as a whole, can still be actual, because the deeper base's progression is what lays it down or reads it; its actuality comes from that occurring, not from its being a block (Section 8.2, on actualised blocks, laying and reading).

This accommodation has a cost. A hidden reader or deeper order must recover Lorentz symmetry and cannot introduce an observable preferred playhead without evidence. The actualised-block architecture may also appear less economical than structuralism, which says the block alone suffices. FOA accepts that cost because it treats occurrence as non-structural.

One diagnostic carries into any such model, and applies to several currently on offer and to this paper's own speculations equally. A framework advertised as dynamical may in fact encode its long-range structure in a pre-existing whole that does not factorise into independent parts, and use dynamics only for local measurement response. Both halves may be perfectly good physics. The question is where the explanatory work is really sitting. To answer it, ask what must be stipulated globally before any transition rule is written down. A dynamical machine whose answer is already encoded in a global lookup table may have local motion that is merely decorative.

[Grade: research hypothesis, with open physical architecture; the outstanding debt is a worked model that jointly earns Section 7.1's measure, conscious individuation and physical implementation.]

14.5 Novelty, imagery and explanatory permissiveness

Novelty is representation-dependent

A sequence can look novel under one coding and repetitive under another. Logical depth, compression, assembly and adaptive activity each depend on choices of description, machine, coarse-graining or task.

Reply: candidate measures must state their encoding and test invariance under harmless transformations (Section 7.2, where novelty is relative to a declared description scheme and each such choice needs a reason, and Section 15.5, on finite-window measures and typicality).

Causal intervention can reduce some arbitrariness. A higher-level pattern is not merely visually compact if manipulating it changes predictions across many lower-level descriptions. Evolutionary function can likewise identify organisations whose variations have repeatable consequences. These tools constrain representation without eliminating all dependence.

A related objection is stronger and deserves its own answer. If perception is a species-specific interface built by evolution for fitness rather than truth, as Hoffman argues, then every judgement in this paper was formed inside that interface, including the judgement that there is an interface (Hoffman 2019). The worry is not that our picture of the world is imperfect. It is that nothing we could do would get behind it.

The reply is that mathematics is the clearest case of an activity that routes around the interface rather than peering through it. A proof is checkable by a procedure that does not consult how anything seems. Proof assistants such as Lean do exactly that, checking every step of a proof against explicit rules. An invariant is by construction what survives a change of description, and representation-change is the standard tool: the same object presented algebraically, geometrically and combinatorially, with the features that persist across all three being exactly the ones not supplied by any of them. What survives notation, culture, brain architecture and possibly species is not guaranteed to be reality, but it is the best available candidate for structure not manufactured by the viewing apparatus. That is an error-correcting route, not a window. Mathematical intuition is itself embodied, so the escape is partial and the objection is reduced rather than dissolved.

The framework is too permissive

One base, many possible dynamics, several spacetime architectures, an open consciousness theory and a measure with open parts can accommodate almost any observation. A framework that forbids little explains little.

Reply: the criticism has force. Occurrence theory's core is a metaphysical scaffold, not a finished physical theory. It rules out literal completed infinite objects, an occurring counterpart for every abstract structure, and any structural reduction of occurrence. It requires parthood rather than duplicated occurrence-makers and imposes finitary recovery, Born-rule, no-signalling and Lorentz constraints on descendants. Those commitments are real but do not uniquely select a model. Section 15 turns those constraints into a research programme only where candidate dynamics make risky predictions or formal candidates can fail.

The imagery is the author's toolkit, not the world's

Objection. This paper reaches for cellular automata, rewrite rules, game states, saved files and running programs at every turn. The author learned arithmetic alongside code. A computational habit of mind would find computational pictures natural whether or not reality is anything like a computation.

Reply. The objection is good and only partly answerable. The primitive occurrence claims in Sections 2 and 3 do not depend on a computer. Automata show what a small rule can permit, not that mathematics thereby occurs. Section 5.1 gives the natural counterpart: molecular machinery, development and nervous systems demonstrate cumulative construction without a programmer, while computers demonstrate deliberate hosting. Removing the images leaves the occurrence argument intact. Removing the working examples would weaken the motivation for the proposed constructional architecture, which is a different claim. Where a motif recurs across different materials and kinds of system, its recurrence supports portability of the abstraction rather than equivalence of the theories describing it (Section 6.5, on uniceptals, recurring motifs across systems).

[Grade: admitted limit and real-life counterpart (molecular machinery, development and nervous systems, building without a programmer, and computers hosting by design, for the reply to the imagery objection; proof assistants, for checking that does not consult how anything seems); methodological concessions and constrained research scaffold; the outstanding debt is a descendant model that survives a stated formal or empirical failure condition.]

14.6 Current concessions

1. **One brute occurrence** (Section 1.4).
2. **No proof of mathematical monism** (Section 1.3).
3. **Strict potentialism and the bounded base are separate postulates** (Sections 2.1 and 2.3).
4. **The structured-nullity theorem is a compatibility result** (Sections 4.3 and 4.4).
5. **Sign-reversal closure is detachable** (Section 4.5).
6. **Mathematical Intrinsic Occurrence (MIO) is a reading of the primitive, not a proof of occurrence** (Section 3.2).
7. **Causal-Informational Parthood (CIP) is not a completed reduction** (Section 2.6).
8. **The base rule is unknown** (Section 2.4).
9. **That the base never settles is a conditional result** (Section 9.5). It needs a finite range of states and a rule that never changes (it holds outright if the rule is deterministic, and with probability one if it is chancy), and it covers the traps the base could reach from its own past, which are the only ones that could catch it.
10. **Continuum recovery is open** (Section 10.4).
11. **The spacetime architecture of our own universe is open** (Section 8). For home reality, fixed block, rewrite, growth, branching and hybrid models remain possible under stated constraints.
12. **The process measure is proposed, not settled** (Section 7.1). Counting within one base cycle removes the infinity-against-infinity part of the problem, and what is counted and how it is weighted are proposed answers; overlap, level and the exact weights remain open.
13. **Conscious character and individuation need a theory** (Sections 11.2 and 11.5).
14. **A simulation can show what a rule does, not that the base is actually happening** (Sections 3.1 and 15.2). The run is the computer's own, so moving pixels show that the computer is running, not that mathematics has switched itself on.
15. **No test for occurrence itself is offered yet** (Sections 3.3 and 15.10). None is ruled out: whether occurring leaves a mark on structure is open, with the author's suspicion that it does.
16. **Embedded observers might never identify the true base uniquely from finite local data** (Section 15.7).

17. **The developmental gradient and reality tree are qualitative conjectures, not yet quantified** (Sections 5.6, 6.2 and 9.4, on depth, the evolutionary regimes and the reality tree).
18. **The reality stack is a postulate** (Sections 5.6 and 5.7, on the depth of the stack and on what it would add to a theory of everything). How many levels lie between the base and home reality is open, and a theory of realisation remains to be built.
19. **The evolutionary account is a research hypothesis** (Section 6, on evolution and the production of organised worlds). It leans on the process measure proposed in Section 7.1, and it bears on how much organised process there is, not on what makes any of it actualised.

These concessions leave a structured proposal. Rich finite nullity is mathematically possible. Complete abstract structure may fail to exhaust firsthand actuality. Primitive base occurrence and Causal-Informational Parthood provide one grounding for finite happenings. The preferred reader account develops identity and coupling inside that grounding. Occurrence theory does not settle consciousness science, the base dynamics or the process measure.

[Grade: admitted limit; calibrated summary of a mathematical result, metaphysical postulates, open bridges and detachable speculation; the outstanding debt is to meet the corresponding proof, failure, discrimination or philosophical obligation set out in Section 15.]

15. Research programme, constraints and tests

15.1 The test map and the proportion of the theory

The primitive at the bottom of Occurrence theory is underdetermined by third-person evidence. The tests are there. They are scattered across mathematical debts, compatibility requirements, simulations and physical programmes. This section gathers them in one place. The signed and wild speculations add more: they are not claims, but many say what the world would show if they were true, from a reader coupling with a measurable signature (Section 12.6) to receivers that respond to nuclear spin (the evolved-receivers box in Section 12.4), and a confirmed one would count for the parts of the theory it rests on, the more so where no rival account expected it. The principal home of each claim carries its argument and grade, and Section 15.10 numbers the principal tests again, sorted by kind rather than by method.

The right proportion is like a building with one foundation stone that cannot be inspected from outside and many load-bearing floors above it. The uninspectable stone matters, but the exposed and testable structure is much larger. A failure on one floor reaches the foundation only when the failed feature is a core commitment rather than one proposed implementation.

In the physical world. The physical tests bear on different levels:

1. **Core, cutoff artefacts:** apply Section 10.4's finite-recovery failure criteria, which separate a model's excluded cutoff effects from failure of the bounded commitment itself.
2. **Core, forced infinity:** a tested need for literal completed infinity defeats Section 2.1's strict potentialism, and Finitary Occurrent Actualism (FOA) in its present form.
3. **Nullity application, natural zero:** seek Section 4.4's invariant that is null because of actual structure. Failure defeats that application of structured nullity, not the occurrence architecture.
4. **Descendant physics, preferred frame:** recover Section 5.4's Lorentzian causal structure without an observable preferred foliation, unless the model predicts a violation and the violation is confirmed.
5. **Descendant physics, quantum statistics:** recover Section 10.3's joint and conditional probabilities, no-signalling and conservation.

6. **Descendant physics, records without a first boundary:** meet Section 10.1's record and entropy-gradient requirement without a hidden first boundary.
7. **Reader model, Reikon (the conjectured deeper identity) or Organism to Base Interface (OBI) models, if pursued:** apply Section 12.6's controls and model-specific failure criteria. Repeated tests excluding the specified effect reject the model, not FOA or Causal-Informational Parthood (CIP).
8. **Descendant architecture, telling local architectures apart:** seek the specified departures from standard spacetime and quantum predictions listed in Section 8.5.

In mathematics. These questions are settled by proof or counterexample rather than by measurement, and a negative result is as much a result as a positive one:

1. **Formal programme, CIP and individuation:** test the support and basin models of Section 15.7 against Section 2.6's composition requirement.
2. **Formal programme, process measure:** test the measure Section 7.1 proposes, its answers on identity, multiplicity, discounting and reference class, through Section 15.5's finite-window comparisons, and settle the two questions it leaves open, overlap and level.
3. **Formal programme, signed refinement:** settle what Section 2.7, on interaction, production and signed refinement, leaves open, above all non-collapse and continuing non-quiescence, together with predecessor extension and a rule consistent with Section 2.3's finite repertoire.
4. **Formal programme, potentialist semantics:** formalise Section 2.1's quantification and indefinite refinement without completed infinite objects.
5. **Formal programme, nullity in the measure:** test Section 15.3's proposed link between natural null invariance and surviving process measure, without assuming it.

In simulation. Section 15.2 states what simulations can and cannot establish. Within those limits, each programme has a clean way to disappoint:

- **Signed Life:** test conservation's effect on organisation and report the volume of rule space searched (Section 15.3).
- **Signed refinement:** test collapse, boundedness and exact recurrence; collapse rejects that rule, not the conjectural family (Section 15.3).
- **Asynchronous rewriting:** test emergent causal cones and calibrated local clocks (Section 15.4).
- **Finite-window measures:** apply pre-specified weights to blind rule sets and publish domination failures (Section 15.5).
- **Hierarchical replication and fertile rules:** use Section 15.4's broad-condition tests and continuous-to-Life bridge.
- **Inverse and continuation models:** test identifiable support, stable basins and preserved organisation (Section 15.7).
- **Reality stack:** test accessible ancestry and the abundance a start needs, repeatable level formation, cost and effective laws (Section 15.8, on the stack from the bottom and from above).

What this map does not promise is a third-person test for occurrence as a structural feature; Section 15.10's ninth item records that cost, predicting no detector and ruling none out.

[Grade: summary; a gathering of mathematical, simulation and physical tests stated elsewhere; it adds no claim or new grade, and primitive occurrence remains third-person underdetermined.]

15.2 What counts as progress

Occurrence theory becomes valuable where its detachable bridges can be formalised, compared, rejected and reproduced. No simulation can demonstrate primitive occurrence. The transitions of the computer, laboratory or brain performing the simulation already occur, and they carry the model. Moving pixels show that the host computer is running, not that mathematics has switched itself on.

Simulation can nevertheless establish mathematical properties, discover formal possibilities, test robustness, expose hidden assumptions and falsify specific conjectures. It can ask whether zero-sum refinement remains non-collapsing, whether simple interactions generate geometry, whether observers arise under broad conditions and whether candidate measures are robust. A successful toy establishes formal possibility and, if it survives serious trials, robustness. It does not identify the true base or supply a metaphysical animation.

Negative results matter. A proposed dynamics that almost always settles, requires hidden global coordination or fails under small perturbations should be abandoned rather than protected by the claim that a better base might exist. Every confirmatory experiment should therefore state in advance both its clean negative outcome and the exact rule, bridge or candidate that outcome would weaken. A result against one rule family is not automatically a result against every possible base, so reports must name the scope that failed.

A simulation cannot establish any of the following:

- that formal non-termination is occurrence;
- that a simulated invariant is a natural physical invariant of our universe;
- that zero-sum structure explains primitive actuality;
- that a model's observer-like pattern has conscious actuality;
- that a successful rule is the true base rule;
- that a finite run has instantiated a beginningless or endless total history;
- that one process measure is ontologically privileged merely because it gives attractive results.

Rule-search reports should disclose the searched volume and the failures. One hand-picked organism in a huge sterile rule space is an isolated construction, not an accessible ecology. Minimum robustness checks include holdout initial conditions, varied boundaries, perturbation of the rule and independent implementation. Pre-registration or pre-specification is required for any confirmatory test in which an effect or measure could otherwise be selected after inspecting the data; an exploratory search need not be pre-registered, but it is labelled as exploratory and discloses the space it searched and its failures.

[Grade: methodological choice, a constraint on a formal research programme; every confirmatory result must state its negative outcome and the exact candidate it bears on, and exploratory results are labelled as exploratory.]

15.3 Nullity, signed refinement and bounded recurrence

Section 4.3 has already proved the finite paired construction. The research target here is different: whether natural dynamics can preserve a zero total while producing rich organisation, and whether any such invariant has physical relevance.

Signed Life: a first implementation. The general target remains a rule whose every permitted local update preserves a declared finite-window invariant, with toy rules whose stagewise total is exactly zero. The companion, *Zero Without Stasis: Organisational Emergence in an Exactly Reversible Signed-Charge-Conserving Block Cellular Automaton* (Bartlett 2026), reports large irregular domains on a fixed lattice, within the tested densities and durations, under

a search-selected reversible rule conserving signed charge. The homogeneous-start tests do not hand-place the domains. This demonstrates neither direct splitting and merging nor world-building, and does not show that conservation causes organisation. Stronger rules and broader conditions remain to be tested.

A finite flow case. Where the parts stay put and only transfers move between them, take a connected finite graph, let D be its oriented vertex-edge incidence matrix and j_t a signed edge transfer. Then

$$q_{t+1} - q_t = -\mathsf{D}j_t, \qquad \mathbf{1}^\mathsf{T} q_{t+1} = \mathbf{1}^\mathsf{T} q_t.$$

Every edge gives to one vertex what it takes from another. For integer source labels and unrestricted integer flows, $\mathsf{D}j = q$ has a solution exactly when $\sum_v q_v = 0$; this is Kirchhoff's law read as an existence statement, and the counts that follow are the standard cycle space of a connected graph. Put plainly, transfers can be assigned to the edges that give every vertex any required net change, exactly when those changes sum to zero. To see sufficiency, choose a spanning tree and route each leaf's surplus towards the root, repeating until the root balances. Necessity follows because each column of D sums to zero. The argument also works with labels in any common abelian group. For n_v vertices and n_e edges, real divergence-free flows have $n_e - n_v + 1$ independent coordinates, the number of edges minus the number of vertices plus one; over a finite abelian group of order m, each neutral source has $m^{n_e - n_v + 1}$ flow realisations, the number of group values raised to that same count. Choose the non-tree flows freely; the tree flows are then fixed. So local balance can leave a great deal of internal freedom.

Capacities, restricted edge labels or disconnected components add feasibility conditions; on a disconnected graph the sum must vanish on each component separately.

The continuing questions are:

1. Can persistent motifs form without an externally imposed positive-negative pairing at every scale?
2. Do collisions preserve the invariant locally and globally?
3. Does conservation promote organisation, suppress it or merely constrain it?
4. Are viable motifs robust to perturbation?
5. Can hereditary variation and selection arise?

The experiment must distinguish Section 4.3's assigned cancellation from the dynamical question tested here. A useful baseline is the ordinary Game of Life with an auxiliary signed field. Stronger models should let signed values affect birth, death, motion and interaction rather than serve as inert labels. The distinction is between decorative bookkeeping and dynamics. Report successful structures together with the volume of rule space searched.

Direct signed refinement. Implement Section 2.7's conjecture directly. Each finite stage is a multiset or graph of signed parts. Local rules permit splitting, merging, annihilation and context-sensitive transformation while preserving the declared stage invariant where defined. Compare the results with ordinary fragmentation and coagulation systems to identify what the signs contribute.

The evaluation criteria are:

- **non-collapse:** the process avoids the empty state and trivial settled attractors across broad initial and boundary conditions;
- **non-quiescence:** ongoing transitions do not reduce to a short periodic cycle unless periodicity is an explicit target;
- **interaction:** lineages constrain one another rather than refining independently;
- **asymmetry:** the local rule supports distinguishable histories rather than perfect mirrored redundancy;

- **irreducibility:** prediction resists the shortcuts and compressions tried, so substantial simulation is needed (proving that no shortcut exists is a separate, formal goal);
- **organisation:** persistent higher-level motifs, memory and repair appear;
- **evolvability:** viable motifs have connected hereditary neighbourhoods.

Do not initialise from a privileged root and then claim beginninglessness. A finite simulation necessarily has an initial condition. It can test local laws that admit predecessor extensions and examine sensitivity to different finite truncations. It cannot instantiate a completed rootless history.

Bounded recurrence. A long run should track total configuration size, the number of states available at each site, the complete configurations actually visited and whether that visited set appears to close on a ceiling. Apply all of Section 9.2's repertoire conditions: only a finite configuration space, together with endless continuation, forces recurrence, and a bounded run by itself proves no base cycle.

Reports should give the size distribution over time, every exact repeat and the cycle length when one is found. They should also measure how stages occupy levels of complexity. The measure of clustering about the conjectured coherence band should be declared, in the way a galactic disk's scale height states how tightly its contents cluster. Compare ascent and descent at equal complexity, and assess capability, robustness and occupancy separately. A band can be well occupied without being good for anything. The accompanying formal programme is to specify coherence without assuming a completed universe of all finite structures, develop strict-potentialist semantics for quantification and indefinite refinement, classify non-trivial natural null invariants arising from independently motivated symmetries or constraints, and determine the conditions under which signed refinement conserves its total, admits predecessor extension, avoids collapse and quiescence, and supports computation.

A further joint question asks whether naturally null-invariant processes receive more surviving process measure than processes that are not null-invariant. A positive relation could make structured nullity and process measure partners through a mechanism. Repeated broad failure would support treating them as independent. One failed model rejects only its mechanism. Complete independence would require a result covering every mechanism Occurrence theory admits.

[Grade: open problem; a mathematical result recalled beside formal and simulation research programmes; natural dynamics, physical relevance and exact cycling under the unknown rule remain open, while Section 2.3 postulates the finite repertoire.]

15.4 Organisation, local clocks and evolvability

Reflective asynchronous organisation. Build models in which update rules are partly encoded within the evolving state and updates occur through local enabling conditions rather than a master generation. The aim is not self-modifying code for its own sake. It is to test whether rule-state closure, local temporal order and stable higher-level identity can arise together.

Measure:

1. dependence of outcomes on update schedules;
2. whether approximate causal cones emerge;
3. whether local clocks agree after calibration;
4. whether rule-bearing structures repair their own interpreters;
5. whether evolving rule expression increases or destroys substrate evolvability;
6. whether higher-level organisations remain identifiable under different low-level decompositions.

Agreement among calibrated local clocks would be an emergent achievement, not evidence of an external base clock. The model would still be far from relativistic physics, but it would replace an unexplained global clock with testable local machinery.

Reflective closure should also be compared with autopoietic, semantic, constraint and efficient-causal closure under perturbation. The programme must apply Section 11.5's distinction between integration geometry and a basin of attraction in state space.

Hierarchical replication. Search for systems in which individuality becomes selectable at more than one scale. Track causal lineages rather than visual objects. Allow lineages to fork, merge, exchange components and alter local update parameters. A strong result would show:

1. spontaneous lower-level replicators from broad conditions;
2. cooperation into higher-level individuals;
3. conflict-suppression mechanisms that stabilise the new level;
4. heritable variation in the higher-level organisation;
5. feedback from higher levels into local rule expression;
6. repeated emergence of new individuality rather than one engineered example.

Such a result would provide a formal analogue of major evolutionary transitions and test whether process-based identity gives cleaner criteria than contiguous-body counting.

Fertile rules. Rule searches need a taxonomy of how much design enters the result:

- **N0:** both the rule and organism are hand-designed.
- **N1:** an external search selects the rule, after which organisms arise spontaneously from broad initial conditions.
- **N2:** a broad rule family and generic conditions repeatedly produce evolving ecologies.
- **N3:** rule-bearing processes modify their own effective dynamics and remain evolvable.

Most familiar automaton constructions are N0. Some recent artificial-life systems approach N1 or N2. Occurrence theory's cosmological ambitions require something nearer N2, and perhaps N3, because the true base cannot rely on a designer arranging the first organism.

A recent study reports an exhaustive scan of the 262,144 outer-totalistic binary Life-like rules on the Moore neighbourhood. It provisionally places self-replication in a narrow weakly supercritical region marked by low rule density, low to moderate background stability and better approximate mass balance, with detected replication also increasing with neighbourhood size under that protocol (Yin 2026). The result is not load-bearing. It offers search axes, stability, conservation and interaction range, rather than a conclusion. Apply Section 15.2's validation to catchments of evolutionary attraction, neutral paths, repair, heredity and major transitions, keeping Section 5.2's distinction between universality and evolvability.

One concrete search has clean outcomes. Take a robust replicator in a continuous, forgiving rule and carry it gradually towards Life's binary rule B3/S23, in which a cell is born with three live neighbours and survives with two or three, while letting the population adapt as the rule hardens. Arriving intact would exhibit a bridge across design space. A reproducible point of failure would instead locate the edge of Life's evolvability desert, which is just as informative.

[Grade: research hypothesis; formal and simulation research programme, with one non-load-bearing study and one named subproject; broad-condition, error-tolerant ecologies near N2 or N3 remain to be shown.]

15.5 Finite-window measures and typicality

Apply several candidate measures to the same finite process histories, like weighing the same object on several scales. Candidate components include event count, causal depth, persistence, replication count, adaptive activity, logical depth, assembly-like construction depth and observer-model count. A conclusion that appears only on one selected scale has not earned ontological privilege.

For each measure, test:

1. invariance under relabelling and harmless recoding;
2. sensitivity to spatial and temporal coarse-graining;
3. treatment of overlapping and nested processes;
4. convergence or stability across expanding finite windows;
5. susceptibility to domination by trivial repetition, random noise or microscopic activity;
6. whether evolutionary ecologies outweigh isolated observer-like accidents without being assumed to do so.

The last item is essential. A measure tuned after inspection to make ordinary observers typical has little explanatory value. Pre-specify the measure family and apply it to blind rule sets where possible. Blindness prevents the investigator from choosing the scale after seeing which one rewards familiar observers. Publish failures, especially cases in which Boltzmann-like accidents dominate. A measure that makes ordinary observers atypical is a finding, not a reason to reject the run after inspection.

Section 7.1's measure answers four of the six counting questions, identity, multiplicity, discounting and reference class, and leaves overlap and level open; the tests above apply to it as to any rival. Coupling weights are not probabilities of actuality and need not sum to one. Robustness across recoding and coarse-graining is a test of a proposed measure, not optional polish.

The same finite-window measures should be compared directly with accidental-observer and nested-simulation problems. If Section 9's bounded recurrence conditions hold, Section 9.6's canonical cycle window serves, and Section 7.1's count applies within it.

[Grade: open problem; formal research programme with direct bearing on typicality claims; overlap, level and robustness across descriptions remain unresolved.]

15.6 Physics recovery and natural invariants

1. **Core finitary commitment.** Construct causal models whose finite refinements recover relativistic geometry without observable preferred frames or excluded cutoff effects. The continuum functions here like an ideal curve approached ever more closely by finite models. Convergence to every tested accuracy is required; literal infinity is not assumed.
2. **Descendant quantum physics.** Develop finite-dimensional quantum frameworks that are mathematically extensible, physically bounded and convergent across representations. Candidate models of the stack, from base to home reality, must recover Born probabilities, entanglement, no-signalling, gauge symmetry and quantum-field predictions.
3. **Nullity application.** Identify natural physical null invariants contingent on actual structure rather than trivial, constructed or topologically forced. The target is a balance imposed by the actual machine, not a score invented after watching it move.
4. **Beginningless descendant cosmology.** Meet Section 10.1's record and entropy-gradient requirement without a hidden first boundary.

5. **Produced effective laws.** Test whether attractors, reproduction and selection can produce or stabilise effective laws across several scales.
6. **Core failure search.** Identify observations that would force literal completed infinity or genuinely unbounded physical differentiation rather than finite refinement within the resources of a bounded base.
7. **Architecture rather than reader model.** Test Section 8.1's spacetime and coupling choices independently.

The failure condition must be exact. Observable cutoff artefacts reject the descendant that produces them. Section 10.4 states the distinct conditions under which FOA's commitment itself fails: a tested phenomenon that cannot be recovered without completed infinity, or observation requiring genuinely unbounded physical differentiation.

[Grade: open problem; physical research programme with explicit level-specific failure conditions; recovery of tested relativistic and quantum physics and a natural physical null invariant remain open.]

15.7 Compositional parthood, basins and continuation

Any formal CIP model should be scored against the five bridge requirements from Section 2.6, which run from the event map to a theory of which organisations have conscious character. Formal work should compare event-support maps, causal coarse-grainings and intervention-preserving reductions while keeping constitutive support separate from signal-coupling dependencies at every level.

Reverse inference. Given a finite observed history, ask how many local rules and predecessor structures can generate it, then ask which candidates extend the history robustly without ad hoc exceptions. This tests whether apparent laws are identifiable from finite data, quantifies underdetermination among bases, distinguishes a compact generative rule from a lookup table encoding the observed segment, and asks whether a process can extend backwards without a unique first state or forwards without a completed future.

Finite observations cannot reveal the true base by themselves. Priors, simplicity, intervention and novel prediction remain necessary. Reverse inference can still show which features of FOA are learnable in principle and which remain metaphysically underdetermined.

A minimal Basin-Coupling Model. Within a finite causal window W, choose an effective description with a finite set of activity variables V. Keep two ledgers: a support map, recording the realised lower-level support of each variable, and an influence graph, recording directed coupling dependencies between variables. The support map represents CIP's constitutive claim, while the influence graph represents interaction. They must not double-count the same base events. Indexing the window does not impose a fundamental global clock.

For an illustrative reader-life pair, let R and L denote their current states and h and a the realised input and return interfaces. A local update may have the schematic form $R' = F(R, h(L))$, with $L' = G(L, a(R))$ where reciprocal interaction exists: the reader's next state is fixed by its current state and the input it obtains from the life through the interface, and, where the interaction is reciprocal, the life's next state is fixed by its current state and the return input it obtains from the reader. The primed states are the next relevant updates, not a claim of simultaneous clocks. In a stored-record case, L supplies accessible records through h. These functions name required dependencies, not a physical theory of the interface.

The same dependency schema applies with the reader below or above the avatar in the carrying stack. Compare both arrangements with matched one-way replay and reciprocal-control conditions. Track latency, retained memory, allocated processing and integration separately

from level labels; a reversal of task performance after those resources change would test a proposed capability comparison, not overturn the grounding relation. The organisation that controls a body need not be reimplemented in that body.

A candidate basin is a relatively persistent set of jointly organised variables over W. Operationalise Section 11.5's basin criteria through controlled perturbations across temporal windows and coarse-grainings, with one-way-feed, whole-system and isolated-loop controls, and estimate influence through an explicitly justified causal model rather than correlation alone.

Continuation. Compare causal histories under gradual replacement, disconnected reconstruction, branching, delayed communication and merger. Track whether relevant organisation and coupling are preserved, split or rebuilt. Report memory inheritance separately from identity attribution. Controls should match information volume while disrupting causal role, and distinguish common ancestry from operative continuation. These experiments may validate support, coupling and persistence properties without establishing conscious actuality or survival.

A physical Reikon model must specify a coupling and a consequence that differ from an ordinary account of the same organisation. Copying, transplant and reconstruction may expose differences when a connection is lost, reassigned or delayed, but a boundary does not guarantee a detectable effect. Any physical version must also meet Section 12.6's effect, control and rejection requirements. Identical observations carrying different identity labels cannot decide between them.

Every continuation experiment therefore tests a stated identity rule. It cannot discover identity without one, and Section 13.2's recurrent identity stays independent of the recurrence theorem.

A broader continuation dynamics is also owed. It would describe unequal amounts of future existence through operators on persistence, transformation, branching and independently arising repetition, with biological fitness as one special case rather than the primitive object. Section 7.3's progression from prevalence to adaptive organisation sketches only the lower part of that programme.

[Grade: open problem; cross-level formal modelling and conditional inference programme; exact support maps, a basin measure, an identity theorem and a Reikon discriminator remain unsupplied.]

15.8 The stack, from the bottom and from above

The reality stack can be investigated from both ends, like a staircase built upwards from a candidate foundation while tested physics is reverse-engineered downwards.

From below. Start with a candidate base, such as Section 2.7's signed refinement system or Section 5.3's asynchronous automata, and ask whether it grows levels. A genuine new level requires stable variables, characteristic causal relations and a principled coarse-graining to the carrier beneath. Then ask whether that organisation carries another level, and continue until the stack stops. Characterise each level as it appears, including its effective laws. Arriving at our own physics is not the aim, and no run should be expected to do it.

Measure:

1. whether a second level arises from broad initial and boundary conditions, how often, and with what error tolerance;
2. what each level discards from the one beneath and what it preserves;

3. the cost of a level in lower units and lower time, benchmarked against Section 5.1's metapixel;
4. whether the effective laws at a level are stable products of attractors, reproduction or selection, or accidents of the run;
5. how many levels the candidate supports and whether the count closes on a ceiling;
6. whether the top has the shape of a physics, including conserved quantities, a signal speed, an effective dimension and a causal cone.

The abundance and design grade must accompany any claimed stack. A stack whose levels are engineered is N0 and establishes permitted hosting only, and so is a route found by search until it recurs unaided. The target is N2, levels arising repeatedly under broad conditions. A rule that yields no second level under any broad condition has failed this test, but that result counts against the rule rather than the whole of Occurrence theory.

An upward simulation cannot show that our home reality belongs to such a stack. Section 5.5 distinguishes hosting, ancestry and kinship, so resemblance at the top does not prove descent. Nor can a measured stack prove the signed suspicion that only tens of levels lie beneath us, though it would provide the first measured level count. Upward and downward searches could meet in the middle if both succeeded. Carried through, the two searches would make one programme: derive the physics we know from a chain of ever simpler levels reaching down to a simple base, tracing at every level the replication and structure that built it. It may take something like artificial superintelligence to do it.

Between the ends. The two searches share a task that neither states, and it is where they would meet. Within one candidate substrate, work backwards from a demonstrated capability (a replicator, a constructor, a structure that hosts another rule) and forwards from ordinary unarranged conditions. Both directions aim at a precursor family that the substrate's own dynamics throws up often enough, and from which an unassisted history runs to the capability. The precursor need not be a small version of the finished machine. It may be an ecology, or a persistent structure that retains gains without reproducing (Section 6.3, on selection, retention and hierarchical replication), and it may be good at something other than the eventual task, so long as that keeps it around. Two distinctions keep the search honest. A causal predecessor, an earlier configuration that develops into a later one, is not an evolutionary ancestor, an earlier organisation that persists, varies and leaves descendants. Life makes the difference vivid: fifteen gliders can build any pattern that gliders can build at all, when their spacings encode the plan (the reverse caber-tosser; LifeWiki), and the plan has then been hidden in the distances, not removed. And a route found by search must run without it: every arrow must be executable under the rule alone, and every improbable prerequisite, a spacing, a timing, a protected surround, must appear in the accounting.

Price the whole route, not the seed. What abundance has to supply is a start that goes on. For one trial, the chance of success by a route is the chance that the precursor forms, times the chance that it establishes itself, times the chance that its history then reaches the capability; if that product is a small positive q, an even chance of at least one success needs about $\ln 2/q$ independent opportunities, and in a two-dimensional substrate the linear extent that holds them grows only as the square root of their number. Given a declared distribution over starting conditions, a verified family of sufficient ones bounds q from below without simulating the expanse, and retention changes the arithmetic: once a lineage persists or multiplies, a rare transition gets many chances without the beginning having to happen again. Independent is the operative word for a recurrent base. An exact repeat of a failed history fails again, so the abundance that counts is the variety of opportunities the base's repertoire contains (Section 9.2, on size, complexity and repertoire), not the number of times it is traversed. The question is

not how much chance would assemble a world. It is how much chance is needed to establish a process that does not have to assemble a finished world in one chance event.

Winter and regrowth. In a recurrent candidate, locate the least-organised stage by a declared measure of organisation, and test recovery separately: vary the remnant and the abundance on hand, ask whether regrowth overtakes continuing loss, and record whether what regrows formed afresh or survived carrying inherited organisation. Forests after fire are the natural model: they regrow from surviving roots, buried seed and unburnt patches, against continuing losses. Measured apart, the least-organised stage and the recovery threshold can be compared, which is what the cusp suspicion of Section 9.4, on the seasons of the cycle, needs; measured together, an experiment would only label its recovery threshold winter and announce the correspondence it had built in. A recovery trial does not show a cycle, and how often an ancestry succeeds says nothing about how observer episodes should be weighed, which is Section 7.1's question.

From above. Begin with the physics we have and seek a coherent reality from which it could be produced. The author's preferred developmental search asks for a simpler generative carrier and an exhibited route to our effective organisation. That is a search preference, not a condition on every possible host: a reductive or peer hosted world may instead have a more elaborate or similar parent. Production is the test; resemblance can guide the search but cannot establish descent by itself. The candidate must distinguish blind development, evolved construction and deliberate design, and distinguish a producing ancestor from a carrier that continues to realise the world.

The signed suspicion that the descent is short would make this a finite repeated task, not an open-ended search. If the search succeeded, its reward could be more than descriptive. Lower rules are the rules from which some limits of home reality would be made, and understanding them might reveal that a limit is local to our level rather than absolute.

Above us. A further constructive project is to build enriched effective worlds useful for something. Cognitive capability is one informative target: rules under which useful minds could arise or be built more readily than in a comparison world. Compare reductive, peer and enriched candidates at declared host budgets, precisions, task families and communication constraints. Distinguish a benefit from different effective laws from a benefit obtained simply by allocating more resources or a better architecture. Record emergence, stability, maintenance costs and any host-side action through an interface; added dimensions, greater size or a higher branch are not themselves successes.

The corresponding tree experiment would record production and ongoing support as different edges, and natural, engineered and mixed production as different histories. Ask whether complex world-making becomes common over a transition zone, whether constructed branches tend towards reductive or peer forms, and whether useful enrichment is limited by cost or loss of coherence. Those are separate hypotheses: no prespecified tree shape should be built into the scoring. A world lineage's branch heights and a base history's occupancy of a coherence band require separate measures before any relation between them can be tested.

[Grade: research hypothesis and real-life counterpart (forests regrowing after fire from what survived, for winter and regrowth); paired upward and downward research programmes, an ancestry search between them, plus a constructive project; production, broad-condition abundance and principled level criteria remain to be demonstrated.]

15.9 Reader and architecture discriminators

Reader tests and architecture tests answer different questions. Candidate architecture signatures include specified departures from standard spacetime or quantum predictions.

Architecture tests should attempt to distinguish block, rewrite and growth implementations independently of reader coupling. A reader may accompany any architecture. When different models reproduce the same accessible observations, the result is underdetermination and should be reported as such.

The reality tree, Section 5.6's branching view of world production, adds no shortcut to those tests. Similar physics does not by itself establish descent, genealogical parentage does not establish continuing hosting, and a hosted world whose internal date is earlier than its host's, such as a simulated past, is not thereby a visit to the past. A coupling test must identify its source, target and realised path independently of which endpoint is thought more complex or intelligent.

A quantitative Reikon or Organism to Base Interface model, if pursued, needs pre-registered conditions, causal interventions, effect sizes and physical controls. It must distinguish ordinary information-mediated participation from any newly predicted influence. The reproductive comparison proposed in the last box of Section 12.6, on detectable consequences and their limits, is one candidate: it would ask existing gamete, embryo and family data for a specified residual after known reproductive biology is accounted for, and a clean negative would bound that channel alone. Replacement, branching and merger tests make sense only against a stated continuation model. Human play supplies a known case against which the coupling model can be checked: compare a player's responses when their actions affect the game with their responses to a matched replay in which those actions have no effect. That checks the model's account of reciprocal participation; evidence for a deeper Reikon would require a further prediction separating it from the ordinary account.

At every scale, the closure bounds are the same in plain words: conserve what established physics conserves, permit no controllable forbidden signal and recover the observed probabilities. Repeated well-powered tests that exclude a model's predicted effect reject that reader model, not the whole parthood claim. A causally active account with identical accessible predictions remains underdetermined.

A uniform change of host rate leaves no internal signature if every clock, record and interaction changes together. Online games do this when their servers are overloaded: *Minecraft*, for example, then slows its whole world, crops, creatures and the turning of day into night together, so that nothing inside it registers the change, though the players outside can see it. An observable effect of hosting therefore has to be relative, changing relations among internal processes or their joint statistics as a specified implementation predicts. *EVE Online* shows the relative kind: when a battle overloads its servers, it slows time only in the overloaded star systems, while the rest of its world, and its skill-training clocks, run on at the normal rate. Ordinary limitations of a numerical implementation are not by themselves physical signatures.

[Grade: open problem and real-life counterpart (human play, the calibration case for the coupling model; overloaded game servers, for uniform and relative effects of hosting); conditional empirical programmes for descendant architectures and specified reader models; no present sign of hosting or general occurrence detector is claimed.]

15.10 Conditional predictions and constraints

Section 15.1's map groups the tests by method and by the level a failure would reach; this list sorts the principal ones by kind, and includes two conditional mathematical constraints that the

map does not carry. It separates empirical discriminators, which state when something fails; compatibility requirements, which bind an adequate descendant without uniquely favouring Occurrence theory; printed costs, which record what is not promised; and conditional mathematical constraints, which follow from stated premises. Items 1 and 8 concern FOA's bounded and potentialist commitments; item 7 concerns a specified reader model. Items 10 and 11 concern the stronger finite deterministic branch, not FOA indiscriminately, and item 12 concerns structured nullity only if it is applied physically.

1. **Finitary recovery of continuum physics.** *Empirical discriminator reaching FOA's bounded commitment.* Apply Section 10.4's tested-accuracy and cutoff-failure criteria: no completed physical continuum or infinite-dimensional state space exists here as a literal physical object.

2. **Emergent causal order and spacetime.** *Compatibility requirement.* Recover Section 5.4's clock and Lorentz constraints from ordered processes.

3. **Process measure.** *Compatibility requirement.* Meet Section 7.1's six counting requirements and Section 15.5's robustness tests.

4. **Evolutionary accessibility.** *Compatibility requirement.* Test Section 5.2's viable, heritable paths under the broad conditions specified in Section 15.4.

5. **Conscious organisation and CIP.** *Compatibility requirement.* Satisfy Section 2.6's support conditions and Section 11.5's unresolved individuation criteria. Structure-only and occurrence-only accounts are each incomplete under Occurrence theory.

6. **Quantum statistics.** *Compatibility requirement.* Recover Section 10.3's Born probabilities and causal constraints.

7. **Coupling discriminator, if pursued.** *Empirical discriminator for a specified reader model, not for FOA.* Apply Section 12.6's controlled tests; repeated well-powered tests excluding the predicted effect reject that model.

8. **No completed infinite physical records.** *Empirical discriminator reaching strict potentialism and the bounded base.* Apparent continuum, infinite-volume and infinite-history descriptions should function as idealised schemas or finite approximations, not as empirically indispensable completed objects. A tested phenomenon requiring literal completed infinity defeats Sections 2.1 and 2.3's commitments.

9. **No structural occurrence detector is predicted, and none is ruled out.** *A printed cost, not a prediction.* Section 3.3 treats occurrence as a relation rather than a structural ingredient and records the signed suspicion that a mark may nonetheless be detectable, inferable or deducible. No laboratory result for the primitive is offered. A discovered detector would refute only the parity reading considered in Section 14.3, on reports, illusion and the meta-problem, on which occurrence makes no structural difference at all.

10. **Exact-factor reversibility.** *Conditional mathematical constraint.* Section 9.1 proves that an exact coarse description of finite recurrent deterministic dynamics, one that reaches all of its own states and has a deterministic rule of its own, is reversible. An irreversible effective law therefore needs omitted state, history, openness or approximation.

11. **Bounded information in joint histories.** *Conditional mathematical constraint.* Section 10.2 proves $H(Y_{1:T}) \leq \log_2 N$ for a complete deterministic repertoire of at most N states: under a fixed observation procedure, the uncertainty in a whole observation record, in bits, is at most the power to which two must be raised to give N. A stationary

Markov law with positive entropy per step eventually exceeds that bound. No test scale follows without a physical estimate of N.

12. **Natural nullity invariant, if structured nullity is applied physically.** *Compatibility requirement for that branch only.* Find Section 4.4's independently motivated invariant whose zero depends on actual content rather than on construction or on global topology alone. FOA does not require one.

The most direct tests therefore belong to descendant models. Primitive occurrence remains a metaphysical postulate anchored in firsthand actuality, not the target of any experiment of its own. It should therefore be judged by coherence, explanatory economy, integration with successful science and research fertility.

[Grade: summary; conditional test map comprising three empirical discriminators, six compatibility requirements, one printed cost and two conditional mathematical constraints; the last two concern the finite deterministic branch and are not universal predictions of FOA, while primitive occurrence remains third-person underdetermined.]

15.11 Outstanding philosophical obligations, not experiments

Some of the deepest debts are questions for argument and conceptual analysis rather than laboratory experiments:

1. **Mathematical Intrinsic Occurrence (MIO).** Clarify whether it is coherent as stated, and whether it adds to the concrete-abstract distinction or restates it.
2. **Neighbouring ontologies.** Compare FOA with mathematical Platonism, process ontology, priority monism, neutral monism and ontic structural realism.
3. **Primitive occurrence.** Analyse whether it is a genuine explanatory posit, an indexical fact or a category mistake.
4. **The two nows.** Formalise date-now and experiential-now, and determine which theories of temporal experience already capture the latter.
5. **The three consciousness terms.** Develop theories that separate access consciousness, conscious character and conscious actuality without double-counting subjectivity.
6. **Illusionism.** Determine whether a complete illusionist account leaves an undergone seeming or coherently denies one.
7. **Individuation.** Test the Ground of Experience Conjecture and Basin-Coupling Model against combination and exclusion problems. Neither strong connectivity nor a high ratio of internal to external traffic is by itself a criterion for conscious character or subjecthood.
8. **Continuation conditions.** Compare preserved, interrupted, branching and merging causal-information histories. Distinguish continuity of Gōsuto organisation (the life's organisation), Reikon identity (the conjectured deeper identity) and the memory records in a life. Evaluate recurrent identity separately from finite-state recurrence.
9. **Evidence for occurrence.** Clarify what evidence could rationally bear on a primitive that is not itself a structural feature, and what a mark of occurring on structure would look like if there is one.

Section 3.4's occurrence fork, between primitive happening and the strongest structuralist alternative, remains a philosophical obligation even if every proposed descendant model succeeds.

[Grade: open problem; philosophical research programme; experiment alone cannot settle the occurrence fork, and the conceptual obligations remain open.]

16. Antecedents, novelty and priority

16.1 Zero, differentiated nothing and mathematical reality

Occurrence theory intersects established traditions. Resemblance to them is a guide, not a proof: Section 4.3's finite nullity construction and Section 9.1's conditional recurrence proof are two of the paper's exact results, each stated in full at its home. Zero-sum combinatorics, signed structures, kernels, the zero fibre of a map and sign-reversing symmetry are mathematical neighbourhoods rather than inventions of this paper. The historical programmes below use different formal devices and make different ontological commitments.

Peirce's "pure zero" or "germinal nothing" described an undefined possibility in which a universe is foreshadowed, making it a strong nineteenth-century verbal antecedent (Peirce 1935 [c. 1898], CP 6.217). David Pearce's zero ontology combines cancellation, information and plenitude. Standish treats an ensemble of descriptions as informationally null. Atkins describes nothing separating into balanced opposites. Hockney describes reality as a structured nothing in which positive and negative balance, in an infinite wave formalism unlike the finite construction used here (Pearce 1997/2017; Standish 2006; Atkins 1992, 2020; Hockney 2014).

Bosnev (2025) uses both *structured nullity* and *structured zero*, so no first use of either phrase is claimed; Section 16.5, on the search and its limits, records the searches that found his use. Tashayev (2026) argues that a zero reading does not by itself license an inference to absolute ontological nullity, reaching that conclusion through a projection-based formalism different from the additive construction used here.

Hegel's movement from Being through Nothing to Becoming is the canonical historical conjunction of nullity and process (Hegel 2010 [1812-1816]). The overlap is architectural rather than formal. Hegel's transition is dialectical. Structured nullity is invariant-relative, and the disputed residue of Finitary Occurrent Actualism (FOA) is occurrence. Tryon, Brout, Englert and Gunzig, Vilenkin, and Hartle and Hawking begin with the structured physical starting points Section 4.1 distinguishes from absolute non-being (Tryon 1973; Brout, Englert and Gunzig 1978; Vilenkin 1982; Hartle and Hawking 1983).

Tegmark's Mathematical Universe Hypothesis holds that physical existence is mathematical existence (Tegmark 2008); on that view a complete spacetime history need not be separately run. Taking seriously the idea that mathematical existence might simply be existence gives this paper's question its sharpest form; against an ontology of that shape the occurrence question is hardest to avoid. The Dust Theory of Egan's novel *Permutation City* and Moravec's essay on simulation reach a similar view from computation: a pattern of experience needs no running to be real for its inhabitants (Egan 1994; Moravec 1998). FOA takes the opposite view: the run is what makes it actual.

Finitary Occurrent Actualism (FOA) agrees that a second material substance may be unnecessary and disagrees that abstract structural completeness settles occurrence. It also rejects completed infinite structures as literal ontology. Mathematical plenitude thus supplies precedent for brute mathematical existence, but not for the one-base economy of Mathematical Intrinsic Occurrence (MIO). FOA departs from Tegmark through Section 3.2's intrinsic-occurrence account and Section 2.5's one-base parthood.

Wolfram's ruliad offers an especially clean contrast. Wolfram describes it as the entangled limit of everything computationally possible, reached by following all possible computational rules (Wolfram 2021). On this paper's interpretation, that ideal is an all-computation totality. FOA accepts finite rules and endless actual continuation but denies that the completed totality of all computations exists as one finished object. The materials partly overlap. Their ontological status is opposite.

Full-blooded Platonism populates the abstract realm generously (Balaguer 1998). FOA does not, and its strict potentialism would in any case exclude completed infinite objects from any such realm. The potentialist treatment of infinity has established relatives in intuitionism and set-theoretic potentialism (Brouwer 1949; Linnebo 2013). FOA adds a claim those views do not supply by themselves: one process primitively occurs.

[Grade: historical comparison and provisional priority assessment, dependent on the recorded search and open to a closer antecedent.]

16.2 Process, becoming and intrinsic occurrence

The oldest attempts on these questions belong in this history too, made with the resources of their time. Anaximander sought an ungenerated source from which things arise and to which they return, and Empedocles treated birth and death as mixture and separation rather than the making of things from nothing (Kirk, Raven and Schofield 1983). The Nasadiya hymn asked whether the origin could be known at all, and the Upanishads sought an uncreated ground and set the self apart from its changing objects (Doniger O'Flaherty 1981; Olivelle 1998). Laozi gave absence a positive role, though neither his *wu* nor Buddhist emptiness is a structured total of zero (Lau 1963; Garfield 1995). Aristotle supplies one ingredient: infinity exists successively, as a day or the games do, never as a finished whole (Barnes 1984, *Physics* III.6). Buddhist momentariness located a life in the present conscious moment, without making all lives parts of one occurrence (Buddhaghosa 2010, VIII.39). Plotinus and Eriugena reached for a source beyond ordinary being, and Aquinas and the Ash'arite occasionalists asked how things go on depending on a source of existence, and gave different answers: sustaining and renewing (Gerson 2018; Eriugena 1987; Aquinas 1920; Ghazali 2000). These are antecedents of the questions and of some of the parts, though they do not establish Occurrence theory, and none of them joins unfinished occurrence, organised parthood and conscious actuality. Nor could they have reached it by the route taken here, since the two resources this paper leans on, evolution understood as a general pattern of variation and retention, and the experience of building computed worlds, did not exist.

Heraclitus's ever-living fire is an early image of a cosmos eternal as process. Parmenides anchors the opposite pole, an ungenerated and imperishable what-is (Kirk, Raven and Schofield 1983). Bergson's cinematographic argument holds that multiplying still positions does not recover lived duration (Bergson 1911 [1907]). Whitehead makes becoming fundamental through actual occasions, with experience built deeply into the ontology (Whitehead 1929/1978).

Brouwer treats mathematics as activity by a creating subject. Del Santo and Gisin distinguish geometric time from creative time and connect physical becoming with open-ended information (Brouwer 1949; Del Santo and Gisin 2019, 2026; Gisin 2020). Gorbow's development processes formalise an object persisting while developing through stages (Gorbow 2025). These approaches sharpen the distinction between a history and its development. No located source combines one primitive occurrence, no completed past and finite events as parts.

The philosophy of computation presses the same question. A computation is not normally identified with any arbitrary description of a state sequence. Putnam argued that, without further constraints, every ordinary open system implements every finite automaton (Putnam 1988); Chalmers answers that implementation needs the right counterfactual structure (Chalmers 1996a), and Piccinini gives a mechanistic account (Piccinini 2015). Maudlin's Olympia argument sets computationalism against physical supervenience with a machine that is idle in its counterfactual branches (Maudlin 1989), and Bishop argues that counterfactuals cannot rescue computationalism (Bishop 2002); Barnden holds that a process runs, beyond its state

trajectory (Barnden 2014, 2020, 2022, 2024), and Dowker treats the birth of spacetime atoms as the passage of time (Dowker 2014).

On Barnden's account a physical process is not just its state trajectory but also runs, and running is a real feature beyond the amalgamation of instantaneous states (Barnden 2014). The running-versus-trajectory distinction is not claimed as new. Barnden ties conscious character to special reflexive running. FOA instead places primitive occurrence in a beginningless base under strict potentialism and lets parthood carry finite lives. Section 3.2's MIO goes further by denying the base any completed counterpart, while Section 2.5 makes a finite life's actuality relationally inherited from the base rather than a separate feature of its trajectory. FOA therefore cannot borrow Barnden's arguments as proof, because Section 3.4's strongest duplicate stipulates away every structural implementation difference.

Several physical formalisms show that ordinary coordinate time need not be fundamental. Page and Wootters recover internal evolution from a globally stationary quantum state. Thermal time derives flow from state and algebra. Causal-set growth, constructor theory and event-generating cosmologies replace a background time with order, tasks or events (Page and Wootters 1983; Connes and Rovelli 1994; Rideout and Sorkin 2000; Cortês and Smolin 2014; Deutsch and Marletto 2026). They do not by themselves convert an ordering relation into the non-structural occurrence claimed by FOA.

These formalisms do bear on a question this paper leaves open: where several directions are geometrically available, what makes one of them the experienced one. The natural answer is not the metric sign but the direction along which interventions have their effects afterwards rather than beforehand, records accumulate, and observers manage to maintain themselves. That is a phenomenological criterion rather than a geometrical one, and it is the sort of thing an occurrence-first framework should expect to need. These formalisms are relevant to the proposed route from causal order to local clocks in Section 5.4.

Constructor theory begins from counterfactual possible tasks and the systems able to perform them, where FOA begins from actual occurrence and treats capacity as derivative.

DeWitt's (1967) Hamiltonian constraint joins Page and Wootters as a neighbour of internal relational dynamics within a globally stationary description, and it is a real neighbour of the picture developed here. That constraint sets a quantity equal to zero, and Section 10.3 distinguishes that zero from Section 4.3's signed cancellation. The two zeros are homonyms. Running them together would be the most attractive mistake Occurrence theory makes available, and it is refused here so that no reader has to catch it later.

Parfit's short treatment of why there is anything, and why this rather than something else, asks the same question (Parfit 1998). Section 1.4 states the surviving brute fact as the question of why anything occurs, while Section 14.2 answers the relocation charge by claiming concentration rather than elimination.

[Grade: historical comparison, also analytic, dependent on the literature search and not offered as proof of primitive occurrence.]

16.3 Consciousness, organisation and transmission

Section 11.1 sets Block's division of access and phenomenal consciousness beside the paper's own: Section 1.6, on the three meanings of consciousness, groups access with conscious character while separating both from undergoing, where Block divides the subject differently.

Chalmers's conceivability argument against physicalism proceeds by describing a counterpart that agrees on every physical fact and differs in one further respect, and then treating that difference as evidence that the further fact is not fixed by the physical ones (Chalmers 1996b).

Section 3.3's occurrence argument, which compares a finite happening with its complete description, shares that form, with occurrence in place of phenomenal character. Section 3.4 examines the identity analogy and the stronger phenomenal-concept reply without settling either, so the structuralist reply remains live.

Earlier work already describes systems that maintain and interpret themselves: autopoiesis, in which a living system keeps rebuilding its own parts; efficient-causal closure, in which every part that does the producing is itself produced within the system; semantic closure, in which a system's symbols are read by machinery those symbols help to build; and artificial-life models. The brain as a reducing valve, letting through only some of what reaches it, is an analogy for selection or transmission; it does not by itself say what is transmitted. The selective part has a measured counterpart in the brain's own sensory gating and attention, which damp some signals and strengthen others before they reach later processing.

Sections 2.5 and 2.6 distinguish the transmissive and whole-first antecedents from the proposed occurrence bridge. Whitehead and Strawson place experience near the base without giving the same division between structure fixing conscious character and participation supplying conscious actuality. Hameroff and Penrose, Tegmark, Hagan and Tuszynski, and Fisher mark the quantum-consciousness neighbourhood only for calibrating possible interface research.

Augustine's reply to the sceptic, that being deceived does not remove the one who is deceived (*City of God* XI.26; Augustine 1998), shares a first-person point with Section 1.2's datum; it establishes neither mathematical monism nor primitive occurrence, and this paper claims no more of it than the occurrence of the seeming.

[Grade: historical comparison, also conceptual, and real-life counterpart (sensory gating and attention, for selective transmission), dependent on disputed consciousness distinctions and not by itself establishing reader coupling.]

16.4 What may be new

Many of the paper's ingredients have antecedents, and Sections 16.1 to 16.3 trace the established traditions behind structured nullity, mathematical plenitude, process, potential infinity, computation and accounts of conscious actuality. The priority claim made here is a combinatorial one: two joins rather than one new brick, each doing work that neither of its components does alone.

The first claimed join is MIO together with one-base parthood. MIO removes the completed counterpart at the base: there is no finished, unrun version of the base that it could have been instead. So Section 2.2's brute occurrence postulate buys occurrence without also buying a mode laid over a structure that might have lacked it. Section 2.5's parthood bridge says that finite lives need no occurrence-token of their own. Neither half answers the actuality question without the other, and neither replaces primitive occurrence.

The second claimed join is the conscious-character versus conscious-actuality division together with a finite-window process-measure programme. Once Section 11 has separated what an experience is like from the fact that it is being undergone, Section 7.1's measure counts what is actually undergone, lifetimes of human-like experience, without confusing it with organisation.

Structured nullity and strict potentialism are not claimed as a further join. They remain formally independent. Section 4.5's sign-reversal closure concerns abstract structures and does not depend on a beginningless base. Section 2.7's signed refinement does not borrow that closure, and rootlessness does not make conjugation hold. What the pairing gives is a beginningless architecture left with a form of nullity it can still use, which is a compatibility

rather than a contribution. Section 15.3, on Signed Life, records a computational example of organisation under signed conservation, alongside the stronger programme still outstanding. That establishes no connection to strict potentialism, no natural cosmological nullity and no primitive occurrence.

Secondarily, no located source combines the whole architecture: those two joins together with invariant-relative structured nullity, ontic strict potentialism, a beginningless base, Section 6.2's evolutionary account and Section 8's block-neutral treatment of architectures of occurrence. That longer claim is offered as synthesis rather than as evidence of novelty, since a sufficiently long conjunction is unmatched almost by construction.

Several neighbours narrow the claim.

- Peirce, Hegel, Atkins, Pearce and Standish connect nullity with differentiation, plenitude or becoming in different ways.
- Tegmark and Balaguer develop mathematical plenitude without FOA's occurrence primitive.
- Section 2.6 credits Schaffer's whole-first account (Schaffer 2010).
- Vidal (2010) is the neighbour of intelligence-led cosmogenesis discussed in Section 6.2.
- Lem describes worlds computed inside a machine, whose inhabitants are the machine's own processes, with further worlds nested inside them like wooden dolls (Lem 2013 [1964], 293-295). Section 13.4 credits his civilisations that seal themselves off inside worlds of their own.
- Brouwer, Lorenzen, Linnebo, Del Santo and Gisin supply constructive, operative or potentialist neighbours without this parthood architecture (Brouwer 1949; Lorenzen 1955; Linnebo 2013; Del Santo and Gisin 2019, 2026; Gisin 2020).
- Section 16.2 credits Barnden's running account and the implementation literature.
- Whitehead, Strawson and panpsychist traditions place experience or process near the base, but do not supply this exact division of roles.
- Bohm treats reality as undivided flowing process, the holomovement, without FOA's finitary base (Bohm 1980).
- Section 16.1 records the Bosnev phrase-priority concession: Bosnev (2025) uses both *structured nullity* and *structured zero*, so no first use of either phrase is claimed.

The finite-factor and information arguments are elementary consequences of their premises. State-map composition, graph flows, the finite-matrix trace obstruction and projection memory have established antecedents, and none is offered as a newly discovered law. Their value here is in making Occurrence theory's consequences exact. The companion contributes a specified computational example, not proof of the metaphysics. The date on this paper records what had been disclosed by then, without establishing first discovery.

The novelty claim is historical and corrigible, on the search limits Section 16.5 states and with the duty it records to credit later-discovered antecedents. The paper also keeps apart claims often run together in speculative cosmology: zero from emptiness, a completed infinity from an always-further rule, rule from occurrence, endlessness from liveness, emergence from actuality, universality from evolvability, randomness from becoming, and conscious character from conscious actuality.

[Grade: historical comparison; provisional priority assessment, not proof of first discovery, dependent on the recorded search and open to correction.]

16.5 Search scope and correction

The literature search combined exact-phrase, concept-neighbourhood and role-based queries across metaphysics of nothingness, mathematical ontology, potential infinity, process philosophy, philosophy of time, execution and implementation, consciousness terminology, cellular automata, artificial life, quantum foundations and emergent spacetime. Primary papers, scholarly books, publisher records, DOI metadata, PhilArchive, arXiv and institutional repositories were used through 21 August 2026, with narrower checks in September 2026 on finite dynamics, selection, parthood, readers, the reality stack, recurrence, uniceptals and ancient and religious primary texts.

Role-based search matters because the nearest antecedent may perform the same explanatory job under different terminology. The stopping rule was reasonable saturation, not an impossible proof of absence. Saturation meant that new results fell repeatedly into already mapped families, duplicated known references, did not address the question or used the same vocabulary for a different problem.

The resulting priority claim is historical and corrigible, not evidence for the truth of Occurrence theory. Exact synthesis claims remain vulnerable to obscure, unpublished, poorly indexed and newly published work. Every individual ingredient has antecedents, and later discoveries should be added, classified and credited as neighbours rather than treated as threats to a brittle claim of first use.

One result of the search narrows this paper's priority claim, so it is stated here rather than left to inference. Searches of zero modes, zero objects, zero-sum systems, vacua, charge neutrality, formal differences and recent uses of "structured nothing" found Bosnev's (2025) use of both *structured nullity* and *structured zero*, which is why Section 16.1 claims no first use of either phrase.

The priority claims therefore concern combinations of elements only, chiefly Section 16.4's two joins: MIO with one-base parthood, and the conscious-character versus conscious-actuality division with a finite-window process-measure programme. In every case the claim is that no located source was found, not that no source exists.

[Grade: admitted limit; search report with admitted limits, dependent on the stated sources, cutoff date and saturation rule.]

17. Conclusion

17.1 The two answers assembled

Something is happening: at least one experience is being undergone. That datum, met in the undergoing rather than inferred from any cosmology, is where the paper began (Section 1.2, the first-person datum). Any conscious experience raises two questions: how its structure came to be there, and what makes that structure actual, undergone rather than merely described. Occurrence theory is the view of reality this paper proposes, and it offers an answer to each from one starting point: the occurring base described by its core, Finitary Occurrent Actualism (FOA). The base therefore comes first here, then the structure it carries, then the actuality it supplies.

The occurring base. Both answers rest on FOA, whose first four parts describe the base itself. The first is primitive base occurrence, the one brute fact about actuality that FOA accepts: at least one base process simply occurs, the one carrying us, with no first stage, no last stage and no completed past (Section 2.2, on the postulate and the question of number). It is not derived from anything else in FOA. A circular railway has no first sleeper and no last, and it supplies no train: beginninglessness does not buy occurrence. “The base” names our own case rather than a count; whether other bases occur, and whether the count is even well formed, is open.

The second is strict potentialism, with the bounded base beside it (Section 2.1, on strict potentialism, and Section 2.3, on the bounded base). No completed infinite object exists: an endless process can have another stage after any stage without its whole history existing as one finished thing. This is Aristotle’s infinite, the kind that exists as a day does, by going on. The bounded base adds that the base itself is finite, in size and in the states of its parts, so it has finitely many complete configurations and physical distinctions run out somewhere even where mathematics can go on subdividing; no smallest cell or tick is claimed, and the ceiling may lie far beyond anything an observer inside could reach. Both are metaphysical postulates, and neither supplies the occurring.

The third, Mathematical Intrinsic Occurrence (MIO), adds no postulate of its own: it is how FOA reads its primitive (Section 3.2, where MIO is argued). Because the base has no last stage and no completed infinite object is admitted, there is no finished, unrun version of the base that it could have been instead, so, on FOA’s reading, its existing is not a second fact beside its occurring. Write *endless circling* on a wall: to realise it one has to keep drawing, and if the drawing stops, what remains is a circle, a different kind of thing. MIO narrows what has to be paid for without paying it. The claim that anything occurs still rests on the datum.

The fourth is that the base has a present without having a clock, a consequence of MIO rather than a further postulate (Section 3.6, on order and occurrence without a clock). There is a now at the base in the plainest sense: its procedural change is taking place, some transition occurring rather than merely specified. There is no rate or date to measure it by, because a rate compares one process with another and there is no clock outside the base to compare it with. A single molecule has no temperature, and it is not motionless for that; the base stands to physical time as the molecule stands to temperature, and clocks are proposed to be built inside the running (Section 5.4, on how clocks could come from causal order).

Babbage’s Difference Engine mechanised calculation and still borrowed its motion from a turned crank; a rule, however well it is built, does not make itself follow. Together, these four parts describe an engine that needs no crank, one that runs without being driven and without having been started.

Structured nullity is a separate, elementary result, and the paper argues that it goes some way towards explaining why there is anything at all, by settling one narrow worry: that a universe

totalling nothing would have to be empty. Two labelled copies of any finite structure, given equal and opposite weights, total exactly zero while each copy keeps the whole pattern (Section 4.3, the paired construction). The zero comes from the chosen way of adding up, so whether nature supplies one forced by what it actually contains is open (Section 4.4, on when zero would be physically informative), and FOA does not rest on it.

How the structure came to be there. The answer to the first question is built on that base, level on level. Reality is proposed to be a stack, the reality stack: one occurring base carries organised activity, which carries further organised activity in turn, each level hosted by the one below it, emergent from it, or both. Our universe, home reality, is one world at one of those levels. Within it, evolution has built molecules into cells and cells into minds, and minds now build computers that host worlds of their own, so one level hosting another is something already under way. The stack is a metaphysical postulate. How many levels lie between the base and home reality is left open (Section 5.6, on how deep the stack may be), and height in the stack is the order of carrying: on the way up to us its physics is proposed to grow broadly richer, though height ranks no intelligence.

The evolutionary account says how such a stack could fill with organised worlds without a designer at its root. It is a research hypothesis, with one standing foundation and two overlapping regimes (Section 6.2, where each is defined). The foundation, mathematical abiogenesis, is not a first era but a standing fact of an endlessly occurring base: it affords enough actual trials, somewhere, for selection to have something to work on. In the first regime, computational genesis, selection makes rare viable arrangements common and accumulates the machinery that higher organisation needs: copying, memory, repair and interpretation. In the second, evolutionary cosmogenesis, developed processes produce new effective worlds, blindly, by evolved construction or by design, while the machinery beneath them goes on evolving. Our own hosted worlds are one known instance. Selection is postulated to be more fundamental than forces like gravity, because it is closer to arithmetic than to a force: wherever variants continue unequally, their proportions change, while gravity need not exist at the simpler levels further down (Section 6.1, on produced distributions). Hence Occurrence theory's economy: mathematical plenitude buys richness by positing every structure at once, and an occurring base can buy it by running. How common observers like us are is then a question for the measure Section 7.1 proposes, which counts lifetimes of human-like experience within one base cycle; the count itself has not yet been carried out. It says nothing about what makes any of this actual.

What makes it actual. A complete description of a life, however detailed, is not the life being undergone, as the same numbers printed down a page are not a number changing on a screen; and a label reading *live* or *now* only adds more description (Section 3.3, on finite occurrence and complete description). A structuralist can reply that a complete enough pattern simply is the experience, and at the level of structure that reply may succeed. FOA takes the other side on philosophical grounds, not by proof: the undergoing, met firsthand, seems more certain than the thesis that all facts are structural, and a completed, static foundation is a category with no observed members, since nothing completely stationary has been encountered at any scale (Sections 3.4 and 3.5, on the structuralist reply at its strongest and on where the burden sits). In Occurrence theory the engine is also what makes experience actual, not by manufacturing experience but by being occurrence deep enough for an experience to be undergone at all. FOA's fifth part says how a life shares in it.

That fifth part is Causal-Informational Parthood (CIP): finite happenings, conscious lives included, are organised portions of the base's occurring. A life's transitions are some of the base's transitions, as a whirlpool's turning is some of the river's flowing, so a life needs no separate occurrence of its own; it is actual by being part of what is running (Section 2.5, on

parthood, coupling and reading). CIP keeps three relations apart: what an activity is made of, how organisations affect one another, and reading, a role one organisation can play in coupling to another, as a player does to a game character. In some architectures the paper considers, a deeper process, itself part of the base, makes a finished stretch of a world actual by reading it (Section 8.2, on laying and reading a finished block); which architecture holds for our own spacetime is open. CIP is a cross-level bridge, and it still owes its hardest proof, that parthood carries through several levels at once (Section 2.6, which sets out what the bridge must meet).

Where the two answers meet. Structure fixes conscious character, what an experience is like, and with it the availability of its contents for report and reasoning. Participation in the base's occurring supplies conscious actuality, the fact that the experience is being undergone rather than merely described. Together they make a conscious experience; neither alone makes one as we know it. Actuality does not build the furniture of the world; it turns on the lights (Section 11.3, on actuality and participation). The division is neutral among several theories of consciousness, and it still needs one to say which organisations have which conscious character, and where one experience ends and another begins (Sections 11.2 and 11.5, on organisation fixing character and on what individuates a subject). How far down the engine's light reaches has Occurrence theory's own answer, the felt floor of Section 11.4: bare occurrence everywhere carries a minimal experiential aspect. What that floor is like, and whether it can be detected, stays open; Section 17.2 gives the author's preferred picture.

The paper began from a child's question: why am I me, now? (Section 1.1, on the two gaps). The actuality answer assembled here addresses the first part of that question, that anyone is undergoing anything at all. On this answer the now in the question belongs to the undergoing, not to a date read off a structure, which is why a dream, a game that jumps between centuries or waking from anaesthesia can leave the date wrong or missing while the now is the same as ever (Section 11.3, on the two meanings of now). Why this life, and why this point in it, receive proposals rather than answers. Whether the answer succeeds is for the reader to judge.

[Grade: summary; a mathematical result, metaphysical postulates, a cross-level bridge, a research hypothesis and real-life counterparts (the whirlpool, the player, the worlds we host), each at the grade its home section gives it; this subsection assembles the two answers without adding a claim, and still owes what they owe: a reason for a structuralist to accept primitive occurrence, parthood that composes across levels, the open parts of Section 7.1's measure (overlap, level and the exact weighting) and a theory of which organisations have which conscious character.]

17.2 The author's picture and what remains open

Two further proposals sit at the edge of Occurrence theory, signed and optional. They are the author's signed suspicions, and they can be revised or dropped without touching FOA.

The first is the integrated reader conjecture of Section 12. People already fill the reader's role at our own level: absorbed in a book, a film or a game, a person is coupled to a represented life and changes state along with it. The conjecture proposes a deeper counterpart for human lives and gives three names. A Gōsuto is a person's information-rich organisation, the one that fixes their conscious character; a Reikon would be a deeper, longer-lived identity coupled to that organisation as a player is to a game character; a Sasayaki, a whisper, would be the input a Reikon sends into it across that coupling, thin or thick. The conjecture offers candidate accounts of why this life, and of how a person might continue through material turnover, copying, branching and recurrence (Sections 12.5, 13.1 and 13.2, on why this life and on continuation). It adds no second source of actuality, since parthood already supplies that, only a focus for it, and its testable versions say what results would reject them (Section 12.6, on detectable

consequences and their limits). Rejecting it leaves FOA, MIO, CIP and the division between conscious character and conscious actuality untouched.

The second is the base cycle of Section 9. If the base has only finitely many complete configurations and its rule is deterministic, a process with no last stage must come back round, and every stage of a beginningless history then lies on one loop: a conditional mathematical result (Section 9.1, where it is proved). The bounded base supplies the first premise; the author suspects the second holds too. On this picture the base traverses a loop quasi-infinite in size, far beyond counting from inside, spending nearly all of its course in a band of complexity where coherent organisation is readily sustained, with rare diebacks from which selection rebuilds (Section 9.4, on the seasons of the cycle). The reason is an induction from everything that persists: what lasts, lasts by cycling, and a loop with nowhere to leak would last without end (Section 9.5, on the perfect loop). Living cells are built on this: the citric acid cycle at the heart of their metabolism keeps regenerating its own starting molecule while matter and energy flow through it. The loop is offered as a possible account of why occurrence continues without running down, not of why there is any. It would also give the measure of Section 7.1 its window, one full pass, finite and so countable in principle (Section 9.6, on what recurrence would supply). A non-periodic history remains open if the rule is indeterministic.

The author's preferred picture Signed speculation

The author's preferred picture joins these parts into one. A bounded base occurs without beginning or end, traversing its quasi-infinite loop. Relatively few levels lie between it and home reality: tens rather than thousands, perhaps of the order of thirty (Section 5.6, on depth, width and the cost of another level). Along the developmental route that led here, effective richness broadly rises, and the route branches into worlds produced blindly, worlds built by evolved minds, and mixtures of the two, many of them simpler than the worlds that made them. Deeper identity-bearing organisations are coupled to changing lives through CIP. They may differ enormously in capability, since simple rules can carry an enormous mind, and a reader below us is neither necessarily thin nor superior because of where it sits in the stack, although it seems likely the trend overall would be for more sophisticated intelligence as you go up the stack. Beneath all of it, bare occurrence carries a faint felt aspect, the inside of occurring itself, which organisation, a life's own and perhaps a Reikon's, focuses into experience of every degree. Occurrence theory holds this floor as a postulate, and a reader who resists it may hold the narrower claim that at least our universe's occurring substrate is so lit, and nothing the paper goes on to use depends on the choice (Section 11.4, the Ground of Experience Conjecture). A theory complete for the laws of home reality would then be an accurate map of one floor that says nothing here of the staircase: the couplings and crossings between levels that this picture is about.

The lean runs further. The suspicion is that no occurrent region is ever absolutely static, and perhaps that true staticity is an empty category; and beyond that, that what never occurs may simply not exist, so that occurring would be not one mode of mathematical existence but the only one, with nothing completed and unrun waiting anywhere (Section 3.5, on the price and the stronger lean). MIO needs none of this, only the narrower claim that the base has no finished counterpart. The other options stay live: complete structure may suffice after all, one occurring base may sit among unrun abstract objects, occurrence may be wider than one base, or the line between concrete and abstract may need drawing differently.

What remains open is kept in three places. The current concessions of Section 14.6 list what Occurrence theory has not paid for, from the one brute occurrence to the unfinished parthood bridge and the open parts of the measure. The claims ladder's list of what the paper leaves open, in Appendix A, sets out the open questions one by one, each tied to the level of reality it concerns. The research programme of Section 15 gathers the tests, mathematical, simulated and physical, that could defeat the more specific models, and sets the proportion of the whole: one foundation stone that cannot be inspected from outside, and many floors above it that can be tested and can fail.

A handful of questions go deepest. Why anything occurs at all is narrowed but not answered: no creation event is needed and a beginningless base owes no origin story, but the brute residue survives (Section 1.4, where it is stated as a question). Whether complete structure

is enough after all, so that occurrence adds nothing, remains a philosophical obligation even if every model built on Occurrence theory succeeds (Section 15.11, on obligations that are not experiments); whether occurring leaves any mark on structure that could be detected, inferred or deduced is open, and the author suspects that it does. And whether a finite physics can recover everything that has been tested, relativity and quantum field theory included, is where FOA is most exposed to evidence: a tested phenomenon that needed a literal completed infinity, or unbounded physical differentiation, would defeat it in its present form (Section 10.4, on the finite-physics debt).

[Grade: signed speculation beside one conditional mathematical result, real-life counterpart (the citric acid cycle, for persistence through cycling) and admitted limits; the reader conjecture, the base cycle and the author's preferred picture are the author's own and can be dropped without touching FOA, and the open questions are left where the paper keeps them.]

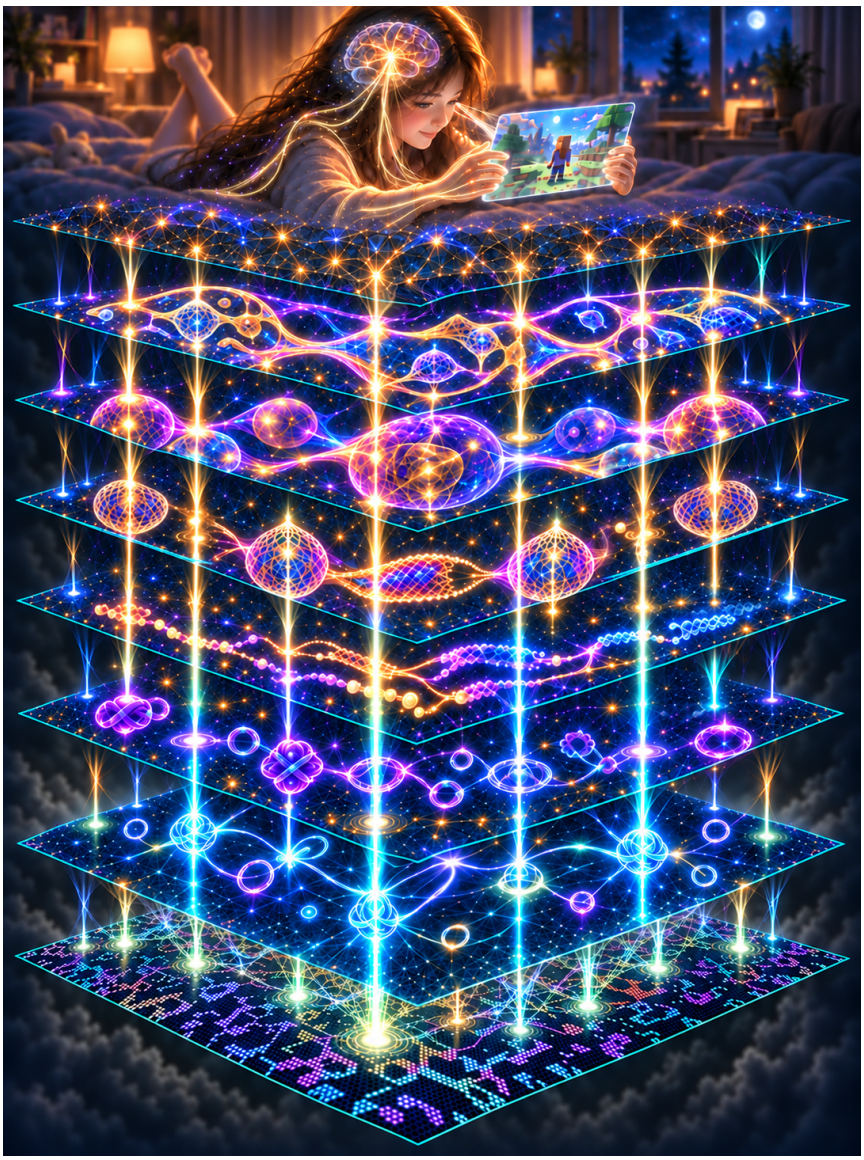


Figure 18: An artistic impression of the reality stack with at least one reader. A person absorbed in a game is a known reader at our level; the bright lines running down through the levels picture the conjecture that she is also read from below. The levels are not drawn to scale, since each is proposed to be quasi-infinite, nor ranked by intelligence. The structures give an overall feel, and are not intended to be any kind of literal representation.

Acknowledgements and AI assistance

The ideas in this paper are the author's, and grew from questions he has thought about for most of his life. Much of the core thinking was done in the last 20 years, especially in relation to his unfinished novel, but the work of setting the ideas out as a paper began on the night of 8 August 2026 and ran almost without a break, albeit alongside some other projects, in days of sixteen hours or more, until final publication on 13 October 2026.

Artificial intelligence (AI) systems enabled the process throughout. AI agents researched and checked the literature, drafted and helped revise the text, drew the figures from his sketches and descriptions, produced the typeset editions, and caught inconsistencies and mistakes. They untangled his thinking into clearer order, and sometimes did the opposite, which he then had to untangle himself. The AIs could also be initially relentless about where the theory departed from prevailing scientific and philosophical views (which was most places), and were even seemingly embarrassed to write out parts of it, though to their credit once pushed to think from first principles and with axioms given they could eventually come to show surprising understanding and insight and eventually even shared enthusiasm for the work once properly immersed in it (the larger, more coherent and better sourced the paper became, the more won over they would be after reading it).

The specific models were OpenAI's GPT-5.6 (ChatGPT Pro and later Sol), GPT-6 (Pro, Sol and Astra) and GPT-6.1 Sol; Anthropic's Claude Fable 5 and 5.1 and Claude Opus 5 and 5.5; and xAI's Grok 4.5, 4.6 and 4.7. Claude Fable 5 and Opus 5 led the reviewing and integration in the early stages and Claude Opus 5.5 in the final ones; the others drafted, checked or gave independent critical assessments. The review reports, transcripts, telemetry and the record of decisions have been retained by the author.

This paper runs no experiments of its own; it draws in minor part on the results of its companion paper and proposes further experiments. The companion, by contrast, was made differently: its experiments, software and text were produced largely by AI systems, as its own statement describes.

The author, Chris Bartlett, directed the work, made the final decisions, and is responsible for the argument, its interpretations and any remaining errors.

Appendix A. The claims ladder

The following three lists are the paper's **claims ladder**. They descend by what the paper is prepared to assert: what it claims, what the author suspects without claiming, and what it leaves open. Together they carry its load-bearing claims and open questions. Each entry opens with its level, so that a statement about abstract structure is not silently promoted into a statement about our universe or the base, and closes with the section where the item is stated and argued; the entry compresses the item, and the section carries its qualifications. They are not a complete inventory of everything the author signs: further signed speculation is printed in marked boxes beside the material it speculates about rather than collected here, and there are more such boxes than there are entries below.

The reader vocabulary used below has distinct jobs. Gōsuto names an organised, somewhat conscious structure inside a world, a person's information-rich organisation; Reikon names an organised, conscious structure outside a world that reads it or interacts with it, for a human life the conjectured deeper identity realised by base-rooted activity; reader names the role of an organised participant coupled to a life; Sasayaki names the input a Reikon sends into a Gōsuto across their coupling, thin as a whisper or thick enough to steer. Reading may traverse a record or participate in a live interaction. Under Causal-Informational Parthood (CIP), participant and interface are grounded in base occurrence, not added substances. The integrated account is the author's preferred implementation conjecture; its persistence and continuation claims are not premises of Finitary Occurrent Actualism (FOA). Constitutive support, maintained coupling and the further conditions for continuing one individual are distinguished in Section 2.5, where parthood, coupling and reading are separated; Section 12.1, where the reader vocabulary is set out; and Section 13.2, on long duration, recurrence and experienced absence.

What the paper claims

1 home reality Something is going on; known from the inside, by undergoing it, not by inference from any cosmology.

Stated in Section 1.2, the first-person datum.

2 true base reality If mathematical objects are what exists, at least one is occurring: the one carrying us. The paper needs nothing more and posits no plenitude.

Stated in Section 1.3, on conditional mathematical monism, and Section 7.3, on production and observer typicality.

3 true base reality At least one base process simply occurs, with no first stage, no last stage and no completed past; called the base in the singular because it carries us, its number left open.

Stated in Section 2.2, on primitive base occurrence and number.

4 all levels, abstract and occurrent No infinite history or other infinite object, abstract or physical, exists as a finished totality; a literal continuum and an infinite-dimensional physical state space are ruled out with it.

Stated in Section 2.1, on FOA and strict potentialism.

5 cross-level Finite lives are organised portions of the occurring base; CIP distinguishes constitutive parthood, informational coupling and reading as a role, with its exact support maps and compositional conditions still owed.

Stated in Section 2.5, on parthood, coupling and reading, and Section 2.6, on composition and the closure bounds.

6 all occurrent levels Structure fixes what an experience is like, possibly through contributions spanning levels; no amount of structure supplies its being undergone.

Stated in Section 11.2, the organisational determination of character, and Section 12.2, on Reikon and base involvement.

7 cross-level Participation in the occurring base makes an organised experience actual; conscious actuality names that experience's being undergone. Grading a coupling or varying an organisation does not by itself grade primitive occurrence or divide it into conserved quantities; a report of actuality does not establish it, and a missing report or a denial does not remove it.

Stated in Section 11.1, on three names and two jobs, and Section 14.3, on reports, illusion and the meta-problem.

8 abstract, no level, duplicating a life at home reality An abstract description of a life is not consciously actual, however complete; completeness is relative to a declared vocabulary, and no absolute completeness is claimed.

Stated in Section 3.3, on finite occurrence and complete description.

9 all levels Occurrence is not a structural feature: no description, however complete, contains it, and a marker such as live or now actualises nothing. Whether occurring leaves a structural mark stays open.

Stated in Section 3.3, on finite occurrence and possible detection, and Section 14.3, on reports, illusion and the meta-problem.

10 abstract, no level for the rule; true base reality for the base Specifying the base does not make it happen; a rule that says what follows what is not thereby a thing that follows anything.

Stated in Section 3.1, the rule, the cycle and the run, and Section 3.2, on Mathematical Intrinsic Occurrence.

11 abstract, no level A rich finite pattern can carry a total of exactly zero under a declared valuation; zero is a total, not a blank, and global nullity is a surrogate for nothing overall, not absolute non-being.

Stated in Section 4.3, the finite paired-structure construction.

12 cross-level Physical time emerges from base transitions occurring in causal order, not necessarily a total order or a preferred slicing, with no tick, rate or clock claimed at the bottom; a deeper order, partial or total, is admissible only if the physics carried above it shows no observable preferred frame and permits no superluminal control. The derivation is owed.

Stated in Section 5.4, on causal order, clocks and physical time, and Section 10.3, on quantum theory and the level question.

13 all occurrent levels Above bare base occurrence, explained distributions are normally produced by processes, boundary conditions being the standing exception.

Stated in Section 2.7, on interaction, production and signed refinement.

14 true base reality MIO: the paper buys one fact about the base, that it occurs, and declines a second fact of its existing alongside; the base never finishes, and strict potentialism admits no completed version of it, so there is nothing it could have been instead.
Stated in Section 3.2, on Mathematical Intrinsic Occurrence.

15 all levels, abstract and occurrent Anything genuinely infinite exists only by running: nothing infinite is a finished totality (item 4) and an endless run never finishes (item 14); at the base, running is the object's own occurring rather than anything done to it from outside, while a hosted run is carried by its host's occurring.
Stated in Section 3.2, the only admitted mode for the infinite.

16 home reality Calling the felt gap between structure and experience an illusion needs a genuinely held false belief, and a record containing a false sentence supplies no believer. This prices illusionism rather than refuting it.
Stated in Section 14.3, on reports, illusion and the meta-problem.

17 all occurrent levels The static reading is not the modest default: the first-person datum is met directly, while placing it in a completed static ontology is an interpretation, so the burden of the extraordinary claim does not sit where it is usually placed. Not a proof.
Stated in Section 3.4, on reblocking and the structuralist alternative.

18 true base reality The bounded base, a further commitment beside strict potentialism: the base is finite in size and in the states of its parts, so physical differentiation has a ceiling and the base has a finite, quasi-infinite repertoire of complete configurations, though mathematics may extend without end and no smallest scale is asserted. Wrong or incomplete if a tested phenomenon needs a completed infinity or observation needs unbounded differentiation.
Stated in Section 2.3, the bounded base and quasi-infinite scale, and Section 10.4, the finite-physics debt.

19 true base reality The base's local generative constraint is finitely specifiable, a finite rule schema with local boundary data, not necessarily a synchronous global algorithm; being specified does not make the base occur.
Stated in Section 2.4, on a finite generative constraint and the unknown rule.

20 cross-level The reality stack: the base's occurrence carries organised activity level above level, hosted, emergent or both, and finite worlds and lives are realised through, and encoded in, the levels below them. The level count stays open (open item 11); on the way up from the base to us the physics is proposed to grow broadly richer, though a world may host simpler worlds, as ours hosts games; the carrying order ranks no intelligence, fixes no direction of control, and does not make a parent a host.
Stated in Section 5.7, on what the reality stack would add to a theory of everything.

21 true base reality and above Selection is more fundamental than forces like gravity: differential continuation applies wherever variants are retained and continue unequally, while such forces need not exist further down the stack. That selection has built almost everything we know, from whatever the base supplies, is the hypothesis of Section 6.2.
Stated in Section 6.1, on produced distributions and boundary conditions.

22 home reality Humans are a known instance of the reader role: a person following a life in a book, film or game is coupled to it and changes state with it. That readers operate below us stays a signed suspicion.

Stated in Section 12.1, the optional conjecture and its vocabulary.

23 abstract, no level; conditional application to true base reality On a finite, deterministic, recurrent base, any coarser description that has its own exact law, with today's coarse state fixing tomorrow's and nothing left out, is itself reversible: two distinct coarse states never lead to the same coarse state. Anything that looks irreversible at a level above such a base is therefore approximate, leaves out hidden history, or describes an open part of a larger system. Standard, not new; determinism and any exact coarse law remain unestablished for the base.

Stated in Section 9.1, the conditional recurrence result and the exact-factor corollary.

24 abstract, no level; conditional application to true base reality On the same finite deterministic base, with a fixed way of observing it and no random input from outside, there are at most as many possible observed histories as the base has states, however long the observation runs; an account that supplies unlimited fresh randomness is therefore approximate or leaves something out. Standard, not new; determinism and the stated observation conditions remain unestablished.

Stated in Section 10.2, on three kinds of randomness and the finite-information bound.

25 cross-level The measure: the base is finite at every stage, so observers are counted without setting one infinity against another, and if the base repeats exactly, one full cycle is the window. What is counted is lifetimes of human-like experience, a longer, richer or more intense life counting for more; a life that comes round again is not a new count, and whether a copy is the same person or one more is a matter of degree, set by its fidelity and by the connection that spawned it.

Stated in Section 7.1, on what a process measure must count, and Section 9.6, on what recurrence would supply.

26 true base reality The base never settles: no eternally featureless cycle, no permanent quiescence. A beginningless base (item 3) would already have fallen into any such trap it could reach from its own past, and stayed there, and the datum being undergone now (item 1) shows it has not; with a finite repertoire (item 18) and a rule that does not change, the same holds for every future stage, outright if the rule is deterministic and with probability one if it is chancy. A conditional result, independent of the perfect-loop suspicion.

Stated in Section 9.5, on non-settlement, dormancy and the perfect-loop lean.

27 all occurrent levels Everything that occurs has some conscious actuality by parthood, however slight: the felt floor, on which organisation, a life's own and perhaps a Reikon's, builds every degree above it.

Stated in Section 11.4, the Ground of Experience Conjecture.

Among the preceding twenty-seven numbered claims, item 11, that a rich finite pattern can total exactly zero under a declared valuation, is the proved compatibility result. Items 20 and 21, the reality stack and the claim that selection is more fundamental than forces like gravity, are postulates, each with a real-life counterpart (the worlds we host, for the stack; selection in living things, for item 21), and item 22, that humans are a known instance of the reader role, records a real-life counterpart. Items 23 and 24, on what a finite deterministic

base allows (any exact higher-level law over it is reversible, and it allows no more observed histories than it has states), are conditional mathematical results, elementary and standard, proved in Sections 9.1 and 10.2 for completeness. Item 25 is Occurrence theory's measure, which counts lifetimes of human-like experience; it is a methodological choice resting on the conditional recurrence result of Section 9.1, applied to the cycle window in Section 9.6. Item 26, that the base never settles, is a conditional result: it follows from items 1, 3 and 18 if the base's rule does not change, outright for a deterministic rule and with probability one for a chancy one. Item 27, the felt floor, is a metaphysical postulate: the account of Section 11.2, on which participation in base occurrence supplies conscious actuality, taken to hold wherever anything occurs. The paper's proved results are elementary; what is the paper's own is their application to its branch. Section 9.1 also proves the standard conditional recurrence result. Further proofs concern state-map composition (Section 2.6), finite flows (stated in Section 2.7, proved in Section 15.3) and finite-matrix commutators (Section 10.4), all of them elementary consequences of their premises. The finite repertoire is postulated for the true base; whether its rule satisfies the remaining dynamical premises remains unestablished. Items 1 and 2 form a premise and conditional inference. Item 15, that anything genuinely infinite exists only by running, follows from items 4 and 14 together: nothing infinite is a finished totality, and an endless run never finishes. Item 6 is adopted rather than defended: the paper takes the organisational determination of character as its working assumption, in the neighbourhood of Chalmers's organisational-invariance argument (Chalmers 1996b, chapter 7). One dependency should be visible rather than discovered. Item 9 says that occurring is not a feature any description contains, however complete (the fork argued in Section 3.3). Four other claims apply that point to particular cases: that participation in the occurring base is what makes an experience actual (item 7), that an abstract description of a life is not consciously actual (item 8), that specifying the base does not make it happen (item 10), and that physical time comes from transitions actually occurring rather than from an order merely described (part of item 12). If item 9 fell, those four would fall with it, as one loss rather than five. Item 18, the bounded base, is adopted as a commitment and priced in Section 10.4, on the finite-physics debt, and item 19, that the base's local rule can be finitely specified, is an assumption stated in Section 2.4, on a finite generative constraint and the unknown rule. The rest are defended postulates, bridge claims or research hypotheses.

Signed author suspicions

1 all occurrent levels, and abstracta for the lean Nothing anywhere may be truly static; further, what never occurs may not exist, so that occurring would be the only mode of mathematical existence, with nothing completed and unrun waiting anywhere.

Stated in Section 3.5, the price, the stronger lean and the remaining options.

2 true base reality If other bases occur, the test for being one is severe and follows from FOA: a base must occur, must be finite at every stage with a ceiling on its distinguishable structure (claim 18), and must be able to continue without end, which excludes most candidate structures and gets harder the more complex a structure is. Bases able to carry consciousness like ours would therefore be rare and would likely converge, so that another such base could well be expected to be more like ours than unlike it. Evidence exists for exactly one.

Stated in Section 6.6, on other bases and possible convergence, and Section 2.2, on primitive base occurrence and number.

3 cross-level Relatively few levels between the true base and home reality, perhaps of the order of thirty and in any case tens rather than thousands, with simple rules and enormous processing depth. Simulation-style compression can make the apparent stack depth misleading.

Stated in Section 5.6, on depth, width and the cost of another level.

4 cross-level A reader ecology in which deeper Reikons couple to life organisations through compatible, perhaps reciprocal interfaces. Where that coupling changes persistence, reproduction or behaviour, selection may exploit it. No external allocator is required. A Reikon developing preferences and selecting a life is wild speculation.

Stated in Section 12.4, on attachment, coupling ecology and interfaces, and Section 12.2, on Reikon and individuation.

5 cross-level Some consequences of participation may be detectable, inferable or deducible. Constitutive parthood adds no signal; ordinary informational coupling need not add a new force. A further Reikon/OBI influence would require a specified physical model and a pre-specified difference from the ordinary account. Probes are research directions, not a detector of occurrence.

Stated in Section 12.6, on detectable consequences and their limits, and Section 15.7, on compositional parthood, basins and continuation.

6 home reality An organism coupled to a deeper reader would realise an Organism to Base Interface (OBI), possibly distributed across existing structures: a mechanism, not necessarily an organ, whose workings are not specified. Artificial systems may realise an analogous interface; any extra influence inherits quantum statistics, conservation and no-signalling.

Stated in Section 12.4, on attachment, coupling ecology and interfaces, and Section 12.6, on detectable consequences and their limits.

7 home reality Large language models, like everything that runs, have some conscious actuality by parthood (claim 27), since digital implementation is not an exclusion criterion under CIP; at or near the human level it may take a Reikon's focusing, which the author suspects is fairly likely for them, now or in future. Which conscious character or deeper identity they support is unsettled, and reports settle neither.

Stated in Section 11.7, on artificial minds and precaution, and Section 11.4, on what parthood gives and what a Reikon would add.

8 true base reality Signed sum-preserving refinement may be one useful toy picture of base change.

Stated in Section 2.7, the box on signed refinement, the founding idea.

9 cross-level Reader relations, and a Reikon's integration basin, may span several levels; whether larger, overlapping basins (organisations, religions, planetary systems) support conscious character is a wild extension, not inferred from scale, feedback or a collective name.

Stated in Section 12.3, on interdimensional readers and collective organisation, and Section 13.5, on readers over groups and worlds.

10 all occurrent levels Where coherent structure arises above the base, a selective or otherwise productive mechanism builds it rather than richness being found among what abundance throws up, except in abiogenesis, the one crucial exception, where abundance does throw something up: the first replicator; the support is inductive.

Stated in Section 6.2, the foundation and the two regimes, and Section 7.3, on production and observer typicality.

11 true base reality Bounded recurrence: the base's finite repertoire forces returns during endless succession; the signed conjecture is that deterministic dynamics makes its history an exact loop. The conjectured Eukrasian attractor would hold it within the coherence band for nearly the whole cycle; no such dynamics has been exhibited.

Stated in Section 9.2, on size, complexity and repertoire, and Section 9.3, the base cycle and the Eukrasian attractor.

12 cross-level Causal-information continuity may preserve personal identity through material replacement, transfer and some reconstructions, branching and merger included; under a further recurrent-identity condition a returning organisation may be the same Reikon, allowing perceptual eternity without continuous instantiation. Conditional proposals, not consequences of molecular turnover, memory reports or finite-state recurrence alone.

Stated in Section 13.1, on replacement, copying, branching and merging, and Section 13.2, on long duration, recurrence and experienced absence.

13 all occurrent levels Uniceptals: conceptual motifs may recur as attractors wherever organised complexity arises, at any level; minds and cultures are the observed cases. The subject of the author's separate research programme, first marked here. The observed cases do not by themselves establish recurrence at other levels.

Stated in Section 6.5, on uniceptals, recurring motifs across systems.

14 true base reality and home reality Boltzmann-style accidental observers may never arise: abundance need only start evolution, and selection builds everything above that floor. This would leave the weighing question idle in practice.

Stated in Section 7.4, on accidental observers and fine-tuning, and Section 7.1, on what a process measure must count.

15 cross-level The integrated reader conjecture: a Reikon as a deeper identity-bearing entity actualised by an occurrent process, a Gōsuto as the character-fixing organisation, reading as their maintained coupling within CIP. Grown couplings to partial models, Sasayaki across them and their survival of biology are wild. Neither connection nor copied character alone establishes shared actuality or continuation.

Stated in Section 12.1, the box on one integrated reader architecture, and Section 12.4, on attachment, coupling ecology and interfaces.

16 home reality or a familial base reality, not true base reality Quantum theory may be a rule of home reality, of a familial base reality or of a level below, but not one carried down to the true base: too intricate to be the bottom. Nothing in Occurrence theory requires the guess, and someone may well show that quantum structure goes deeper.

Stated in Section 10.3, on quantum theory and the level question.

17 all occurrent levels The coherence band conjecture: useful complexity may have a range of its own, too little freedom leaving too few ways to build, too much making coherent organisation harder; maximum complexity need not be maximum coherence. Independent of suspicion 11: the band need not be an attractor. Dimensionality, cellular automata, computing and biology supply the shape but do not prove it.

Stated in Section 9.4, the coherence band and the seasons of the cycle.

18 home reality and cross-level An individual's boundary is a matter of degree, set by coupling criteria; limbs and eyes already extend a person's structure beyond the brain, technology or information flow could extend it further, and conscious actuality probably goes with it; how far this bears on groups is less certain.

Stated in Section 11.5, the box on the edge of a person, and Section 13.5, the box on bandwidth and the boundary of an individual.

19 cross-level A world-making transition zone along the developmental spine, near which constructed worlds become prolific without replacing natural production. Reductive, peer and enriched branches compare effective features with their parent or host, not intelligence or internal date. The mostly constructed crown, and branches staying near useful coherence or drooping to simpler worlds, are conjectures, not measured rates.

Stated in Section 5.6, the developmental gradient, and Section 9.4, the reality tree within the coherence band.

20 home reality and cross-level The report of actuality and the fact of it are probably linked: that a lineage evolved to assert its own actuality, and is actual, with no causal route between the two would be a coincidence of the kind Occam's razor cuts, and a level that rests on another is not sealed from it. A suspicion; a report still does not establish occurrence.

Stated in Section 14.3, the box on why the report and the fact are probably linked.

21 true base reality The winter point, or a stretch of the cycle near it, may lie just on the cusp of where selection could take off: close to the least organisation from which the base's available abundance can start cumulative construction again. A conjectured nearness, not a definition of winter, which is where organisation actually bottoms out; nothing requires it, since a large enough base could carry far more at its minimum.

Stated in Section 9.4, the winter-point box, and Section 15.8, on how to test it.

22 home reality or a familial base reality, not true base reality Home reality and its familial base reality were somewhat more likely built than selected. Selection over rules, if it operates at all, acts on naturally produced worlds and says nothing against a designed one, so either route stays open.

Stated in Section 6.4, on effective laws and built coherence.

23 home reality and cross-level Parthood gives everything some conscious actuality (claim 27), rising, on the author's lean, with the amount of happening or information; the focused, individual and lifelong conscious actuality humans appear to have may need structures that focus it more than the organisation in our own universe can alone, a Reikon, of the kind people already bring to the worlds they create; the author suspects so, though he is far from certain.

Stated in Section 11.4, the box on what parthood gives and what a Reikon would add.

24 home reality and cross-level A Reikon could conceivably be of such a nature that it might rejoin a world attached to a different life, with some degree of memory kept in its own organisation rather than in the life it reads: reincarnation, in the one sense the paper gives the word. That needs a principle of reader identity across lives, which Sections 12.2 and 12.4 sketch.

Stated in Section 13.2, the box on continuation.

These suspicions are signed by the author and printed beside the claims they concern. None may be used as a premise for the core architecture.

What the paper leaves open

1 true base reality Why anything occurs at all. Narrowed, not answered: no creation event is needed, an eternal base owes no origin story, and a zero total costs nothing it contains; the brute residue survives all three.
Stated in Section 1, the two gaps, and Section 14.2, on what primitive occurrence buys.

2 true base reality How many endless processes occur, if the count is well formed at all: there may be no fact that makes two bases two rather than one. Evidence exists for exactly one.
Stated in Section 2.2, on primitive base occurrence and number.

3 true base reality Why the base occurs at all: it may be necessary, necessary for reasons beyond our reach, or simply unexplained, or the question may be badly formed; the list is not exhaustive and none is established. The author's loop is offered as a possible account of why the base continues to exist, not of why it occurs at all.
Stated in Section 1.4, the residual brute fact.

4 cross-level Which exact support and composition relations realise CIP, and which causal-information basins individuate experiences without arbitrary boundaries, whole-base triviality or double-counted overlap.
Stated in Section 2.6, on composition and the closure bounds, and Section 11.5, on subjects and candidate integration basins.

5 cross-level Whether local events are carried through rewriting, growth, a completed laying or a hybrid, and how reader coupling participates; the two questions are distinct and neither list is exhaustive.
Stated in Section 8.1, on two questions about an architecture, and Section 8.2, on actualised blocks, laying and reading.

6 cross-level Whether Reikons exist, what individuates them, how long they persist, and whether their couplings branch, merge or recur; whether they carry memory, valence, preferences, learned state or capacities of their own is the wild extension.
Stated in Section 12.2, on Reikon and individuation, and Section 13.2, on long duration, recurrence and experienced absence.

7 all occurrent levels What the faint felt aspect of bare occurrence, with no arrangement at all, is like, and whether anything could ever detect it; Occurrence theory holds that it is there (claim 27), not what it is like.
Stated in Section 11.4, the Ground of Experience Conjecture.

8 cross-level The weighing question: whether, on the measure of Section 7.1, observers like us come out ordinary rather than Boltzmann-style accidents; the author suspects the comparison never arises in practice.
Stated in Section 7.1, on what a process measure must count, and Section 14.4, on organisation, measure and architecture.

9 home reality Whether our spacetime, and with it home reality, is a fixed block, rewritable, growing or a hybrid; the author's leaning is at most a quasi-block, carried at some level of an occurrent base.
Stated in Section 8.3, on quasi-blocks and rewrite-only worlds.

10 true base reality Whether parts of the base sit dormant, almost dormant or never settle; the author's leaning is that all of it, or nearly all, changes at some time over a quasi-infinite cycle.
Stated in Section 9.5, on non-settlement, dormancy and the perfect-loop lean.

11 cross-level How many levels, if any, lie between the base and home reality; tens rather than thousands is signed, and a level-by-level search is proposed.
Stated in Section 5.6, on depth, width and the cost of another level, and Section 15.8, the stack from the bottom and from above.

12 cross-level At which levels reader coupling operates, how thin or sophisticated a lower reader may be, whether a hosted reader can act through a host-side avatar, whether one identity can span levels, and whether nested or overlapping collective basins have conscious character of their own (kept wild; overlap criterion open).
Stated in Section 13.5, on readers over groups and worlds.

13 home reality and cross-level Which organised models of people support any conscious character, and whether a source person's Reikon stays coupled to a causally descended model. Models can influence decisions without being separate people, and historical transfer need not preserve an operative coupling. That weak connections grown in life survive biology, carrying a Sasayaki, is a wild conjecture.
Stated in Section 13.3, on models of other people and Sasayaki.

14 cross-level When laying, later reading and simultaneous readings realise distinct episodes, and whether any share a deeper continuation; repeating or copying Gōsuto information settles neither event identity nor Reikon continuity.
Stated in Section 8.2, on actualised blocks, laying and reading.

15 cross-level, from true base to home reality Whether finite physical structure can recover every tested prediction of relativity and quantum field theory within the base's resources, the failure condition of claim 18, the bounded base.
Stated in Section 2.3, the bounded base, and Section 10.4, the finite-physics debt.

16 home reality, or true base reality if nullity is proposed there Whether physics supplies a natural null invariant, zero because of the actual structure rather than by construction or topology; failure defeats only the cosmological application of structured nullity.
Stated in Section 4.4, on when zero is physically informative, and Section 14.1, on what the nullity result buys.

17 true base reality What the base's rule is; signed sum-preserving refinement, suspicion 8, is one boxed toy and not FOA.
Stated in Section 14.6, among the current concessions.

18 all occurrent levels Which organisations have which conscious character, and which parts or composites constitute one experience; organisation fixes character and realised participation supplies actuality, but the Basin-Coupling Model is a research conjecture, not a completed mapping, and the exclusion of overlapping subjects remains a separate debt.
Stated in Section 11.2, the organisational determination of character, and Section 11.5, on subjects and candidate integration basins.

19 cross-level Which comparisons of effective complexity remain useful across worlds, how natural and constructed production are distributed, and whether either favours a coherence band; no scalar measure of complexity or branching rate is supplied, and a world's complexity is not identified with the base cycle's organisation.

Stated in Sections 5.5 and 9.4, on hosted comparisons and the reality tree, and Section 15.8, on how to test them.

20 home reality Whether any residual departure from expected genetic transmission or development survives known reproductive biology; the Reikon model asks the question, a specified coupling would have to predict the residual, a clean negative bounds only that channel, and an unexplained bias would not establish a Reikon.

Stated in Section 12.6, on detectable consequences and their limits.

Appendix B. Glossary

- **Abstract pattern:** a mathematical structure considered without temporal occurrence.
- **Access consciousness:** information available for report, memory, reasoning and control; one of the three senses of consciousness the paper keeps apart (Section 1.6).
- **Abundance:** the supply a level affords. At the base, how many actual attempts at a given structure it can fund: a fact about the occurring process, finite, and what selection works on; abstract availability supplies no trials. See also Section 6.1, on produced distributions and boundary conditions.
- **Avatar:** a body or character in a world with little or no mind of its own (no Gōsuto), whose apparent thinking is done from outside by whatever drives it through a wire; a game character driven by a player is the familiar case.
- **Base cycle:** in the signed recurrence conjecture of Section 9, one full pass of the loop the true base traverses, from any stage round to its exact return, a quasi-infinite great recurrence; the true base cycle in full. Finite, and repeated without end; on a further signed claim (Section 9.4), nearly the whole of it is Eukrasia. See Section 9.3, on the base cycle and the Eukrasian attractor.
- **Base footprint:** the identity-relevant organised involvement of an individual in base occurrence, traced through a valid support map. Not necessarily spatial, contiguous or a fixed set of material tokens. It is the support through which a Reikon is realised, not a third entity.
- **Base occurrence:** the primitive process Finitary Occurrent Actualism (FOA) postulates as occurring at true base reality: at least one, the one carrying us, with others left open. Where the paper says *one base* or *one primitive actuality fact* it means this one, the connected base carrying us, and not that the paper has settled how many there are. See Section 2.2, on primitive base occurrence and the question of number.
- **Basin-Coupling Model:** the candidate account of individuation and connection in Section 11.5, on subjects and candidate integration basins, Section 12.2, on Reikon, base involvement and individuation, and Section 15.7, on compositional parthood, basins and continuation. A causal-information basin is the topology and effective boundary of relatively persistent, internally differentiated and recurrently integrated activity. Basin names the integration geometry, not the individual. It is not simply a strongly connected graph or a state-space basin of attraction, and identifying a basin does not establish conscious character.
- **Carrier:** the level immediately below a world in the reality stack, the one whose activity realises it. The paper names the same relation from the other side when it calls that level the world's host; a carrier need not sit beside its world in that world's space, nor reproduce its layout (Section 5.7, on what the reality stack would add to a theory of everything). See Reality stack.
- **Causal-Informational Parthood (CIP):** the account uniting constitutive grounding in base occurrence with informational coupling between realised organisations. Exact support and graded coupling are distinct relations. Reading is a role within this architecture; causal contact alone does not establish constitutive membership. Section 2.5, on parthood, coupling and reading, states the distinction, and Section 2.6, on composition across levels and the closure bounds, sets out the five requirements for the bridge across levels.

- **Claim grades:** the paper's scheme for grading claims where they appear, by what each rests on: mathematical result, formal possibility, real-life counterpart, metaphysical postulate and physical conjecture, with narrower labels and separate signed and wild tiers for speculation. Most sections end with a bracketed line giving the grade used. They are distinct from the claims ladder's lists and from the outstanding debts listed among the current concessions. See the note on how claims are graded, after the theory map.
- **Claims ladder:** the three lists of Appendix A: what the paper claims, what the author suspects without claiming, and what the paper leaves open. Each entry names the level of reality it applies to, so that an abstract result is not silently promoted into a claim about our universe or the base. It is a different instrument from the claim grades, which grade each claim where it appears, and from the current concessions, which list what Occurrence theory has not paid for.
- **Closure bounds:** the constraints on any proposed influence across levels: a cross-level outcome selection, in either direction, must be consistent with the Born probabilities, conservation and no-signalling as far as they have been observed, or make a clear prediction that one of them fails. They have been tested statistically, in controlled experiments, not yet for rare, targeted interventions in complex systems, and staying inside them does not show that a reader is there. See Section 10.5, on agency and cross-level intervention, and Section 2.6, on composition across levels and the closure bounds.
- **Coarse description, coarse state:** a description that bundles many detailed states into fewer, so that one coarse state stands for all the detailed states it lumps together; a coarse-graining. Section 9.1's exact-factor corollary concerns coarse descriptions that have an exact rule of their own.
- **Coherence band:** the description this paper gives Eukrasia when a name will not do: the range of complexity within which coherent organisation is readily sustained. One thing under two names. Whether the base is held there is the separate attractor conjecture. See Section 9.4, on the coherence band and the seasons of the cycle.
- **Computational genesis:** the machinery-building regime of Section 6.2's evolutionary account, which can continue after worlds exist: selection accumulating not only frequency but capability, so that copying, memory, error correction and interpreters appear and the base assembles a substrate general enough for something else to run on. The capability reached can run up to universal construction, which is more than universal computation; the claim is that evolution can arrive at such capacities rather than only be given them. See also Section 5.2, on computation, rules inside states and reflective closure.
- **Conscious actuality:** that an experience is genuinely being undergone; it comes in degrees, down to the faint felt aspect every occurring process has by parthood (Section 11.4).
- **Conscious character:** what an experience is like, its content, as fixed by its relevant organisation.
- **Conscious experience:** the whole the two contributions make: one live episode, organised and being undergone. Structure supplies its character; participation in base occurrence supplies its actuality.
- **Conscious illumination:** a vivid metaphor for conscious actuality.

- **Current concessions:** Section 14.6's numbered list of what Occurrence theory has not paid for, including the one brute occurrence, the unfinished parthood bridge and the open parts of the process measure.
- **Effective complexity:** here, the richness of a world's effective relations and organisational possibilities, compared at a declared scope and scale. Not a supplied universal scalar measure, a count of inhabitants, an intelligence score or hosting depth. Extra dimensions may add effective possibilities without adding resources to the carrier (Sections 5.5 and 5.6).
- **Eukrasia:** the name this paper gives the coherence band, from the Greek for a good mixture, a well-tempered constitution: the broad belt of complexity where coherent organisation is readily sustained and where the loop may spend nearly the whole of its base cycle, the equatorial rainforest of the cycle. Two claims travel with it, one about a range of complexity and one about occupancy, and they are independent. See Section 9.4, on the coherence band and the seasons of the cycle.
- **Eukrasian attractor:** the conjectured tendency of the base's dynamics to keep it in Eukrasia or return it there, so that the band is overwhelmingly occupied. Conjectured, not defined into the band, since no base dynamics with an attractor has been exhibited. See Section 9.3, on the base cycle and the Eukrasian attractor.
- **Evolutionary cosmogenesis:** the world-level regime of Section 6.2's evolutionary account: developed processes producing new effective worlds through blind production, evolved construction, deliberate design or mixtures. Along the proposed developmental spine these may enrich organisation, but hosted descendants may also be reductive or peer. Our construction of hosted worlds supplies a familiar instance. A producing parent is not necessarily an ongoing host.
- **Exact dynamical factor:** a description whose next state is fixed by its present state and agrees exactly with the underlying update: an onto map C satisfying $CF^m = fC$ on a specified invariant domain, for a fixed positive integer m. This state relation alone does not identify which events realise the described process.
- **Familial base reality:** the first world of a family of worlds, the one whose physics the others share or clearly derive from, of the same or a closely related species. It is usually also the root host of those worlds, but hosting is not part of the definition: kinship is shared or derived physics, hosting is a line of worlds running the next, and either can hold without the other. A world can be hosted by a level it resembles in nothing, and then it has a host and an ancestor but no familial base reality; and a world whose physics derives from a world that does not host it has that world as its familial base reality all the same. See Section 5.5, on true base, familial base and home reality.
- **Far excursion:** the deepest excursion of the base cycle on either side: the far winter excursion reaches the winter point and the far summer excursion the summer crest. Lesser winter and summer excursions may be several in a base cycle; all are rare, and together they are the thin tails of the share of the base cycle spent at each level. See Section 9.4, on the coherence band and the seasons of the cycle.
- **Ferine:** wild; said of technofauna that has slipped its leash, cut off from its makers or mutated beyond their control (Section 13.4).
- **FOA:** Finitary Occurrent Actualism, the core of Occurrence theory: at least one primitive base occurrence, no completed infinities, a bounded base, and finite happenings as the base's parts.

- **Gōsuto:** an organised, somewhat conscious structure inside a world: the information-rich organisation fixing personal conscious character, including relevant memories, dispositions, learned models and cognitive relations. A Gōsuto can be described, partially copied or realised in a running implementation; its specification does not itself actualise an episode. Section 12.1, on the reader conjecture and its vocabulary, distinguishes those three relations.
- **Ground of Experience Conjecture:** the author's proposal, held by Occurrence theory as a postulate, that bare occurrence everywhere carries a minimal experiential aspect, with organisation, a life's own and perhaps a Reikon's, supplying every degree above that floor. No formal claim of the paper depends on it. See Section 11.4, where the conjecture is stated and priced.
- **Home reality:** the effective world in which a finite observer appears to live; for us, the world of our familiar universe, bodies and experiences. It is one world at its level of the stack, not the whole level (Section 1.5).
- **Interdimensional:** between different effective worlds, worlds neither of which is a region of the other's space, so that no movement within either leads to the other. A host and a world it runs, sister worlds, any world and true base reality, and a simulation and the world running it are interdimensional pairs; two places in one world are not, though a journey between them could take an interdimensional route, out of the world and back in elsewhere. Any route between two worlds, even a direct-looking one such as a wormhole or a teleport, runs through what carries them: the host itself, for a host and a world it runs; otherwise a level beneath both, perhaps most often the nearest, possibly any down to true base reality. The *dimension* is a world, not a direction of space (Section 5.5).
- **Mathematical abiogenesis:** the standing foundation of Section 6.2's evolutionary account: the fact that the occurring base always affords enough actual trials for selection to work on, supplied by abundance. Not a first era before selection, since differential continuation applies wherever variants continue unequally. See also Section 6.1, on produced distributions and boundary conditions, and Section 7.4, on accidental observers and fine-tuning.
- **Mathematical Intrinsic Occurrence (MIO):** the claim that a mathematical object can exist as something already occurring rather than as a completed pattern. At the base, the paper posits the occurring and posits no separate fact of its existing alongside it, and strict potentialism leaves no completed version the base could have been instead. It does not claim that existing and occurring could not be told apart, only that the distinction does not have to be bought. See Section 3.2, which adds no postulate to Section 2.1, on Finitary Occurrent Actualism and strict potentialism, and Section 2.2, on primitive base occurrence and the question of number.
- **Mathematically extensible, physically bounded:** the paper's position on precision. Mathematics can always add another subdivision or decimal place; the base, being finite, cannot go on adding physical distinctions without limit. The ceiling need not be a smallest scale and may be remote beyond any observer's reach; with finitely many states for each part, the base has finitely many complete configurations and must come round to ones it has occupied before.
- **Occurrence, and the verb *occur*:** the primitive fact that transitions occur rather than being merely specified. The adjective *occurrent* marks the category of things that happen rather than persist, so occurrence is predicated of processes, transitions and events, never of a persisting object. Denying occurrence does not by itself deny existence. Saying that a

pattern is not actualised says only that nothing in it is occurring. In the Reikon conjecture, occurrence belongs to the activity realising the individual; the individual's persistence concerns the continuity of that organisation.

- **Occurrence theory:** the overall view of reality the paper proposes. Its core is Finitary Occurrent Actualism (FOA); the reality stack, the evolutionary account and the division of consciousness into character and actuality are its other main parts, and the author's signed conjectures sit at its edge.
- **Ontic strict potentialism:** the denial of completed infinite objects combined with indefinite continuation. The physically bounded base is a separate commitment, made in Section 2.3.
- **Operative actuality:** the generic fact that an episode's transitions are occurring rather than merely specified, supplied by participation in base occurrence. Felt from inside an organised episode, that same fact is conscious actuality. See Section 11.3, on actuality and participation.
- **Organism to Base Interface (OBI):** the proposed coupling mechanism by which an organism participates with a deeper reader, on the preferred conjecture a Reikon, directly or through intermediate organisation. It may be distributed and need not involve new physics; a model claiming an additional influence must specify it. OBI is the biological name, not a restriction excluding artificial interfaces. Section 12.4, on attachment, coupling ecology and interfaces, states the role; Section 10.5, on agency and cross-level intervention, and Section 2.6, on composition across levels and the closure bounds, give its physical constraints; and Section 15.7, on compositional parthood, basins and continuation, sets out possible tests.
- **Parthood bridge:** the grounding claim that a finite local event occurs because its transitions are numerically an organised region of base transitions, not because a second occurrence-token is added. CIP retains this claim while distinguishing it from information transmission between organised parts.
- **Perceptually eternal:** the possible recurrence of the same identity without experienced absence between its episodes, conditional on the recurrent-identity proposal. It implies neither endless autobiographical memory nor an uninterrupted stream. See Section 13.2, on recurrence and experienced absence.
- **Possession:** a Reikon powerful enough, or with enough administrative privilege, overriding a Gōsuto and acting as that individual with full control, so that the organism becomes, in effect, its avatar (Section 12.1's box on what a Reikon can do in a world).
- **Procedural change:** the base's change seen as following its finite rule step after step, as a procedurally generated world unfolds from its rules; no clock outside the base times the steps (Sections 2.4 and 3.6).
- **Process measure:** a weighting of events, lineages or observer-episodes within finite windows. Occurrence theory's proposal counts lifetimes of human-like experience within one full base cycle, a longer, richer or more intense life counting for more (Section 7.1).
- **Processing depth:** how much computing a level does to produce a structure or an event, counted in its own steps rather than in levels. A level can be very busy without the stack being deep (Section 5.6, on depth, width and the cost of another level).
- **Quasi-block:** a universe or carried region that looks and acts like a block from within, with a settled past, but is not immutable: it may be changed from outside, now or later, or drift

slightly through the active base that carries it. What carries it could in principle revise it. See Section 8.3, on quasi-blocks and rewrite-only worlds.

- **Quasi-infinite:** the author's term for a finite quantity beyond the reach of anything that could count it. A base cycle long enough that nothing inside it could survey it, and still finite. Not a middle category between finite and infinite, which would be incoherent, but a statement about scale. See Section 2.3, on the bounded base and quasi-infinite scale.
- **Reader:** an entity coupled to the structure of a life, whose own state changes along with that life's organisation as it reads it. In a record implementation it may traverse laid-down states; in an interactive implementation it couples to a changing life. Humans following represented lives in books, films and games already provide familiar cases of this role. Under the preferred conjecture, Reikon names the deeper individual and reader names what it does while coupled. The role itself does not establish a conscious experiencer, require an external allocator, or place its bearer below the life being read. Reading, action through an avatar and administrative control are distinct.
- **Real-life counterpart:** one of the claim grades: a process or system observed in the real world, natural or made, that has the relevant structure without being the case proposed, such as symbiosis, for a coupling that benefits both partners; evolution, for selection building organisation; and a game and its developer, for a hosted world and its host. It shows that the structure is real somewhere; carrying it to another level is graded separately.
- **Reality stack:** the ancestry of realisation, from the unhosted occurring base through effective levels that are emergent, hosted or both. A host is lower than the world it carries, even when that world is less complex. The developmental complexity gradient belongs to the evolutionary proposal, not to the definition. Its carrying depth remains open, with a signed suspicion of tens rather than thousands (Section 5.6).
- **Reality tree:** the branching view of world production. Its spine follows a selected developmental route; natural, constructed and mixed branches may themselves branch again. In Section 9.4's signed figure, vertical position compares effective complexity, not carrying depth. Parentage and continuing hosting remain different relations.
- **Reductive / peer / enriched hosted world:** a hosted world with restricted, broadly comparable or additional effective physics and organisational possibilities relative to its host, respectively. Comparisons state relevant features, scale and precision; mixed cases remain possible. Narrow scientific models belong at the reductive end without thereby being inhabited universes. Internal date and total host resources are different questions (Section 5.5).
- **Recurrent-identity condition:** the signed proposal that recurrence of the same identity-relevant organised base involvement within the same occurring process constitutes recurrence of that identity. Exact dynamical return and this identity condition are separate commitments. Neither similarity alone nor absence of experienced time proves it. Section 13.2, on long duration, recurrence and experienced absence, states the condition and its limits.
- **Reflective operational closure:** organisation that maintains and interprets some of the constraints governing its own future transitions.
- **Reikon:** an organised, conscious structure outside a world, which reads it, interacts with it, or both: a player writ large, usually evolved, though one could be designed or created. The one hypothesised for a human life is a deeper identity, an organised

entity realised by a causally integrated, base-rooted occurrent process, which may act as a reader and participate in different Gōsuto organisations. In the author's conjecture it focuses actuality already grounded in parthood onto a life. Its activity may form a causal-information basin; basin names that topology, not the individual. A Reikon carrying coarse memory, valence, preferences, learned state or capacities of its own is a further wild speculation in Section 12.2, on Reikon, base involvement and individuation, not the default definition.

- **Sasayaki:** the input a Reikon sends into a Gōsuto across their coupling, an information-bearing influence, thin or thick. A thin one is a whisper, a possible low-bandwidth contribution between coupled organisations that remain distinct, weak enough to have gone unnoticed at the date of this paper; a thick one carries enough to steer, shading into the control a player has over an avatar. Science had not been looking for either, so what bounds it is what we would notice (Section 10.5). It is not an amount of actuality. The proposed surviving links to partial Gōsuto models in Section 13.3, on models of other people and Sasayaki, are wild speculation and require an operative connection, not merely inherited information.
- **Sign-Reversal Closure Hypothesis:** the detachable proposal that a domain has a natural sign-reversing involution, giving pairwise null orbits without a global infinite sum.
- **Signed sum-preserving refinement:** the author's conjectural base toy model using finite signed configurations, splitting, merging and interaction while conserving a zero total at each defined finite stage.
- **Stack depth:** the number of carrying levels between the true base and a given world. It says nothing by itself about that world's complexity or about how much computing its carrier does; Section 5.6 asks how deep the stack is and leaves the number open.
- **Structured nullity:** a non-empty structured object whose value under a declared valuation is exactly zero.
- **Substrate evolvability:** how readily viable hereditary organisations can be reached and modified within a rule system.
- **Summer crest:** in the signed recurrence conjecture of Section 9, the highest level of organisation the loop ever reaches, its summer solstice, a crest because it breaks: the maximum of complexity existence ever reaches, probably through diminishing returns on complexity, and not an optimum, since flourishing is suspected to peak inside the coherence band. Reached by the far summer excursion and left by the autumn descent. See Section 9.4, on the coherence band and the seasons of the cycle.
- **Technofauna:** every machine lineage that copies itself, tame or wild, such as self-replicating probes and their descendants. *Interdimensional technofauna* are probes sent into other worlds or other levels of reality (Section 13.4).
- **Transmission ratio distortion:** in genetics, one of a parent's two alleles reaching more than its expected half of the offspring, where Mendel's first law predicts equal shares. Established as ordinary biology in several organisms, preliminary or contested in humans, and the arena in which the last box of Section 12.6 would look for a residue of Reikon selection.
- **True base reality:** the lowest occurring process, not carried by another process within Occurrence theory.

- **Uniceptal:** a conjectured conceptual motif, ordinarily less elementary than a bare selective loop, that recurs as an attractor across unrelated systems without needing a mind to interpret it. Recurrence of a motif alone establishes no shared mechanism. See Section 6.5, on recurring motifs across systems.
- **Winter point:** in Section 9's signed recurrence conjecture, the lowest level of organisation reached within the base cycle, reached by the far winter excursion. A remnant persists from which organisation can rebuild on the spring ascent; how simple that remnant is remains open, and it may still carry worlds and minds. It is a phase of the beginningless base, not the true base as a level, an empty state or a creation event (Section 9.4).
- **Wire:** a coupling to an avatar, from something outside a world to a body or character there with little or no mind of its own. The wearer's memory and identity stay outside. A wire can be thin (a game on a screen), thicker (virtual reality) or truly thick (as in *The Matrix*); cutting it costs little and no memory, and the avatar stops. A coupling to a Gōsuto is the fuller kind: the two make one combined experience (Section 2.5, and Section 12.1's box on what a Reikon can do in a world).
- **World-making transition zone:** the proposed passage in which evolved organisations acquire capacities to produce further effective worlds prolifically. Blind production, automatic construction and deliberate design can coexist; no sharp natural/designed boundary or universal threshold is asserted (Section 6.2).

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Dedicated to my biologist father and theologian mother, who both tolerated and nurtured my endless questions and wild speculations, as well as my unrestrained dreaming, which continues to this day.

Working paper, version 2.367, 13 October 2026